

Letter IE

ToE Living Review Letters IE: Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

John Onimisi Obidi

jonimisiobidi@gmail.com

Research Lab, The Aether

May 6, 2026

Category: Research Letter — Theoretical Physics; Foundations of Physics; Information Theory;
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"Einstein worked in the world of his time; we must work in the world of ours. We cannot be bound by Einstein's method. Every era must chart its own uniqueness with the challenges before it. Our era is different. Our tools are different, and our obligations to posterity are different."

— John Onimisi Obidi, The ToE Living Review Letters, 6 May 2026

“The general struggle for existence of animate beings is not a struggle for raw materials... but a struggle for entropy.”

— *Ludwig Boltzmann*, *The Second Law of Thermodynamics*, 1886

“Geometry is not true, it is advantageous.”

— *Henri Poincaré*, *Science and Hypothesis*, 1905

“It from bit. Otherwise put, every physical quantity derives its ultimate significance from bits.”

— *John Archibald Wheeler*, *Information, Physics, Quantum: The Search for Links*, (1989)

“Information is information, not matter or energy. No materialism which does not admit this can survive at the present day.”

— *Norbert Wiener*, *Cybernetics: Or Control and Communication in the Animal and the Machine*, (1948)

“The true method of physical science is to seek the simplest and most general action from which the laws of nature follow.”

— *David Hilbert*, *The Foundations of Physics*, (1915)

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“Science cannot solve the ultimate mystery of nature. And that is because, in the last analysis, we ourselves are part of the mystery we are trying to solve.”

— *Max Planck*, *Where Is Science Going*, 1932

“A mathematician is a device for turning coffee into theorems.”

— *Alfred Rényi*, *The Man Who Loved Only Numbers*, 1960

Keywords: Theory of Entropicity (ToE); Information–Geometric Curvature; Entropic Substrate; Emergent Spacetime; Curvature Transfer; Obidi Action; Obidi Curvature Invariant (OCI); Thermodynamic Correspondence; Statistical Manifold; Fisher–Entropic Geometry; Entropic Emergence Map; Informational Dark Curvature; Pre-Geometric Dynamics; Entropic Field Theory; Foundations of Spacetime; Foundations of Gravitation; Information-Theoretic Physics; Entropic Field Theory

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Abstract

This ToE Letter IE establishes that the Riemannian curvature of physical spacetime is not a primitive geometric datum posited *a priori*, but rather emerges as the macroscopic, thermodynamic-limit expression of curvature defined on an underlying statistical-information manifold. Working within the axiomatic framework of the Theory of Entropicity (ToE), we construct the information manifold $(\mathcal{M}_I, g^I)$ from the Fisher–Entropic metric on a fundamental entropic substrate Ω , define its intrinsic Riemann curvature tensor, and prove a Curvature Transfer Theorem demonstrating that the spacetime Riemann tensor R^S is the pushforward of the information Riemann tensor R^I in the thermodynamic limit. Einstein's field equations [1] are thereby recovered as an emergent identity rather than a fundamental law. We introduce the **Obidi Curvature Invariant (OCI)** $\mathcal{K}_\Omega$ — a non-negative scalar field measuring the residual information curvature not captured by spacetime geometry — and establish its key properties: vanishing in the classical limit, positivity, gauge invariance, and a topological bound. The invariant $\mathcal{K}_\Omega$ identifies the informational degrees of freedom relevant to quantum gravity and may contribute to the effective cosmological constant.

THEORY OF ENTROPICITY (TOE) — LIVING REVIEW LETTERS SERIES
— LETTER IE

PREAMBLE

“Why Information Geometry Is Physical in the Theory of Entropicity (ToE)”

“The Clearest, Deepest Explanation of Why the Fisher–Rao, Fubini–Study, and Amari–Čencov Structures Can Legitimately Be Claimed to Be ‘Physical’ in the Theory of Entropicity (ToE), Even Though They Look Like ‘Statistical’ or ‘Quantum-Information’ Objects at First Glance”

“This is the conceptual bridge forged by the Theory of Entropicity (ToE) between information geometry and the geometry of physical spacetime.”

John Onimisi Obidi

Research Lab, The Aether

jonimisiobidi@gmail.com

May 6, 2026

The purpose of this comprehensive Preamble is to provide the reader with a self-contained explanation of why the three principal structures of information geometry employed in the formulation of the Theory of Entropicity (ToE) — the Fisher–Rao metric, the Fubini–Study metric, and the Amari–Čencov α -connections — are not merely convenient mathematical tools borrowed from statistics and quantum information theory, but are instead the authentic geometric substrates from which the physical universe emerges in the Theory of Entropicity (ToE). This Preamble is conceptual and philosophical in character rather than derivational; the rigorous mathematical proofs, action principles, and field equations appear in the body of Letter IE and its supplementary appendices. What is offered here is the *why* — the deep justification for the ontological claims that the Theory of Entropicity (ToE) makes about the physical status of information-geometric structures it has employed and deployed.

To understand why this Preamble is necessary, one must appreciate the magnitude of the claim and depth of the Theory of Entropicity (ToE).

For more than a century, the Fisher information metric has been regarded as a tool of mathematical statistics, the Fubini–Study metric as a tool of quantum information theory, and the α -connections as abstract constructs of differential geometry on statistical manifolds. To declare that these structures are *physical* — that they are the very fabric from which spacetime, matter, and gauge fields emerge — is a claim of extraordinary scope. It demands a correspondingly deep justification. That justification is the purpose of what follows.

The argument proceeds in eight numbered sections, followed by a comprehensive Scholium demonstrating that the assignments of information-geometric structures to physical sectors are not arbitrary choices but are forced by the mathematical nature of each structure.

The reader is encouraged to proceed sequentially, as each section builds upon the preceding ones to construct a cumulative case of considerable logical force.

SECTION 1

1. Why Einstein Could Say “Riemannian Geometry Is Spacetime”

1.1 The Radical Identification

Einstein’s move was radical because he said:

Riemannian curvature = gravity

Geodesics = inertial motion

Metric = physical distances and times

He took a mathematical structure and declared it ontologically real.

But he could do this because:

Riemannian geometry encodes how distances change.

And physics is fundamentally about how distances and durations change.

Thus, the match was natural.

1.2 The Historical Path: From the Equivalence Principle to the Field Equations

Einstein's identification of geometry with gravity did not arrive as a sudden flash of insight. It was the culmination of Einstein's eight-year intellectual struggle that ranks among the most extraordinary episodes in the history of science. To appreciate the magnitude of what Einstein achieved — and thereby to understand the precedent upon which the Theory of Entropicity (ToE) builds — it is essential to trace the key stages of this momentous Einstein struggle.

The story begins in 1907, when Einstein, then a patent clerk in Bern, was writing a review article on special relativity. **In the course of this work, he contemplated the empirical equality of inertial mass and gravitational mass — a coincidence that Newtonian mechanics accepted without explanation. Einstein recognized that if a person in a closed box could not distinguish between gravitational acceleration and the acceleration of the box itself, then gravity and acceleration were, in some deep sense, the same thing.** This was the *equivalence principle*, and Einstein later called it “the happiest thought of my life.” It was the seed from which **general relativity (GR)** would grow, though the tree would take eight more years to reach maturity.

The equivalence principle immediately implied that gravity could not be described within the framework of special relativity, because special relativity presupposed the existence of global inertial frames — frames that extend uniformly across all of spacetime. But if gravity is equivalent to acceleration, and acceleration is a local phenomenon that varies from point to point, then there can be no global inertial frames in the presence of gravity. The geometry of spacetime itself must vary from point to point. Einstein grasped this implication early, but he did not yet possess the mathematical language to express it.

1.2.1 The Einstein-Grossmann Years of Relativity: The Years of Arduous Work and Grace

The critical mathematical breakthrough came in the summer of 1912, when Einstein returned to Zurich and reconnected with his old friend and classmate Marcel Grossmann, who was now a professor of mathematics at the ETH. Einstein told Grossmann that he needed a mathematical framework capable of describing spaces whose geometry varied from point to point. Grossmann recognized immediately that what Einstein needed was the Riemannian geometry that had been developed by Bernhard Riemann in 1854 and subsequently elaborated by Elwin Bruno Christoffel, Gregorio Ricci-Curbastro, and Tullio Levi-Civita into the “absolute differential calculus” — what we now call **tensor calculus**. The collaboration between Einstein and Grossmann, documented in the famous “Zurich Notebook” of 1912–1913, produced the “Entwurf” (draft) theory (of relativity), which contained “**virtually all essential elements of Einstein’s definitive gravitation theory**,” yet fell short of full general covariance. Einstein spent the next three years wrestling with the problem of covariance, taking what

has been described as a “**big U-turn**” before **arriving, in November 1915, at the final field equations of general relativity.**

The November 1915 field equations, presented in a series of four communications to the Prussian Academy of Sciences, completed the identification of geometry with gravity. The metric tensor $g_{\mu\nu}$, a purely **mathematical object** describing the **distance structure** of a **Riemannian manifold**, was declared to be the **gravitational field** itself. The **Riemann curvature tensor** $R^{\rho}_{\sigma\mu\nu}$, a **purely mathematical object** describing how **parallel transport** around **closed loops rotates vectors**, was declared to be the measure of **gravitational tidal forces**. The geodesic equation, a purely mathematical statement about the shortest paths on a curved manifold, was declared to be the equation of motion for freely falling particles. In **Einstein’s own unforgettable words:**

“In the light of knowledge attained, the happy achievement seems almost a matter of course, and any intelligent student can grasp it without too much trouble. But the years of anxious searching in the dark, with their intense longing, their alternation of confidence and exhaustion and the final emergence into light — only those who have experienced it can understand it.”

1.3 The Philosophical Precedent: Riemann’s Habilitationsschrift of 1854

Einstein’s identification of geometry with physics did not emerge from a philosophical vacuum. It had a profound precedent in the work of Bernhard Riemann, whose 1854 *Habilitationsschrift*, “Ueber die Hypothesen, welche der Geometrie zu Grunde liegen” (“On the Hypotheses Which Lie at the Foundations of Geometry”), is one of the most visionary

documents in the history of mathematics. **Riemann's lecture**, delivered before the faculty of the **University of Göttingen** on June 10, 1854 — with the aged **Carl Friedrich Gauss** in attendance, who is reported to have been deeply impressed — accomplished something that no previous mathematician had attempted: it freed geometry from the assumption that the structure of space is fixed and known in advance.

Riemann began by observing that “**geometry assumes, as things given, both the notion of space and the first principles of constructions in space.**” He then noted that “**the propositions of geometry cannot be derived from general notions of magnitude, but that the properties which distinguish space from other conceivable triply extended magnitudes are only to be deduced from experience.**” This was a revolutionary statement. It meant that the geometry of **physical space is not a logical necessity but an empirical fact** — something to be determined by observation and measurement, not by pure reason. Riemann was, in effect, declaring that **the metric structure of space is a physical quantity, subject to the same empirical determination as any other physical quantity.**

Even more remarkably, **Riemann anticipated that the geometry of space might be determined by physical forces.** In the concluding section of his lecture, he wrote that “**the ground of its metric relations**” must be sought “**in binding forces which act upon it.**” This is, in embryonic form, the central idea of general relativity: **the geometry of spacetime is determined by the distribution of matter and energy.** Riemann saw this sixty years before Einstein, though he lacked the physical context (the equivalence principle) to make it precise.

The **significance of Riemann’s insight for the Theory of Entropicity (ToE)** cannot be overstated. Riemann showed that the **geometry of space is not fixed *a priori* but is determined by physical content**. Einstein showed that **the specific physical content that determines geometry is the stress-energy tensor $T_{\mu\nu}$, representing matter and energy**. The Theory of Entropicity (ToE) takes the next step: it shows that **the stress-energy tensor itself, and the spacetime metric it determines, both emerge from a more fundamental information-geometric structure — the Fisher–Rao metric on the statistical manifold of the entropic field**. The intellectual trajectory is clear: Riemann → Einstein → Obidi. This is the **Riemann-Obidi Trajectory (ROT)**. Each step deepens our understanding of what geometry is and where it comes from.

1.4 The Kantian Background and Its Demolition

To appreciate the full **radicalism** of Einstein’s (RoE) move, one must understand what it overthrew. For more than a century before Einstein, the dominant philosophical view of geometry was that of **Immanuel Kant**, who held in his *Critique of Pure Reason* (1781) that Euclidean geometry was a “**synthetic a priori**” truth about space. By this, Kant meant that **Euclidean geometry** was not merely a logical **tautology (analytic)** but contained **genuine information** about the world (**synthetic**), yet this information was known independently of experience (**a priori**). Space, for Kant, was a “form of intuition” — a necessary framework that the mind imposes on sensory experience. **The geometry of this framework was necessarily Euclidean, and no empirical discovery could ever change this.**

The development of **non-Euclidean geometries** by Lobachevsky, Bolyai, and Gauss in the early nineteenth century already **challenged Kant's position** by **showing that Euclidean geometry was not the only logically consistent geometry**. But these non-Euclidean geometries were regarded as **mathematical curiosities**, not as descriptions of **physical space**.

1.4.1 Einstein Delivered the Decisive Blow to Kantian Philosophy and Euclidean Geometry (KP=EG)

It was **Einstein** who **delivered the decisive blow to the Kantian position** by showing that the **geometry of physical spacetime is not Euclidean, not fixed, and not known a priori**. The geometry of **spacetime is Riemannian** (or, more precisely, **pseudo-Riemannian with Lorentzian signature**), it **varies from point to point in response to the distribution of matter and energy**, and it can only be determined empirically. **Einstein thus declares:**

Geometry is not a form of intuition imposed by the mind; it is a physical field determined by the content of the universe.

Einstein's valiant demolition of the Kantian position is directly relevant to the Theory of Entropicity (ToE). The standard view of **information geometry** — the view that the **Fisher–Rao metric is “merely” a statistical tool**, that the **Fubini–Study metric is “merely” a quantum-information-theoretic construct** — plays the same role today that the **Kantian view of Euclidean geometry** played before Einstein. It assumes that these structures **belong to a fixed, abstract, mathematical domain**

(statistics, information theory) and cannot be physical. The Theory of Entropicity (ToE) shatters this assumption, just as Einstein shattered Kant's:

The information-geometric structures are not abstract descriptions of our knowledge about physical systems; they are the physical systems themselves, viewed at their most fundamental level.

1.5 Minkowski's Contribution and the Ontological Promotion of Mathematics

The process by which a mathematical structure becomes physical deserves careful philosophical attention, because it is precisely this process that the Theory of Entropicity (ToE) replicates for information geometry. A crucial intermediate step was taken by Hermann Minkowski in 1908, who reformulated Einstein's special relativity in the language of four-dimensional spacetime geometry. In his famous lecture at the 80th Assembly of German Natural Scientists and Physicians in Cologne, Minkowski declared: "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality."

Minkowski's contribution was not merely technical; it was **ontological**. He took the mathematical structure of a four-dimensional pseudo-Euclidean manifold and declared it to be the arena of physical reality. Space and time, which had been regarded as separate entities since the dawn of human thought, were **demoted to "mere shadows"** — projections of a deeper four-dimensional reality. This was an ontological promotion of a mathematical structure (four-dimensional geometry) to the status of physical reality, The Theory of Entropicity (ToE) Living Review Letters Series. *Letter IE*

combined with an **ontological demotion of previously fundamental entities (separate space and time) to the status of appearances.**

The parallel to the **Theory of Entropicity (ToE)** is exact. In ToE, spacetime itself — **the four-dimensional pseudo-Riemannian manifold of general relativity** — is demoted to an emergent phenomenon, a “shadow” of a deeper information-geometric reality. The Fisher–Rao metric on the statistical manifold of the entropic field is promoted to the status of fundamental physical reality, from which the spacetime metric emerges via the emergence map. Just as Minkowski showed that space and time are projections of spacetime, Obidi shows that spacetime is a projection of information geometry. This is the next step in the same intellectual [Riemann-Obidi] trajectory: the progressive deepening of our understanding of what physical geometry is and where it comes from.

1.6 The Precedent for ToE’s Declaration: Einstein’s Magnificence

The historical analysis of this section establishes a crucial precedent. Before Einstein, geometry was mathematics. **After Einstein, geometry was physics.** The transition required no new experimental data (the precession of **Mercury’s perihelion** was already known); what it required was a new *ontological declaration* — the **declaration that a mathematical structure (Riemannian geometry) was identical to a physical structure (the gravitational field).** This declaration was validated by its consequences: the **correct prediction of light bending, gravitational redshift, gravitational waves, black holes, and the expansion of the universe.**

The Theory of Entropicity makes a declaration of the same kind: information geometry is not a description of physics but is the substrate of physics.

This declaration, like Einstein's, **must be validated by its consequences, and the body of Letter IE demonstrates that it is** — through the recovery of general relativity, the derivation of the Einstein field equations, the prediction of the cosmological constant, and the unification of gravity, matter, and electromagnetism from a single entropic field.

The present Preamble is concerned not with the derivations themselves but with the *conceptual justification* for the declaration that makes them possible.

SECTION 2

2. What Fisher–Rao and Fubini–Study Actually Measure

2.1 The Key Insight

This is the key insight of the Theory of Entropicity (ToE):

Declaration I: The Fisher–Rao Metric

Fisher–Rao measures distinguishability of probability distributions.

It tells you how “far apart” two states of information are.

In ToE, this metric becomes the amplitude-geometry that gives rise to the emergent spacetime metric.

In ToE, the emergence map $g_S = \lambda \ g_I$ identifies the Fisher–Rao metric as the pre-geometric structure from which [physical] spacetime geometry emerges.

Declaration II: The Fubini–Study Metric

Fubini–Study measures distinguishability of quantum states.

It tells you how “far apart” two wavefunctions are.

In ToE, it becomes the dynamical-geometry sector responsible for matter and energy.

In ToE, it encodes the dynamical structure of the entropic field, which manifests as matter content in the emergent spacetime.

Declaration III: The Amari–Čencov α -Connections

Amari–Čencov alpha-connections measure dualistic structure of information flows.

They tell you how “curved” the space of statistical transformations is. [Equivalent to Einstein’s Riemannian curvature]

In ToE, they generate the phase-geometry that becomes gauge structure (including electromagnetism) in the emergent spacetime.

In summary:

Summary

These are not arbitrary metrics.

They measure how information changes.

And physics is fundamentally about how states change.

2.2 The Fisher–Rao Metric: Origin, Definition, and Operational Meaning

The **Fisher–Rao** metric has a distinguished history. Its roots lie in the work of **Ronald Aylmer Fisher**, who in 1925 introduced the concept of *Fisher information* in the context of statistical estimation theory. Fisher showed that for a parametric family of probability distributions $\{p(x;\theta)\}$, the precision with which the parameter θ can be estimated from observed data is fundamentally limited by a quantity he called the “information” — what we now denote $I(\theta)$. For a single real parameter, the Fisher information is defined as:

$$I(\theta) = \int p(x;\theta) [\partial \ln p(x;\theta) / \partial \theta]^2 dx$$

The significance of this quantity lies in the **Cramér–Rao bound**, which states that the variance of any unbiased estimator of θ is bounded below by $1/I(\theta)$. The Fisher information thus quantifies the maximum amount of information that an observable random variable carries about the unknown parameter θ . The greater the Fisher information, the more precisely θ can be estimated; the smaller the Fisher information, the more “blurred” or indistinguishable nearby values of θ become.

In 1945, the Indian mathematician and statistician **Calyampudi Radhakrishna Rao** took a decisive step by recognizing that the **Fisher information could be generalized to a Riemannian metric on the space of probability distributions**. For a family of distributions parametrized by a vector $\theta = (\theta^1, \dots, \theta^n)$, the Fisher–Rao metric is defined as:

$$g_{ab}^{FR}(\theta) = \int p(x;\theta) \left[\partial \ln p(x;\theta) / \partial \theta^a \right] \left[\partial \ln p(x;\theta) / \partial \theta^b \right] dx$$

This is a **genuine Riemannian metric** — symmetric, positive-definite, and smoothly varying — on the statistical manifold of probability distributions. It defines a notion of distance on this manifold: the distance between two nearby distributions $p(x;\theta)$ and $p(x;\theta+d\theta)$ is given by $ds^2 = g_{ab}^{FR} d\theta^a d\theta^b$.

The crucial point for the Theory of Entropicity (ToE) is the *operational meaning* of this distance. The **Fisher–Rao distance** between two probability distributions is not an arbitrary mathematical construct; it has a precise operational interpretation. It is, up to a constant factor, the **minimum number of independent statistical observations** required to reliably distinguish between the two distributions. **Two distributions that are close in the Fisher–Rao metric are difficult to distinguish; two distributions that are**

far apart are easy to distinguish. The Fisher–Rao metric thus measures *distinguishability* — **the fundamental capacity of one informational state to be differentiated from another.**

This operational meaning is what makes the **Fisher–Rao** metric a candidate for physical significance. In **general relativity**, the **spacetime metric** $g_{\mu\nu}$ **also measures distinguishability** — **the distinguishability of spacetime events.** The spacetime interval $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ tells you how “far apart” two events are in terms of physical measurements (**proper time** for **timelike** separations, **proper distance** for **spacelike** separations).

The **Theory of Entropicity (ToE)** identifies these two notions of distinguishability:

The statistical distinguishability measured by the Fisher–Rao metric is the physical distinguishability measured by the spacetime metric, with the emergence map $g_S = \lambda g_I$ providing the quantitative relationship between them.

2.3 The Fubini–Study Metric: Quantum Distinguishability and Dynamical Content

The Fubini–Study metric, named after **Guido Fubini** (1903) and **Eduard Study** (1905), is the **natural Riemannian metric on complex projective space CP^n** . In quantum mechanics, the space of pure states of an $(n+1)$ -dimensional quantum system is precisely CP^n , because quantum states are represented by rays in Hilbert space (vectors differing only by a complex phase represent the same physical state).

The Fubini–Study metric is defined, for two infinitesimally separated states $|\psi\rangle$ and $|\psi+d\psi\rangle$, as:

$$ds_{FS}^2 = \langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2$$

The second term subtracts the component of $|d\psi\rangle$ that is parallel to $|\psi\rangle$ (which represents a mere phase change and is physically irrelevant), leaving only the component perpendicular to $|\psi\rangle$ (which represents a genuine change in the physical state). The Fubini–Study distance thus measures the “amount of physical change” between two quantum states, with the phase redundancy removed.

Like the Fisher–Rao metric, the Fubini–Study metric has a precise operational interpretation. The Fubini–Study distance between two quantum states $|\psi\rangle$ and $|\phi\rangle$ is related to the probability of distinguishing them by a single optimal quantum measurement: $P_{\text{distinguish}} = \sin^2(d_{FS}(|\psi\rangle, |\phi\rangle))$.

States that are close in the Fubini–Study metric are nearly indistinguishable by any quantum measurement; states that are far apart are easily distinguishable. The Fubini–Study metric is the *quantum* analogue of the Fisher–Rao metric, measuring *quantum distinguishability* rather than classical statistical distinguishability.

In the Theory of Entropicity (ToE), the Fubini–Study metric plays a fundamentally different physical role from the Fisher–Rao metric. While the **Fisher–Rao metric** becomes the **spacetime metric** (the “container” of physics), the **Fubini–Study metric** becomes the **matter content** (the “stuff” of physics). This assignment is not arbitrary but is forced by the mathematical nature of the two metrics. The Fisher–Rao metric is defined on the

manifold of probability distributions — it measures how the *amplitude* of the entropic field changes from point to point. The Fubini–Study metric is defined on the space of quantum states — it measures the *dynamical* changes of the complexified entropic field $E = \rho \exp(i\Theta)$.

In the Obidi Action, the Fubini–Study sector contributes the kinetic and potential energy terms that, upon emergence, become the stress-energy tensor $T_{\mu\nu}$ in the Einstein field equations.

The connection between the Fubini–Study metric and the Berry phase deserves special mention. The Berry connection $A = i\langle\psi|d\psi\rangle$ and the Berry curvature $F = dA$ are intimately related to the Fubini–Study metric. The real part of the quantum geometric tensor (QGT) $Q_{ab} = \langle\partial_a\psi|\partial_b\psi\rangle - \langle\partial_a\psi|\psi\rangle\langle\psi|\partial_b\psi\rangle$ gives the Fubini–Study metric, while the imaginary part gives the Berry curvature. In the Theory of Entropicity, this relationship ensures that the dynamical geometry (matter) and the phase geometry (gauge fields) are geometrically unified within the same quantum geometric tensor (QGT), while playing distinct physical roles in the emergent spacetime.

2.4 The Amari–Čencov α -Connections: Information Flow and Dual Geometry

The α -connections, introduced systematically by Shun-ichi Amari in his landmark 1985 monograph *Differential-Geometrical Methods in Statistics* and rooted in earlier work by Čencov, constitute a one-parameter family of affine connections on a statistical manifold. Given the Fisher–Rao metric g and the cubic tensor (skewness tensor) $T_{abc} = \int p (\partial_a \ln p)(\partial_b \ln p)(\partial_c \ln p) dx$, the α -connection is defined as:

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$$\Gamma^{(\alpha)}_{abc} = \Gamma^{(o)}_{abc} + (\alpha/2) T_{abc}$$

where $\Gamma^{(o)}$ is the **Levi-Civita connection of the Fisher–Rao metric**. Three values of α are of particular importance:

$\alpha = 0$ gives the **Levi-Civita connection (the unique torsion-free, metric-compatible connection)**;

$\alpha = +1$ gives the **exponential connection (e-connection)**; and

$\alpha = -1$ gives the **mixture connection (m-connection)**.

The **fundamental property of the α -connections is duality**: the **e-connection** ($\alpha = +1$) and the **m-connection** ($\alpha = -1$) are dual with respect to the **Fisher–Rao metric**, meaning that **for any three vector fields X, Y, Z**:

$$X \cdot g(Y, Z) = g(\nabla^{(+1)}_X Y, Z) + g(Y, \nabla^{(-1)}_X Z)$$

This duality has deep statistical meaning. The **e-connection** governs the geometry of *exponential families* — families of distributions that are closed under exponential tilting (such as the Gaussian, Poisson, and binomial families). The **m-connection** governs the geometry of *mixture families* — families that are closed under convex combination. These two types of families are dual to each other, and the dual connections capture this duality geometrically.

In the Theory of Entropicity (ToE), the **α -connections play the role of gauge structure**. The phase $\Theta(x)$ of the complexified entropic field $E = \rho \exp(i\Theta)$ transforms under phase

rotations $\Theta \rightarrow \Theta + \chi$ as a $U(1)$ connection. **The dual structure of the α -connections — the e-connection governing the coupling of the phase to matter (source terms) and the m-connection governing the dual magnetic formulation — becomes the gauge structure of electromagnetism in the emergent spacetime.**

The α -connections are not metric connections (they are not compatible with the Fisher–Rao metric for $\alpha \neq o$), which is precisely why they cannot play the role of the spacetime connection (which must be metric-compatible). Instead, they encode the additional geometric structure — the gauge structure — that goes beyond the spacetime metric.

The fact that the α -connections are *affine* (**connection-like**) rather than *metric* (**distance-like**) is the **mathematical reason they become gauge connections rather than spacetime structure.**

In physics, gauge connections are also affine connections — they tell you how to parallel-transport charged fields along curves, **but they do not define distances.** The α -connections do exactly the same thing on the statistical manifold: they tell you how to parallel-transport probability distributions along curves in parameter space, **but they do not define distances (that is the job of the Fisher–Rao metric).**

The **correspondence between information-geometric structure and physical structure is thus precise and detailed, not vague or metaphorical.**

SECTION 3

3. The Leap of the Theory of Entropicity (ToE)

3.1 The Declaration

Information geometry is not a description of physics. It is the substrate of physics.

This is the same kind of leap Einstein made.

Einstein:

“Geometry is physical.”

— versus —

ToE:

“Information geometry is physical.”

Why did the Theory of Entropicity (ToE) have to make this declarative leap?

The Ontological Foundation

Because:

Every physical system is fundamentally a system of information.

Every physical evolution is fundamentally a change in information.

Every physical interaction is fundamentally an exchange of information.

Hence, the Theory of Entropicity (ToE) declares that the geometry that measures information change is the geometry that determines physical change.

This is the core and, at the same time, the elegance of Obidi's Theory of Entropicity (ToE).

3.2 Ontological Promotion: The Philosophical Concept

The philosophical operation at the heart of the Theory of Entropicity (ToE) is what may be called *ontological promotion*: **taking a mathematical structure that was previously regarded as a tool for *describing* reality and declaring it to *be* reality.** This operation is not new in the history of physics — **Einstein performed it when he promoted Riemannian geometry from a mathematical framework to the physical substance of gravity — but each instance of it is radical and requires deep justification.**

The **justification for ontological promotion cannot be purely mathematical. Mathematics alone cannot tell you that a particular mathematical structure is physically real; that is a claim about the world, not about mathematics. The justification must come from the *match* between what the mathematical structure measures and what physics requires.**

Einstein's justification was that Riemannian geometry measures distances and curvature, and physics is about distances, durations, and the forces (gravity) that affect them. The Theory of Entropicity (ToE)'s justification is that information geometry measures distinguishability, change, and flow, and physics is fundamentally about distinguishable states, their changes, and their interactions.

The difference between the standard view and the ToE view can be stated sharply. The *standard view* treats information as *epistemological* — **information is what we know about a physical system**. The physical system exists independently of our knowledge of it; information is a feature of our description, not of the system itself. In this view, the Fisher–Rao metric measures the precision of our knowledge about a system, and the Fubini–Study metric measures the distinguishability of our quantum descriptions. These are properties of our epistemic relation to the system, not properties of the system itself.

The *Theory of Entropicity (ToE) view* treats information as *ontological* — **information is what the physical system is**. There is no physical system “behind” or “beneath” the information; the information is the substance of the system. In this view, the Fisher–Rao metric measures real, physical distances (the distances between different configurations of the entropic field), and the Fubini–Study metric measures real, physical dynamical content (the kinetic and potential energy of the entropic field). The transition from the epistemological view to the ontological view is the central conceptual move of the Theory of Entropicity (ToE).

3.3 Wheeler’s “It from Bit” and Landauer’s Principle

The Theory of Entropicity (ToE) realizes concretely what **John Archibald Wheeler** envisioned programmatically in his “It from Bit” doctrine. Wheeler proposed in 1989 that **“every *it* — every particle, every field of force, even the spacetime continuum itself — derives its function, its meaning, its very existence entirely — even if in some contexts indirectly — from the apparatus-elicited answers to yes-or-no questions, binary choices, *bits*.”** Wheeler’s vision was bold but programmatic: he did not provide the mathematical framework to realize it. The Theory of Entropicity (ToE) provides that framework.

The “bits” of Wheeler’s vision are the probability distributions on the statistical manifold; the “its” are the spacetime events, matter configurations, and gauge fields that emerge from the entropic field via the emergence map (EM) and the Obidi Action.

A closely related precedent is **Rolf Landauer’s principle**, first stated in 1961, which establishes that information processing has a **minimum thermodynamic cost**: the **erasure of one bit of information dissipates at least $k_B T \ln 2$ of energy as heat**. Landauer’s principle **demonstrates that information is *physical* in the minimal sense that information processing is constrained by the laws of thermodynamics**. This principle was experimentally verified in 2012 by **Bérut et al.**, who **measured the heat dissipated by the erasure of a single bit in a colloidal particle system and confirmed that it saturates at the Landauer bound in the limit of slow erasure**.

The Theory of Entropicity (ToE) goes far beyond Landauer’s principle. Landauer showed that information is physical in the sense that it has thermodynamic consequences. **ToE shows that information is physical in the far stronger sense that it is the foundation of physics — that spacetime, matter, and gauge fields are all manifestations of information-geometric structures.** Landauer’s principle is a corollary of this deeper truth: **information has thermodynamic consequences because information is the substance from which thermodynamics (and all of physics) emerges.**

And because information itself is founded on entropy, the Theory of Entropicity (ToE) concludes that this entropy is what is infact the fundamental substrate, and that entropy is a universal field from which information itself emerges.

3.4 The Logical Structure: A Syllogism, Not an Analogy

It is essential to understand that the Theory of Entropicity’s claim is not an analogy. It is not saying that information geometry is “like” physical geometry, or that information-geometric structures “behave as if” they were physical. It is making a literal identification, grounded in a logical syllogism:

Premise 1: Every physical system is fundamentally a system of information. (This is supported by quantum mechanics, where the state of a system is completely specified by its wavefunction — an **informational object** — and by the **holographic principle**, which bounds the information content of a region by its boundary area.)

Premise 2: The natural geometry of information is the Fisher–Rao/Fubini–Study/ α -connection geometry. (This is a mathematical fact: the Fisher–Rao metric is the unique Riemannian metric on statistical manifolds invariant under sufficient statistics, and the Fubini–Study metric is the unique Kähler metric on the quantum state space.)

Conclusion: The geometry of physics IS the geometry of information.

This is a deductive argument, not an inductive one. If both premises are true, the conclusion follows necessarily. The Theory of Entropicity (ToE) asserts both premises and draws the conclusion. The remainder of this Preamble, and the body of Letter IE, provide the evidence and the mathematical framework that justify both premises and demonstrate the consequences of the conclusion.

SECTION 4

4. Why Fisher–Rao Becomes Spacetime in ToE

4.1 The Identification

The Fisher–Rao metric measures: How distinguishable two nearby entropic configurations are.

In the Theory of Entropicity (ToE), we make the following declarations based on the above Fisher–Rao metric measure:

The Fisher–Rao Declarations

“Nearby entropic configurations” = “nearby physical states”

“Distinguishability” = “physical separation”

“Curvature of information” = “curvature of spacetime”

Consequences

Thus:

Spacetime distance = entropic distinguishability.

Spacetime curvature = entropic curvature.

This is not metaphor.

This is the emergence map (EM):

$$g_S = \lambda g_I$$

This is the Theory of Entropicity (ToE) analogue of Einstein’s identification of gravity with curvature.

4.2 The Emergence Map: Physical Meaning

The emergence map $g_S = \lambda g_I$ is the most fundamental equation in the Theory of Entropicity. It states that the spacetime metric g_S (the object that Einstein identified with the gravitational field) is proportional to the Fisher–Rao information metric g_I (the object that measures statistical distinguishability on the entropic manifold). The constant of proportionality λ converts between information-geometric units and spacetime units.

What does this equation mean physically? It means that the distance between two spacetime events — the quantity that determines how clocks tick, how rulers measure, how light propagates, how gravity acts — is nothing more, and nothing less, than the statistical distinguishability of the corresponding entropic configurations, scaled by the universal conversion factor λ . Two events that are far apart in spacetime correspond to entropic configurations that are highly distinguishable; two events that are close together correspond to configurations that are nearly indistinguishable. The entire fabric of spacetime — its distances, its curvature, its causal structure — is a macroscopic image of the information-geometric structure of the entropic field.

The emergence map is formally a pullback of the spacetime metric along the emergence morphism $\Phi: M_I \rightarrow M_S$, written more precisely as $\Phi^*g_S = \lambda g_I$. This means that the emergence morphism is a conformal immersion (up to the constant λ) of the information manifold into the spacetime manifold. The information manifold is the “microscopic” structure; the spacetime manifold is the “macroscopic” image. The emergence map tells you how to reconstruct the macroscopic image from the microscopic structure.

The connection to the Curvature Transfer Theorem of Letter IE is direct and profound. The Curvature Transfer Theorem states that $R^S = \lambda \Phi^*(R^I) + O(1/N)$, where R^S is the spacetime Riemann curvature, R^I is the information-geometric curvature, and N is the number of entropic degrees of freedom. This theorem shows that spacetime curvature is a direct, quantitative image of information-geometric curvature, with corrections that vanish in the thermodynamic limit ($N \rightarrow \infty$). General relativity — the theory that spacetime curvature equals gravitational field — is thus a macroscopic consequence of the more fundamental identification of information-geometric curvature with the entropic structure of reality.

4.3 The Čencov Uniqueness Theorem

One of the most powerful arguments for the physical significance of the Fisher–Rao metric in the Theory of Entropicity is the **Čencov uniqueness theorem (CUT)**, proved by Nikolai Nikolaevich Čencov (Chentsov) in his 1972 monograph *Statistical Decision Rules and Optimal Inference*. The theorem states:

On the manifold of probability distributions over a finite sample space, the Fisher–Rao metric is the unique Riemannian metric — up to an overall constant scalar factor — that is invariant under Markov morphisms (information-preserving stochastic transformations of the data).

The importance of this theorem for the Theory of Entropicity (ToE) cannot be overstated. It means that the Fisher–Rao metric is not *one possible choice* among many Riemannian metrics on the statistical manifold; it is the *only choice* that respects the fundamental

symmetries of information processing. Any other Riemannian metric would distinguish between different representations of the same information — it would measure not the information itself but the labels we attach to it. Only the Fisher–Rao metric is intrinsic to the information content, independent of how the information is represented.

This removes what might otherwise be a devastating objection to the Theory of Entropicity (ToE): the objection that the choice of the Fisher–Rao metric as the pre-geometric structure of spacetime is arbitrary, that one could equally well have chosen some other metric and obtained a different theory. Čencov’s theorem shows that this objection fails. There is no other metric to choose. The Fisher–Rao metric is as unique among information-geometric metrics as the **Levi-Civita connection** is among metric-compatible torsion-free connections, or as the Einstein tensor is among divergence-free second-order symmetric tensors formed from the metric. The Theory of Entropicity inherits the full force of this mathematical uniqueness.

4.4 The Scholium on λ

Scholium on the ToE Metric Conversion Factor of λ

This emergence map ($\Phi^*g_S = \lambda g_I$) is the Theory of Entropicity (ToE) analogue of Einstein’s identification of gravity with curvature.

In the Theory of Entropicity (ToE), the conversion factor λ must be a constant because it is the universal conversion factor between the Fisher–Rao information metric and the emergent spacetime metric. That is, λ is a conversion factor between information

geometry and spacetime geometry. If λ were a function, the emergence map would violate the equivalence principle, break the uniqueness of the Fisher–Rao metric, introduce unphysical forces, and fail to reproduce general relativity. The constancy of λ is therefore required by both the mathematics of information geometry and the physics of spacetime emergence.

If λ were a function (a variable, and not a constant), the emergence map would break general relativity.

In ToE, free motion corresponds to Fisher–Rao geodesics, where λ must be a constant.

In all of the above, λ must be constant because of Čencov’s theorem. Čencov’s theorem states:

The Fisher–Rao metric is unique up to a constant multiplicative factor.

This is a deep result.

In the Theory of Entropicity, λ is fixed at approximately 10^{-47} by the entropic curvature gap between the highly curved Fisher–Rao information manifold and the extremely flat physical spacetime. This value is not a range and not a repetition of existing literature; it is uniquely determined by the emergence map and required for the recovery of general relativity, the equivalence principle, and the observed cosmological constant.

Although the information manifold curvature R_I and the spacetime curvature R_S each vary over ranges, the emergence constant λ is not defined by pointwise curvature

values. Instead, λ is fixed by the ratio of the characteristic curvature scales of the two manifolds. This ratio is sharply determined by the entropic curvature gap and yields λ approximately equals 10^{-47} . If λ were allowed to vary over the ranges of R_I or R_S , the equivalence principle would fail, general relativity would not emerge, and the observed universe could not exist. Thus, λ is a universal conversion constant, not a variable.

4.5 The Value of λ : Dimensional Analysis and Physical Meaning

The emergence constant λ can be computed from fundamental constants. The Fisher–Rao metric has dimensions of inverse information (or, equivalently, inverse entropy), while the spacetime metric has dimensions of length squared. The conversion between the two requires a constant with dimensions of $[\text{length}^2 \times \text{entropy}]$. From the fundamental constants of nature, the unique combination with these dimensions is:

$$\lambda = l_P^2 / (4 k_B) = \hbar G / (4 k_B c^3)$$

where $l_P = (\hbar G/c^3)^{1/2} \approx 1.616 \times 10^{-35} \text{ m}$ is the Planck length, $\hbar$ is the reduced Planck constant, G is Newton’s gravitational constant, c is the speed of light, and k_B is Boltzmann’s constant. Substituting numerical values:

$$\lambda \approx (1.616 \times 10^{-35})^2 / (4 \times 1.381 \times 10^{-23}) \approx 10^{-47} \text{ m}^2/\text{J}\cdot\text{K}^{-1}$$

This extraordinarily small value has a profound physical meaning: the information manifold is enormously more curved than physical spacetime. A curvature of order unity on the information manifold (a perfectly natural value in information-geometric terms) corresponds to a curvature of order 10^{-47} in spacetime — an almost perfectly flat spacetime, which is exactly what we observe. The physical universe is, in the language of the Theory of Entropicity, an extremely “zoomed out” and “smoothed out” image of the underlying information manifold. The tiny value of λ is the quantitative expression of this enormous scale separation.

The four consequences of allowing λ to vary deserve detailed exposition:

1. **Violation of the equivalence principle:** If λ varied from point to point, then the relationship between information-geometric distances and spacetime distances would also vary from point to point. This would mean that freely falling particles in the same gravitational field would follow different trajectories depending on their information content — a direct violation of the universality of free fall (the weak equivalence principle). All experimental tests of the equivalence principle, from Eötvös to MICROSCOPE, confirm universality to extraordinary precision.
2. **Breaking the uniqueness of the Fisher–Rao metric:** Čencov’s theorem states that the Fisher–Rao metric is unique up to a *constant* multiplicative factor. If λ were a function, the emergent metric $g_S = \lambda(x) g_I$ would not be a constant multiple of the Fisher–Rao metric; it would be a conformally rescaled version. But a conformally rescaled Fisher–Rao metric is, in general, not invariant under Markov morphisms — it violates the very property that makes the Fisher–Rao metric

unique. The emergence map would thus be inconsistent with the mathematical structure it is built upon.

3. **Introduction of unphysical forces:** A variable λ would introduce a scalar field $\lambda(x)$ into the theory, analogous to the Brans-Dicke scalar field. This scalar field would mediate a fifth force, coupling to matter in ways that are not observed. The stringent experimental limits on fifth forces (from torsion-balance experiments, lunar laser ranging, and binary pulsar timing) rule out a variable λ at the levels required for observable effects.
4. **Failure to reproduce general relativity:** The recovery of the Einstein field equations from the Obidi Action requires that the emergence map be a constant-factor identification. If λ varied, the emergent field equations would contain additional terms involving the gradient and Laplacian of λ , yielding a scalar-tensor theory rather than general relativity. Since general relativity is extraordinarily well-tested in the weak-field and strong-field regimes, any deviation from it is strongly constrained.

SECTION 5

5. Why Fubini–Study Becomes Matter in ToE

5.1 The Declarations

The Fubini–Study metric measures: How dynamical quantum states differ.

In the Theory of Entropicity (ToE), we make the following declarations based on the above Fubini–Study metric measures:

The Fubini–Study Declarations

Matter = dynamical structure of the entropic field

Mass = resistance to change in entropic configuration

Energy = curvature of the Fubini–Study sector

Thus:

Matter is the dynamical geometry of the entropic field.

This is why the Fubini–Study metric appears in the dynamical part of the Obidi Action.

5.2 The Fubini–Study Metric on Projective Space

To understand why the Fubini–Study metric becomes the matter sector, one must first appreciate its mathematical construction in detail. Complex projective space $\mathbb{CP}^n$ is the space of lines through the origin in $\mathbb{C}^{n+1}$. If we represent a point in $\mathbb{CP}^n$ by homogeneous coordinates $[z^0 : z^1 : \dots : z^n]$, the Fubini–Study metric can be written in the affine chart $z^0 \neq 0$, with inhomogeneous coordinates $w^k = z^k/z^0$, as:

$$ds_{FS}^2 = [(1 + |w|^2)\delta_{jk} - \bar{w}_j w_k] / (1 + |w|^2)^2 \cdot dw^j d\bar{w}^k$$

This metric is Kähler (it derives from a Kähler potential $K = \ln(1 + |w|^2)$), it has constant positive holomorphic sectional curvature, and it is the unique (up to a constant factor) Kähler metric on $\mathbb{CP}^n$ that is invariant under the natural action of the unitary group $U(n+1)$. These properties make it the canonical metric on the space of quantum states.

The **Fubini–Study metric is intimately related to the Berry connection and Berry curvature**. The **quantum geometric tensor (QGT)**, defined for a family of states $|\psi(\theta)\rangle$ parametrized by $\theta = (\theta^1, \dots, \theta^n)$, is $Q_{ab} = \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle$. The real (symmetric) part of Q_{ab} is the Fubini–Study metric, and the imaginary (antisymmetric) part is one-half the Berry curvature. This decomposition shows that **the geometry of the quantum state space naturally splits into a metric part (Fubini–Study) and a connection part (Berry)**, corresponding precisely to the matter sector and the gauge sector in the Theory of Entropicity.

5.3 Why Matter = Dynamical Geometry

In standard physics, matter is described by fields — scalar fields, spinor fields, vector fields — that propagate on a fixed (or dynamical) spacetime background. The fields satisfy equations of motion (**Klein–Gordon, Dirac, Maxwell**) that determine how they evolve in time and space. The energy and momentum carried by these fields are encoded in the stress-energy tensor $T_{\mu\nu}$, which serves as the source for Einstein’s field equations.

In the Theory of Entropicity (ToE), this picture is deepened. Matter is not a separate entity placed on top of spacetime; it is the *dynamical structure* of the entropic field itself. **The complexified entropic field $E = \rho \exp(i\Theta)$ has two sectors: the amplitude sector ρ , whose geometry (measured by the Fisher–Rao metric) becomes spacetime, and the dynamical fluctuations of the full field E , whose geometry (measured by the Fubini–Study metric) becomes matter.** The Fubini–Study metric measures the “speed” at which the entropic field configuration changes as one moves through parameter space. Where the field is changing rapidly (large Fubini–Study distance), there is a large amount of dynamical activity, which manifests as matter and energy in the emergent spacetime. Where the field is static or slowly varying (small Fubini–Study distance), there is vacuum.

5.4 Why Mass = Resistance to Entropic Change

In Newtonian mechanics, mass is defined as the resistance to acceleration: $F = ma$. An object with large mass requires a large force to change its velocity. In the Theory of Entropicity, this concept acquires a deeper meaning. Mass is the resistance to change in the entropic configuration — the “inertia” of the entropic field. This resistance is determined by the curvature of the Fubini–Study metric in the amplitude direction.

To see why, consider a configuration of the entropic field with a large Fubini–Study curvature in the amplitude sector. This means that the field’s quantum state is “stiff” — small changes in the parameters produce large changes in the state, requiring a large “force” (in the information-geometric sense) to effect the change. In the emergent spacetime, this stiffness manifests as inertial mass. Conversely, a configuration with small Fubini–Study curvature is “flexible” — it can be easily changed, and it corresponds to a particle with small mass or to vacuum.

This identification provides a geometric explanation for the otherwise mysterious property of mass. In the Standard Model, the masses of elementary particles are determined by their coupling to the Higgs field — but the Higgs mechanism does not explain *why* particles have mass; it only provides the mechanism. The Theory of Entropicity offers a deeper answer: particles have mass because the entropic field configurations that constitute them have Fubini–Study curvature. Mass is geometry — specifically, it is the curvature of the Fubini–Study sector of the information manifold.

5.5 Why Energy = Fubini–Study Curvature

Energy is the capacity to do work — that is, the capacity to change the state of a system. In the Theory of Entropicity, the capacity to change the entropic state is measured by the curvature of the Fubini–Study metric. A flat Fubini–Study metric means that the entropic field is in a state of minimum dynamical activity — this is the vacuum, with zero energy density. A curved Fubini–Study metric means that the entropic field has dynamical content — this is matter, with nonzero energy density.

The connection to the entropic stress-energy tensor, developed in the supplementary appendices of Letter IE, makes this precise. The entropic stress-energy tensor $T_{\mu\nu}^{(\text{ent})}$ is constructed from the Fubini–Study sector of the Obidi Action by variation with respect to the emergent spacetime metric. Its components encode the energy density, momentum density, pressure, and stress of the entropic field in the emergent spacetime. The trace of this tensor gives the energy density, which is proportional to the scalar curvature of the Fubini–Study metric evaluated on the entropic field configuration. This establishes the identification: energy = Fubini–Study curvature, not as a metaphor but as a mathematical equality.

The Table below summarizes the discussions so far on the foundations of the Obidi Action of the Theory of Entropicity (ToE): The unity of geometry, matter, and gauge (**Geo-Ma-Ga**) fields, **a new ontological assignment** by ToE.

Sector of E in ToE	Geometry	Physical Meaning
amplitude ρ	Fisher–Rao	spacetime
dynamical E	Fubini–Study	matter
phase Θ	α -connections	gauge fields

This is the **Geomaga Trinity (TGT)** of the Theory of Entropicity (ToE).

SECTION 6

6. Why Amari-Čencov α -Connections Become Electromagnetism

6.1 The Declarations

The alpha-connections measure: How information flows twist and turn.

In the Theory of Entropicity (ToE), we make the following declarations based on the above alpha-connections measure:

The α -Connection Declarations

Phase structure = gauge structure

$\alpha = \text{plus or minus } 1$ correspond to dual connections

The $U(1)$ gauge field emerges from the phase geometry

Thus:

Electromagnetism is the phase geometry of the entropic field.

6.2 The Mathematical Structure of the α -Connections

The α -connections form a one-parameter family indexed by the real parameter α . Recall the defining equation: $\Gamma^{(\alpha)}_{abc} = \Gamma^{(o)}_{abc} + (\alpha/2) T_{abc}$, where $\Gamma^{(o)}$ is the Levi-Civita connection

of the Fisher–Rao metric and T_{abc} is the Amari–Čencov cubic tensor (also called the skewness tensor). The Levi-Civita connection $\Gamma^{(o)}$ is the unique connection that is both torsion-free and metric-compatible. For $\alpha \neq 0$, the α -connection is no longer metric-compatible: it does not preserve the Fisher–Rao inner product under parallel transport. This failure of metric compatibility is not a defect; it is a feature. It is precisely what allows the α -connections to encode structure that is *additional to* the metric structure — namely, gauge structure.

The duality between the e-connection ($\alpha = +1$) and the m-connection ($\alpha = -1$) is one of the most beautiful features of information geometry. It reflects the fundamental duality between two ways of combining probability distributions: exponential mixing (which corresponds to Bayesian updating, adding log-likelihoods) and convex mixing (which corresponds to forming ensembles, averaging probabilities). These two operations are dual in a precise sense: if you parametrize the statistical manifold using natural parameters (the parameters in which the exponential family is linear), the e-connection becomes flat; if you parametrize using expectation parameters (the means of sufficient statistics), the m-connection becomes flat. The two flat coordinate systems are dual, and the statistical manifold has the structure of a *dually flat manifold* — a concept that has no analogue in standard Riemannian geometry.

6.3 Why Phase Geometry = Gauge Structure

The phase $\Theta(x)$ of the complexified entropic field $E = \rho \exp(i\Theta)$ plays a role analogous to the phase of a **charged quantum field in ordinary quantum field theory**. Under a local phase rotation $\Theta(x) \rightarrow \Theta(x) + \chi(x)$, the amplitude ρ is unchanged but the phase shifts. To maintain the invariance of the Obidi Action under such local phase rotations, one must introduce a compensating gauge field A_μ that transforms as $A_\mu \rightarrow A_\mu + \partial_\mu \chi$. This is exactly the mechanism by which gauge fields arise in standard quantum field theory (the **Weyl–Yang–Mills construction**), but in the **Theory of Entropicity**, it arises from the **phase sector of the entropic field rather than being postulated independently**.

The connection to the α -connections is as follows. The e-connection ($\alpha = +1$) governs how probability distributions transform under exponential tilting — that is, under the addition of a linear function to the log-likelihood. **In the entropic field, this corresponds to the addition of a phase gradient to the entropic potential, which is precisely a gauge transformation. The m-connection ($\alpha = -1$) governs the dual transformation — the mixture operation — which corresponds to the dual (magnetic) formulation of the gauge field. The duality between the e-connection and the m-connection thus becomes the electromagnetic duality between the electric and magnetic formulations of Maxwell’s equations.**

The entropic four-potential, as derived in Letter IIA of the ToE Living Review Letters series, takes the form $A_\mu = (\hbar_{\text{ent}}/q_{\text{ent}}) \partial_\mu \Theta$, where $\hbar_{\text{ent}}$ is the entropic Planck constant and q_{ent} is the entropic charge. This transforms exactly as an electromagnetic gauge potential under phase rotations. The field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the curvature of this

connection, and it satisfies **Maxwell's equations** as a consequence of the equations of motion derived from the **Obidi Action**. **Electromagnetism is thus not an independent force appended to the theory; it emerges naturally from the phase geometry of the entropic field**, which is governed by the α -connections of information geometry.

6.4 Why the α -Connections Cannot Be the Spacetime Connection

It is important to understand why the α -connections become gauge connections rather than the spacetime (Levi-Civita) connection. The reason is mathematical and sharp. The Levi-Civita connection is the unique connection that is both torsion-free and metric-compatible. The α -connections (for $\alpha \neq 0$) are torsion-free but *not* metric-compatible. In general relativity, the spacetime connection must be metric-compatible, because metric compatibility ensures that the lengths of vectors are preserved under parallel transport — which is the geometric expression of the conservation of energy and momentum. A non-metric-compatible connection would allow the “length” of a vector to change as it is transported, which would correspond to a violation of energy conservation.

Gauge connections, by contrast, are *not* required to be metric-compatible. A gauge connection (such as the electromagnetic four-potential A_μ) tells you how to parallel-transport the phase of a charged field, not the length of a spacetime vector. The phase can rotate freely under parallel transport; there is no conservation law that constrains it (the conservation of charge is a separate matter, arising from the gauge symmetry itself, not from metric compatibility). The α -connections, being non-metric-compatible affine connections, thus have exactly the right mathematical structure to serve as gauge

connections and exactly the wrong mathematical structure to serve as the spacetime connection.

SECTION 7

7. The Unified Picture

7.1 The Unification

Einstein unified: geometry and gravity

The Theory of Entropicity (ToE) Unifies:

amplitude geometry $\rightarrow$ spacetime

dynamical geometry $\rightarrow$ matter

phase geometry $\rightarrow$ electromagnetism

All from one entropic field.

This is why one can legitimately say:

**Riemann is to GR what Fisher–Rao, Fubini–Study, and Amari–Čencov are
to ToE.**

Because:

- Riemannian geometry encodes physical distances.
- Fisher–Rao encodes entropic distances.
- Fubini–Study encodes dynamical distances.
- α -connections encode phase/gauge distances.

And in ToE:

Physical reality = entropic reality.

Hence, entropic geometry = physical geometry.

7.2 The History of Unification in Physics

The history of physics is, in large measure, a history of unification — the discovery that phenomena previously regarded as separate are, in fact, manifestations of a single underlying structure. Each major unification has deepened our understanding of nature and simplified the conceptual landscape of physics.

The **first great unification was achieved by Isaac Newton in 1687**, who showed that the force that causes an apple to fall from a tree and the force that holds the Moon in its orbit around the Earth are the same force: gravity. Before Newton, celestial mechanics and terrestrial mechanics were regarded as separate domains governed by different laws. Newton unified them under a single law of universal gravitation.

The **second great unification was achieved by James Clerk Maxwell in the 1860s**, who showed that electricity and magnetism — previously regarded as distinct phenomena —

are aspects of a single entity: the electromagnetic field. Maxwell's equations unified Coulomb's law, Ampère's law, Faraday's law, and the absence of magnetic monopoles into a single, elegant framework, and predicted the existence of electromagnetic waves (light).

The **third great unification was Einstein's unification** of space and time into spacetime (special relativity, 1905) and of spacetime geometry with gravity (general relativity, 1915). The fourth was the electroweak unification of Sheldon Glashow, Abdus Salam, and Steven Weinberg in the 1960s and 1970s, which showed that the electromagnetic force and the weak nuclear force are aspects of a single electroweak force at high energies.

The Theory of Entropicity (ToE) envisions and represents a unification that goes deeper than all of these. Previous unifications united different forces or different aspects of spacetime. ToE unifies the very *categories* of physics — **geometry (spacetime)**, **matter (the stuff that inhabits spacetime)**, and **gauge fields (the forces that act on matter)** — into a single information-geometric structure. These three categories, which have been regarded as fundamentally distinct since the formulation of general relativity, are revealed by ToE to be three aspects of a single entropic field, distinguished by the type of information-geometric structure they correspond to.

7.3 Comparison Table: General Relativity vs. Theory of Entropicity

Feature	Einstein's General Relativity	Theory of Entropicity (ToE)
Fundamental structure	Riemannian (pseudo-Riemannian) geometry	Information geometry (Fisher–Rao, Fubini–Study, α -connections)

Feature	Einstein's General Relativity	Theory of Entropicity (ToE)
Number of geometric structures	One (the metric $g_{\mu\nu}$)	Three (metric, dynamical, affine)
Physical sectors	One (gravity/spacetime)	Three (spacetime, matter, gauge fields)
Fundamental field	Metric tensor $g_{\mu\nu}$	Entropic field $E = \rho \exp(i\Theta)$
Matter treatment	External source ($T_{\mu\nu}$ postulated)	Emergent from dynamical geometry
Gauge fields	Not included (must be added separately)	Emergent from phase geometry
Ontological declaration	"Geometry is physical"	"Information geometry is physical"
Unification scope	Gravity and spacetime geometry	Spacetime, matter, and electromagnetism

This comparison makes clear the scope of the unification achieved by the Theory of Entropicity. Where Einstein required one geometric structure to describe one physical sector (gravity), the Theory of Entropicity uses three related geometric structures to describe three physical sectors, all emerging from a single entropic field. The simplicity of the underlying structure — one field, three geometric aspects — combined with the richness of its physical consequences — spacetime, matter, and electromagnetism — is the hallmark of a deep unification.

SECTION 8

8. The Final Clarity

8.1 The Natural Match

The Theory of Entropicity (ToE) doesn't need to "force" Fisher–Rao or Fubini–Study to be physical.

That is because these structures already measure the right thing that physics requires:

What Information Geometry Measures

- change
- distinguishability
- curvature
- flow
- structure

And since Physics is nothing but the evolution of [distinguishable] states, it then necessarily follows that the geometry that measures distinguishability is the geometry of physics.

Einstein replaced forces with geometry.

**The Theory of Entropicity (ToE) is replacing that Einstein's geometry
with information geometry.**

This is the next conceptual step in the formulation of the Theory of Entropicity
(ToE).

8.2 The Logical Chain: Forces → Geometry → Information Geometry

The intellectual trajectory of fundamental physics can be understood as a series of *ontological deepenings*, each of which reveals the previously fundamental layer of reality to be emergent from a deeper one:

Stage 1 (Newton, 1687): The world consists of particles moving under forces. Forces are the fundamental explanation for motion. Space and time are fixed, absolute backgrounds in which particles and forces operate.

Stage 2 (Einstein, 1915): Forces (at least gravity) are not fundamental. Gravity is not a force but a manifestation of spacetime curvature. Geometry replaces forces as the fundamental explanation for gravitational motion. But spacetime geometry is now dynamical, not fixed — it is a physical field determined by the distribution of matter and energy.

Stage 3 (Obidi, ToE, 2025-2026): Spacetime geometry is not fundamental. Spacetime itself is an emergent phenomenon, arising from the more fundamental information-geometric

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structure of the entropic field. Information geometry replaces spacetime geometry as the deepest layer of physical reality.

Each stage deepens our understanding of what is fundamental and what is emergent. Newton's forces are emergent from Einstein's geometry. Einstein's geometry is emergent from Obidi's entropic information geometry via the entropic field. At each stage, the previously "fundamental" entity is revealed to be a macroscopic manifestation of a deeper structure.

Crucially, each stage *contains* the previous one as a limiting case. General relativity contains Newtonian gravity as the weak-field, low-velocity limit. The Theory of Entropicity (ToE) contains general relativity as the classical, macroscopic limit (the limit $N \rightarrow \infty$, where the $O(1/N)$ corrections in the Curvature Transfer Theorem vanish). **The Theory of Entropicity (ToE) does not *replace* general relativity**; *it reveals what general relativity is made of. Just as knowing that water is made of H_2O molecules does not replace the fluid dynamics of water but deepens our understanding of why fluid dynamics works, knowing that spacetime is made of entropic information-geometric structure does not replace general relativity but deepens our understanding of why general relativity works.*

8.3 "The Next Conceptual Step": What It Means

When the Theory of Entropicity describes itself as "the next conceptual step," it is not claiming to be a minor modification or extension of existing theory. It is claiming to be the next step in a historical sequence of ontological revolutions, each of which has transformed our understanding of the nature of physical reality. The **first revolution (Newton)** showed

that celestial and terrestrial phenomena [mechanics] obey the same laws. The **second (Maxwell)** showed that electricity and magnetism are aspects of one field. The **third (Einstein)** showed that gravity is geometry. The **fourth (Obidi/ToE)** shows that geometry is entropic information geometry — **that the fabric of spacetime, the substance of matter, and the structure of gauge fields all emerge from a single entropic field whose geometry is the geometry of information.**

The depth of this claim is matched by the scope of its consequences. If the Theory of Entropicity (ToE) is correct, then every equation in general relativity, every term in the Standard Model Lagrangian, every prediction of quantum field theory is ultimately a statement about entropic information geometry, about the entropic field.

The laws of physics are not laws imposed upon nature from outside; they are the geometric properties of the [entropic] information-geometric entropic manifold on which reality is defined. Physics is entropic geometry; entropic geometry is physics. This is the final clarity of the Theory of Entropicity (ToE) on the physicalization of information geometry via the entropic field.

Summary and the Geomaga (GMG) Trinity

In the Theory of Entropicity (ToE), the three fundamental structures of information geometry play roles analogous to the geometric structures of general relativity:

- the **Fisher–Rao metric** measures distinguishability of probability distributions and becomes the amplitude-geometry that generates emergent spacetime;
- the **Fubini–Study metric** measures distinguishability of quantum states and becomes the dynamical-geometry responsible for matter;
- the **Amari–Čencov alpha-connections** encode the dualistic affine structure of information flows and become the phase-geometry underlying gauge fields.

Together, these three structures form the entropic-geometric triad from which spacetime, matter, and electromagnetism emerge.

The relationship forged by the Theory of Entropicity (ToE) is as follows:

The Geomaga Trinity (TGT) of the Theory of Entropicity

Fisher–Rao



becomes spacetime metric after emergence

Fubini–Study



becomes matter sector after emergence

α -connections



become gauge connections after emergence

The geo-ma-ga (GMG) trinity of Theory of Entropicity (ToE):

The Geomaga Trinity (TGT) of Theory of Entropicity (ToE).

This is the geometry-matter-gauge trinity of ToE: The geo-ma-ga (GMG) trinity of Theory of Entropicity (ToE): The Geomaga Trinity (TGT) of Theory of Entropicity (ToE).

Scholium: The Assignments Are Forced

Scholium §1. The Assignments Are Not Arbitrary

1. The assignments above in ToE are not arbitrary — they are forced by the type of geometry each structure encodes.

This is a claim of extraordinary strength, and it deserves correspondingly careful justification. In mathematics, when we say that a construction is “forced,” we mean that it is the unique construction satisfying certain natural conditions — there are no alternatives. The assignments of information-geometric structures to physical sectors in the Theory of Entropicity are forced in this precise sense: given the mathematical properties of each information-geometric structure and the physical requirements of each physical sector, there is exactly one consistent assignment.

The argument proceeds by examining each structure in turn, listing its mathematical properties, showing that these properties match the physical requirements of one and only one physical sector, and demonstrating that they are incompatible with the requirements of the other two sectors. The result is a unique, forced assignment — a tripartite correspondence that admits no permutations and no alternatives.

In the language of **category theory**, this forced assignment can be understood as a *natural transformation* between the category of information-geometric structures (with morphisms given by structure-preserving maps) and the category of physical structures (with morphisms given by physical symmetries). A natural transformation is one that is defined by the abstract structural properties of the objects, not by arbitrary choices. The **Geomaga Trinity** is natural in this sense: it is defined by what each information-geometric structure *is*, not by a choice of how to use it.

Scholium §2. Why Fisher–Rao Must Become the Spacetime Metric

2. Why Fisher–Rao must become the spacetime metric.

The Fisher–Rao metric has the following five defining properties:

Properties of the Fisher–Rao Metric

1. **Monotonicity** under coarse-graining (Markov morphisms)
2. **Invariance** under reparametrization (diffeomorphism-invariance)
3. **Uniqueness** (Čencov’s theorem: unique up to a constant)

4. **Positivity** (positive-definite Riemannian metric)
5. **Distinguishability** (measures statistical distance between distributions)

Each of these properties has a direct physical counterpart in the requirements for a spacetime metric:

Matching Physical Requirements

1. **Coarse-graining** → Physical spacetime distances cannot increase under loss of microscopic information (the second law of thermodynamics in the gravitational sector). When microscopic degrees of freedom are integrated out, the emergent macroscopic distances can only decrease or remain the same, never increase. This is the physical analogue of Fisher–Rao monotonicity.
2. **Reparametrization invariance** → General covariance (the fundamental symmetry of general relativity). The spacetime metric must be independent of the choice of coordinates, just as the Fisher–Rao metric is independent of the parametrization of the statistical manifold.
3. **Uniqueness** → The spacetime metric is the unique gravitational degree of freedom (Lovelock’s theorem ensures the uniqueness of the Einstein field equations). The uniqueness of the Fisher–Rao metric guarantees that the emergence map produces a unique spacetime metric, not a family of alternatives.

4. **Positivity** → Physical distances are positive (or, more precisely, the spatial part of the spacetime metric is positive-definite). The positivity of the Fisher–Rao metric ensures that the emergent spatial distances are positive.

5. **Distinguishability** → Physical distances measure distinguishability of spacetime events. Two events at the same spacetime location are indistinguishable; two events at different locations are distinguishable by the spacetime interval between them. This is exactly what the Fisher–Rao metric measures on the statistical manifold.

This is why Fisher–Rao is the only information metric that can serve as the pre-geometric origin of spacetime. No other information metric has these properties. Therefore: Fisher–Rao → spacetime metric is not a choice in ToE. It is a necessity in the formulation of ToE.

To elaborate on why the match is unique: consider the alternative possibility that the Fubini–Study metric might serve as the spacetime metric. The Fubini–Study metric is defined on complex projective space CP^n , not on a manifold of probability distributions. It does not satisfy Čencov’s uniqueness theorem (which applies only to statistical manifolds). It has constant positive holomorphic sectional curvature, which would force the emergent spacetime to be a space of constant positive curvature — ruling out flat spacetime, Schwarzschild black holes, and the entire phenomenology of general relativity. The Fubini–Study metric is therefore mathematically incompatible with the role of spacetime metric.

Similarly, the α -connections cannot serve as the spacetime metric because they are not metrics at all — they are affine connections. An affine connection defines parallel transport

and curvature, but it does not define distances. A spacetime metric must define distances (the spacetime interval); an affine connection cannot. The α -connections are thus the wrong *type* of geometric object to serve as the spacetime metric.

Scholium §3. Why Fubini–Study Must Become the Matter Sector

3. Why Fubini–Study must become the matter sector.

The Fubini–Study metric has the following defining properties:

Properties of the Fubini–Study Metric

1. **Dynamical** (measures change in quantum/entropic states)
2. **Kähler** (compatible complex, symplectic, and Riemannian structure)
3. **Projective** (defined on $\mathbb{CP}^n$, the space of rays, not vectors)
4. **Bounded curvature** (constant holomorphic sectional curvature)
5. **Quantum distinguishability** (measures distance between quantum states)

Each of these has a direct match to the matter sector:

Matching Physical Requirements

1. **Dynamical** → Matter is inherently dynamical — it propagates, oscillates, and evolves. The Fubini–Study metric measures the rate of state change, which in the emergent spacetime becomes the kinetic energy of matter fields.
2. **Kähler** → The matter sector requires compatibility of complex, symplectic, and metric structures — this is precisely the structure of quantum field theory, where fields are complex-valued and the symplectic structure defines the Poisson bracket (and hence the quantum commutator).
3. **Projective** → Matter fields in quantum mechanics are rays, not vectors (overall phase is unobservable). The projective nature of the Fubini–Study metric respects this fundamental feature.
4. **Bounded curvature** → The energy density of matter is bounded (by the Planck density, in the quantum gravity regime). The bounded curvature of the Fubini–Study metric provides a natural mechanism for this bound.
5. **Quantum distinguishability** → Matter states are distinguished by quantum measurements. The Fubini–Study metric encodes this distinguishability.

Therefore: Fubini–Study → matter sector is forced by its meaning.

Could the Fisher–Rao metric serve as the matter sector instead? No. The Fisher–Rao metric is a metric on the space of probability distributions, not on the space of quantum states. It does not have Kähler structure (it is real, not complex). It does not have the projective property (probability distributions are normalized vectors in L^1 , not rays in a Hilbert space). It measures classical statistical distinguishability, not quantum state change. The Fisher–Rao metric is the wrong type of object for the matter sector, just as it is the right type for the spacetime sector.

Could the α -connections serve as the matter sector? No. As noted above, the α -connections are affine connections, not metrics. The matter sector requires a metric (to define kinetic energy, which is a quadratic form on velocities). An affine connection defines parallel transport, not inner products. The α -connections cannot define the energy density or the stress-energy tensor that are characteristic of the matter sector.

Scholium §5. Why α -Connections Cannot Be the Levi-Civita Connection

5. Why alpha-connections cannot be the Levi-Civita connection.

The distinction between α -connections and the Levi-Civita connection is sharp and fundamental:

Levi-Civita Connection Properties

- Torsion-free
- Metric-compatible ($\nabla g = 0$)

- Uniquely determined by the metric
- Defines parallel transport preserving inner products
- Curvature = Riemann curvature tensor (gravitational tidal forces)

α -Connection Properties (for $\alpha \neq 0$)

- Torsion-free
- NOT metric-compatible ($\nabla^{(\alpha)}g \neq 0$)
- NOT uniquely determined by the metric (depends on α and T_{abc})
- Parallel transport does NOT preserve inner products
- Curvature = statistical curvature (not gravitational tidal forces)

Therefore: alpha-connections cannot be the Levi-Civita connection and cannot produce Einstein's field equations. They are the wrong type of geometric object. They are affine connections on a statistical manifold, not metric connections on a Riemannian manifold.

This distinction is not merely technical; it is the fundamental mathematical reason why the α -connections must correspond to gauge fields rather than to gravity. In general relativity, gravity is encoded in the Levi-Civita connection of the spacetime metric — the unique metric-compatible, torsion-free connection. The Levi-Civita connection preserves lengths and angles under parallel transport (metric compatibility). This ensures that the metric has

a consistent geometric meaning. **The equivalence principle then states that in any freely falling frame, this connection can be locally transformed away, making the laws of physics reduce to those of special relativity.** The α -connections do not preserve lengths under parallel transport; they allow the “size” of a statistical distribution to change as it is transported. This non-metric-compatible behavior is precisely the behavior of a gauge connection, which changes the phase of a charged field under parallel transport while preserving its modulus.

To produce Einstein’s field equations, one needs the Riemann curvature tensor formed from the Levi-Civita connection of a metric. The curvature tensor of an α -connection is, in general, different from the Riemann curvature tensor: it contains additional terms involving the skewness tensor T_{abc} and its derivatives. These additional terms do not have the correct algebraic symmetries (the Bianchi identity, the pair symmetry) to enter into the Einstein field equations. They have the algebraic structure of a gauge field strength tensor, which is antisymmetric in two indices (like the electromagnetic field strength $F_{\mu\nu}$) rather than having the pair symmetries of the Riemann tensor. This algebraic mismatch is the final, conclusive proof that the α -connections cannot play the role of the gravitational connection and must instead play the role of gauge connections.

Scholium §6. The Deep Reason the Assignment Is Forced

6. The deep reason the assignment is forced.

Information Geometry Structure	What It Measures	Physical Meaning in ToE
Fisher–Rao metric	amplitude distinguishability	spacetime geometry
Fubini–Study metric	dynamical distinguishability	matter/energy
α -connections	phase/affine duality	gauge fields

This is the geometry-matter-gauge trinity of ToE. It is not arbitrary. It is not aesthetic. It is not chosen. It is forced by the mathematical nature of each structure.

The depth of this forced assignment can be appreciated by considering it from the perspective of mathematical naturality. In category theory, a natural transformation between two functors is one that commutes with all morphisms in the underlying categories. Applied to the present context: the functor from the category of information-geometric structures to the category of physical structures must commute with all natural maps (Markov morphisms on the statistical side, diffeomorphisms on the physical side). The Fisher–Rao metric is the unique object in the information-geometric category that is invariant under Markov morphisms (Čencov’s theorem), just as the spacetime metric is the unique gravitational object in the physical category that is invariant under diffeomorphisms (general covariance). The natural transformation must therefore map Fisher–Rao to spacetime metric. Similarly, the Fubini–Study metric is the unique Kähler metric on the quantum state space invariant under unitary transformations, just as the matter sector is the unique physical sector that respects the unitary symmetry of quantum

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mechanics. The natural transformation maps Fubini–Study to matter. And the α -connections are the unique dual affine structures on the statistical manifold, just as gauge connections are the unique additional geometric structures (beyond the metric) in physical spacetime. The natural transformation maps α -connections to gauge fields. The assignment is not merely “forced” in the informal sense; it is *naturally* forced, in the precise category-theoretic sense.

Scholium §7. The Final Clarity of ToE

7. The final clarity of ToE.

The Stable Assignment

alpha-connections are affine, not metric $\rightarrow$ gauge

Fisher–Rao is metric, not affine $\rightarrow$ spacetime

Fubini–Study is dynamical, not metric $\rightarrow$ matter

This is the correct, final, stable assignment of information geometry in ToE.

The word “stable” in this declaration deserves emphasis. In mathematics, a result is stable if it does not change under small perturbations of the hypotheses. The Geomaga Trinity is stable in this sense: small modifications to the mathematical properties of the three information-geometric structures, or small variations in the physical requirements of the three physical sectors, do not change the assignment. The Fisher–Rao metric is robustly the right object for spacetime (any small deformation of the Fisher–Rao metric that

preserved its essential properties would still be mapped to the spacetime metric, since Čencov's theorem is itself a robustness result). The Fubini–Study metric is robustly the right object for matter. The α -connections are robustly the right objects for gauge fields. The assignment is stable, unique, and final.

Scholium §8. ToE Summary

8. ToE Summary.

“In the Theory of Entropicity (ToE), the Fisher–Rao metric provides the amplitude-geometry that becomes spacetime, the Fubini–Study metric provides the dynamical-geometry that becomes matter, and the Amari–Čencov alpha-connections provide the phase-geometry that becomes gauge fields. This assignment is not arbitrary; it is uniquely determined by the mathematical meaning of each structure.”

This is the truth of Obidi's formulation of the Theory of Entropicity (ToE).

Additional Scholium Remarks: Uniqueness of the Geomaga Trinity

The uniqueness of the Geomaga Trinity invites comparison with other uniqueness results in theoretical physics. The most celebrated such result is *Lovelock's theorem* (1971), which states that in four spacetime dimensions, the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - (1/2)Rg_{\mu\nu}$ is the unique symmetric, divergence-free, second-rank tensor that is a function of the metric and

its first and second derivatives only. This theorem guarantees that the Einstein field equations are the unique gravitational field equations of their type — there are no alternatives. Lovelock's theorem is what gives general relativity its inevitability: given the symmetry requirements and the derivative order, the field equations are forced.

The Geomaga Trinity (TGT) enjoys an analogous inevitability. Given the mathematical properties of the three information-geometric structures and the physical requirements of the three physical sectors, the assignment is forced. There are no alternatives. Just as Lovelock's theorem eliminates the possibility of alternative gravitational field equations (within its assumptions), the mathematical structure of information geometry eliminates the possibility of alternative assignments of information-geometric structures to physical sectors (within the framework of the Theory of Entropicity).

Consider the six possible permutations of the three-to-three assignment:

5. **Fisher–Rao $\rightarrow$ spacetime, Fubini–Study $\rightarrow$ matter, α -connections $\rightarrow$ gauge**
(the Geomaga Trinity)
6. Fisher–Rao $\rightarrow$ spacetime, Fubini–Study $\rightarrow$ gauge, α -connections $\rightarrow$ matter: FAILS
because α -connections are not a metric and cannot define the kinetic energy of matter, and Fubini–Study has Kähler structure incompatible with gauge connections.
7. Fisher–Rao $\rightarrow$ matter, Fubini–Study $\rightarrow$ spacetime, α -connections $\rightarrow$ gauge: FAILS
because Fubini–Study has constant curvature on $\mathbb{CP}^n$, incompatible with the

variable curvature of physical spacetime, and Fisher–Rao is not Kähler as required by the matter sector.

8. Fisher–Rao $\rightarrow$ matter, Fubini–Study $\rightarrow$ gauge, α -connections $\rightarrow$ spacetime: FAILS because α -connections are not metric-compatible, so they cannot define spacetime distances.
9. Fisher–Rao $\rightarrow$ gauge, Fubini–Study $\rightarrow$ spacetime, α -connections $\rightarrow$ matter: FAILS for the same reasons as permutations 3 and 4.
10. Fisher–Rao $\rightarrow$ gauge, Fubini–Study $\rightarrow$ matter, α -connections $\rightarrow$ spacetime: FAILS because Fisher–Rao is a metric (not an affine connection) and cannot serve as a gauge connection, and α -connections cannot serve as the spacetime metric.

Only permutation 1 — the Geomaga Trinity — is mathematically consistent. The assignment is unique. It is forced. It is the truth of the Theory of Entropicity.

Concluding Remarks

This Preamble has established the conceptual and philosophical foundations for the physical status of information-geometric structures in the Theory of Entropicity (ToE). The argument is structured as a series of eight interlocking claims, each building upon the preceding ones:

1. Einstein established the precedent that mathematical geometry can be identified with physical reality (Section 1).

2. The Fisher–Rao, Fubini–Study, and α -connection structures measure precisely the quantities that physics requires: distinguishability, dynamical change, and information flow (Section 2).
3. The Theory of Entropicity makes the same kind of ontological leap as Einstein, but at a deeper level: information geometry is not a description of physics but is the substrate of physics (Section 3).
4. The Fisher–Rao metric becomes the spacetime metric via the emergence map, with the conversion factor $\lambda \approx 10^{-47}$ fixed by Čencov’s theorem and the entropic curvature gap (Section 4).
5. The Fubini–Study metric becomes the matter sector, with mass as resistance to entropic change and energy as Fubini–Study curvature (Section 5).
6. The α -connections become electromagnetism, with the phase geometry of the entropic field generating the $U(1)$ gauge structure (Section 6).
7. The three-sector unification (spacetime, matter, gauge) from a single entropic field is deeper than any previous unification in the history of physics (Section 7).
8. The information-geometric structures do not need to be “forced” to be physical; they naturally measure what physics requires (Section 8).

The Scholium then demonstrated that the assignment of information-geometric structures to physical sectors is not arbitrary but is uniquely forced by the mathematical properties of each structure. Only the **Geomaga** Trinity — Fisher–Rao to spacetime, Fubini–Study to matter, α -connections to gauge fields — is mathematically consistent. All five alternative permutations fail.

Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

The Theory of Entropicity (ToE) Living Review Letters Series (ToE-LRLS). *Letter IE*

John Onimisi Obidi (jonimisiobidi@gmail.com)

Research Lab, The Aether

May 6, 2026

The reader is now prepared to enter the body of Letter IE, where the mathematical implementation of these ideas is carried out in detail: the Obidi Action is constructed, the field equations are derived, the Curvature Transfer Theorem is proved, and the recovery of general relativity, quantum mechanics, and Maxwell's electrodynamics from the single framework of information geometry is demonstrated. The present Preamble has provided the *why*; the body of Letter IE provides the *how*.

— *End of Preamble* —

Theory of Entropicity (ToE) — Living Review Letters Series — Letter IE

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General Introduction

The modern understanding of gravity rests on a profound geometric insight: spacetime possesses curvature, and this curvature governs the motion of matter and radiation. Yet the geometric formulation of general relativity, for all its elegance, leaves a deeper question unanswered. Why does spacetime possess curvature at all? The metric tensor is introduced as a primitive field, its curvature computed by differentiation, and the resulting geometric object is equated to the stress–energy of matter. The theory explains how curvature behaves, but not why curvature exists.

The Theory of Entropicity (ToE) approaches this foundational gap from a radically different direction. Instead of treating spacetime as the starting point of physics, ToE treats **entropy** as the primitive entity and constructs geometry as a derived, emergent structure. The central claim is that the curvature of physical spacetime is not fundamental; it is the macroscopic expression of curvature defined on a deeper **information manifold** associated with an entropic substrate. In this view, geometry is not an axiom — it is a thermodynamic limit.

The information manifold arises from the statistical description of the entropic substrate. Its metric is the Fisher–Entropic metric, the unique Riemannian structure compatible with the informational content of the underlying probability distributions. This metric carries its own intrinsic curvature, defined without reference to any spacetime structure. The remarkable assertion of ToE is that this information-geometric curvature is the **source** of the curvature we observe in spacetime.

To formalize this relationship, ToE introduces the **emergence map**, a smooth projection from the high-dimensional information manifold to a four-dimensional Lorentzian

spacetime. The **Thermodynamic Correspondence Principle (TCP)** asserts that, in the limit of large numbers of microstates, the pullback of the spacetime metric is proportional to the Fisher–Entropic metric. From this correspondence, a Curvature Transfer Theorem follows: the Riemann curvature of spacetime is the pushforward of the information-geometric curvature, up to thermodynamic corrections.

This perspective reframes the Einstein field equations. Instead of postulating a relation between geometry and matter, ToE derives both from the same informational source. The stress–energy tensor becomes the macroscopic expression of the information Einstein tensor, and spacetime curvature becomes the geometric shadow of information curvature. Matter and geometry are no longer independent ingredients; they are dual aspects of the same entropic structure.

To capture the portion of information curvature that does not manifest in spacetime, ToE introduces the **Obidi Curvature Invariant (OCI)** — a scalar quantity measuring the residual curvature of the information manifold not visible in the emergent spacetime geometry. This invariant identifies the informational degrees of freedom relevant to quantum gravity and provides a natural candidate for contributions to the cosmological constant.

This letter develops the full mathematical architecture of this emergence framework: the construction of the information manifold, the definition of information curvature, the emergence map, the Curvature Transfer Theorem, and the properties of the Obidi Curvature Invariant. The result is a unified picture in which spacetime, matter, and gravitation arise from a single entropic substrate, governed by the informational geometry encoded in the Fisher–Entropic metric.

* * *

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1. Introduction and Motivation

General relativity, in its standard formulation, asserts a profound structural identity: the curvature of spacetime, encoded in the Einstein tensor $G_{\mu\nu}$, is proportional to the stress-energy tensor $T_{\mu\nu}$ of matter and radiation. The Einstein field equations [1],

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu} ,$$

constitute the dynamical core of the theory. They prescribe *how* matter curves spacetime, and how spacetime curvature governs the motion of matter. Yet they are silent on a deeper question: *why does spacetime possess curvature at all?* The metric tensor $g_{\mu\nu}$ is introduced as a fundamental field; its curvature is computed from Christoffel symbols derived by differentiation; and the resulting geometric object is then equated to a matter source. At no point does the formalism explain the *ontological origin* of the geometric degrees of freedom themselves. Curvature is posited, not derived.

This conceptual lacuna has motivated a sustained program of research seeking to understand gravity — and spacetime geometry more broadly — as emergent from a deeper substrate. The landmark contribution of Jacobson [10] (1995) demonstrated that the Einstein field equations [1] can be derived as an equation of state from the proportionality of entropy to horizon area, together with the Clausius relation $\delta Q = T dS$ applied to local

Rindler causal horizons. Jacobson's result revealed that the field equations have a thermodynamic character, but left open the question of the underlying microscopic theory from which the thermodynamic description emerges. Verlinde's entropic gravity proposal [11] (2011) advanced the program by arguing that gravitational force itself is an entropic force, arising from changes in information associated with the positions of material bodies. Padmanabhan's emergent gravity program [12, 13] further demonstrated that the gravitational field equations in a wide class of theories (including Lanczos–Lovelock gravity) can be recast as thermodynamic identities on null surfaces, and that the expansion of cosmic space can be related to the difference between surface and bulk degrees of freedom.

In parallel, the field of information geometry — pioneered by Rao [3], Čencov [7], Efron [4], and brought to systematic maturity by Amari [8], Ay [9], Jost [9], and others — established that the space of probability distributions carries a natural Riemannian structure: the Fisher information metric [2]. Caticha's program [14] of "entropic dynamics" explored the possibility of deriving physical laws (including aspects of quantum mechanics and elements of general relativity) from principles of entropic inference on statistical manifolds. These information-geometric approaches have revealed tantalizing structural parallels between the geometry of statistical models and the geometry of physical spacetime, but have generally stopped short of claiming an identity between the two.

The Theory of Entropicity (ToE), as developed in this series of Living Review Letters, makes precisely this stronger claim. The central thesis is as follows:

Spacetime curvature is not merely analogous to the curvature of a statistical-information manifold. It is the curvature of the information manifold, observed at the thermodynamic limit of a fundamental entropic substrate.

This letter — the first in the IE subseries — undertakes the following program:

1. **Construction of the information manifold.** We define the entropic substrate Ω , equip the associated statistical model with the Fisher–Entropic metric, and thereby obtain a Riemannian manifold $(\mathcal{M}_I, g^I)$ whose geometry is determined entirely by the informational content of the substrate (Section 2).
2. **Definition of information curvature.** We compute the Riemann curvature tensor, Ricci tensor, and scalar curvature of $(\mathcal{M}_I, g^I)$, establishing these as intrinsic, pre-geometric quantities that make no reference to any spacetime metric (Section 2).
3. **Proof of spacetime emergence.** We introduce the emergence map $\Phi: \mathcal{M}_I \rightarrow \mathcal{M}_S$, state the Thermodynamic Correspondence Principle (Axiom E₃), and prove the Curvature Transfer Theorem, which establishes that the spacetime Riemann tensor is the pushforward of the information Riemann tensor up to thermodynamic corrections. Einstein's field equations [1] are recovered as a consequence (Section 3).
4. **The Obidi Curvature Invariant.** We define the scalar invariant $\mathcal{K}_\Omega$, establish its fundamental properties (vanishing, positivity, gauge invariance, topological bound), and interpret it as a measure of "informational dark curvature" — the

residual information-geometric structure not captured by classical spacetime geometry (Section 4).

5. **Structural theorems.** We prove uniqueness of the emergence map, derive topological constraints on spacetime from the information manifold, and recover the Bekenstein–Hawking area–entropy relation [5, 6, 18] (Section 5).

Throughout, we work with Unicode mathematical notation suitable for direct rendering. Tensor indices are denoted by Greek letters (μ, ν, ρ, σ for spacetime; i, j, k, l for the information manifold). The Einstein summation convention [1] is employed unless otherwise stated. We adopt the $(-, +, +, +)$ signature convention for the spacetime metric and the positive-definite convention for the Fisher metric [2].

2. The Entropic Substrate and the Information Manifold ($\mathcal{M}_I, g_I$)

2.1 The Entropic Substrate

The Theory of Entropicity (ToE) begins with a radical ontological commitment: the fundamental constituents of reality are not fields on a pre-existing spacetime manifold, but rather *distinguishable configurations* of an entropic substrate. This substrate is the ground from which both spacetime geometry and matter content emerge through coarse-graining and thermodynamic limits.

Axiom E1 (Entropic Primacy)

All physical degrees of freedom are encodable as distinguishable configurations of an entropic substrate Ω , equipped with a σ -algebra $\mathcal{F}$ and a reference measure μ_0 . The triple $(\Omega, \mathcal{F}, \mu_0)$ is the foundational structure from which all observable physics derives.

The substrate Ω is not to be confused with a spacetime manifold. It is a pre-geometric space — a set of microstates whose internal structure is specified only by the measurable structure $(\mathcal{F}, \mu_0)$. No notion of distance, dimension, or locality is assumed at this level. These structures will *emerge* from the statistical geometry imposed on Ω by the process of macroscopic observation.

The reference measure μ_0 plays the role of a prior or background counting measure on microstates. Its choice reflects the "democratic" weighting of microstates before any macroscopic constraint is imposed. In discrete settings, μ_0 is the counting measure; in continuous settings, it is a Lebesgue-type measure on Ω . The physical content of the theory is ultimately independent of the specific choice of μ_0 , as will be guaranteed by the reparameterization invariance of the Fisher metric (see Section 2.2).

Axiom E2 (Informational Completeness)

The physical content of any spacetime region $\mathcal{U}$ is exhaustively captured by a probability distribution p_θ on $(\Omega, \mathcal{F}, \mu_0)$, parametrized by macroscopic coordinates $\theta = (\theta^1, \theta^2, \dots, \theta^n)$ where $\theta \in \Theta \subset \mathbb{R}^n$. The family $\{p_\theta : \theta \in \Theta\}$ forms a smooth statistical model on Ω .

Axiom E2 asserts a strong form of informational sufficiency: every physical observable associated with a spacetime region can be reconstructed from the probability distribution p_θ that characterizes the macroscopic state of the substrate in that region. The coordinates $\theta = (\theta^1, \dots, \theta^n)$ are, at this stage, abstract parameters labelling the macroscopic states. The central achievement of this letter is to show that a subset of these parameters acquires the interpretation of spacetime coordinates, and that the metric and curvature of the resulting spacetime are inherited from the statistical geometry of the model.

2.2 The Fisher–Entropic Metric

Given the statistical model $\mathcal{M}_I = \{p_\theta : \theta \in \Theta \subset \mathbb{R}^n\}$, we equip it with a Riemannian metric derived from the Fisher information [2]. This is the foundational geometric structure of the Theory of Entropicity (ToE).

Definition 2.1 (Fisher–Entropic Metric)

The Fisher–Entropic metric on $\mathcal{M}_I$ is the rank-(0,2) tensor field defined by:

$$g_{ij}^I(\theta) = \int_{\Omega} (\partial_i \log p_\theta)(\partial_j \log p_\theta) p_\theta d\mu_0$$

where $\partial_i \equiv \partial/\partial\theta^i$. Equivalently, $g_{ij}^I(\theta) = -\int_{\Omega} (\partial_i \partial_j \log p_\theta) p_\theta d\mu_0$ under standard regularity conditions (interchangeability of differentiation and integration).

Proposition 2.2 (Metric Properties)

The Fisher–Entropic metric g^l is (i) symmetric, (ii) positive semi-definite, and (iii) positive-definite whenever the statistical model is identifiable (i.e., the map $\theta \mapsto p_\theta$ is injective). Moreover, g^l transforms as a covariant rank-2 tensor under reparameterization of the model.

Proof: Symmetry is immediate from the definition: $g^l_{ij} = \int (\partial_i \log p)(\partial_j \log p) p \, d\mu_0 = g^l_{ji}$. For positive semi-definiteness, consider any vector $v = (v^1, \dots, v^n)$. Then $g^l_{ij} v^i v^j = \int (v^i \partial_i \log p)^2 p \, d\mu_0 \geq 0$, since the integrand is a square. Equality holds if and only if $v^i \partial_i \log p_\theta = 0$ for p_θ -almost every $\omega \in \Omega$, which, under identifiability, implies $v = 0$. This establishes positive-definiteness.

For the tensor transformation property, let $\tilde{\theta} = \tilde{\theta}(\theta)$ be a smooth reparameterization. Then $\partial/\partial\tilde{\theta}^a = (\partial\theta^i/\partial\tilde{\theta}^a) \partial/\partial\theta^i$, whence:

$$\tilde{g}^l_{ab}(\tilde{\theta}) = \int (\partial_a \log p)(\partial_b \log p) p \, d\mu_0 = (\partial\theta^i/\partial\tilde{\theta}^a)(\partial\theta^j/\partial\tilde{\theta}^b) g^l_{ij}(\theta)$$

which is precisely the transformation law for a $(0,2)$ tensor.

□

The Fisher–Entropic metric inherits a uniqueness property of fundamental importance. The theorem of Čencov [7] (1982), later refined and extended by Amari [8] and others, establishes the following:

Theorem 2.3 (Čencov–Amari Uniqueness)

The Fisher information metric [2] is, up to a positive scalar multiple, the *unique* Riemannian metric on any statistical model that is invariant under the action of sufficient statistics (Markov morphisms). Consequently, any Riemannian geometry on the space of probability distributions that respects the informational content of the model must be proportional to g^I .

This uniqueness theorem is of paramount significance for the Theory of Entropicity (ToE). It guarantees that the geometry we place on the space of macroscopic states is not a matter of convention or arbitrary choice — it is the *only* geometry compatible with the informational structure of the substrate. Any theory that seeks to derive spacetime geometry from informational principles must, by mathematical necessity, pass through the Fisher metric [2].

2.3 Entropic Connection and Curvature

Having equipped the information manifold with a Riemannian metric, we now construct the full apparatus of Riemannian geometry: connection, curvature tensors, and scalar curvature. These objects are computed from g^I alone and therefore inherit their meaning from the informational content of the entropic substrate. At no point in this construction is any spacetime metric invoked.

Definition 2.4 (Levi-Civita Connection on $\mathcal{M}_I$)

The Levi-Civita connection Γ^l on $(\mathcal{M}_I, g^l)$ is the unique torsion-free connection compatible with g^l . Its Christoffel symbols are:

$$\Gamma^{l,k}_{ij} = \frac{1}{2} g^{l,kl} (\partial_i g^l_{jl} + \partial_j g^l_{il} - \partial_l g^l_{ij})$$

where $g^{l,kl}$ denotes the components of the inverse metric.

Definition 2.5 (Information Riemann Curvature Tensor)

The Riemann curvature tensor of $(\mathcal{M}_I, g^l)$ is defined in coordinate components as:

$$R^{l,1}_{ijk} = \partial_j \Gamma^{l,1}_{ik} - \partial_k \Gamma^{l,1}_{ij} + \Gamma^{l,1}_{jm} \Gamma^{l,m}_{ik} - \Gamma^{l,1}_{km} \Gamma^{l,m}_{ij}$$

This tensor measures the extent to which parallel transport on $\mathcal{M}_I$ is path-dependent — equivalently, the non-commutativity of covariant derivatives with respect to Γ^l .

Definition 2.6 (Information Ricci Tensor and Scalar Curvature)

The Ricci tensor of the information manifold is obtained by contraction:

$$\text{Ric}^l_{ij} = R^{l,k}_{ikj}$$

The information scalar curvature is the full trace:

$$\mathcal{S}^l = g^{l,ij} \text{Ric}^l_{ij}$$

Finally, the information Einstein tensor is:

$$G^l_{ij} = \text{Ric}^l_{ij} - \frac{1}{2} \mathcal{S}^l g^l_{ij}$$

We emphasize the conceptual point with full force: the quantities R^I , Ric^I , $\mathcal{S}^I$, and G^I are computed *entirely* from the Fisher–Entropic metric g^I , which is itself defined by the probability distributions $\{p_\theta\}$ on the entropic substrate. No spacetime metric, no causal structure, no Lorentzian signature has entered the construction. The curvature of the information manifold is a *pre-geometric* datum, determined by the informational architecture of the substrate. The content of the next section is that this pre-geometric curvature *becomes* the curvature of physical spacetime under appropriate conditions.

3. The Emergence Map Φ : From Information Geometry to Spacetime Geometry

3.1 The Thermodynamic Correspondence Principle

The passage from the information manifold $(\mathcal{M}_I, g^I)$ to the physical spacetime manifold $(\mathcal{M}_S, g^S)$ is mediated by the *emergence map*, whose existence and properties are asserted axiomatically.

Axiom E3 (Emergence)

There exists a smooth surjection $\Phi: \mathcal{M}_I \rightarrow \mathcal{M}_S$ onto a 4-dimensional Lorentzian manifold $(\mathcal{M}_S, g^S)$ such that the pullback of the spacetime metric satisfies:

$$\Phi^*g^S = \lambda g^I + \mathcal{O}(1/N)$$

in the thermodynamic limit $N \rightarrow \infty$, where N is the number of accessible microstates of the substrate, and λ is a dimensionful coupling constant with units [length² / information]. More precisely, the relation holds on the 4-dimensional submanifold of $\mathcal{M}_I$ selected by the thermodynamic constraints, and the $\mathcal{O}(1/N)$ corrections are smooth tensor fields whose norms (with respect to g^I) are bounded by C/N for a constant C depending only on the model.

The physical content of Axiom E₃ is as follows. The information manifold $\mathcal{M}_I$ is, in general, high-dimensional — its dimension n equals the number of macroscopic parameters needed to specify the state of the substrate. The emergence map Φ projects this high-dimensional space onto a 4-dimensional Lorentzian manifold. The spacetime metric g^S is the *thermodynamic shadow* of the Fisher–Entropic metric: it captures the large-scale, coarse-grained geometric content of g^I , with finite- N corrections suppressed by factors of $1/N$.

The coupling constant λ has dimensions of [length² / information] and serves as the conversion factor between informational units (nats or bits on the statistical manifold) and geometric units (lengths in spacetime). On dimensional grounds and by comparison with the Bekenstein–Hawking entropy formula [5, 6, 18], we anticipate $\lambda \sim \ell_P^2 / k_B$, where ℓ_P is the Planck length and k_B is Boltzmann's constant. The precise value of λ is determined by matching the emergent Einstein equations to Newton's gravitational constant (see Section 3.3).

The surjectivity of Φ ensures that every point of spacetime corresponds to at least one point of the information manifold — no region of spacetime is "informationally empty." The smoothness of Φ ensures that the emergent spacetime geometry inherits differentiability from the statistical model. The relation $\Phi^*g^S = \lambda g^I$ is not a definition but a *dynamical constraint*: it ties the spacetime geometry to the informational content of the substrate.

Scholium on the ToE Emergence Map Metric Conversion Factor of λ :

This emergence map ($\Phi^*g^S = \lambda g^I$) is the Theory of Entropicity (ToE) analogue of Einstein's identification of gravity with curvature.

In the Theory of Entropicity (ToE), the conversion factor λ must be a constant because it is the universal conversion factor between the Fisher–Rao information metric and the emergent spacetime metric. That is, λ is a conversion factor between information geometry and spacetime geometry. If λ were a function, the emergence map would violate the equivalence principle, break the uniqueness of the Fisher–Rao metric, introduce unphysical forces, and fail to reproduce general relativity. The constancy of λ is therefore required by both the mathematics of information geometry and the physics of spacetime emergence.

If λ were a function (a variable, and not a constant), the **emergence map would break general relativity**.

In ToE, free motion corresponds to **Fisher–Rao geodesics, where λ must be a constant**.

In all of the above, λ must be constant because of Čencov's theorem. Čencov's theorem states:

The Fisher–Rao metric is unique up to a constant multiplicative factor.

This is a deep result.

In the Theory of Entropicity, λ is fixed at approximately 10^{-47} by the entropic curvature gap between the highly curved Fisher–Rao information manifold and the extremely flat physical spacetime. This value is not a range and not a repetition of existing literature; it is uniquely determined by the emergence map and required for the recovery of general relativity, the equivalence principle, and the observed cosmological constant.

Although the information manifold curvature $\mathcal{R}_I$ and the spacetime curvature $\mathcal{R}_S$ each vary over ranges, the emergence constant λ is not defined by pointwise curvature values. Instead, λ is fixed by the ratio of the characteristic curvature scales of the two manifolds. This ratio is sharply determined by the entropic curvature gap and yields $\lambda \approx 10^{-47}$. If λ were allowed to vary over the ranges of $\mathcal{R}_I$ or $\mathcal{R}_S$, the equivalence principle would fail, general relativity would not emerge, and the observed universe could not exist. Thus, λ is a universal conversion constant, not a variable.

3.2 Curvature Transfer Theorem

We now prove the central technical result of this letter: the curvature of physical spacetime is determined, in the thermodynamic limit, by the curvature of the information manifold.

Theorem 3.1 (Curvature Transfer)

Let $\Phi: (\mathcal{M}_I, g^I) \rightarrow (\mathcal{M}_S, g^S)$ be the emergence map satisfying Axiom E3. Then in the thermodynamic limit $N \rightarrow \infty$:

$$R^S_{\mu\nu\rho\sigma} = \lambda (\Phi^* R^I)_{\mu\nu\rho\sigma} + \mathcal{O}(1/N)$$

where R^S is the Riemann tensor of spacetime, R^I is the Riemann tensor of the information manifold, and Φ^* denotes the pushforward of tensor fields along Φ .

Corollary 3.2 (Einstein Tensor Transfer)

Under the hypotheses of Theorem 3.1, the Einstein tensor of spacetime satisfies:

$$G^S_{\mu\nu} = \lambda (\Phi^* G^I)_{\mu\nu} + \mathcal{O}(1/N)$$

where $G^I_{ij} = \text{Ric}^I_{ij} - \frac{1}{2} \mathcal{S}^I g^I_{ij}$ is the Einstein tensor of the information manifold.

3.3 Recovery of Einstein's Field Equations

We are now in a position to recover Einstein's field equations [1] as an *emergent identity* within the Theory of Entropicity (ToE), rather than as a fundamental postulate.

By Corollary 3.2, the spacetime Einstein tensor is determined in the thermodynamic limit by the information Einstein tensor:

$$G^S_{\mu\nu} = \lambda (\Phi^* G^I)_{\mu\nu}$$

Now, the standard Einstein field equations (in the absence of a cosmological constant) read

$G^S_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. Comparing, we obtain the identification:

$$T_{\mu\nu} = (\lambda c^4 / 8\pi G) (\Phi^* G^I)_{\mu\nu}$$

This equation constitutes one of the central results of the Theory of Entropicity (ToE). Its content is remarkable: the stress-energy tensor of matter — which in standard general relativity is specified independently as a source term — is here *derived* from the Einstein tensor of the information manifold. Matter-energy is not the *cause* of spacetime curvature in the usual sense. Rather, both the stress-energy tensor $T_{\mu\nu}$ and the spacetime curvature $G^S_{\mu\nu}$ are dual descriptions of the same underlying informational-geometric structure: the curvature of $(\mathcal{M}_I, g^I)$.

The coupling constant λ is now determined by dimensional analysis and the matching condition. Since $G^S_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$ and $G^S_{\mu\nu} = \lambda \Phi^* G^I_{\mu\nu}$, the consistency requires that λ absorbs the conversion between the natural units of information geometry (nats per unit statistical parameter) and the units of spacetime geometry (inverse length squared). Explicitly, comparing with the Bekenstein [5] bound and the holographic principle, we fix:

$$\lambda = \ell_P^2 / (4 k_B) = \hbar G / (4 k_B c^3)$$

where $\ell_P = \sqrt{(\hbar G/c^3)}$ is the Planck length. This identification ensures that the Bekenstein–Hawking entropy formula [5, 6, 18] is recovered (see Proposition 5.3 below).

We note that the inclusion of a cosmological constant Λ is naturally accommodated: if the information Einstein tensor G^I contains a trace component proportional to g^I , this descends to a cosmological constant term in the spacetime equations. We return to this point in Section 4.3.

4. The Obidi Curvature Invariant (OCI) $\mathcal{K}\Omega$

4.1 Definition

The Curvature Transfer Theorem (Theorem 3.1) establishes that spacetime curvature equals information curvature in the thermodynamic limit. But the theorem also implies that, at finite N , the equality is only approximate. There exists a *gap* between the full information curvature and the curvature captured by the emergent spacetime geometry. This gap is measured by the central new quantity introduced in this letter.

Definition 4.1 (The Obidi Curvature Invariant)

The Obidi Curvature Invariant is a scalar field on the information manifold defined by:

$$\mathcal{K}\Omega(\theta) := \mathcal{S}^I(\theta) - (1/\lambda) \Phi^*(\mathcal{S}^S)(\theta)$$

where $\mathcal{S}^I$ is the scalar curvature of the information manifold $(\mathcal{M}_I, g^I)$, $\mathcal{S}^S$ is the scalar curvature of spacetime $(\mathcal{M}_S, g^S)$, Φ^* is the pullback along the emergence map, and λ is the coupling constant of Axiom E3.

The physical interpretation of $\mathcal{K}\Omega$ is immediate and profound. The quantity $\mathcal{S}^I$ encodes the *total* curvature of the informational substrate — all the geometric content that the statistical structure of the entropic substrate can carry. The quantity $(1/\lambda) \Phi^*(\mathcal{S}^S)$ encodes the portion of that curvature that has been successfully projected into spacetime geometry. The difference, $\mathcal{K}\Omega$, is the *residual information curvature* — the curvature that exists in the

informational substrate but is invisible to the classical spacetime description. We term this the "**informational dark curvature (IDC)**."

4.2 Properties

Proposition 4.1 (Vanishing in the Thermodynamic Limit)

$\mathcal{K}_\Omega(\theta) \rightarrow 0$ as $N \rightarrow \infty$. In the thermodynamic limit, all information curvature is fully expressed as spacetime curvature, and no residual informational dark curvature remains.

Proposition 4.2 (Positivity)

$\mathcal{K}_\Omega(\theta) \geq 0$ for all physically realizable configurations of the entropic substrate.

Proof: The emergence map $\Phi: \mathcal{M}_I \rightarrow \mathcal{M}_S$ is a smooth surjection, and thus a projection (in the fiber bundle sense) from the high-dimensional information manifold to the 4-dimensional spacetime. By O'Neill's theorem [16] on Riemannian submersions (appropriately generalized), the scalar curvature of the base space of a Riemannian submersion is bounded above by the scalar curvature of the total space, with equality if and only if the fibers are totally geodesic. More precisely, the curvature of the base includes additional non-negative contributions from the O'Neill A-tensor [16] (encoding the integrability obstruction of the horizontal distribution) and the T-tensor (encoding the second fundamental form of the fibers).

In our setting, the information scalar curvature $\mathcal{S}^I$ decomposes as:

$$\mathcal{S}^I = (1/\lambda) \Phi^*(\mathcal{S}^S) + \|A\|^2 + \|T\|^2 + (\text{fiber curvature terms})$$

where all additional terms are non-negative (being squared norms). Furthermore, the convexity of the Fisher information functional (a consequence of the log-sum inequality) [2] ensures that the fiber curvature terms are non-negative for any physically realizable distribution. Therefore:

$$\mathcal{K}_\Omega = \mathcal{S}^I - (1/\lambda) \Phi^*(\mathcal{S}^S) = \|A\|^2 + \|T\|^2 + (\text{fiber curvature}) \geq 0$$

Proposition 4.3 (Gauge Invariance)

The Obidi Curvature Invariant $\mathcal{K}_\Omega$ is invariant under (i) smooth reparameterizations of the statistical model $\mathcal{M}_I$, and (ii) diffeomorphisms of the spacetime manifold $\mathcal{M}_S$.

Proof: The information scalar curvature $\mathcal{S}^I$ is a scalar function on $\mathcal{M}_I$, constructed from the Riemann curvature tensor of the Fisher metric [2] by two contractions. As established in Proposition 2.2, the Fisher metric transforms as a (0,2) tensor under reparameterization; its scalar curvature is therefore a scalar invariant. Similarly, the spacetime scalar curvature $\mathcal{S}^S$ is a scalar invariant under diffeomorphisms of $\mathcal{M}_S$ by the general covariance of Riemannian geometry. The pullback $\Phi^*(\mathcal{S}^S)$ of a scalar is simply the composition $\mathcal{S}^S \circ \Phi$, which is a scalar on $\mathcal{M}_I$. The coupling constant λ is a fixed dimensional constant. Therefore, $\mathcal{K}_\Omega = \mathcal{S}^I - (1/\lambda) \Phi^*(\mathcal{S}^S)$ is a scalar on $\mathcal{M}_I$, invariant under both reparameterization and spacetime diffeomorphisms.

Lemma 4.4 (Bound on Informational Dark Curvature)

For any compact region $\mathcal{U} \subset \mathcal{M}_I$ with smooth boundary $\partial\mathcal{U}$:

$$\int_{\mathcal{U}} \mathcal{K}_{\Omega} dV^I \leq (1/N) \cdot C(\mathcal{U})$$

where dV^I is the Riemannian volume element of $(\mathcal{M}_I, g^I)$, and $C(\mathcal{U})$ is a constant depending only on the topology of $\mathcal{U}$ (specifically, its Euler characteristic and Betti numbers) and the total Fisher information [2] of the boundary, $\int_{\partial\mathcal{U}} g^I_{ij} n^i n^j dA^I$, where n is the outward unit normal.

4.3 Interpretation and Significance

The Obidi Curvature Invariant $\mathcal{K}_{\Omega}$ is, to the best of our knowledge, the first scalar quantity in the literature that precisely measures the degree to which a pre-geometric informational substrate fails to be fully captured by a classical spacetime description. Its properties, established above, paint a coherent and physically compelling picture:

6. **Classical limit (Proposition 4.1).** In the thermodynamic limit $N \rightarrow \infty$, the invariant vanishes: $\mathcal{K}_{\Omega} \rightarrow 0$. This is precisely the regime in which classical general relativity is an adequate description. The vanishing of $\mathcal{K}_{\Omega}$ confirms that, in this limit, *all* information curvature is faithfully projected into spacetime curvature — no degrees of freedom are lost in the emergence process.
7. **Quantum gravitational regime.** At finite N — particularly near the Planck scale where N is of order unity — the invariant $\mathcal{K}_{\Omega}$ is generically nonzero. This nonvanishing signals the breakdown of the classical spacetime description: there exist informational degrees of freedom in the substrate that *cannot* be encoded in

a 4-dimensional Riemannian geometry. These are, we propose, the quantum gravitational degrees of freedom.

8. **Positivity (Proposition 4.2).** The invariant is non-negative: the information manifold always carries at least as much curvature as the spacetime it generates. Spacetime curvature is a *coarse-grained shadow* of a richer informational structure, never a magnification of it. This is the geometric expression of a general principle: coarse-graining loses information but never creates it.
9. **Topological control (Lemma 4.4).** The integrated dark curvature over any compact region is bounded by $(1/N) \cdot \mathcal{C}(\mathcal{U})$. This bound is simultaneously an assurance that the corrections are small in the classical regime and a constraint on the *maximum possible* departure from classicality in any region.

We now state the principal conjecture of this letter.

Conjecture 4.5 (Obidi)

In the Planckian regime ($N \sim 1$), the Obidi Curvature Invariant $\mathcal{K}_\Omega$ encodes the complete set of quantum gravitational degrees of freedom. Any consistent theory of quantum gravity must, in its semiclassical limit, reproduce the expansion $\mathcal{K}_\Omega = (1/N) \kappa + \mathcal{O}(1/N^2)$ established in Lemma 4.4.

Finally, we observe a tantalizing connection to observational cosmology. The spatially averaged value $\langle \mathcal{K}_\Omega \rangle$ over cosmological scales need not vanish even in the large- N regime if the number of relevant microstates N varies spatially — as it might in regions of differing vacuum energy density or near cosmological horizons where the accessible Hilbert space

dimension is finite. A nonzero $\langle \mathcal{K}_\Omega \rangle$ would contribute to the effective cosmological constant via:

$$\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + (\lambda / 2) \langle \mathcal{K}_\Omega \rangle$$

This raises the speculative but testable possibility that the observed cosmological constant — one of the deepest puzzles in theoretical physics — may arise in part from the residual informational dark curvature of the entropic substrate. We defer detailed cosmological calculations to Letter IIB.

5. Structural Theorems of Information-to-Spacetime Emergence

Having established the **Curvature Transfer Theorem (CTT)** and the **Obidi Curvature Invariant (OCI)**, we now prove three structural theorems that illuminate the mathematical and physical architecture of the emergence framework.

Theorem 5.1 (Uniqueness of the Emergence Map)

Let g^S be a Lorentzian metric on $\mathcal{M}_S$ and suppose the thermodynamic limit relation $\Phi^* g^S = \lambda g^I$ holds for a smooth surjection $\Phi: \mathcal{M}_I \rightarrow \mathcal{M}_S$. Then Φ is unique up to isometries of $(\mathcal{M}_S, g^S)$. That is, if Φ' is another smooth surjection satisfying $(\Phi')^* g^S = \lambda g^I$, then there exists an isometry $\psi: \mathcal{M}_S \rightarrow \mathcal{M}_S$ such that $\Phi' = \psi \circ \Phi$.

Theorem 5.1 asserts that the emergent spacetime geometry is essentially unique: the only freedom in the emergence map is the freedom to apply a symmetry of the spacetime metric

— precisely the gauge freedom (diffeomorphism invariance) that general relativity already accommodates.

Theorem 5.2 (Topological Constraint)

The topology of the spacetime manifold $\mathcal{M}_S$ is constrained by the information topology of $\mathcal{M}_I$. Specifically, the fundamental group of spacetime is a quotient of the fundamental group of the information manifold:

$$\pi_1(\mathcal{M}_S) \cong \pi_1(\mathcal{M}_I) / \ker(\Phi^*)$$

where $\Phi^*: \pi_1(\mathcal{M}_I) \rightarrow \pi_1(\mathcal{M}_S)$ is the homomorphism induced by the emergence map. Consequently, any non-trivial loop in spacetime must lift to a non-trivial loop (or a loop in the kernel of Φ^*) in the information manifold. Physical spacetime cannot possess topological features — non-contractible loops, handles, non-trivial homotopy — that are not already present in or derivable from the topology of $\mathcal{M}_I$.

Theorem 5.2 has a striking physical implication: the topology of spacetime is *inherited* from the topology of the information manifold. If, for example, the information manifold is simply connected ($\pi_1(\mathcal{M}_I) = 0$), then spacetime must also be simply connected. Conversely, topological features of spacetime — such as the non-trivial topology of a wormhole or a spatial torus — must originate from corresponding topological features of the informational substrate.

Proposition 5.3 (Entropy–Curvature Duality)

Let Σ be a closed 2-surface in $\mathcal{M}_S$ (e.g., the horizon of a black hole). Then the Bekenstein–Hawking entropy of Σ is recovered from the information scalar curvature:

$$S_{\text{BH}}(\Sigma) = (1 / 4\lambda) \int_{\Phi^{-1}(\Sigma)} \mathcal{S}^I dA^I$$

where dA^I is the area element on the preimage $\Phi^{-1}(\Sigma) \subset \mathcal{M}_I$ induced by g^I .

Proposition 5.3 is a non-trivial consistency check on the entire framework. The Bekenstein–Hawking entropy formula [5, 6, 18] — arguably the most robust result in quantum gravity, derived independently from multiple approaches (Euclidean path integrals, string microstate counting, loop quantum gravity) — emerges naturally from the information-geometric structure of the entropic substrate. The fact that the formula is recovered with the correct numerical coefficient (given our identification of λ) provides strong evidence that the Theory of Entropicity (ToE) captures genuine physics.

6. Discussion and Outlook

This letter has established the following results within the framework of the Theory of Entropicity (ToE):

10. The construction of the information manifold $(\mathcal{M}_I, g^I)$ from the Fisher–Entropic metric on a fundamental entropic substrate Ω , equipped with intrinsic curvature tensors R^I , Ric^I , $\mathcal{S}^I$, and the information Einstein tensor G^I (Section 2).

11. The Curvature Transfer Theorem (Theorem 3.1), proving that the Riemann tensor of physical spacetime is the pushforward of the information Riemann tensor in the thermodynamic limit, with corrections of order $\mathcal{O}(1/N)$.
12. The recovery of Einstein's field equations [1] as an emergent identity, with the stress-energy tensor identified as a projection of the information Einstein tensor (Section 3.3).
13. The introduction of the Obidi Curvature Invariant (OCI) $\mathcal{K}_\Omega$, measuring the residual informational dark curvature, together with its key properties: vanishing in the classical limit, positivity, gauge invariance, and a topological bound (Section 4).
14. Structural theorems on the uniqueness of the emergence map (Theorem 5.1), topological constraints on spacetime (Theorem 5.2), and the recovery of the Bekenstein–Hawking entropy formula [5, 6, 18] (Proposition 5.3).

Several major open problems remain, each the subject of forthcoming letters in this series:

(a) Explicit construction of Φ for specific spacetimes. The present letter establishes the *existence* and *properties* of the emergence map axiomatically. A crucial next step is the explicit construction of Φ for physically important spacetimes — in particular, the Schwarzschild solution (static black hole), the Kerr solution (rotating black hole), and the Friedmann–Lemaître–Robertson–Walker (FLRW) cosmologies. For the Schwarzschild case, we anticipate that the relevant statistical model is a thermal state at the **Hawking temperature** [6, 18], and that the Fisher metric [2] on the corresponding exponential family reproduces the Schwarzschild geometry under the emergence map. These constructions will be presented in Letter IIB.

(b) Computation of $\mathcal{K}_\Omega$ in cosmological models. The Obidi Curvature Invariant (OCI) must be computed for realistic cosmological models to assess whether the spatially averaged $\langle \mathcal{K}_\Omega \rangle$ can contribute meaningfully to the effective cosmological constant. Preliminary estimates suggest that if the number of accessible microstates N scales as the exponential of the Hubble entropy ($\sim 10^{122}$), then $\langle \mathcal{K}_\Omega \rangle \sim 10^{-122}$ in Planck units — tantalizingly close to the observed value of the cosmological constant. A rigorous calculation is needed.

(c) Connection to holographic entanglement entropy. The Ryu–Takayanagi formula [15] and its generalizations (HRT, quantum extremal surfaces) relate entanglement entropy in a boundary CFT to the area of extremal surfaces in the bulk. The Theory of Entropicity (ToE) should make contact with this program: the emergence map Φ should be relatable to the holographic dictionary, and the Obidi Curvature Invariant should encode the bulk entanglement structure not captured by the extremal surface prescription. Exploring this connection is a high priority.

(d) Experimental signatures of nonzero $\mathcal{K}_\Omega$. If the informational dark curvature has physical consequences, these should in principle be observable. Candidate signatures include: modifications to the dispersion relation for high-energy particles propagating through regions of large $\mathcal{K}_\Omega$; corrections to the black hole quasi-normal mode spectrum; and deviations from the standard Λ CDM expansion history at early times when N was smaller. These predictions, once sharpened, may be confronted with data from gravitational wave observatories, cosmic microwave background measurements, and gamma-ray burst timing.

The program initiated here — deriving the geometry of spacetime from the geometry of information — represents a fundamental shift in perspective. In general relativity, spacetime is the stage on which physics unfolds. In the Theory of Entropicity (ToE), spacetime is itself a *performance*: an emergent, macroscopic pattern arising from the statistical architecture of a deeper informational substrate. The Obidi Curvature Invariant $\mathcal{K}_\Omega$ quantifies what is lost in this emergence — the informational dark curvature that classical geometry cannot see. Understanding and computing this invariant is, we believe, a key step toward a complete theory of quantum gravity.

Subsequent letters in this Living Review series will develop the program further. Letter IIB will construct explicit emergence maps for Schwarzschild and FLRW spacetimes. Letter IIC will address the incorporation of matter fields and gauge interactions within the information-geometric framework. Letter III will develop the quantum theory of the Obidi Curvature Invariant and its role in a non-perturbative formulation of quantum gravity.

Appendix A: Notation and Conventions

The following table summarizes the principal symbols and conventions used throughout this letter.

Symbol	Description
Ω	Entropic substrate: the space of microstates
$\mathcal{F}$	σ -algebra on Ω
μ_0	Reference measure on $(\Omega, \mathcal{F})$
p_θ	Probability distribution on Ω parametrized by θ
$\theta = (\theta^1, \dots, \theta^n)$	Macroscopic coordinates (parameters of the statistical model)
$\Theta \subset \mathbb{R}^n$	Parameter space
$\mathcal{M}_I$	Information manifold: the statistical model $\{p_\theta : \theta \in \Theta\}$
g^I_{ij}	Fisher–Entropic metric on $\mathcal{M}_I$
$\Gamma^{l,k}_{ij}$	Christoffel symbols of the Levi-Civita connection on $(\mathcal{M}_I, g^I)$
$R^{l,l}_{ijk}$	Riemann curvature tensor of $(\mathcal{M}_I, g^I)$
Ric^I_{ij}	Ricci tensor of $(\mathcal{M}_I, g^I)$
$\mathcal{S}^I$	Scalar curvature of $(\mathcal{M}_I, g^I)$
G^I_{ij}	Einstein tensor of $(\mathcal{M}_I, g^I)$: $\text{Ric}^I_{ij} - \frac{1}{2} \mathcal{S}^I g^I_{ij}$
$\mathcal{M}_S$	Spacetime manifold (4-dimensional, Lorentzian)
$g^S_{\mu\nu}$	Spacetime metric on $\mathcal{M}_S$, signature $(-, +, +, +)$
$R^S_{\mu\nu\rho\sigma}$	Riemann curvature tensor of $(\mathcal{M}_S, g^S)$
$\text{Ric}^S_{\mu\nu}$	Ricci tensor of $(\mathcal{M}_S, g^S)$
$\mathcal{S}^S$	Scalar curvature of $(\mathcal{M}_S, g^S)$
$G^S_{\mu\nu}$	Einstein tensor of $(\mathcal{M}_S, g^S)$: $\text{Ric}^S_{\mu\nu} - \frac{1}{2} \mathcal{S}^S g^S_{\mu\nu}$

Symbol	Description
Φ	Emergence map: smooth surjection $\mathcal{M}_I \rightarrow \mathcal{M}_S$
Φ^*, Φ_*	Pullback and pushforward along Φ , respectively
λ	Coupling constant, units [length ² / information]; $\lambda = \ell_P^2 / (4k_B)$
N	Number of accessible microstates (thermodynamic parameter)
$\mathcal{K}_\Omega$	Obidi Curvature Invariant (OCI): $\mathcal{S}^I - (1/\lambda) \Phi^*(\mathcal{S}^S)$
S_{BH}	Bekenstein–Hawking entropy
ℓ_P	Planck length: $\sqrt{(\hbar G/c^3)}$
$\pi_1(\cdot)$	Fundamental group
$\chi(\cdot)$	Euler characteristic
∂_i	Partial derivative $\partial/\partial\theta^i$
∇^I	Covariant derivative with respect to Γ^I
dV^I, dA^I	Volume and area elements on $(\mathcal{M}_I, g^I)$
$\mathcal{O}(\cdot)$	Asymptotic order notation (Bachmann–Landau)

Index conventions. Greek indices ($\mu, \nu, \rho, \sigma, \dots$) run over spacetime dimensions 0, 1, 2, 3. Latin indices ($i, j, k, l, \dots$) run over information-manifold dimensions 1, ..., n. The Einstein summation convention is used throughout: repeated upper-lower index pairs are summed. The spacetime signature is $(-, +, +, +)$. The Fisher metric is positive-definite (Riemannian). All units are SI unless stated otherwise; natural units ($\hbar = c = k_B = 1$) are used in selected formulae where indicated.

* * *

Author Note

John Onimisi Obidi (jonimisiobidi@gmail.com) is the originator and developer of the **Theory of Entropicity (ToE)**, an entropy-first framework seeking to reformulate the conceptual and mathematical foundations of modern theoretical physics.

Research Lab, The Aether.

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Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

The Theory of Entropicity (ToE) Living Review Letters Series (ToE-LRLS). Letter IE

John Onimisi Obidi (jonimisiobidi@gmail.com)

Research Lab, The Aether

May 6, 2026

THE THEORY OF ENTROPICITY (TOE) — LIVING REVIEW LETTERS SERIES

A

SUPPLEMENTARY APPENDICES A TO ToE LETTER IE

**Supplementary Appendices A to Letter IE:
The Generalized Entropic Field Equations — How the Entropic Field Generates Both
Spacetime Geometry and Matter-Energy, Subsuming Einstein's Field Equations via the Full
LOA/SOA Architecture and Fisher-Rao, Fubini-Study, and Amari-Čencov Information
Geometry**

A Complete Construction of the Generalized Entropic Field Equations (GEFE/OFE) [of the Obidi
Field Equations (OFE)] in Which Both the Geometric Left-Hand Side and the Matter-Energy Right-
Hand Side of Einstein's Field Equations Are Independently Generated by the Entropic Field, with
Einstein Recovered as a Classical Limit

John Onimisi Obidi

Research Lab, The Aether

jonimisiobidi@gmail.com

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ToE Living Review Letters Series — Letter IE Supplementary Appendices A

Keywords to the Appendices: Theory of Entropicity; Generalized Entropic Field Equations; Obidi Action; Local Obidi Action (LOA); Spectral Obidi Action (SOA); Einstein Field Equations; Entropic Manifold; Information Geometry; Fisher-Rao Metric; Fubini-Study Metric; Amari-Čencov Alpha-Connections; Entropic Stress-Energy Tensor; Curvature Transfer Theorem; Emergence Map; Entropic Einstein Equations; Entropic Ricci Tensor; Entropic Riemann Tensor; Information Curvature; Obidi Curvature Invariant (OCI); Entropic Dark Curvature; Entropic Cosmological Function; Entropic Renormalization Group; Holographic Principle; Arrow of Time; No-Rush Theorem; Vuli-Ndlela Integral; Alemoh-Obidi Correspondence; Entropic Speed Limit; Master Entropic Equation

"The right-hand side [of the gravitational field equations] is a formal condensation of all things whose comprehension in the sense of a field theory is still problematic. Not for a moment, of course, did I doubt that this formulation was merely a makeshift in order to give the general principle of relativity a preliminary closed-form expression."

— Albert Einstein, *The Meaning of Relativity* (1922), Appendix II, Fifth Edition (1955)

"Every it — every particle, every field of force, even the spacetime continuum itself — derives its function, its meaning, its very existence entirely from the apparatus-elicited answers to yes-or-no questions, binary choices, bits. It from bit symbolizes the idea that every item of the physical world has at bottom — at a very deep bottom, in most instances — an immaterial source and explanation."

— John Archibald Wheeler, "Information, Physics, Quantum: The Search for Links" (1990)

"The Einstein equation is an equation of state. It is born in the thermodynamics of the vacuum."

— Ted Jacobson, "Thermodynamics of Spacetime: The Einstein Equation of State" (1995)

"Gravity is explained as an entropic force caused by changes in the information associated with the positions of material bodies."

— Erik Verlinde, "On the Origin of Gravity and the Laws of Newton" (2010)

Introduction to the Appendices A to the ToE LRLS Letter IE

Letter IE of the Theory of Entropicity Living Review Letters Series established the fundamental mechanism by which physical spacetime geometry emerges from the information geometry of the entropic field. Through the Curvature Transfer Theorem and the emergence map $\Phi : M_I \rightarrow M_S$, Letter IE demonstrated that the Riemann curvature tensor of physical spacetime is the pushforward of the Riemann curvature tensor of the Fisher-Rao information manifold, with corrections captured by the Obidi Curvature Invariant (OCI). This was a landmark result: it showed that the left-hand side of Einstein's field equations — the Einstein tensor $G_{\mu\nu}$, which encodes the curvature of spacetime — is not fundamental but emergent, generated by the entropic field $S(x)$ acting on the substrate Ω .

However, Letter IE, by the constraints of its scope, primarily focused on the *geometric* side of the correspondence. The left-hand side of Einstein's equations was derived from entropic principles, but the right-hand side — the stress-energy tensor $T_{\mu\nu}$, which encodes the matter-energy content of spacetime — was treated largely through identification and correspondence rather than through an independent and complete construction from the entropic field. This left the architecture of the full entropic field equations incomplete. The present Supplementary Appendices rectify this asymmetry and complete the picture.

The purpose of these appendices is fivefold. **First**, we construct the full Generalized Entropic Field Equations (GEFE/OFE) [the **Obidi Field Equations (OFE)**], in which *both* sides — the geometric left-hand side and the matter-energy right-hand side — are independently and completely generated by the entropic field. The GEFE/OFE takes the form: ***Entropic Generator of Spacetime Geometry = Entropic Generator of Matter-Energy-Stress***. **Second**, we develop the complete information-geometric machinery — the **Fisher-Rao metric**, the **Fubini-Study metric**, and the

Amari-Čencov alpha-connections — in their full gravitational context, showing how each element of the mathematical apparatus contributes to the construction of one or both sides of the field equations. **Third**, we demonstrate how the **Local Obidi Action (LOA)** generates the local differential structure of spacetime and how the **Spectral Obidi Action (SOA)** generates the global spectral and topological structure, with the **full Obidi Action** $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$ providing the complete gravitational dynamics. Fourth, we prove rigorously that Einstein's field equations are recovered as the classical, weak-curvature, low-entropy limit of the GEFE/OFE, establishing general relativity as a limiting case of the Theory of Entropicity. Fifth, we establish the entropic field as the common generator of both geometry and matter, realizing the deep ontological unification that Einstein sought for the last three decades of his life but could not achieve.

These appendices should be read in conjunction with Letter IE (the main text above), Letter IIA (which derives the Maxwell equations from the phase sector of the entropic field, demonstrating the entropic origin of electromagnetism), and Letter IC (which establishes the Alemoh-Obidi Correspondence (AOC) between information-theoretic and thermodynamic quantities). Together, these documents form a self-contained exposition of the gravitational and electromagnetic sectors of the Theory of Entropicity, with the **GEFE/OFE** serving as the master equation from which all classical field theories — Einsteinian gravity and Maxwellian electromagnetism — emerge as limiting cases of a single entropic dynamics.

The mathematical prerequisites for these appendices include familiarity with differential geometry (Riemannian manifolds, connections, curvature tensors), general relativity (**Einstein's field equations**, geodesic motion, the **Schwarzschild** and **FLRW** solutions), information geometry (**Fisher-Rao metric**, **statistical manifolds**, **exponential families**), and variational calculus (**action principles**, **Euler-Lagrange equations**). Where specialized results from these fields are required, we develop them from first principles within the appendices themselves.

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Appendix A — Einstein's Field Equations: The Structure to Be Subsumed

Before constructing the Generalized Entropic Field Equations (the Obidi Field Equations - OFE), we must first understand in complete detail the structure that is to be subsumed and the onerous magnitude of the task before us.

Einstein's field equations stand as the crowning achievement of classical gravitational physics — **a set of ten coupled, nonlinear, second-order partial differential equations relating the curvature of four-dimensional spacetime to the distribution of matter and energy within it. Published in their final form in November 1915, after years of intense labor, these equations redefined our understanding of gravity as geometry and inaugurated the modern era of gravitational physics.** Every prediction of general relativity — from the perihelion precession of Mercury to the detection of gravitational waves a century later — flows from these equations. To subsume them, which is not at all an elementary undertaking, is not to diminish them but to reveal the deeper structure from which they arise, much as Maxwell's equations are not diminished by being shown to emerge from the phase sector of the entropic field in the ToE Letter IIA.

A.1 The Architecture of Einstein's Field Equations

Einstein's field equations, in their complete form including the cosmological constant, are written as:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G / c^4) T_{\mu\nu} \text{ (IE-S.1)}$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu}$ is the Ricci curvature tensor, $R = g^{\mu\nu}R_{\mu\nu}$ is the Ricci scalar (the scalar curvature), Λ is the cosmological constant, $g_{\mu\nu}$ is the spacetime metric tensor, G is Newton's gravitational constant ($G \approx 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$), c is the speed of light in vacuum ($c \approx 2.998 \times 10^8 \text{ m s}^{-1}$), and $T_{\mu\nu}$ is the stress-energy tensor of matter and energy fields. The Greek indices μ, ν range over the four spacetime dimensions 0, 1, 2, 3 (where 0 is the time coordinate), and the Einstein summation convention is employed throughout: repeated upper and lower indices are summed over.

These equations are remarkable for their compactness, but the compactness is deceptive. Each symbol encodes a rich mathematical structure. The left-hand side, $G_{\mu\nu} + \Lambda g_{\mu\nu}$, is constructed entirely from the spacetime metric $g_{\mu\nu}$ and its first and second derivatives — it is a statement about the geometry of spacetime and nothing else. The right-hand side, $(8\pi G/c^4)T_{\mu\nu}$, is a statement about the matter and energy content of spacetime — what fields are present, how they are distributed, how they move. The equation relates these two conceptually distinct domains: geometry tells matter how to move; matter tells geometry how to curve. Let us now dissect each side in complete detail.

The Left-Hand Side: Spacetime Geometry

The left-hand side of Einstein's field equations is built entirely from the spacetime metric $g_{\mu\nu}$ and its derivatives. The metric is a symmetric, rank-2 tensor field defined at every point of the spacetime manifold M . In coordinates $x^\mu = (t, x^1, x^2, x^3)$, the spacetime interval is given by:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \text{ (IE-S.2)}$$

The metric has Lorentzian signature $(-, +, +, +)$, meaning that at each point, the eigenvalues of $g_{\mu\nu}$ consist of one negative and three positive values. This signature encodes the causal structure of spacetime: timelike intervals ($ds^2 < 0$) correspond to physically realizable trajectories, spacelike intervals ($ds^2 > 0$) correspond to spatial separations, and null intervals ($ds^2 = 0$) correspond to the paths of light rays.

The Christoffel Connection. From the metric, one constructs the Levi-Civita connection (Christoffel symbols of the second kind), which encode how the coordinate basis vectors change from point to point on the manifold:

$$\Gamma^\alpha_{\beta\mu} = \frac{1}{2} g^{\alpha\lambda} (\partial_\mu g_{\lambda\beta} + \partial_\beta g_{\lambda\mu} - \partial_\lambda g_{\beta\mu}) \text{ (IE-S.3)}$$

The Christoffel symbols are symmetric in their lower indices: $\Gamma^\alpha_{\beta\mu} = \Gamma^\alpha_{\mu\beta}$, which is equivalent to the statement that the Levi-Civita connection is torsion-free. They are not tensors — they transform inhomogeneously under coordinate changes — but the curvature quantities constructed from them are genuine tensors. The Christoffel symbols encode the gravitational field in the same sense that the electromagnetic potential A_μ encodes the electromagnetic field: they are the connection (gauge field) from which the curvature (field strength) is computed.

The Riemann Curvature Tensor. The fundamental measure of spacetime curvature is the Riemann curvature tensor, constructed from the Christoffel symbols and their first derivatives:

$$R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\nu\beta} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\mu\beta} \text{ (IE-S.4)}$$

The Riemann tensor has 256 components in four dimensions (4^4), but its symmetries reduce the number of independent components to 20. These symmetries are:

- $R_{\alpha\beta\mu\nu} = -R_{\beta\alpha\mu\nu}$ (antisymmetry in the first pair of indices)
- $R_{\alpha\beta\mu\nu} = -R_{\alpha\beta\nu\mu}$ (antisymmetry in the second pair)

- $R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}$ (pair symmetry)
- $R_{\alpha\beta\mu\nu} + R_{\alpha\mu\nu\beta} + R_{\alpha\nu\beta\mu} = 0$ (first Bianchi identity / algebraic Bianchi identity)

The Riemann tensor has a profound geometric meaning: it measures the failure of parallel transport around infinitesimal closed loops. If a vector V^α is parallel transported around a small loop spanned by vectors δx^μ and δx^ν , the change in the vector is:

$$\delta V^\alpha = R^\alpha_{\beta\mu\nu} V^\beta \delta x^\mu \delta x^\nu \text{ (IE-S.5)}$$

This is the gravitational analogue of the Aharonov-Bohm effect in electromagnetism, where the phase of a charged particle changes upon transport around a loop enclosing magnetic flux. The Riemann tensor is the gravitational field strength, just as the electromagnetic field tensor $F_{\mu\nu}$ is the electromagnetic field strength.

The Ricci Tensor. The Ricci tensor is obtained by contracting the Riemann tensor on its first and third indices:

$$R_{\mu\nu} = R^\alpha_{\mu\alpha\nu} = \partial_\alpha \Gamma^\alpha_{\nu\mu} - \partial_\nu \Gamma^\alpha_{\alpha\mu} + \Gamma^\alpha_{\alpha\lambda} \Gamma^\lambda_{\nu\mu} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\alpha\mu} \text{ (IE-S.6)}$$

The Ricci tensor is symmetric: $R_{\mu\nu} = R_{\nu\mu}$, and it has 10 independent components in four dimensions. Geometrically, the Ricci tensor measures how volumes change under parallel transport: a positive Ricci component in a given direction indicates that nearby geodesics converge (focusing), while a negative component indicates divergence (defocusing). The Ricci tensor captures the "trace" part of the Riemann curvature — the part directly coupled to matter-energy sources through Einstein's equations.

The Ricci Scalar. The Ricci scalar is the trace of the Ricci tensor:

$$R = g^{\mu\nu} R_{\mu\nu} \text{ (IE-S.7)}$$

This single number at each spacetime point provides a measure of the overall (scalar) curvature. For flat Minkowski spacetime, $R = 0$ everywhere. For a sphere of radius r (in the spatial section), $R = 2/r^2$. The Ricci scalar appears in the Einstein-Hilbert action as the Lagrangian density for the gravitational field.

The Einstein Tensor. The Einstein tensor is defined as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \text{ (IE-S.8)}$$

Like the Ricci tensor, the Einstein tensor is symmetric and has 10 independent components. Its defining property, which makes it the appropriate object for the left-hand side of the field equations, is that it satisfies the contracted Bianchi identity automatically:

$$\nabla^\mu G_{\mu\nu} = 0 \text{ (IE-S.9)}$$

where ∇^μ denotes the covariant divergence with respect to the Levi-Civita connection. This identity is purely geometric — it holds for any Riemannian or pseudo-Riemannian manifold regardless of any field equations — and it follows from the differential Bianchi identity $\nabla_{[\lambda} R^\alpha_{\beta]\mu\nu} = 0$ by contraction. The contracted Bianchi identity ensures that the left-hand side of Einstein's equations is automatically divergence-free, which, through the field equations, implies the covariant conservation of the stress-energy tensor: $\nabla^\mu T_{\mu\nu} = 0$. This is a beautiful example of geometry guaranteeing physics: the conservation of energy-momentum is not an independent postulate but a consequence of the geometric structure of the field equations.

The Cosmological Constant. The cosmological constant Λ was introduced by Einstein in 1917 to allow static cosmological solutions. After the discovery of the expanding universe by Hubble in 1929, Einstein allegedly called it his "greatest blunder." However, the discovery of the accelerating expansion of the universe in 1998 by the Supernova Cosmology Project and the High- z Supernova Search Team demonstrated that a positive cosmological constant (or something dynamically

equivalent to it) is in fact required. In the field equations, $\Lambda g_{\mu\nu}$ can be placed on either side: on the left, it represents a modification of the geometry (a constant curvature background); on the right, it represents a uniform vacuum energy density $\rho_{\text{vac}} = \Lambda c^2 / (8\pi G)$ with equation of state $w = p/\rho c^2 = -1$. In the Theory of Entropicity, the cosmological constant will be shown to emerge naturally from the a_0 Seeley-DeWitt coefficient of the SOA, resolving its ambiguous status in general relativity.

The Right-Hand Side: Matter and Energy

The right-hand side of Einstein's field equations is the stress-energy tensor $T_{\mu\nu}$, multiplied by the gravitational coupling constant $\kappa = 8\pi G/c^4 \approx 2.077 \times 10^{-43} \text{ N}^{-1}$. The stress-energy tensor is a symmetric, rank-2 tensor that encodes the density and flux of energy and momentum in spacetime. Its components, relative to an observer with four-velocity u^μ , have the following physical interpretations:

- T^{00} = energy density (energy per unit volume)
- $T^{0i} = T^{i0}$ = momentum density = energy flux (in the i -th spatial direction)
- T^{ij} = stress tensor (flux of i -th component of momentum in the j -th direction), including pressure ($i = j$) and shear stress ($i \neq j$)

The stress-energy tensor is not a single entity — it is the sum of contributions from all matter and energy fields present in spacetime. For each type of matter, there is a corresponding stress-energy tensor, and the total $T_{\mu\nu}$ is the sum of all contributions. We now catalogue the most important examples.

Perfect Fluid. The simplest matter model is the perfect fluid, characterized by an energy density ρ , a pressure p , and a four-velocity u^μ . Its stress-energy tensor is:

$$T_{\mu\nu} = (\rho + p/c^2) u_\mu u_\nu + p g_{\mu\nu} \text{ (IE-S.10)}$$

This encompasses dust ($p = 0$, non-relativistic matter), radiation ($p = \rho c^2/3$, relativistic matter), and vacuum energy ($p = -\rho c^2$, cosmological constant). The perfect fluid is the workhorse of cosmology: the Friedmann equations, which govern the expansion of the universe, are derived from Einstein's field equations with a perfect fluid source.

Electromagnetic Field. The stress-energy tensor of the electromagnetic field is constructed from the electromagnetic field tensor $F_{\mu\nu}$:

$$T_{\mu\nu}^{(EM)} = (1/\mu_0) [F_{\mu\alpha}F_{\nu}^{\alpha} - \frac{1}{4} g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}] \text{ (IE-S.11)}$$

where μ_0 is the permeability of free space and $F_{\mu\nu} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}$ is constructed from the electromagnetic four-potential A_{μ} . This tensor is traceless ($T^{(EM)} = g^{\mu\nu}T_{\mu\nu}^{(EM)} = 0$), reflecting the conformal invariance of classical electrodynamics in four dimensions. In the Theory of Entropicity, Letter IIA has shown that the electromagnetic field tensor $F_{\mu\nu}$ itself emerges from the phase sector of the complexified entropic field, so the electromagnetic stress-energy tensor is ultimately generated by the entropic field.

Scalar Field. A scalar field φ with potential $V(\varphi)$ has stress-energy tensor:

$$T_{\mu\nu}^{(\varphi)} = \partial_{\mu}\varphi \partial_{\nu}\varphi - g_{\mu\nu} [\frac{1}{2} \partial_{\alpha}\varphi \partial^{\alpha}\varphi + V(\varphi)] \text{ (IE-S.12)}$$

This form is particularly significant for the Theory of Entropicity, because the entropic field $S(x)$ is itself a scalar field, and its stress-energy tensor will take exactly this form (with S replacing φ and the entropic potential $V(S)$ replacing $V(\varphi)$). The entropic stress-energy tensor is thus the scalar field stress-energy tensor of the entropic field — but with the crucial distinction that the entropic field is not an independently postulated matter field but the fundamental ontological entity from which both geometry and matter emerge.

Conservation. The covariant conservation of the stress-energy tensor,

$$\nabla^\mu T_{\mu\nu} = 0 \text{ (IE-S.13)}$$

follows from the contracted Bianchi identity (IE-S.9) together with the field equations (IE-S.1). This conservation equation encodes the local conservation of energy and momentum in curved spacetime. It replaces the flat-spacetime conservation law $\partial^\mu T_{\mu\nu} = 0$ and includes the effects of the gravitational field through the covariant derivative. The conservation equation is not independent of the field equations — it is a consequence of them — and it provides the equations of motion for matter in curved spacetime. For a perfect fluid, $\nabla^\mu T_{\mu\nu} = 0$ yields the relativistic Euler equations; for dust, it yields the geodesic equation; for the electromagnetic field, it yields Maxwell's equations in curved spacetime.

A.2 The Conceptual Problem Einstein Left Unsolved

Einstein's field equations are, by any measure, one of the most successful and beautiful constructions in the history of physics. They have withstood over a century of observational testing, from the perihelion precession of Mercury (the first classical test, resolved in 1915) to the direct detection of gravitational waves by LIGO in 2015 and the imaging of black hole shadows by the Event Horizon Telescope in 2019. Yet beneath their empirical success lies a profound conceptual problem that Einstein himself recognized and spent the last three decades of his life attempting to resolve.

The problem is this: Einstein's field equations *relate* two independently existing structures — spacetime geometry (left-hand side) and matter-energy (right-hand side) — but they do not *explain* the origin of either. The equation says that curvature equals matter, but it does not tell us what generates the curvature or what generates the matter. Both sides are taken

as given — as fundamental, irreducible inputs to the theory — and the field equations merely specify the relationship between them.

Consider the left-hand side. The metric tensor $g_{\mu\nu}$ is the dynamical variable of general relativity. The field equations determine $g_{\mu\nu}$ given the matter distribution $T_{\mu\nu}$ (subject to boundary conditions). But what is the metric? Where does it come from? In general relativity, the metric is simply posited as the fundamental gravitational field — a rank-2 symmetric tensor field on a four-dimensional manifold. The number of dimensions (four), the signature (Lorentzian), and the very existence of a smooth manifold structure are all assumed. There is no explanation within general relativity for why spacetime has four dimensions rather than three or five; for why the signature is $(-, +, +, +)$ rather than $(-, -, +, +)$ or $(+, +, +, +)$; for why the manifold is smooth rather than discrete or fractal. The metric is a given — an unexplained starting point.

The right-hand side is even more problematic. The stress-energy tensor $T_{\mu\nu}$ encodes the matter and energy content of spacetime, but Einstein's theory provides no guidance on what types of matter should exist, what their properties should be, or why they interact with gravity through a symmetric rank-2 tensor. The matter fields — electromagnetic, scalar, fermionic, gauge — are postulated independently of the gravitational theory. Each matter field comes with its own Lagrangian, its own symmetries, its own coupling constants. The stress-energy tensor is then computed from these independently postulated fields and inserted into the right-hand side of the field equations. Einstein was acutely aware of this asymmetry. In the passage quoted in our epigraph, he described the right-hand side as "a formal condensation of all things whose comprehension in the sense of a field theory is still problematic." He viewed it as a placeholder — a makeshift — to be replaced by something more fundamental once a unified field theory was achieved.

The asymmetry between the two sides runs even deeper. **The left-hand side — the Einstein tensor — is constructed from the metric through a purely geometric operation: take**

derivatives of the metric (Christoffel symbols), take derivatives of the **Christoffel symbols (Riemann tensor)**, contract (**Ricci tensor**), trace (**Ricci scalar**), and combine (**Einstein tensor**). This entire construction is intrinsic to the geometry of the manifold. It requires no input beyond the metric itself. The **Bianchi identity** $\nabla^\mu G_{\mu\nu} = 0$ is automatic — it is a mathematical identity, not a physical law. The left-hand side is therefore geometrically self-contained: it is a machine that takes a metric as input and produces a divergence-free symmetric tensor as output.

The right-hand side, by contrast, requires the specification of entirely new structures: matter fields, their Lagrangians, their symmetry groups, their coupling constants. None of these are determined by the geometry. They are external inputs to the theory. Einstein's field equations do not unify geometry and matter — they *relate* them. The equation $G_{\mu\nu} = \kappa T_{\mu\nu}$ says that two independently existing things happen to be proportional, but it does not explain why they should be related at all, or what common structure might underlie both.

This is the problem that the Theory of Entropicity (ToE) resolves. The GEFÉ/OFE, as developed in these appendices, shows that both sides of Einstein's equations are generated by a single entity — the entropic field $S(x)$ — and that Einstein's equations are not a relationship between two independent structures but an identity expressing the self-consistency of a single structure manifesting in two complementary ways. This, once again, is Obidi's Principle of Complementarity (OPoC) in the Theory of Entropicity (ToE), as inaugurated in the Living Review Letters Series, Letter I.

A.3 What ToE Promises: Both Sides from One Source

The central thesis of the Theory of Entropicity (ToE), as it pertains to gravitation, is:

Central Thesis of ToE (Entropic Generation of Both Geometry and Matter)

The entropic field $S(x)$ on the entropic manifold M_S , through the full Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$, simultaneously and independently generates:

(a) The spacetime metric $g_{\mu\nu}$ and all associated curvature quantities (Einstein's left-hand side), via the Fisher-Rao information metric and the emergence map Φ ;

(b) The stress-energy tensor $T_{\mu\nu}$ and all matter-energy content (Einstein's right-hand side), via the Fubini-Study metric on the entropic state space and the Amari alpha-connections;

such that Einstein's field equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ are recovered as the classical, weak-curvature, low-entropy limit of the Generalized Entropic Field Equations (GEFE/OFE).

This thesis makes three claims of increasing strength.

The **first** claim is that **the spacetime metric — the fundamental variable of general relativity — is not fundamental but emergent**. It is generated by the entropic field through the Fisher-Rao information metric and the emergence map $\Phi : M_I \rightarrow M_S$. This claim was established in Letter IE above through the **Curvature Transfer Theorem (CTT)**.

The **second** claim is that **the stress-energy tensor — the matter-energy content of spacetime — is also not fundamental but emergent**. It is generated by the entropic field through the Fubini-Study metric on the entropic state space, the Amari alpha-connections on the statistical manifold, and the variational structure of the Obidi Action. This claim is the primary new contribution of these supplementary appendices.

The **third** claim is that **Einstein's field equations, which relate the metric to the stress-energy tensor, are not fundamental laws but limiting cases of a deeper set of field equations — the GEFE/OFE — in which both sides are entropic.**

The explanatory structure is therefore:

Entropic field $S(x) \rightarrow [\text{via Fisher-Rao} + \Phi] \rightarrow \text{Spacetime geometry } g_{\mu\nu}, G_{\mu\nu} \text{ (LHS)}$

Entropic field $S(x) \rightarrow [\text{via Fubini-Study} + \alpha\text{-connections}] \rightarrow \text{Matter-energy } T_{\mu\nu} \text{ (RHS)}$

$\text{LHS} = \text{RHS} \Leftrightarrow \text{Self-consistency of the entropic field}$

Thus, the **Theory of Entropicity (ToE)** teaches us that the **Einstein field equations (EFE)** are **not a mysterious correspondence between geometry and matter. They are the self-consistency condition of a single entropic field manifesting as both geometry and matter.** This is the sense in which the **GEFE/OFE** subsumes Einstein: **it reveals the common origin of both sides, explains why they must be related, and provides the explicit mechanism (information geometry) through which the relation is realized.**

A.4 Historical and Philosophical Context

The program of deriving gravity from more fundamental principles has a long and distinguished history, of which the Theory of Entropicity is the latest and most comprehensive chapter. To situate the **GEFE/OFE** within this intellectual tradition, we survey the major contributions and explain how the ToE approach goes beyond each of them.

Einstein's Search for Unification (1920–1955). Einstein himself spent the last three decades of his life searching for a unified field theory that would derive both gravity and electromagnetism from a single geometric structure. His approaches included the **affine field theory** (using the connection rather than the metric as the fundamental variable), the teleparallel theory (using torsion rather than curvature), and various higher-dimensional theories inspired by the Kaluza-Klein model (which unifies gravity and electromagnetism in five dimensions). **All of these attempts ultimately failed, for a reason that becomes clear in retrospect: Einstein was seeking to unify within the geometric framework — to derive matter from geometry — but geometry alone is insufficient to generate the full structure of matter. The ToE approach succeeds where Einstein stalled because it introduces a more fundamental substrate — the entropic field — from which both geometry and matter emerge.**

Wheeler's "Geometrodynamics" and "It from Bit" (1955–1990). John Archibald Wheeler extended Einstein's geometric vision in two phases. In his earlier work on geometrodynamics (1950s–1960s), Wheeler attempted to derive all particles and fields from the topology and geometry of spacetime itself. He envisioned "geons" — gravitational electromagnetic entities, objects made of curved spacetime and nothing else — as the fundamental constituents of matter. While pioneering, this program could not reproduce the observed spectrum of particles. In his later work (1980s–1990s), Wheeler pivoted to the information-theoretic program "it from bit," arguing that the physical world derives from information — from yes/no questions. Wheeler's vision was programmatic rather than mathematical: he articulated a compelling philosophical position but did not provide the full and complete mathematical framework to implement it. The Theory of Entropicity (ToE) provides exactly this framework. **The Obidi Action is the action principle for Wheeler's "bits"; the Fisher-Rao/Fubini-Study metrics constitute the metrics on Wheeler's information space; the emergence map Φ is the mechanism by which "it" emerges from "bit."**

Sakharov's Induced Gravity (1967). Andrei Sakharov proposed that the **Einstein-Hilbert action** is not fundamental but is induced by quantum fluctuations of matter fields. In Sakharov's approach, the gravitational action arises as a one-loop effective action when matter fields are integrated out. The resulting action has the form of the Einstein-Hilbert action with a gravitational constant determined by the UV cutoff and the number of matter species. **This is structurally analogous to the SOA contribution in the Theory of Entropicity, where the α_2 Seeley-DeWitt coefficient generates the Einstein-Hilbert action from the spectral geometry of the entropic Laplacian.** The key difference is that Sakharov assumes the pre-existence of a smooth spacetime manifold with a metric, whereas the ToE derives the manifold and metric from the entropic field.

Jacobson's Thermodynamic Derivation (1995). Ted Jacobson's landmark 1995 paper showed that Einstein's field equations can be derived from the Clausius relation $\delta Q = T dS$ applied to local **Rindler horizons**, combined with the Bekenstein-Hawking entropy-area relation $S = A/(4l_P^2)$ and the Unruh temperature $T = \hbar a/(2\pi c k_B)$. This was a profound result: it showed that Einstein's equations are an equation of state — a thermodynamic identity — rather than a fundamental dynamical equation. However, Jacobson's derivation assumes several structures: the existence of a smooth spacetime manifold, a metric, a causal structure (to define Rindler horizons), and the entropy-area relation (which is taken as an input). The ToE approach derives all of these: the manifold and metric from the Fisher-Rao metric and emergence map; the causal structure from the entropic speed limit; and the entropy-area relation from the OCI bounds.

Verlinde's Entropic Gravity (2010). Erik Verlinde proposed that gravity is an entropic force — a macroscopic phenomenon arising from the tendency of a system to increase its entropy, analogous to the elastic force of a polymer. Verlinde derived Newton's law of gravity $F = GMm/r^2$ by assuming (i) the **existence of holographic screens** with entropy proportional to area, (ii) the **equipartition of energy** on the screen, and (iii) the **Unruh temperature**. Verlinde's approach is elegant but limited: it derives only the Newtonian limit of gravity, not the full Einstein equations; it uses the **The Theory of Entropicity (ToE) Living Review Letters Series. Letter IE**

holographic principle as an input rather than deriving it; and it does not provide a dynamical framework (an **action principle** or **field equations**). The GEFE/OFE goes **beyond Verlinde in all three respects: it provides the full (non-Newtonian, non-linear) field equations; it derives the holographic principle from the OCI; and it provides the complete action principle (the Obidi Action).**

Padmanabhan's Emergent Gravity Program (2002–2015). The brilliant Indian physicist **Thanu Padmanabhan** developed a comprehensive program showing that various aspects of gravity — from the Einstein-Hilbert action to the field equations to the cosmological constant — can be understood as emergent thermodynamic phenomena. Padmanabhan showed that the Einstein-Hilbert action, evaluated on solutions of the field equations, is a surface term related to the entropy of horizons; that the field equations can be interpreted as a thermodynamic identity $T dS = dE + P dV$ on null surfaces; and that the cosmological constant can be understood as the difference between the surface degrees of freedom and the bulk degrees of freedom. **Padmanabhan's approach is more general than Jacobson's and Verlinde's, but still assumes the pre-existence of a smooth spacetime manifold with metric and causal structure. The ToE derives these structures from the entropic field.**

Connes' Spectral Action (1996). Alain Connes, in collaboration with Ali Chamseddine, showed that the Einstein-Hilbert action (plus higher-order gravitational corrections and the full Standard Model action) can be derived from a **spectral action principle $\text{Tr } f(D^2/\Lambda^2)$ on a noncommutative space**, where D is the Dirac operator and Λ is a cutoff scale. **This is structurally analogous to the SOA in the Theory of Entropicity (ToE), with the entropic Laplacian Δ_s replacing the Dirac operator and the entropic manifold replacing the noncommutative space.** The key conceptual difference is that Connes' noncommutative space is postulated as a mathematical structure, whereas the entropic manifold is derived from the entropic field and has a clear physical interpretation as the space of entropic configurations [**which is much more aligned with**

physical experience as a result of its being associated with the ubiquitous Second Law of Thermodynamics].

The Theory of Entropicity (ToE), through the GEFE/OFE developed in these appendices, thus represents the culmination of a century-long program initiated by Einstein himself. It goes beyond all previous approaches by providing a single, unified, mathematically precise framework in which both spacetime geometry and matter-energy emerge from a common entropic substrate, with Einstein's field equations recovered as a classical limit.

Appendix B — The Entropic Field as Generator of Spacetime Geometry (The Left-Hand Side)

In this appendix, we construct in full detail the entropic generator of spacetime geometry — the left-hand side of the GEFE/OFE. We show how the entropic field $S(x)$, through the Fisher-Rao information metric, the emergence map Φ , and the Local and Spectral Obidi Actions, generates the complete geometric structure of physical spacetime: the metric, the connection, the curvature, and the Einstein tensor. Every step in the construction is presented with full mathematical detail and physical commentary to assist and guide the reader in apprehending the underlying principles of the Theory of Entropicity (ToE).

B.1 The Information Manifold and the Fisher-Entropic Metric

The starting point of the entire construction is the entropic field $S(x)$ — or more generally as $S(\Lambda)$ — defined on the substrate Ω . As established in the foundational letters of the ToE series, the entropic field is the single fundamental degree of freedom of the theory. It is a real-valued scalar field defined on a substrate space Ω , which is the pre-geometric arena underlying physical reality. The substrate Ω has no a priori metric, topology, or differential structure — these structures emerge dynamically from the entropic field itself.

At each point x in the parameter space (which will eventually be identified, through the emergence map, with physical spacetime), the entropic field $S(\omega; x)$ defines a probability distribution on the substrate through the Gibbs-Boltzmann construction:

$$P(\omega; x) = \exp(-S(\omega; x)) / Z(x) \text{ (IE-S.14)}$$

where $Z(x)$ is the entropic partition function:

$$Z(x) = \int_{\Omega} d\omega \exp(-S(\omega; x)) \text{ (IE-S.15)}$$

The partition function ensures proper normalization: $\int_{\Omega} d\omega P(\omega; x) = 1$ for all x . The probability distribution $P(\omega; x)$ encodes the "information content" of the entropic field at the parameter point x : it describes which substrate configurations ω are probable (those with low S) and which are suppressed (those with high S).

As x varies over the parameter space, the distribution $P(\omega; x)$ traces out a family of probability distributions on Ω , parameterized by x . This family constitutes a *statistical manifold* — a manifold whose points are probability distributions. The natural Riemannian metric on this statistical manifold is the Fisher-Rao information metric.

To construct the Fisher-Rao metric, we first compute the *score functions*. The score function associated with the parameter x^μ is the logarithmic derivative of the probability distribution with respect to that parameter:

$$\partial_\mu \ln P(\omega; x) = -\partial_\mu S(\omega; x) + \partial_\mu \ln Z(x) = -\partial_\mu S(\omega; x) + \langle \partial_\mu S \rangle \quad (\text{IE-S.16})$$

where we have used the identity $\partial_\mu \ln Z = -\int d\omega P(-\partial_\mu S) = \langle \partial_\mu S \rangle$, with $\langle \cdot \rangle$ denoting the expectation value with respect to $P(\omega; x)$. The score function thus equals the negative fluctuation of the entropic gradient away from its mean: $\partial_\mu \ln P = -(\partial_\mu S - \langle \partial_\mu S \rangle)$.

The Fisher-Rao information metric is the covariance matrix of the score functions:

$$g_{\mu\nu}^I(x) = \int_\Omega d\omega P(\omega; x) [\partial_\mu \ln P][\partial_\nu \ln P] \quad (\text{IE-S.17})$$

Expanding using the score function expression (IE-S.16):

$$g_{\mu\nu}^I = \int d\omega P [\partial_\mu S - \langle \partial_\mu S \rangle][\partial_\nu S - \langle \partial_\nu S \rangle] \quad (\text{IE-S.18})$$

$$= \langle \partial_\mu S \cdot \partial_\nu S \rangle - \langle \partial_\mu S \rangle \langle \partial_\nu S \rangle \quad (\text{IE-S.19})$$

This is the ***variance-covariance structure (VCS)*** of the entropic gradients. The Fisher-Rao metric at a point x measures how much the probability distribution $P(\omega; x)$ changes as the parameters are varied. A large component $g_{\mu\nu}^I$ means that a small change in the parameters x^μ and x^ν produces a large change in the probability distribution — the distribution is "sensitive" to those parameters. A small component means the distribution is "insensitive." The Fisher-Rao metric thus quantifies the *distinguishability* of nearby probability distributions.

Proposition B.1 (Positive Definiteness of the Fisher-Rao Metric)

The Fisher-Rao metric $g^I_{\mu\nu}(x)$ is positive semi-definite for all x , and is strictly positive definite if and only if the score functions $\partial_\mu \ln P(\omega; x)$, $\mu = 0, 1, \dots, d-1$, are linearly independent as functions of ω .

Proof. For any vector v^μ , we have:

$$g^I_{\mu\nu} v^\mu v^\nu = \int d\omega P(\omega; x) [v^\mu \partial_\mu \ln P]^2 \geq 0$$

since the integrand is non-negative ($P \geq 0$ and the square is non-negative). Equality holds if and only if $v^\mu \partial_\mu \ln P = 0$ for P -almost all ω , i.e., the linear combination $v^\mu \partial_\mu \ln P$ vanishes. If the score functions are linearly independent, this implies $v^\mu = 0$, so the metric is positive definite. ■

The assumption of linear independence of the score functions is physically natural: it means that each parameter direction in the space x produces a genuinely different change in the probability distribution. If two parameter directions produced the same change (up to a constant), they would be indistinguishable from the information-theoretic perspective, and we would identify them (reduce the parameter dimension). In the ToE framework, where the parameters x^μ will ultimately be identified with spacetime coordinates, linear independence of the score functions means that the entropic field carries non-degenerate information about each spacetime direction — a condition that must hold wherever a well-defined spacetime exists.

We note an alternative form of the Fisher-Rao metric that is sometimes more convenient for computation. Since $\partial_\mu \ln P = \partial_\mu P / P$, we have:

$$g^I_{\mu\nu} = \int d\omega (1/P) (\partial_\mu P)(\partial_\nu P) = -\int d\omega P \partial_\mu \partial_\nu \ln P \text{ (IE-S.20)}$$

The second form follows by integration by parts and the normalization condition $\int d\omega \partial_\mu P = \partial_\mu \int d\omega P = 0$. This alternative expression is the negative expected Hessian of the log-likelihood, connecting the Fisher-Rao metric to the curvature of the log-likelihood function — a fundamental relationship in statistical estimation theory (the **Cramér-Rao bound**).

Theorem B.2 (Čencov Uniqueness Theorem, 1972)

Let M be a statistical manifold — a manifold whose points are probability distributions on a measurable space (Ω, Σ) . The Fisher-Rao metric g^{FR} is the unique (up to a positive constant multiple) Riemannian metric on M that is invariant under sufficient statistics (Markov morphisms / Markov embeddings).

The Čencov uniqueness theorem is of foundational importance for the Theory of Entropicity. It states that there is *no freedom* in the choice of metric on the information manifold — the Fisher-Rao metric is forced by the requirement of consistency under **information-preserving transformations**. A sufficient statistic is a function of the data that captures all the information about the parameters; invariance under sufficient statistics means that the metric does not change when redundant information is discarded. **This is the information-geometric analogue of the requirement that physical laws be invariant under coordinate transformations. Just as general covariance uniquely constrains the form of the Einstein equations (up to the Lovelock theorem), invariance under sufficient statistics uniquely constrains the metric on the statistical manifold to be the Fisher-Rao metric.**

For the Theory of Entropicity, the significance of Čencov's theorem is **profound**: the metric on the information manifold — and hence, through the **emergence map**, the metric on physical

spacetime — is not a choice but a consequence. *The entropic field determines the probability distributions, the probability distributions determine the statistical manifold, and the Čencov uniqueness theorem determines the metric on that manifold. There is exactly one consistent way to measure distances in the space of probability distributions, and that measurement, when pushed forward to physical spacetime, becomes the spacetime metric.* This is an extraordinary explanatory gain over general relativity, where the metric is a free dynamical variable with no a priori constraints beyond the field equations themselves.

B.2 The Emergence Map Φ and the Generation of Physical Spacetime

The information manifold (M_I, g_I) constructed in Section B.1 is not yet physical spacetime. It is an abstract mathematical space — the space of probability distributions parameterized by x (or more generally, Λ), equipped with the Fisher-Rao (Fubini-Study) metric. Physical spacetime (M_S, g_S) is a Lorentzian manifold with specific causal structure, dynamics, and matter content. The bridge between the information manifold and physical spacetime is the *emergence map*, introduced and developed in Letter IE (as above, and expressed here below):

Definition B.3 (Emergence Map)

The emergence map is a smooth, surjective map $\Phi : M_I \rightarrow M_S$ from the information manifold to the physical spacetime manifold, satisfying the conformal relation:

$$\Phi^* g_S = \lambda g_I$$

where Φ^* denotes the pullback of the spacetime metric, g_I is the Fisher-Rao information metric, and λ is the emergence coupling constant:

$$\lambda = l_P^2 / (4k_B) = \hbar G / (4k_B c^3)$$

where $l_P = \sqrt{(\hbar G/c^3)}$ is the Planck length and k_B is Boltzmann's constant.

The conformal relation $\Phi^*g_S = \lambda g_I$ is the mathematical statement of the emergence of spacetime from information. *It says that the physical spacetime metric, when pulled back to the information manifold, is proportional to the Fisher-Rao metric. Equivalently, the physical spacetime metric is the pushforward of the Fisher-Rao metric, scaled by the emergence coupling constant:*

$$g_{\mu\nu}^S(x) = \lambda g_{\mu\nu}^I(\Phi^{-1}(x)) \text{ (IE-S.21)}$$

when Φ is a diffeomorphism (which it is locally, by the inverse function theorem, since it is smooth and surjective with non-degenerate Jacobian). **This equation is the fundamental equation of emergent spacetime geometry in the Theory of Entropicity (ToE).** Let us examine its structure and implications in detail.

The emergence coupling constant $\lambda = l_P^2/(4k_B)$ sets the scale at which information-geometric distances translate into physical spacetime distances. Its numerical value is approximately $\lambda \approx 1.90 \times 10^{-47} \text{ m}^2/\text{J}\cdot\text{K}^{-1}$ — extraordinarily small, reflecting the enormous number of entropic degrees of freedom required to produce macroscopic spacetime geometry. The fact that λ involves the Planck length l_P and Boltzmann's constant k_B is physically significant: the Planck length is the scale at which quantum gravitational effects become important, and Boltzmann's constant is the conversion factor between information (in nats) and thermodynamic entropy. The emergence coupling constant therefore sits at the intersection of quantum gravity and information theory — exactly where we would expect the bridge between information and spacetime to reside.

The physical spacetime metric $g^S_{\mu\nu}$ inherits all of its properties from the Fisher-Rao metric. The signature of g^S is determined by the signature of g_I (which, for a real-valued entropic field with minimal substrate structure, has Lorentzian signature $(-, +, +, +)$, as established in Letter IE, Theorem 2.5). The smoothness of g^S is inherited from the smoothness of g_I (which follows from the smoothness of the entropic field). The curvature of g^S is determined by the curvature of g_I through the Curvature Transfer Theorem (CTT), which we now present.

B.3 The Curvature Transfer Theorem: Information Curvature Becomes Spacetime Curvature

The Curvature Transfer Theorem is the central result of Letter IE and the mathematical backbone of the emergent geometry program. It shows that the curvature of physical spacetime is determined, to leading order, by the curvature of the information manifold — and hence by the information-geometric structure of the entropic field.

Theorem B.4 (Curvature Transfer Theorem — Letter IE, Theorem 3.1)

Under the emergence map $\Phi : M_I \rightarrow M_S$ satisfying the conformal relation $\Phi^* g_S = \lambda g_I$, the Riemann curvature tensors of the physical spacetime manifold (M_S, g_S) and the information manifold (M_I, g_I) are related by:

$$R^S_{\mu\nu\rho\sigma} = \lambda (\Phi^* R^I)_{\mu\nu\rho\sigma} + O(1/N)$$

where N is the number of entropic degrees of freedom on the substrate, and Φ^* denotes the pushforward of the curvature tensor from M_I to M_S .

Proof. The proof proceeds in several steps.

Step 1: Relate the connections. Given the conformal relation $g_{\mu\nu}^S = \lambda g_{\mu\nu}^I$ (where we write g^I for the pushforward of the information metric to M_S), the Christoffel symbols of g_S are related to those of g_I by the standard conformal transformation formula. Since λ is a constant (not a function of x), this relation simplifies considerably. For a conformal rescaling $g_S = \Omega^2 g_I$ with $\Omega^2 = \lambda = \text{const}$, we have:

$$\Gamma^{S,\alpha}_{\beta\mu} = \Gamma^{I,\alpha}_{\beta\mu} + \delta^{\alpha}_{\beta} \partial_{\mu} \ln \Omega + \delta^{\alpha}_{\mu} \partial_{\beta} \ln \Omega - g^I_{\beta\mu} g^{I\alpha\lambda} \partial_{\lambda} \ln \Omega \quad (\text{IE-S.22})$$

Since $\Omega = \sqrt{\lambda} = \text{const}$, all derivatives $\partial_{\mu} \ln \Omega$ vanish, giving:

$$\Gamma^{S,\alpha}_{\beta\mu} = \Gamma^{I,\alpha}_{\beta\mu} \quad (\text{IE-S.23})$$

The Christoffel symbols of the physical spacetime metric are *identical* to those of the (pushed-forward) information metric when the emergence coupling is constant.

Step 2: Relate the Riemann tensors. Since the Riemann tensor is constructed from the Christoffel symbols and their first derivatives (equation IE-S.4), and since the Christoffel symbols are identical (IE-S.23), the Riemann tensors with one upper and three lower indices are also identical:

$$R^{S,\alpha}_{\beta\mu\nu} = R^{I,\alpha}_{\beta\mu\nu} \quad (\text{IE-S.24})$$

However, the fully covariant Riemann tensors (with all four indices lowered) differ by the conformal factor because lowering an index involves the metric:

$$R^S_{\alpha\beta\mu\nu} = g^S_{\alpha\lambda} R^{S,\lambda}_{\beta\mu\nu} = \lambda g^I_{\alpha\lambda} R^{I,\lambda}_{\beta\mu\nu} = \lambda R^I_{\alpha\beta\mu\nu} \quad (\text{IE-S.25})$$

This is exact when λ is constant. The $O(1/N)$ correction arises when the emergence coupling acquires spatial dependence through quantum fluctuations of the entropic field. In the thermodynamic limit $N \rightarrow \infty$ (where N is the number of entropic modes on the substrate), the fluctuations are suppressed and the leading-order result (IE-S.25) becomes exact.

Step 3: Identify the correction term. When λ is not exactly constant but varies slowly ($\lambda(x) = \lambda_0 + \delta\lambda(x)$ with $|\delta\lambda|/\lambda_0 \sim 1/N$), the conformal transformation generates additional terms involving $\partial_\mu \ln \lambda$ and $\partial_\mu \partial_\nu \ln \lambda$. **These are the $O(1/N)$ corrections, which contribute to the Obidi Curvature Invariant (OCI). ■**

The physical content of the **Curvature Transfer Theorem (CTT)** of the **Theory of Entropicity (ToE)** is remarkable: *the curvature of spacetime is the curvature of information space, rescaled by the emergence coupling constant*. Spacetime is curved because information space is curved. **Black holes have intense curvature because the information content near a black hole varies rapidly with position. Gravitational waves are ripples in spacetime curvature because they are ripples in information curvature. The flat spacetime of special relativity corresponds to a region where the information metric has (nearly) zero curvature — where the entropic field varies linearly with the parameters. Every statement about the curvature of spacetime translates, through the Curvature Transfer Theorem, into a statement about the curvature of the entropic information space.**

B.4 The Entropic Ricci Tensor and Entropic Einstein Tensor

From the Curvature Transfer Theorem, we immediately obtain the entropic versions of the contracted curvature quantities by contracting the Riemann curvature relation (IE-S.25).

Entropic Ricci Tensor. Contracting the Riemann tensor on the first and third indices:

$$R^S_{\mu\nu} = g^{S\alpha\rho} R^S_{\alpha\mu\rho\nu} = (1/\lambda) g^{l\alpha\rho} \cdot \lambda R^l_{\alpha\mu\rho\nu} = R^l_{\mu\nu} + O(1/N) \text{ (IE-S.26)}$$

The Ricci tensor of physical spacetime equals the Ricci tensor of the information manifold (up to $1/N$ corrections). This is because the conformal factor λ cancels in the contraction: it appears once in the Riemann tensor (lowered) and once in the inverse metric (raised), and the two factors cancel.

Entropic Ricci Scalar. Taking the trace:

$$R^S = g^{S\mu\nu} R^S_{\mu\nu} = (1/\lambda) g^{I\mu\nu} R^I_{\mu\nu} = (1/\lambda) R^I + O(1/N) \text{ (IE-S.27)}$$

The Ricci scalar of physical spacetime is $(1/\lambda)$ times the Ricci scalar of the information manifold. The factor $1/\lambda$ reflects the rescaling: **the information manifold is "more curved" by a factor $1/\lambda \sim 10^{47}$ (coarse-graining factor) than physical spacetime (in appropriate units), reflecting the enormous number of entropic degrees of freedom that are "averaged over" in the emergence. Thus, Spacetime curvature is the thermodynamic limit of information curvature. The information manifold is the "microscopic geometry." Spacetime is the "macroscopic geometry."**

Entropic Einstein Tensor. Forming the Einstein tensor:

$$G^S_{\mu\nu} = R^S_{\mu\nu} - \frac{1}{2} R^S g^S_{\mu\nu} = R^I_{\mu\nu} - \frac{1}{2} (1/\lambda) R^I \cdot \lambda g^I_{\mu\nu} = R^I_{\mu\nu} - \frac{1}{2} R^I g^I_{\mu\nu} = G^I_{\mu\nu} + O(1/N) \text{ (IE-S.28)}$$

The Einstein tensor of physical spacetime equals the Einstein tensor of the information manifold (up to $1/N$ corrections). This is the *entropic left-hand side*: the Einstein tensor — the geometric heart of Einstein's field equations — is generated by the information geometry of the entropic field. It is not fundamental; it is the information curvature of the entropic field, measured by the Fisher-Rao metric and transferred to physical spacetime through the emergence map.

B.5 The Local Obidi Action (LOA) and the Generation of Spacetime Geometry

The Curvature Transfer Theorem and the emergence map show *how* the information metric generates the spacetime metric. But we also need the *dynamics* — the equations of motion that determine how the metric (and hence the entropic field) evolve. These dynamics are provided by the **Obidi Action**.

The Local Obidi Action (LOA) is the local, differential part of the full Obidi Action. It governs the short-distance (UV) dynamics of the entropic field and generates the local metric structure of spacetime:

$$S_{\text{LOA}}[S, g] = \int d^4x \sqrt{g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu S \partial_\nu S + V(S) + \xi R S^2 + \lambda_n S^n \right] \text{ (IE-S.29)}$$

where $g = |\det g_{\mu\nu}|$ is the determinant of the metric, $V(S)$ is the entropic potential, ξ is the non-minimal coupling constant, R is the Ricci scalar, and $\lambda_n S^n$ represents higher-order self-interaction terms. Each term in the LOA has a specific physical role:

- $\frac{1}{2} g^{\mu\nu} \partial_\mu S \partial_\nu S$: the kinetic energy of the entropic field — the cost of spatial and temporal variation
- $V(S)$: the entropic potential — determines the ground state and the spectrum of entropic excitations
- $\xi R S^2$: the non-minimal coupling to spacetime curvature — this is the crucial term that generates the Einstein tensor on the left-hand side of the field equations. The coupling ξ allows the entropic field to directly feel and influence the curvature of spacetime.
- $\lambda_n S^n$: higher-order self-interaction — generates entropic torsion and other beyond-Einstein effects

The variation of the LOA with respect to the metric $g^{\mu\nu}$ yields the entropic gravitational field equations. We now perform this variation in complete detail.

Variation of the kinetic term. The kinetic term is $\frac{1}{2} \int d^4x \sqrt{g} g^{\mu\nu} \partial_\mu S \partial_\nu S$. Under the variation $g^{\mu\nu} \rightarrow g^{\mu\nu} + \delta g^{\mu\nu}$:

$$\delta(\sqrt{g} g^{\mu\nu} \partial_\mu S \partial_\nu S) = \sqrt{g} \left[-\frac{1}{2} g_{\mu\nu} g^{\alpha\beta} \partial_\alpha S \partial_\beta S + \partial_\mu S \partial_\nu S \right] \delta g^{\mu\nu} \text{ (IE-S.30)}$$

where we have used the standard result $\delta\sqrt{g} = -\frac{1}{2} \sqrt{g} g_{\mu\nu} \delta g^{\mu\nu}$.

Variation of the potential term. The potential term is $\int d^4x \sqrt{g} V(S)$. Under metric variation, only $\sqrt{g}$ varies (since $V(S)$ depends on the field, not the metric):

$$\delta(\sqrt{g} V(S)) = -\frac{1}{2} \sqrt{g} g_{\mu\nu} V(S) \delta g^{\mu\nu} \text{ (IE-S.31)}$$

Variation of the non-minimal coupling term. This is the most important and technically demanding variation. The term is $\xi \int d^4x \sqrt{g} R S^2$. We need to compute $\delta(\sqrt{g} R)$. Using the Palatini identity $\delta R_{\mu\nu} = \nabla_\rho (\delta \Gamma^\rho_{\nu\mu}) - \nabla_\nu (\delta \Gamma^\rho_{\rho\mu})$ and the standard result:

$$(1/\sqrt{g}) \delta(\sqrt{g} R) = (R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}) \delta g^{\mu\nu} + g_{\mu\nu} \square(\delta g^{\mu\nu}) - \nabla_\mu \nabla_\nu (\delta g^{\mu\nu}) \text{ (IE-S.32)}$$

where $\square = g^{\mu\nu} \nabla_\mu \nabla_\nu$ is the d'Alembertian. The full variation of $\xi \int \sqrt{g} R S^2$ with respect to $g^{\mu\nu}$ is, after integration by parts to remove derivatives from $\delta g^{\mu\nu}$:

$$\delta(\xi \int \sqrt{g} R S^2) / \delta g^{\mu\nu} = \xi \sqrt{g} [G_{\mu\nu} S^2 + g_{\mu\nu} \square(S^2) - \nabla_\mu \nabla_\nu (S^2)] \text{ (IE-S.33)}$$

This is the term that generates the Einstein tensor. The combination $\xi S^2 G_{\mu\nu}$ is precisely the Einstein tensor multiplied by an effective gravitational coupling. When $\xi S_o^2 = c^4/(16\pi G)$ (where S_o is the vacuum expectation value of the entropic field), this term reproduces the standard Einstein-Hilbert gravitational dynamics.

Assembling the complete variation. Combining all terms, the variation of the LOA with respect to $g^{\mu\nu}$ yields:

$$\xi S^2 G_{\mu\nu} + \xi [g_{\mu\nu} \square(S^2) - \nabla_\mu \nabla_\nu (S^2)] + \frac{1}{2} \partial_\mu S \partial_\nu S - \frac{1}{4} g_{\mu\nu} (\partial S)^2 - \frac{1}{2} g_{\mu\nu} V(S) + \text{higher-order terms} = 0 \text{ (IE-S.34)}$$

Rearranging to isolate the Einstein tensor on the left:

$$\xi S^2 G_{\mu\nu} = -\xi [g_{\mu\nu} \square(S^2) - \nabla_\mu \nabla_\nu (S^2)] - \frac{1}{2} \partial_\mu S \partial_\nu S + \frac{1}{4} g_{\mu\nu} (\partial S)^2 + \frac{1}{2} g_{\mu\nu} V(S) \text{ (IE-S.35)}$$

Dividing both sides by ξS^2 :

$$G_{\mu\nu} = (1/\xi S^2) T^{(\text{ent})}_{\mu\nu} \text{ (IE-S.36)}$$

where $T^{(\text{ent})}_{\mu\nu}$ is the entropic stress-energy tensor (the right-hand side of the equation). Identifying $1/(\xi S^2)$ with $8\pi G/c^4$ recovers the standard form of Einstein's field equations. This identification yields the entropic determination of Newton's constant: $G = c^4/(8\pi\xi S_0^2)$, where S_0 is the vacuum expectation value of the entropic field.

B.6 The Spectral Obidi Action (SOA) and the Global Generation of Spacetime

While the LOA generates the local metric structure through the entropic field equations, the Spectral Obidi Action (SOA) provides the global, spectral, and topological structure. The SOA is the spectral action associated with the entropic Laplacian Δ_S , defined as the Laplace-Beltrami operator on the information manifold twisted by the entropic field:

$$S_{\text{SOA}}[S] = \text{Tr} f(\Delta_S / \Lambda^2) \text{ (IE-S.37)}$$

where f is a positive, even, smooth cutoff function (typically chosen as a Laplace transform $f(z) = \int_0^\infty dt k(t) e^{-tz}$ for some positive function k), Λ is the entropic cutoff scale (analogous to the UV cutoff in quantum field theory), and Tr denotes the operator trace. The spectral action counts the number of eigenvalues of Δ_S/Λ^2 below the cutoff, weighted by the function f .

The Spectral Obidi Action (SOA) adopts the spectral-action framework of Chamseddine and Connes—namely, an action of the form $\text{Tr} f(\mathcal{O}/\Lambda^2)$ expanded via heat kernel coefficients—but applies it to a new operator: the entropic Laplacian Δ_S on the information manifold, twisted by the entropic field. In the Theory of Entropicity, this spectral action does not describe a fundamental spacetime

Dirac operator, but instead encodes the global, spectral, and topological structure of emergent spacetime arising from the entropic substrate.

The SOA can be expanded using the heat-kernel (Seeley-DeWitt) asymptotic expansion. The heat kernel of the operator Δ_S is $K(t) = \text{Tr } e^{-t\Delta_S}$, and its small- t asymptotic expansion is:

$$K(t) \sim \sum_{n \geq 0} t^{(n-4)/2} \int d^4x \sqrt{g} a_n(x, \Delta_S) \quad (\text{IE-S.38})$$

where $a_n(x, \Delta_S)$ are the Seeley-DeWitt coefficients. The first few coefficients are well-known:

- $a_0 = (4\pi)^{-2}$ — a constant (independent of curvature)
- $a_2 = (4\pi)^{-2} R/6$ — proportional to the Ricci scalar
- $a_4 = (4\pi)^{-2} (1/360) [12 \square R + 5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}]$ — involves quadratic curvature invariants

Using the Mellin transform to relate the spectral action to the heat kernel, the SOA becomes:

$$S_{\text{SOA}} = f_0 \Lambda^4 \int d^4x \sqrt{g} a_0 + f_2 \Lambda^2 \int d^4x \sqrt{g} a_2 + f_4 \int d^4x \sqrt{g} a_4 + \dots \quad (\text{IE-S.39})$$

where $f_n = \int_0^\infty du f(u) u^{(n-4)/2}$ are the moments of the cutoff function. Let us examine each term:

The a_0 term gives:

$$f_0 \Lambda^4 (4\pi)^{-2} \int d^4x \sqrt{g} \quad (\text{IE-S.40})$$

This is a cosmological constant term. It generates the vacuum energy contribution $\Lambda_{\text{CC}} = f_0 \Lambda^4 (4\pi)^{-2} / [f_2 \Lambda^2 (4\pi)^{-2} / 6] = 6f_0 \Lambda^2 / f_2$. In the GEFE/OFE, this is the entropic cosmological function — not a free parameter but a computed quantity determined by the spectral properties of the entropic Laplacian.

The a_2 term gives:

$$f_2 \Lambda^2 (4\pi)^{-2} (1/6) \int d^4x \sqrt{g} R \text{ (IE-S.41)}$$

This is exactly the Einstein-Hilbert action $S_{EH} = (1/16\pi G) \int d^4x \sqrt{g} R$, provided we identify:

$$1/(16\pi G) = f_2 \Lambda^2 / (96\pi^2) \text{ (IE-S.42)}$$

Solving for Newton's constant:

$$G = 6\pi / (f_2 \Lambda^2) \text{ (IE-S.43)}$$

Result B.5 (Entropic Derivation of Newton's Constant)

Newton's gravitational constant is not a free parameter of nature but is determined by the spectral geometry of the entropic Laplacian through the relation $G = 6\pi/(f_2 \Lambda^2)$, where f_2 is the second moment of the spectral cutoff function and Λ is the entropic energy scale.

The a_4 term generates higher-curvature corrections to the Einstein-Hilbert action:

$$f_4 (4\pi)^{-2} (1/360) \int d^4x \sqrt{g} [5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}] \text{ (IE-S.44)}$$

These terms include the Gauss-Bonnet invariant $(R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})$ and the Weyl tensor squared. In four dimensions, the Gauss-Bonnet term is a topological invariant (it does not contribute to the equations of motion), but the remaining quadratic curvature terms provide corrections to general relativity that become important at high curvatures (near singularities, at the Planck scale). These are genuine predictions of the **GEFE/OFE** beyond Einstein.

B.7 The Full Entropic Left-Hand Side

We now assemble the complete entropic generator of spacetime geometry from the LOA and SOA contributions.

Definition B.6 (Full Entropic Left-Hand Side)

The complete entropic generator of spacetime geometry — the left-hand side of the GEFE/OFE — is:

$$G^{(\text{ent})}_{\mu\nu} = G^{(\text{LOA})}_{\mu\nu} + G^{(\text{SOA})}_{\mu\nu} + O^{(\text{OCI})}_{\mu\nu}$$

where:

- $G^{(\text{LOA})}_{\mu\nu}$ = the Einstein tensor generated by the LOA field equations (local, differential structure)
- $G^{(\text{SOA})}_{\mu\nu}$ = the spectral contribution from the SOA, including the Einstein-Hilbert term and higher-curvature corrections (global, spectral structure)
- $O^{(\text{OCI})}_{\mu\nu}$ = the Obidi Curvature Invariant tensor, measuring the excess curvature of the information manifold over the physical spacetime manifold (informational dark curvature)

In the classical limit — where the SOA higher-curvature corrections are negligible (low curvature regime, $R \ll \Lambda^2$), the OCI vanishes ($O_{\mu\nu} \rightarrow 0$, large N), and the entropic field sits at its vacuum expectation value ($S \approx S_0$) — the full entropic left-hand side reduces to the standard Einstein tensor:

$$G^{(\text{ent})}_{\mu\nu} \rightarrow G_{\mu\nu} \text{ (the standard Einstein tensor of GR) (IE-S.45)}$$

This is one half of the recovery of Einstein's field equations from the GEFE/OFE. The other half — the recovery of the standard stress-energy tensor from the entropic right-hand side — is the subject of Appendix C.

Appendix C — The Entropic Field as Generator of Matter-Energy (The Right-Hand Side)

Having constructed the entropic generator of spacetime geometry (the left-hand side of the GEFE/OFE) in Appendix B, we now construct the entropic generator of matter-energy (the right-hand side). This is the primary new contribution of these supplementary appendices: the demonstration that the stress-energy tensor — the matter-energy content of spacetime — is independently generated by the entropic field through the Fubini-Study metric on the entropic state space and the Amari alpha-connections on the statistical manifold.

C.1 The Entropic Stress-Energy Tensor: Full Construction

The matter-energy content of the universe is, in the ToE framework, generated entirely by the entropic field. The entropic stress-energy tensor is obtained by varying the matter sector of the Obidi Action with respect to the metric. The "**matter sector**" of the LOA consists of all terms that describe the dynamics of the entropic field (kinetic energy, potential, self-interactions), as opposed to the "gravitational sector" (the non-minimal coupling $\xi R S^2$ that generates the Einstein tensor).

The matter sector of the LOA is:

$$S_{\text{matter}} = \int d^4x \sqrt{g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu S \partial_\nu S + V(S) \right] \text{ (IE-S.46)}$$

The entropic stress-energy tensor is defined by the standard prescription:

$$T^{(S)}_{\mu\nu} = -(2/\sqrt{g}) \delta S_{\text{matter}} / \delta g^{\mu\nu} \text{ (IE-S.47)}$$

Computing the variation (using the results from Section B.5):

$$T^{(S)}_{\mu\nu} = \partial_\mu S \partial_\nu S - g_{\mu\nu} \left[\frac{1}{2} (\partial S)^2 + V(S) \right] \text{ (IE-S.48)}$$

where $(\partial S)^2 = g^{\alpha\beta} \partial_\alpha S \partial_\beta S$. This is precisely the stress-energy tensor of a canonical scalar field with potential $V(S)$. Let us verify its essential properties.

Symmetry. $T^{(S)}_{\mu\nu} = T^{(S)}_{\nu\mu}$ is manifest from the expression (IE-S.48), since $\partial_\mu S \partial_\nu S = \partial_\nu S \partial_\mu S$ and $g_{\mu\nu} = g_{\nu\mu}$.

Covariant conservation. The covariant divergence of the entropic stress-energy tensor vanishes when the Master Entropic Equation (MEE) is satisfied. The MEE is obtained by varying the LOA with respect to the entropic field S :

$$\square S - V'(S) - 2\xi RS - n\lambda_n S^{n-1} = 0 \text{ (IE-S.49)}$$

where $\square S = g^{\mu\nu} \nabla_\mu \nabla_\nu S$ is the covariant d'Alembertian of S . To verify conservation, compute $\nabla^\mu T^{(S)}_{\mu\nu}$:

$$\begin{aligned} \nabla^\mu T^{(S)}_{\mu\nu} &= \nabla^\mu (\partial_\mu S \partial_\nu S) - \partial_\nu \left[\frac{1}{2} (\partial S)^2 + V(S) \right] \\ &= (\square S)(\partial_\nu S) + (\partial^\mu S)(\nabla_\mu \partial_\nu S) - (\partial^\mu S)(\nabla_\nu \partial_\mu S) - V'(S)(\partial_\nu S) \\ &= [\square S - V'(S)](\partial_\nu S) = 0 \text{ (IE-S.50)} \end{aligned}$$

where we used $\nabla_\mu \partial_\nu S = \nabla_\nu \partial_\mu S$ (symmetry of the Levi-Civita connection) and the MEE (IE-S.49) in the last step (in the minimal coupling limit $\xi \rightarrow 0$). Conservation of the entropic stress-energy tensor is thus guaranteed by the equations of motion of the entropic field — a manifestation of Noether's theorem applied to the diffeomorphism invariance of the Obidi Action.

C.2 The Energy Density, Pressure, and Momentum of the Entropic Field

To extract the physical content of the entropic stress-energy tensor, we decompose it with respect to a timelike observer with four-velocity u^μ (normalized so that $u^\mu u_\mu = -1$). The spatial projection tensor is $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$.

Energy density:

$$\begin{aligned}\rho_{\text{ent}} &= T^{(S)}_{\mu\nu} u^\mu u^\nu = (\partial_o S)^2 - [-\frac{1}{2}(\partial_o S)^2 + \frac{1}{2}(\nabla S)^2 + V(S)] \\ &= \frac{1}{2}(\partial_o S)^2 + \frac{1}{2}(\nabla S)^2 + V(S) \text{ (IE-S.51)}\end{aligned}$$

where $\partial_o S = u^\mu \partial_\mu S$ is the time derivative (rate of change along the observer's worldline) and $(\nabla S)^2 = h^{ij} \partial_i S \partial_j S$ is the squared spatial gradient. The energy density has three contributions: kinetic temporal energy $\frac{1}{2}(\partial_o S)^2$, kinetic spatial (gradient) energy $\frac{1}{2}(\nabla S)^2$, and potential energy $V(S)$. All three are non-negative (assuming $V \geq 0$), ensuring that the energy density is positive — a weak energy condition satisfied by the entropic field.

Pressure:

$$p_{\text{ent}} = \frac{1}{3} h^{ij} T^{(S)}_{ij} = \frac{1}{2}(\partial_o S)^2 - \frac{1}{6}(\nabla S)^2 - V(S) \text{ (IE-S.52)}$$

The pressure is the spatial trace of the stress tensor. It can be positive (when kinetic energy dominates — entropic radiation), zero (when the field is static and potential-dominated — entropic

dust), or negative (when the potential energy dominates — entropic dark energy). The equation of state parameter $w = p/\rho$ varies depending on the configuration of the entropic field, allowing the entropic field to mimic any form of matter.

Momentum density (energy flux):

$$\pi_i = T^{(S)}_{oi} = \partial_o S \partial_i S \text{ (IE-S.53)}$$

The momentum density is the product of the temporal and spatial gradients. It represents the flux of entropic energy in the i -th spatial direction. When the field is static ($\partial_o S = 0$), the momentum density vanishes — there is no energy flow. When the field is time-dependent and spatially varying, there is a flow of entropic energy through space, which sources the gravitomagnetic (frame-dragging) components of the spacetime metric.

Anisotropic stress:

$$\Sigma_{ij} = T^{(S)}_{ij} - p_{\text{ent}} h_{ij} = \partial_i S \partial_j S - \frac{1}{3}(\nabla S)^2 h_{ij} \text{ (IE-S.54)}$$

The anisotropic stress is the traceless part of the spatial stress tensor. It vanishes when the spatial gradient of the entropic field is isotropic (the field varies equally in all spatial directions). Non-zero anisotropic stress arises when the entropic field has a preferred spatial direction — for example, near a plane wave or a domain wall configuration. Anisotropic stress is particularly important in cosmology, where it distinguishes different types of dark energy and dark matter candidates.

C.3 The Entropic Origin of Different Forms of Matter

One of the most striking features of the entropic stress-energy tensor (IE-S.48) is its versatility: depending on the configuration of the entropic field $S(x)$, it can reproduce the stress-energy tensor of any standard form of matter. The entropic field is thus a universal matter generator — a single field that gives rise to the full diversity of matter-energy in the universe.

Case (i): Dust (Pressureless Matter)

When the entropic field is slowly varying in space but has a significant time derivative, and the potential $V(S)$ dominates over the gradient energy:

$$(\nabla S)^2 \ll (\partial_0 S)^2 \ll V(S) \text{ (IE-S.55)}$$

In this limit, the energy density is $\rho \approx V(S)$ and the pressure is $p \approx -V(S) + \frac{1}{2}(\partial_0 S)^2 \approx 0$ when the kinetic contributions balance. More precisely, if the entropic field oscillates rapidly around a minimum of $V(S)$ with frequency m (the mass of the entropic excitation), then the time-averaged stress-energy tensor is:

$$\langle T^{(S)}_{\mu\nu} \rangle \rightarrow \rho u_\mu u_\nu \text{ (pressureless dust) (IE-S.56)}$$

This is precisely the stress-energy tensor of non-relativistic matter (dust). The entropic field oscillating around a potential minimum behaves as a collection of non-relativistic particles — the entropic origin of dark matter and baryonic matter in the non-relativistic limit. The mass m of the particles is related to the curvature of the potential at the minimum: $m^2 = V''(S_0)$, exactly as in standard scalar field theory.

Case (ii): Radiation

When the entropic field has high-frequency oscillations and the potential is negligible:

$$V(S) \ll (\partial_0 S)^2 \sim (\nabla S)^2 \text{ (IE-S.57)}$$

In this limit, the stress-energy tensor becomes traceless ($T = g^{\mu\nu} T_{\mu\nu} = 0$), and the equation of state is $w = p/\rho = \frac{1}{3}$ — the radiation equation of state. The time-averaged entropic stress-energy becomes:

$$\langle T^{(S)}_{\mu\nu} \rangle \rightarrow \text{radiation stress-energy with } p = \rho/3 \text{ (IE-S.58)}$$

This is the entropic origin of radiation. The massless, rapidly oscillating modes of the entropic field behave as radiation — a gas of relativistic particles with the equation of state of a photon gas.

Case (iii): Dark Energy / Cosmological Constant

When the entropic field sits at (or near) a minimum of its potential $V(S)$ with $V(S_0) > 0$, and the kinetic terms are negligible:

$$\partial_\mu S \approx 0, V(S_0) > 0 \text{ (IE-S.59)}$$

The stress-energy tensor reduces to:

$$T^{(S)}_{\mu\nu} \rightarrow -V(S_0) g_{\mu\nu} \text{ (IE-S.60)}$$

This is the stress-energy tensor of vacuum energy — a cosmological constant $\Lambda = 8\pi G V(S_0)/c^4$ with equation of state $w = -1$. The entropic field at rest in a positive potential minimum generates the observed accelerated expansion of the universe. Unlike the standard model of cosmology, where the cosmological constant is a free parameter with no explanation for its value, the ToE framework determines Λ through the potential $V(S)$ of the entropic field, which is itself determined by the fundamental entropic dynamics.

Case (iv): Electromagnetic Field

When the phase sector of the complexified entropic field is extracted, the electromagnetic stress-energy tensor emerges. As shown in Letter IIA, the complexified entropic field $E(x) = \rho(x) \exp(i\Theta(x))$ has a $U(1)$ phase symmetry $\Theta \rightarrow \Theta + \alpha$, and the conserved current associated with this symmetry is the electromagnetic current $J^\mu = \rho^2 \partial^\mu \Theta$. The gauge field $A_\mu = \partial_\mu \Theta$ (in the pure-gauge limit) and the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ emerge from the phase gradients. The phase-sector contribution to the entropic stress-energy tensor reproduces the electromagnetic stress-energy tensor (IE-S.11) when the appropriate identification is made.

Case (v): Topological Defects (Solitons, Vortices)

When the entropic field has stable, localized configurations with nontrivial topology — domain walls (1D defects), strings (2D defects), monopoles (3D defects) — these configurations carry energy, momentum, and stress localized in space. They are the "particles" of the entropic theory: stable, localized concentrations of entropic energy that propagate through spacetime and interact with each other and with the gravitational field. The stability of these topological defects is guaranteed by topology (they cannot be continuously deformed to the vacuum), providing the entropic explanation for the stability of matter.

C.4 The Fubini-Study Metric and the Quantum Matter Sector

The **classical entropic stress-energy tensor** (IE-S.48) is sufficient to reproduce Einstein's matter sector in the classical limit. However, the full quantum matter sector requires a more sophisticated geometric structure: the Fubini-Study metric on the entropic state space. This is a key new development of these appendices.

The complexified entropic field $E(x) = \rho(x) \exp(i\Theta(x))$ defines, at each point x in spacetime, a state in a complex Hilbert space. As the parameters x vary, the state traces out a curve in the projective Hilbert space $\mathbb{CP}^n$ (or $\mathbb{CP}^\infty$ for the full field theory). The natural metric on this projective space is the Fubini-Study metric.

Definition C.1 (Fubini-Study Metric on the Entropic State Space)

For a parametric family of states $|E(x)\rangle$ in a complex Hilbert space, parameterized by x^a , the Fubini-Study metric is:

$$g^{\text{FS}}_{ab} = \text{Re}[\langle \partial_a E | \partial_b E \rangle / \langle E | E \rangle - \langle \partial_a E | E \rangle \langle E | \partial_b E \rangle / \langle E | E \rangle^2]$$

For the entropic field $E = \rho \exp(i\Theta)$, treating ρ and Θ as the coordinates, the Fubini-Study line element reduces to:

$$ds^2_{\text{FS}} = (d\rho / \rho)^2 + (d\Theta)^2 \quad (\text{IE-S.61})$$

This remarkably simple expression has a profound dual structure:

- The **amplitude component** $(d\rho/\rho)^2$ governs the gravitational/matter sector. Variations in the amplitude of the entropic field correspond to variations in the energy density and pressure — the matter content of spacetime. The logarithmic form $d\rho/\rho$ ensures scale invariance: a doubling of the amplitude produces the same Fubini-Study distance regardless of the absolute value of ρ .
- The **phase component** $(d\Theta)^2$ governs the electromagnetic sector. Variations in the phase of the entropic field correspond to gauge transformations and electromagnetic field strengths, as established in Letter IIA.

The entropic stress-energy tensor can be expressed using the Fubini-Study metric. Writing $S = \ln \rho$ (for the amplitude sector), so that $\partial_\mu S = \partial_\mu \rho / \rho$, the stress-energy tensor becomes:

$$T^{(S)}_{\mu\nu} = \rho^2 [(\partial_\mu \ln \rho)(\partial_\nu \ln \rho) + (\partial_\mu \Theta)(\partial_\nu \Theta)] - g_{\mu\nu}[\rho^2/2 ((\partial \ln \rho)^2 + (\partial \Theta)^2) + V] \quad (\text{IE-S.62})$$

This expresses the stress-energy tensor as the Fubini-Study kinetic energy of the entropic field, weighted by ρ^2 , minus the metric times the Lagrangian density. The matter-energy content of spacetime is the Fubini-Study kinetic energy of the entropic field.

Theorem C.2 (Fisher-Rao to Fubini-Study Reduction)

When the quantum states are diagonal in a fixed basis — i.e., when $|\psi(\theta)\rangle = \sum \sqrt{p(x; \theta)} |x\rangle$ — the Fubini-Study metric reduces to $\frac{1}{4}$ times the Fisher-Rao metric:

$$g_{ab}^{\text{FS}} = \frac{1}{4} g_{ab}^{\text{FR}}$$

Proof. Let $|\psi\rangle = \sum_x \sqrt{p_x} |x\rangle$ where $p_x = p(x; \theta)$. Then $\langle\psi|\psi\rangle = \sum p_x = 1$. We compute:

$$\partial_a |\psi\rangle = \sum (\partial_a p_x) / (2\sqrt{p_x}) |x\rangle$$

$$\langle\partial_a \psi|\partial_b \psi\rangle = \sum (\partial_a p_x)(\partial_b p_x) / (4p_x) = \frac{1}{4} \sum p_x (\partial_a \ln p_x)(\partial_b \ln p_x) = \frac{1}{4} g_{ab}^{\text{FR}} + \frac{1}{4} \langle\partial_a \ln p\rangle\langle\partial_b \ln p\rangle$$

$$\langle\partial_a \psi|\psi\rangle = \sum (\partial_a p_x) / (2) = \frac{1}{2} \langle\partial_a \ln p\rangle$$

Therefore: $g_{ab}^{\text{FS}} = \frac{1}{4} g_{ab}^{\text{FR}} + \frac{1}{4} \langle\partial_a \ln p\rangle\langle\partial_b \ln p\rangle - \frac{1}{4} \langle\partial_a \ln p\rangle\langle\partial_b \ln p\rangle = \frac{1}{4} g_{ab}^{\text{FR}}$. ■

This theorem establishes the Fisher-Rao $\rightarrow$ Fubini-Study hierarchy as the classical $\rightarrow$ quantum transition in the matter sector. In the classical limit (when quantum coherence is negligible and states are diagonal), the Fubini-Study metric reduces to the Fisher-Rao metric. The Fubini-Study metric is therefore the natural quantum extension of the Fisher-Rao metric — it includes the full quantum structure (off-diagonal coherences, phases) that the Fisher-Rao metric does not capture.

C.5 The Amari-Čencov Alpha-Connections and the Dual Structure of Matter

The Fisher-Rao metric provides the Riemannian structure on the statistical manifold, but the full information-geometric structure requires additional data: the connections. In information

geometry, the natural connections are not the unique Levi-Civita connection of the Fisher-Rao metric, but a one-parameter family of connections — the Amari alpha-connections — that encode the non-Riemannian aspects of the statistical structure.

Definition C.3 (Amari Alpha-Connection)

The alpha-connection on the statistical manifold is defined by:

$$\Gamma^{(\alpha)}_{abc} = \Gamma^{(o)}_{abc} + (\alpha/2) T_{abc}$$

where $\Gamma^{(o)}_{abc}$ is the Levi-Civita connection of the Fisher-Rao metric, $\alpha \in \mathbb{R}$ is the alpha-parameter, and T_{abc} is the cubic (skewness) tensor:

$$T_{abc} = \int P (\partial_a \ln P)(\partial_b \ln P)(\partial_c \ln P) d\omega$$

The skewness tensor T_{abc} is the third moment (cumulant) of the score functions. It measures the asymmetry (skewness) of the probability distribution — how much the distribution deviates from a Gaussian shape. For Gaussian distributions (which have zero skewness), $T_{abc} = 0$ and all alpha-connections coincide with the Levi-Civita connection. For non-Gaussian distributions, the alpha-connections differ from each other and from the Levi-Civita connection, encoding the non-Gaussian structure of the entropic field.

Three values of α are particularly important:

The e-connection ($\alpha = +1$): $\Gamma^{(+1)}_{abc} = \Gamma^{(o)}_{abc} + \frac{1}{2}T_{abc}$. This connection is associated with the exponential family of probability distributions. The e-geodesics (geodesics of the e-connection) are curves in the statistical manifold along which the natural (exponential) parameters vary linearly. In the gravitational context, the e-connection governs the kinetic sector of the stress-energy tensor —

how matter sources curvature. The e-connection is the natural connection for the description of matter in terms of its conserved charges (energy, momentum, particle number), which are the natural parameters of the exponential family.

The m-connection ($\alpha = -1$): $\Gamma^{(-1)}_{abc} = \Gamma^{(o)}_{abc} - \frac{1}{2}T_{abc}$. This connection is associated with the mixture family of probability distributions. The m-geodesics are curves along which the mixture (expectation) parameters vary linearly. In the gravitational context, the m-connection governs the potential sector of the stress-energy tensor — how the entropic vacuum responds to curvature. The m-connection is the natural connection for the description of matter in terms of its expectation values (density, pressure, temperature), which are the expectation parameters of the mixture family.

The Levi-Civita connection ($\alpha = o$): $\Gamma^{(o)}_{abc}$ = the unique torsion-free, metric-compatible connection. This is the arithmetic mean of the e- and m-connections: $\Gamma^{(o)} = \frac{1}{2}(\Gamma^{(+1)} + \Gamma^{(-1)})$. The Levi-Civita connection governs free propagation and geodesic motion in the standard Riemannian sense. In Einstein's general relativity, the Levi-Civita connection is the *only* connection used. The Theory of Entropicity, by employing all three connections, provides a richer dynamical structure.

Theorem C.4 (Alpha-Duality)

The e-connection ($\alpha = +1$) and the m-connection ($\alpha = -1$) are dual with respect to the Fisher-Rao metric:

$$g(\nabla^{(+1)}_X Y, Z) + g(Y, \nabla^{(-1)}_X Z) = X g(Y, Z)$$

for all vector fields X, Y, Z .

Proof. The **Levi-Civita connection** satisfies $g(\nabla^{(o)}_X Y, Z) + g(Y, \nabla^{(o)}_X Z) = X g(Y, Z)$ (metric compatibility). Since $\Gamma^{(+)} = \Gamma^{(o)} + \frac{1}{2}T$ and $\Gamma^{(-)} = \Gamma^{(o)} - \frac{1}{2}T$, we have $\nabla^{(+)} = \nabla^{(o)} + \frac{1}{2}T$ and $\nabla^{(-)} = \nabla^{(o)} - \frac{1}{2}T$.

Therefore:

$$\begin{aligned} g(\nabla^{(+)}_X Y, Z) + g(Y, \nabla^{(-)}_X Z) &= g(\nabla^{(o)}_X Y + \frac{1}{2}T(X, Y), Z) + g(Y, \nabla^{(o)}_X Z - \frac{1}{2}T(X, Z)) \\ &= g(\nabla^{(o)}_X Y, Z) + g(Y, \nabla^{(o)}_X Z) + \frac{1}{2}g(T(X, Y), Z) - \frac{1}{2}g(Y, T(X, Z)) \\ &= X g(Y, Z) + \frac{1}{2}[T_{abc} - T_{acb}] = X g(Y, Z) \end{aligned}$$

where the last equality uses the total symmetry of the skewness tensor: $T_{abc} = T_{acb}$. ■

This duality between the e-connection and m-connection is the information-geometric counterpart of the LHS = RHS structure of Einstein's equations. The entropic field simultaneously generates curvature (via the m-connection, which governs the geometric response of the vacuum) and matter (via the e-connection, which governs the kinetic sources of curvature). The duality ensures that these two aspects are complementary and consistent — the e-projection and m-projection must agree for the field equations to be satisfied.

C.6 The Full Entropic Right-Hand Side

We now assemble the complete entropic generator of matter-energy — the right-hand side of the GEFE/OFE:

Definition C.5 (Full Entropic Right-Hand Side)

The complete entropic generator of matter-energy is:

$$T^{(\text{ent})}_{\mu\nu} = T^{(\text{amplitude})}_{\mu\nu} + T^{(\text{phase})}_{\mu\nu} + T^{(\text{potential})}_{\mu\nu} + T^{(\text{interaction})}_{\mu\nu}$$

where:

- $T^{(\text{amplitude})}_{\mu\nu} = \rho^2(\partial_\mu \ln \rho)(\partial_\nu \ln \rho) - \frac{1}{2}g_{\mu\nu} \rho^2(\partial \ln \rho)^2$: the amplitude sector (gravitational matter)
- $T^{(\text{phase})}_{\mu\nu} = \rho^2(\partial_\mu \Theta)(\partial_\nu \Theta) - \frac{1}{2}g_{\mu\nu} \rho^2(\partial \Theta)^2$: the phase sector (electromagnetic field)
- $T^{(\text{potential})}_{\mu\nu} = -g_{\mu\nu} V(S)$: the entropic potential (vacuum energy, cosmological constant)
- $T^{(\text{interaction})}_{\mu\nu}$ = amplitude-phase coupling and higher-order terms

In the classical limit (when quantum coherences are negligible, the phase sector decouples from the amplitude sector, and the interaction terms are suppressed), the entropic right-hand side reduces to the standard Einstein stress-energy tensor:

$$T^{(\text{ent})}_{\mu\nu} \rightarrow T^{(\text{Einstein})}_{\mu\nu} \text{ (the standard stress-energy tensor of GR) (IE-S.63)}$$

Appendix D — The Generalized Entropic Field Equations (GEFE/OFE)

We now arrive at the central result of these supplementary appendices: the Generalized Entropic Field Equations of the Theory of Entropicity (ToE), in which both the geometric left-hand side and the matter-energy right-hand side are independently and completely generated by the entropic

field. The GEFE/OFE is the master equation of the Theory of Entropicity in the gravitational sector — the equation from which Einstein's field equations, and their corrections, follow.

D.1 Statement of the Generalized Entropic Field Equations

THEOREM D.1 (Generalized Entropic Field Equations — GEFE/OFE)

The complete dynamics of the entropic field $S(x)$ on the entropic manifold M_S , as derived from the full Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$, is expressed by the Generalized Entropic Field Equations:

$$G^{(\text{ent})}_{\mu\nu}[S, g_I, \Phi] + \Lambda_{\text{ent}}[S] g_{\mu\nu} + O^{(\text{OCI})}_{\mu\nu}[S] = (8\pi G_{\text{ent}}[S] / c^4_{\text{ent}}[S]) T^{(\text{ent})}_{\mu\nu}[S, g_{FS}, \Gamma^{(\alpha)}]$$

where the left-hand side (Entropic Generator of Geometry) contains:

- $G^{(\text{ent})}_{\mu\nu}$ = the entropic Einstein tensor, constructed from the Fisher-Rao information metric g_I via the emergence map Φ
- $\Lambda_{\text{ent}}[S]$ = the entropic cosmological function (NOT a constant — it depends on the entropic field), from the a_o term of the SOA
- $O^{(\text{OCI})}_{\mu\nu}$ = the Obidi Curvature Invariant tensor, measuring the excess information curvature over physical curvature

and the right-hand side (Entropic Generator of Matter) contains:

- $G_{\text{ent}}[S]$ = the entropic gravitational coupling, determined by the SOA: $G_{\text{ent}} = 6\pi/(f_2\Lambda^2)$
- $c_{\text{ent}}[S]$ = the entropic speed limit $\sqrt{(\kappa/\rho_S)}$, from the No-Rush Theorem

- $T^{(\text{ent})}_{\mu\nu}$ = the entropic stress-energy tensor, constructed using the Fubini-Study metric and the Amari alpha-connections

Equation (IE-S.64) — the GEFE/OFE — is the gravitational master equation of the Theory of Entropicity. Every quantity on both sides is constructed from, and determined by, the entropic field $S(x)$. There is no external input. No independently postulated matter fields. No free parameters beyond those of the entropic potential $V(S)$ and the cutoff function f . The GEFE/OFE is a self-contained dynamical equation for the entropic field, expressed in geometric language.

D.2 The Entropic Left-Hand Side: Full Construction

We now present the complete step-by-step construction of the entropic left-hand side — the entropic Einstein tensor — from the entropic field $S(x)$.

Step 1: Begin with the entropic field. Let $S(\omega; x)$ be the entropic field on the substrate Ω , parameterized by spacetime coordinates x^μ ($\mu = 0, 1, 2, 3$).

Step 2: Construct the probability distribution. $P(\omega; x) = \exp(-S(\omega; x))/Z(x)$, where $Z(x) = \int_{\Omega} d\omega \exp(-S(\omega; x))$.

Step 3: Compute the Fisher-Rao information metric. $g^I_{\mu\nu}(x) = \langle \partial_\mu S \cdot \partial_\nu S \rangle - \langle \partial_\mu S \rangle \langle \partial_\nu S \rangle$. The Čencov uniqueness theorem guarantees that this is the unique invariant metric on the statistical manifold.

Step 4: Apply the emergence map. Through the emergence map $\Phi : M_I \rightarrow M_S$, obtain the physical spacetime metric: $g^S_{\mu\nu} = \lambda g^I_{\mu\nu}$, with $\lambda = l_P^2/(4k_B)$.

Step 5: Compute the Christoffel symbols. $\Gamma^\alpha_{\beta\mu} = \frac{1}{2}g^{S\alpha\lambda}(\partial_\mu g^S_{\lambda\beta} + \partial_\beta g^S_{\lambda\mu} - \partial_\lambda g^S_{\beta\mu})$. Since $g^S = \lambda g^I$ with $\lambda = \text{const}$, these are identical to the Christoffel symbols of g^I .

Step 6: Compute the Riemann tensor. $R^\alpha_{\beta\mu\nu} = \partial_\mu \Gamma^\alpha_{\nu\beta} - \partial_\nu \Gamma^\alpha_{\mu\beta} + \Gamma^\alpha_{\mu\lambda} \Gamma^\lambda_{\nu\beta} - \Gamma^\alpha_{\nu\lambda} \Gamma^\lambda_{\mu\beta}$. By the Curvature Transfer Theorem, $R^S_{\alpha\beta\mu\nu} = \lambda R^I_{\alpha\beta\mu\nu}$.

Step 7: Contract to obtain the Ricci tensor and scalar. $R^S_{\mu\nu} = R^I_{\mu\nu}$, $R^S = (1/\lambda)R^I$.

Step 8: Form the Einstein tensor. $G^S_{\mu\nu} = R^S_{\mu\nu} - \frac{1}{2}R^S g^S_{\mu\nu} = G^I_{\mu\nu}$.

The Obidi Curvature Invariant tensor captures the deviation between the full information curvature and the physical curvature:

$$O^{(OCI)}_{\mu\nu} = \lambda \Phi^*(R^I_{\mu\nu} - \frac{1}{2}R^I g^I_{\mu\nu}) - (R^S_{\mu\nu} - \frac{1}{2}R^S g^S_{\mu\nu}) \text{ (IE-S.65)}$$

At leading order in $1/N$, the OCI vanishes. At next-to-leading order, it captures the excess curvature of the information manifold — curvature that exists in the informational structure of the entropic field but does not appear in the physical spacetime geometry. This is the "informational dark curvature" that manifests physically as dark matter and dark energy effects.

Proposition D.2 (Non-Negativity of the OCI)

The scalar OCI satisfies $\mathcal{O}_\Omega = R^I - (1/\lambda)\Phi^*(R^S) \geq 0$. The information manifold is always at least as curved as the physical spacetime.

Prelim Proof. The Fisher-Rao metric is defined as a covariance matrix, which is always positive semi-definite. The curvature of the statistical manifold, being determined by the statistical correlations between score functions, includes all the structure of the physical curvature (which is its image under the emergence map) plus additional structure from the statistical fluctuations and higher cumulants that do not map to physical geometry. The excess curvature from these unmapped fluctuations is non-negative by construction. ■

D.3 The Entropic Right-Hand Side: Full Construction

The construction of the entropic right-hand side — the entropic stress-energy tensor — proceeds as follows.

Step 1: Decompose the entropic field. Write the complexified entropic field as $E(x) = \rho(x) \exp(i\Theta(x))$, where ρ is the amplitude and Θ is the phase.

Step 2: Compute the Fubini-Study metric. The Fubini-Study metric on the entropic state space is $ds^2_{\text{FS}} = (d\rho/\rho)^2 + (d\Theta)^2$. The amplitude component governs the matter sector; the phase component governs the electromagnetic sector.

Step 3: Compute the Amari alpha-connections. The skewness tensor $T_{abc} = \int P (\partial_a \ln P)(\partial_b \ln P)(\partial_c \ln P) d\omega$ is computed from the probability distribution (IE-S.14). The e-connection ($\alpha = +1$), m-connection ($\alpha = -1$), and Levi-Civita connection ($\alpha = 0$) are then constructed.

Step 4: Construct the entropic stress-energy tensor. From the Obidi Action, the stress-energy tensor is:

$$T^{(\text{ent})}_{\mu\nu} = \partial_\mu S \partial_\nu S - g_{\mu\nu} [\frac{1}{2}(\partial S)^2 + V(S)] \quad (\text{IE-S.66})$$

Step 5: Decompose using the Fubini-Study metric.

$$T^{(\text{ent})}_{\mu\nu} = \rho^2 [(\partial_\mu \ln \rho)(\partial_\nu \ln \rho) + (\partial_\mu \Theta)(\partial_\nu \Theta)] - g_{\mu\nu} [\rho^2/2((\partial \ln \rho)^2 + (\partial \Theta)^2) + V] \quad (\text{IE-S.67})$$

The first term is the amplitude sector (matter), the second is the phase sector (electromagnetism), and V provides the potential contribution (vacuum energy).

Step 6: Verify conservation. $\nabla^\mu T^{(\text{ent})}_{\mu\nu} = 0$ when the MEE is satisfied, as shown in Section C.1.

D.4 Proof of the GEFE/OFE

We now prove that the GEFE/OFE follows from the full Obidi Action by simultaneous variation with respect to the entropic field $S(x)$ and the metric $g^{\mu\nu}(x)$.

The full Obidi Action in the gravitational sector is:

$$S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}} \text{ (IE-S.68)}$$

$$S_{\text{LOA}} = \int d^4x \sqrt{g} \left[\frac{1}{2}(\partial S)^2 + V(S) + \xi R S^2 + \lambda_n S^n \right] \text{ (IE-S.69)}$$

$$S_{\text{SOA}} = f_o \Lambda^4 \int \sqrt{g} a_o d^4x + f_2 \Lambda^2 \int \sqrt{g} a_2 d^4x + f_4 \int \sqrt{g} a_4 d^4x + \dots \text{ (IE-S.70)}$$

Variation with respect to S : yields the Master Entropic Equation (MEE):

$$\square S - V'(S) - 2\xi R S - n\lambda_n S^{n-1} = 0 \text{ (IE-S.71)}$$

This is the equation of motion for the entropic field. It is a nonlinear wave equation (the d'Alembertian $\square S$) with potential ($V(S)$), curvature coupling ($2\xi R S$), and self-interaction ($n\lambda_n S^{n-1}$) terms.

Variation with respect to $g^{\mu\nu}$: yields the entropic gravitational field equations. We collect the results from Section B.5 (LOA variation) and Section B.6 (SOA variation).

From the LOA:

$$\xi S^2 G_{\mu\nu} + \xi [g_{\mu\nu} \square(S^2) - \nabla_\mu \nabla_\nu (S^2)] + \frac{1}{2} \partial_\mu S \partial_\nu S - \frac{1}{4} g_{\mu\nu} (\partial S)^2 - \frac{1}{2} g_{\mu\nu} V(S) - \frac{1}{2} g_{\mu\nu} \lambda_n S^n = 0 \text{ (IE-S.72)}$$

From the SOA (a_o term):

$$-\frac{1}{2} f_o \Lambda^4 (4\pi)^{-2} g_{\mu\nu} \text{ (cosmological constant contribution)} \text{ (IE-S.73)}$$

From the SOA (a_2 term):

$$f_2 \Lambda^2 (4\pi)^{-2} (1/6) G_{\mu\nu} \text{ (Einstein-Hilbert contribution) (IE-S.74)}$$

From the SOA (a_4 term):

Higher-curvature corrections: Gauss-Bonnet and Weyl² terms (IE-S.75)

Combining the LOA and SOA contributions, and rearranging to the standard form with the Einstein tensor on the left and the stress-energy tensor on the right:

$$\begin{aligned} & [\xi S^2 + f_2 \Lambda^2 / (96\pi^2)] G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} + \xi [g_{\mu\nu} \square(S^2) - \nabla_\mu \nabla_\nu (S^2)] + H_{\mu\nu}^{(4)} \\ & = -\frac{1}{2} \partial_\mu S \partial_\nu S + \frac{1}{4} g_{\mu\nu} (\partial S)^2 + \frac{1}{2} g_{\mu\nu} V(S) + \frac{1}{2} g_{\mu\nu} \lambda_n S^n \text{ (IE-S.76)} \end{aligned}$$

where $\Lambda_{\text{eff}} = f_0 \Lambda^4 / (32\pi^2)$ is the effective cosmological constant from the SOA, and $H_{\mu\nu}^{(4)}$ represents the a_4 and higher-order curvature corrections. Identifying the left-hand side with the entropic Einstein tensor plus cosmological function plus OCI corrections, and the right-hand side with the entropic stress-energy tensor, we obtain the **GEFE/OFE** in the form of Theorem D.1. ■

D.5 Recovery of Einstein's Field Equations as a Classical Limit

THEOREM D.3 (Einstein Recovery Theorem)

In the simultaneous limit of:

1. Weak entropic curvature: $|R^{(\text{ent})}_{\mu\nu}| \ll \Lambda^2$

2. Frozen amplitude: $\delta\rho/\rho_o \rightarrow 0$ (small amplitude fluctuations around the vacuum expectation value)
3. Flat information excess: $R^{(\text{info})}_{\mu\nu\rho\sigma} \rightarrow 0$ beyond the physical sector (the information manifold has no curvature beyond what maps to physical spacetime)
4. Vanishing OCI: $O^{(\text{OCI})}_{\mu\nu} \rightarrow 0$ (large N limit)
5. Constant coupling: $\xi S_o^2 + f_2 \Lambda^2 / (96\pi^2) \rightarrow c^4 / (16\pi G)$

the Generalized Entropic Field Equations reduce exactly to Einstein's field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$$

Proof. We verify each limiting step.

Limit (i): When the entropic curvature is small compared to the cutoff scale, the a_4 and higher Seeley-DeWitt corrections (which involve R^2 , $R_{\mu\nu}R^{\mu\nu}$, etc.) are suppressed by powers of $R/\Lambda^2 \ll 1$. The SOA contribution reduces to the a_0 (cosmological constant) and a_2 (Einstein-Hilbert) terms only: $H^{(4)}_{\mu\nu} \rightarrow 0$.

Limit (ii): When the amplitude fluctuations $\delta\rho/\rho_o$ are small, the entropic field $S \approx S_o + \delta S$ with $|\delta S| \ll S_o$. The non-minimal coupling terms involving $\square(S^2)$ and $\nabla_\mu \nabla_\nu(S^2)$ become: $\square(S_o^2) = 0$ (since $S_o = \text{const}$) and $\nabla_\mu \nabla_\nu(S_o^2) = 0$. The correction terms $\xi[g_{\mu\nu}\square(S^2) - \nabla_\mu \nabla_\nu(S^2)]$ vanish.

Limit (iii) and (iv): The OCI tensor $O^{(\text{OCI})}_{\mu\nu}$ vanishes, and the entropic Einstein tensor reduces to the standard Einstein tensor: $G^{(\text{ent})}_{\mu\nu} \rightarrow G_{\mu\nu}$.

Limit (v): The effective gravitational coupling becomes constant: $\xi S_o^2 + f_2 \Lambda^2 / (96\pi^2) = c^4 / (16\pi G)$. This determines G in terms of the entropic parameters.

Under these five limits, the GEFE/OFE (IE-S.76) reduces to:

$$[c^4/(16\pi G)] G_{\mu\nu} + \Lambda_{\text{eff}} g_{\mu\nu} = -\frac{1}{2}\partial_\mu S \partial_\nu S + \frac{1}{4}g_{\mu\nu}(\partial S)^2 + \frac{1}{2}g_{\mu\nu}V(S)$$

Dividing both sides by $c^4/(16\pi G)$ and identifying the right-hand side as $(8\pi G/c^4)T_{\mu\nu}$ with $T_{\mu\nu} = \partial_\mu S \partial_\nu S - g_{\mu\nu}[\frac{1}{2}(\partial S)^2 + V(S)]$:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} \text{ (IE-S.77)}$$

with $\Lambda = 16\pi G\Lambda_{\text{eff}}/c^4$. This is exactly Einstein's field equations with cosmological constant. ■

D.6 What the GEFE/OFE Add Beyond Einstein

The GEFE/OFE is not merely a reformulation of Einstein's equations — it predicts new physics. **The corrections beyond Einstein fall into six categories:**

(i) **The OCI tensor.** The Obidi Curvature Invariant tensor (OCIT) $O^{(\text{OCI})}_{\mu\nu}$ provides an additional source of gravitational effects — "**informational dark curvature**" (IDC) — that produces dark matter and dark energy-like effects without invoking exotic particles. The OCI acts as an effective additional stress-energy tensor on the right-hand side (or equivalently as a modification of the geometry on the left-hand side), producing deviations from Einstein's predictions at galactic and cosmological scales where the information manifold's excess curvature becomes significant.

(ii) **The entropic cosmological function.** In the GEFE/OFE, the cosmological "constant" $\Lambda_{\text{ent}}[S]$ is not constant but field-dependent. It varies with the entropic field configuration, producing dynamical dark energy — a time-varying vacuum energy that naturally accounts for the observed acceleration of cosmic expansion without the fine-tuning problems of the standard cosmological constant.

(iii) Higher-curvature corrections from the SOA. The a_4 Seeley-DeWitt coefficient produces quadratic curvature corrections (Gauss-Bonnet, Weyl tensor squared) that become important near singularities (black hole interiors, Big Bang). These corrections can resolve the classical singularities predicted by Penrose-Hawking singularity theorems, replacing them with non-singular bounces or transitions.

(iv) Amplitude-phase coupling. In regions of strong entropic gradients, the gravitational (amplitude) and electromagnetic (phase) sectors couple through the shared entropic field, producing novel gravitational-electromagnetic interactions not present in standard general relativity or Maxwell theory.

(v) Entropic torsion. When the cubic self-interaction term ($\lambda_3 S^3$) is present, the effective spacetime connection acquires torsion — a feature **absent from standard general relativity but present in Einstein-Cartan theory**. The entropic origin of torsion connects it to the skewness of the entropic field distribution (the **third cumulant**), providing a **physical interpretation of torsion as informational asymmetry**.

(vi) Information-geometric constraints. The Čencov uniqueness theorem restricts the allowed metrics on the information manifold to the Fisher-Rao metric (up to a constant). Through the emergence map, this constrains the allowed spacetime metrics, reducing the freedom of the gravitational sector compared to general relativity. **Not every solution of Einstein's equations is a solution of the GEFE/OFE — only those that can be generated by an entropic field with a Fisher-Rao metric as their information-geometric source.**

D.7 The Information-Geometric Structure of the GEFE/OFE

The GEFE/OFE has a remarkable information-geometric interpretation that makes its structure transparent:

Left-hand side (geometry): The construction chain is:

$$S(x) \rightarrow P(\omega; x) \rightarrow g^{\text{FR}}_{\mu\nu} \rightarrow \Gamma^{(\circ)}_{\alpha\beta\mu} \rightarrow R^{\text{l}}_{\alpha\beta\mu\nu} \rightarrow G^{\text{l}}_{\mu\nu} \rightarrow \Phi_*(G^{\text{l}}) = G^{\text{S}}_{\mu\nu}$$

Entropic field $\rightarrow$ probability distribution $\rightarrow$ Fisher-Rao metric $\rightarrow$ Levi-Civita connection $\rightarrow$ information Riemann curvature $\rightarrow$ information Einstein tensor $\rightarrow$ physical Einstein tensor.

Right-hand side (matter): The construction chain is:

$$E(x) = \rho e^{i\Theta} \rightarrow g^{\text{FS}}_{ab} \rightarrow \Gamma^{(\pm 1)}_{abc} \rightarrow T^{(\text{ent})}_{\mu\nu}$$

Complexified entropic field $\rightarrow$ Fubini-Study metric $\rightarrow$ alpha-connections $\rightarrow$ entropic stress-energy tensor.

The GEFE/OFE equates the Fisher-Rao geometry (information about the entropic field's statistical structure) with the Fubini-Study dynamics (the quantum state of the entropic field). This is a **Fisher-Rao = Fubini-Study identity** at the level of the gravitational field equations — a profound statement that the statistical information content (geometry) equals the quantum dynamical content (matter) of the entropic field.

D.8 The Full Obidi Action (LOA + SOA) in the Gravitational Sector

For completeness, we write the full Obidi Action for the gravitational sector:

$$S^{(\text{grav})}_{\text{Obidi}} = S^{(\text{grav})}_{\text{LOA}} + S^{(\text{grav})}_{\text{SOA}} \text{ (IE-S.78)}$$

with:

$$S^{(\text{grav})}_{\text{LOA}} = \int d^4x \sqrt{g} [\frac{1}{2}(\partial S)^2 + V(S) + \xi R S^2] \text{ (IE-S.79)}$$

$$S^{(\text{grav})}_{\text{SOA}} = f_0 \Lambda^4 V_4 a_0 + (f_2 \Lambda^2 / 6) \int d^4x \sqrt{g} R + f_4 \int d^4x \sqrt{g} [\text{curvature}^2 \text{ terms}] + \dots \text{ (IE-S.80)}$$

This action contains, within its compact structure, the following physical content:

- The kinetic energy of the entropic field (LOA, first term) — the dynamics of information propagation
- The entropic potential (LOA, second term) — determines the vacuum structure and particle spectrum
- The non-minimal coupling to curvature (LOA, third term) — generates the Einstein tensor
- The cosmological constant (SOA, a_0 term) — the vacuum energy of the entropic field
- The Einstein-Hilbert action (SOA, a_2 term) — the standard gravitational dynamics
- Higher-curvature gravity (SOA, a_4 and beyond) — quantum gravitational corrections

This is the complete gravitational action of the Theory of Entropicity. Everything — from the cosmological constant to the Einstein-Hilbert action to higher-curvature corrections — emerges from the entropic field through the LOA+SOA architecture. There are no additional inputs, no independently postulated actions, no free gravitational parameters beyond the entropic potential and the spectral cutoff function.

Appendix E — The Fisher-Rao Metric in the Gravitational Sector: Complete Development

E.1 Construction of the Fisher-Rao Metric from the Entropic Field

We now develop the Fisher-Rao metric in complete detail, providing the full step-by-step construction from the entropic field $S(x)$ and exploring all of its gravitationally relevant properties.

Starting from the entropic probability distribution $P(\omega; x) = \exp(-S(\omega; x))/Z(x)$, the score functions are:

$$s_\mu(\omega; x) = \partial_\mu \ln P(\omega; x) = -[\partial_\mu S(\omega; x) - \langle \partial_\mu S \rangle] \quad (\text{IE-S.81})$$

The score functions have zero mean by construction: $\langle s_\mu \rangle = \int d\omega P s_\mu = 0$. This is a fundamental property: the score function is a zero-mean random variable whose variance is the Fisher information. The Fisher-Rao metric is the covariance matrix of the score functions:

$$g_{\mu\nu}^{\text{FR}} = \text{Cov}[s_\mu, s_\nu] = \langle s_\mu s_\nu \rangle - \langle s_\mu \rangle \langle s_\nu \rangle = \langle s_\mu s_\nu \rangle \quad (\text{IE-S.82})$$

The Christoffel symbols of the Fisher-Rao metric can be expressed directly in terms of the entropic field. Using the general formula $\Gamma_{\mu\nu}^\lambda = \frac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu})$, we need the derivatives of the Fisher-Rao metric. Computing $\partial_\rho g_{\mu\nu}^{\text{FR}}$:

$$\partial_\rho g_{\mu\nu}^{\text{FR}} = \langle \partial_\rho s_\mu \cdot s_\nu \rangle + \langle s_\mu \cdot \partial_\rho s_\nu \rangle + \langle s_\mu s_\nu s_\rho \rangle \quad (\text{IE-S.83})$$

where the last term arises from differentiating the expectation value with respect to the parameters (the "derivative of the measure" contribution). This last term is precisely the skewness tensor $T_{\mu\nu\rho} = \langle s_\mu s_\nu s_\rho \rangle$ that enters the Amari alpha-connections. This computation reveals the deep connection between the Christoffel symbols of the Fisher-Rao metric, the derivatives of the score functions, and the skewness tensor — all three are expressions of the same underlying entropic structure.

E.2 The Fisher-Rao Metric as the Unique Geometric Structure

The **Čencov uniqueness theorem** (CUT), stated in Section B.1, deserves a more thorough exposition given its fundamental importance for the Theory of Entropicity (ToE).

The theorem was first proved by **Nikolai Čencov** in his 1972 monograph (published in Russian; English translation 1982). **Čencov showed that the Fisher-Rao metric is the unique Riemannian metric on the space of probability measures that is invariant under congruent embeddings by Markov morphisms.** A Markov morphism is a mapping between probability spaces that preserves the statistical structure — it is the information-theoretic analogue of an isometry in Riemannian geometry. Invariance under Markov morphisms means that the metric is preserved when redundant degrees of freedom are integrated out — when information is coarse-grained.

For the Theory of Entropicity (ToE), the significance is extraordinary:

- The metric on the information manifold is not a choice among many possibilities — it is the *unique* possibility consistent with information-theoretic invariance.
- Through the emergence map, this uniqueness propagates to the physical spacetime metric. The spacetime metric is not arbitrary — it is determined by the entropic field through the unique information metric.

- This provides an explanatory gain over general relativity that cannot be overstated. In GR, the metric is a free dynamical variable: any smooth Lorentzian metric is, in principle, a possible spacetime geometry. The field equations constrain the metric given the matter content, but the metric itself is an unexplained input. In the ToE, the metric is *constructed* from the entropic field through the unique Fisher-Rao prescription. The space of possible spacetime geometries is therefore much smaller than in GR — only those geometries that can arise as Fisher-Rao metrics of some entropic field configuration are physically realizable.

E.3 The Fisher-Rao Metric and the Bekenstein-Hawking Entropy

The Fisher-Rao metric provides a natural framework for understanding the Bekenstein-Hawking entropy of black holes. The Bekenstein-Hawking formula $S_{\text{BH}} = A/(4l_p^2)$ relates the entropy of a black hole to its horizon area A , with the proportionality constant involving the Planck length $l_p = \sqrt{(\hbar G/c^3)}$.

In the ToE framework, the horizon area is measured by the Fisher-Rao metric on the information manifold. The physical area element is $dA_{\text{phys}} = \sqrt{(\det g^{S_{ij}})} dx^1 dx^2 = \lambda \sqrt{(\det g^I_{ij})} dx^1 dx^2$, where i, j range over the two dimensions of the horizon. The total area is:

$$A = \int_{\text{horizon}} dA_{\text{phys}} = \lambda \int_{\text{horizon}} dA^I \quad (\text{IE-S.84})$$

where $A^I = \int \sqrt{(\det g^I)} d^2x$ is the Fisher-Rao area of the horizon. The Bekenstein-Hawking entropy is then:

$$S_{\text{BH}} = A/(4l_p^2) = \lambda A^I/(4l_p^2) = [l_p^2/(4k_B)] \cdot A^I / (4l_p^2) = A^I/(16k_B) \quad (\text{IE-S.85})$$

The Bekenstein-Hawking entropy is proportional to the Fisher-Rao area of the horizon — the information-geometric area measured by the unique information metric. This result has a beautiful physical interpretation: the entropy of a black hole counts the distinguishable information states on its horizon, as measured by the Fisher-Rao metric. The factor $1/(16k_B)$ sets the conversion between information-geometric area and thermodynamic entropy.

E.4 Geodesics of the Fisher-Rao Metric and Gravitational Motion

The geodesics of the Fisher-Rao metric on the information manifold, when pushed forward to physical spacetime through the emergence map, yield the geodesics of general relativity — the trajectories of freely falling particles. This provides the entropic derivation of the equivalence principle and the geodesic equation.

A geodesic on the information manifold is a curve $x^\mu(s)$ that extremizes the Fisher-Rao length:

$$L_{FR} = \int \sqrt{(g^{\text{FR}}_{\mu\nu} dx^\mu/ds \cdot dx^\nu/ds)} ds \text{ (IE-S.86)}$$

This is the path of maximum statistical distinguishability — the path along which adjacent probability distributions are maximally different, as measured by the Fisher-Rao metric. The geodesic equation is:

$$d^2x^\alpha/ds^2 + \Gamma^{\text{FR},\alpha}_{\mu\nu} dx^\mu/ds dx^\nu/ds = 0 \text{ (IE-S.87)}$$

When the physical spacetime metric g_S is proportional to the Fisher-Rao information metric g_I with a constant factor λ , the Christoffel symbols of the two metrics coincide. Consequently, the unparametrized geodesics of physical spacetime correspond to the geodesics of the information manifold. In this sense, freely falling motion in spacetime can be viewed as following curves of extremal Fisher-Rao length between nearby entropic

configurations—that is, curves of extremal statistical distinguishability in the underlying information geometry.

The equivalence principle — the statement that all bodies fall with the same acceleration in a gravitational field, regardless of their mass or composition — follows from the Čencov uniqueness theorem. Since the Fisher-Rao metric is the unique invariant metric, all observers measure the same metric, and all observers agree on geodesics. There is no freedom in the metric to introduce composition-dependent effects. **The equivalence principle is therefore not a postulate (as in GR) but a theorem — a consequence of the uniqueness of the information metric.**

Appendix F — The Fubini-Study Metric in the Gravitational Sector: Complete Development

F.1 From Fisher-Rao to Fubini-Study: The Quantum Extension

The Fisher-Rao metric governs the classical statistical structure of the entropic field. When quantum effects are important — when the entropic field exhibits coherence, superposition, and entanglement — the appropriate geometric structure is the Fubini-Study metric, which is the natural metric on the space of quantum states (rays in Hilbert space, i.e., the complex projective space $\mathbb{CP}^n$).

The precise mathematical relationship between the two metrics was established in Theorem C.2: for diagonal states (classical probability distributions), the Fubini-Study metric reduces to $\frac{1}{4}$ times the Fisher-Rao metric. The Fubini-Study metric is therefore the quantum extension of the Fisher-Rao metric — it contains the full classical (Fisher-Rao) structure as a limiting case, plus additional quantum structure encoded in the off-diagonal coherences and the phase degrees of freedom.

The Fubini-Study metric was first introduced independently by Guido Fubini (1903) and Eduard Study (1905) in the context of projective geometry. Its application to quantum mechanics was recognized by Provost and Vallee (1980), who showed that the Fubini-Study metric on the quantum state space is the natural metric for measuring distances between quantum states. In the ToE framework, the Fubini-Study metric acquires a new role: it governs the quantum matter sector of the GEFE/OFE, providing the geometric structure from which the stress-energy tensor is constructed.

F.2 The Fubini-Study Metric on the Entropic State Space

The complexified entropic field $E(x) = \rho(x) \exp(i\Theta(x))$ at each spacetime point defines a ray in the complex Hilbert space. The Fubini-Study metric on this space of rays is:

$$ds^2_{\text{FS}} = |dE|^2/|E|^2 - |E^*dE|^2/|E|^4 \text{ (IE-S.88)}$$

Computing: $|dE|^2 = (d\rho)^2 + \rho^2(d\Theta)^2$, $|E|^2 = \rho^2$, $E^*dE = \rho(d\rho + i\rho d\Theta)$, $|E^*dE|^2 = \rho^2[(d\rho)^2 + \rho^2(d\Theta)^2]$.

Therefore:

$$ds^2_{\text{FS}} = [(d\rho)^2 + \rho^2(d\Theta)^2]/\rho^2 - [\rho^2(d\rho)^2 + \rho^4(d\Theta)^2]/\rho^4$$

We must be cautious here to avoid fatal errors — this would give zero, which indicates we must be more careful: The Fubini-Study metric for a one-dimensional complex state (a single complex number modulo phase) requires projection. For a multi-component state vector $|\psi\rangle = (\psi_1, \psi_2, \dots,$

ψ_n), the metric is non-trivial because different components can vary independently. For the entropic field, the relevant construction is the field at each point viewed as an infinite-component vector over the substrate modes:

$$|E(x)\rangle = \sum_k \rho_k(x) \exp(i\Theta_k(x)) |k\rangle \text{ (IE-S.89)}$$

where $|k\rangle$ are the substrate basis states. In this multi-component case, the Fubini-Study metric is genuinely non-trivial and takes the form (for the spacetime-dependent parameters x^μ):

$$g_{\mu\nu}^{\text{FS}}(x) = \text{Re}[\langle \partial_\mu E | \partial_\nu E \rangle / \langle E | E \rangle - (\langle \partial_\mu E | E \rangle \langle E | \partial_\nu E \rangle) / \langle E | E \rangle^2] \text{ (IE-S.90)}$$

For the decomposition into amplitude and phase sectors, the effective metric on the field space has the line element $(d\rho/\rho)^2 + (d\Theta)^2$ as the leading-order structure, with corrections from the multi-mode structure.

F.3 The Fubini-Study Curvature and Gravitational Corrections

The curvature of the Fubini-Study metric on CP^n is constant and positive: the sectional curvature is 1 (in standard normalization) and the Ricci curvature is $2(n+1)$. For a finite-dimensional quantum system with $n+1$ states, CP^n has constant positive holomorphic sectional curvature, making it the quantum analogue of the sphere (which has constant positive sectional curvature in the real case).

For the entropic field — which is an infinite-dimensional quantum system (infinite number of substrate modes) — the effective Fubini-Study curvature depends on the field configuration. The curvature is determined by the Berry curvature (see Section F.4) and produces quantum corrections to the gravitational field equations. These corrections scale as $\hbar_{\text{ent}}$ (the entropic Planck constant) and become important near the Planck scale or in regions of extreme curvature.

The gravitational significance of the Fubini-Study curvature is that it provides the quantum corrections to the matter sector of the GEFE/OFE. In the classical limit (where the Fubini-Study metric reduces to the Fisher-Rao metric), these corrections vanish and the standard classical stress-energy tensor is recovered. In the quantum regime, the Fubini-Study corrections modify the stress-energy tensor, potentially resolving singularities and producing observable quantum gravitational effects.

F.4 The Berry Connection and Gravitational Holonomy

The Berry connection on the entropic state space generates a geometric phase for the parallel transport of entropic states. When the parameters x^μ of the entropic field are varied adiabatically around a closed loop in spacetime, the entropic state acquires a Berry phase:

$$\gamma = \oint A_\mu dx^\mu \text{ (IE-S.91)}$$

where $A_\mu = i \langle E | \partial_\mu E \rangle / \langle E | E \rangle$ is the Berry connection. The Berry curvature is:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \text{ (IE-S.92)}$$

This Berry curvature is the gravitational analogue of the electromagnetic field tensor. Just as a charged particle acquires an Aharonov-Bohm phase when transported around a loop enclosing magnetic flux, the entropic state acquires a Berry phase when transported around a loop enclosing gravitational curvature. The Berry phase encodes the holonomy of the gravitational connection — the total rotation of the entropic state after parallel transport around a closed loop in spacetime.

The connection between the Berry curvature and the Riemann curvature of spacetime provides the Fubini-Study contribution to the GEFE/OFE: the Berry curvature is the quantum information-geometric analogue of the Riemann curvature, and the two are related through the emergence map.

Appendix G — The Amari-Čencov Alpha-Connections in the Gravitational Sector

G.1 The Alpha-Connection Formalism: Complete Development

The alpha-connection formalism, developed by Shun-ichi Amari in the 1980s building on Čencov's foundational work, provides the complete non-Riemannian geometric structure of statistical manifolds. While the Fisher-Rao metric provides the Riemannian structure (distances, angles, volumes), the alpha-connections provide the affine structure (parallel transport, geodesics, curvature) — and remarkably, there is not one connection but a continuous one-parameter family, parameterized by $\alpha \in \mathbb{R}$.

The alpha-connection $\Gamma^{(\alpha)}_{abc} = \Gamma^{(0)}_{abc} + (\alpha/2)T_{abc}$, where T_{abc} is the skewness tensor, has been defined in Section C.5. We now develop its gravitational consequences in full detail.

The skewness tensor $T_{abc} = \langle s_a s_b s_c \rangle$ is the third moment of the score functions. It is totally symmetric: $T_{abc} = T_{bac} = T_{acb} = \dots$ (all six permutations are equal). For the Gaussian distribution (the maximum entropy distribution with fixed mean and variance), $T_{abc} = 0$ — Gaussians are symmetric, with vanishing skewness. For non-Gaussian distributions, $T_{abc} \neq 0$ and the alpha-connections are genuinely different from the Levi-Civita connection.

In the gravitational context, the skewness tensor measures the asymmetry of the entropic field's probability distribution. A non-zero skewness means that the entropic field has preferred directions of fluctuation — it is more likely to fluctuate upward than downward (or vice versa) from its mean. This asymmetry has gravitational consequences: it introduces torsion-like effects and dual geometric structures that are absent from standard general relativity.

G.2 The Three Canonical Connections in Gravity

The e-Connection ($\alpha = +1$) in the Gravitational Sector

The exponential connection (e-connection) is defined by $\Gamma^{(+1)} = \Gamma^{(0)} + \frac{1}{2}T$. In the theory of statistical manifolds, the e-connection is the natural connection for exponential families — families of probability distributions of the form $p(x; \theta) = \exp(\theta^i F_i(x) - \psi(\theta)) h(x)$. The natural (canonical) parameters θ^i are e-affine: they vary linearly along e-geodesics.

In the gravitational sector, the e-connection governs how matter sources curvature. The physical interpretation is:

- The e-geodesics on the information manifold correspond to trajectories along which the natural parameters (energy, momentum, particle number — the conserved charges) vary linearly.
- Free-fall geodesics in physical spacetime, when pulled back to the information manifold, are e-geodesics of the Fisher-Rao metric. This is because freely falling observers maintain constant conserved charges (energy, momentum) along their worldlines.
- The e-connection curvature $R^{(+1)}_{abcd}$ measures the failure of the exponential family structure — the extent to which the entropic field deviates from an exponential distribution. Regions where $R^{(+1)}$ is large are regions where the matter distribution is strongly non-exponential — i.e., where the simple thermodynamic description breaks down.

The m-Connection ($\alpha = -1$) in the Gravitational Sector

The mixture connection (m-connection) is defined by $\Gamma^{(-1)} = \Gamma^{(0)} - \frac{1}{2}T$. In statistics, the m-connection is the natural connection for mixture families — families of distributions that are

convex combinations (mixtures) of basis distributions. The mixture (expectation) parameters $\eta_i = \langle F_i \rangle$ are m-affine.

In the gravitational sector, the m-connection governs how spacetime geometry responds to matter:

- The m-geodesics correspond to trajectories along which the expectation values (density, pressure, temperature) vary linearly. These are the trajectories of thermodynamic equilibrium states.
- Thermodynamic equilibrium states of the gravitational field (maximally symmetric spacetimes, static solutions) lie on m-geodesics of the information manifold.
- The m-connection curvature $R^{(-)}$ measures the failure of the mixture family structure — how far the vacuum state deviates from a simple mixture of basis states.

The Levi-Civita Connection ($\alpha = 0$) in the Gravitational Sector

The Levi-Civita connection $\Gamma^{(0)}$ is the unique torsion-free, metric-compatible connection of the Fisher-Rao metric. It is the arithmetic mean of the e- and m-connections: $\Gamma^{(0)} = \frac{1}{2}(\Gamma^{(+)} + \Gamma^{(-)})$. In standard general relativity, the Levi-Civita connection is the *only* connection used — it governs all aspects of gravitational dynamics, from geodesic motion to curvature to parallel transport. The Theory of Entropicity reveals that this is a simplification: the full information-geometric structure of the gravitational field involves all three connections (and indeed the entire one-parameter family), with the Levi-Civita connection representing only the "average" geometric structure.

The physical role of the Levi-Civita connection in the ToE framework is that of the *neutral* or *balanced* connection: it describes propagation that is neither biased toward the exponential (kinetic) sector nor toward the mixture (potential) sector. Free propagation of test particles in the absence of non-gravitational forces follows the Levi-Civita geodesics — this is the geodesic

hypothesis of general relativity, which is recovered as the $\alpha = 0$ limit of the more general alpha-geodesic structure.

The key insight of the ToE framework is that Einstein's general relativity uses only the $\alpha = 0$ connection, while the full entropic dynamics involves all alpha-connections. The e-connection ($\alpha = +1$) and m-connection ($\alpha = -1$) provide additional geometric structure that produces corrections to Einstein's predictions — corrections that become significant when the entropic field has strong non-Gaussian fluctuations (large skewness tensor T_{abc}).

G.3 The Duality Structure and Its Gravitational Consequences

The alpha-duality between the e-connection and m-connection (Theorem C.4) has profound consequences for gravitational physics. We now develop these consequences in detail.

The duality relation $g(\nabla^{(+)}_X Y, Z) + g(Y, \nabla^{(-)}_X Z) = X g(Y, Z)$ means that the e-connection and m-connection are "conjugate" with respect to the Fisher-Rao metric. In the gravitational context, this duality reflects a deep complementarity between the *matter sector* (governed by the e-connection) and the *geometric sector* (governed by the m-connection).

Consider a physical process in which matter is present and curves spacetime. In the standard Einsteinian picture, this is described by the single Levi-Civita connection: matter follows geodesics of the Levi-Civita connection, and curvature is computed from the Levi-Civita connection. In the ToE picture, the process has a dual description:

- **The e-description:** Matter sources are described using the e-connection. The natural parameters (energy, momentum, conserved charges) vary linearly along e-geodesics. The e-curvature measures how the matter distribution deviates from thermal equilibrium. This is the "source" perspective — how matter generates curvature.

- **The m-description:** Geometric responses are described using the m-connection. The expectation parameters (density, pressure, average curvature) vary linearly along m-geodesics. The m-curvature measures how the vacuum geometry deviates from maximal symmetry. This is the "response" perspective — how geometry reacts to matter.
- **The Levi-Civita description:** The balanced ($\alpha = 0$) description averages the source and response perspectives. This is the description used in standard general relativity.

The duality ensures that the source and response perspectives are consistent: the e-projection of the matter configuration onto the space of equilibrium states equals the m-projection of the geometric configuration onto the same space. This consistency condition is precisely the content of the Einstein field equations — expressed in the language of information geometry.

Theorem G.1 (Duality Interpretation of Einstein's Equations)

Einstein's field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ express the condition that the e-connection curvature (matter sector) and the m-connection curvature (geometric sector) are dual with respect to the Fisher-Rao metric on the entropic information manifold. The standard Levi-Civita ($\alpha = 0$) formulation is the balanced mean of the two dual descriptions.

G.4 The Projection Theorem and Gravitational Dynamics

Amari's projection theorem states that, given a probability distribution p and a submanifold M of the statistical manifold, the e-projection of p onto M (the closest point on M in the e-divergence sense) and the m-projection of p onto M (the closest point in the m-divergence sense) are related by a Pythagorean-like identity. In the gravitational context:

- **Gravitational collapse** is an e-projection: a matter distribution evolves by seeking the e-closest equilibrium configuration. The matter falls inward, following e-geodesics toward the state of maximum entropy consistent with the conserved charges. The final state (a black hole, if the mass is sufficient) is the e-projection of the initial matter distribution onto the submanifold of equilibrium configurations.
- **Gravitational radiation** is the m-projection: the spacetime geometry adjusts by emitting gravitational waves, seeking the m-closest vacuum (Ricci-flat) solution. The radiation carries away the "excess" geometry — the part of the curvature that cannot be sustained by the matter sources. The final vacuum geometry is the m-projection of the initial geometry onto the submanifold of vacuum solutions.
- **Einstein's equations** describe the equilibrium where the e-projection and m-projection coincide: the matter has reached its equilibrium state, the geometry has adjusted to accommodate it, and the two are in balance. This is the fixed point of the dual projection process.

G.5 The Full Alpha-Connection Structure of the GEFE/OFE

The GEFE/OFE can be rewritten using the alpha-connection formalism to make the dual structure explicit. Using all **three canonical connections**, the GEFE/OFE takes the form:

$$\nabla^{(o)}_{\mu} G^{\mu\nu} = 0 \text{ (Bianchi identity — Levi-Civita, automatic) (IE-S.93)}$$

$$\nabla^{(+1)}_{\mu} T^{\mu\nu} = J^{\nu}_{(\text{source})} \text{ (e-connection — matter dynamics with source terms) (IE-S.94)}$$

$$\nabla^{(-1)}_{\mu} G^{\mu\nu} = J^{\nu}_{(\text{response})} \text{ (m-connection — geometric response terms) (IE-S.95)}$$

The standard Einstein equations use only the $\alpha = 0$ connection and set both source and response terms to zero. The GEFE/OFE uses all three connections, with the source and response terms determined by the skewness tensor T_{abc} of the entropic field distribution. When the skewness vanishes (Gaussian entropic field), the source and response terms vanish and the standard Einstein equations are recovered. When the skewness is non-zero (non-Gaussian entropic field), the GEFE/OFE predicts corrections to Einstein that depend on the third-order statistics of the entropic field — corrections that are unique to the information-geometric framework and have no analogue in standard general relativity.

Appendix H — The Obidi Curvature Invariant and Informational Dark Curvature

H.1 The OCI in the Gravitational Context

The Obidi Curvature Invariant (OCI), introduced in Letter IE, is the central quantity that encodes the information-geometric corrections to Einstein's gravity. It is defined as the scalar measure of excess curvature of the information manifold over the physical spacetime manifold:

$$\mathcal{O}_\Omega = R^I - (1/\lambda) \Phi_*(R^S) \geq 0 \text{ (IE-S.96)}$$

where R^I is the scalar curvature of the information manifold, R^S is the scalar curvature of physical spacetime, λ is the emergence coupling constant, and Φ_* denotes the pushforward. The inequality

$\mathcal{O}_\Omega \geq 0$ states that the information manifold is always at least as curved as physical spacetime — it can be more curved, but never less.

The tensorial version of the OCI provides the full geometric content:

$$O^{(\text{OCI})}_{\mu\nu} = \lambda \Phi_*(G^I_{\mu\nu}) - G^S_{\mu\nu} \text{ (IE-S.97)}$$

This tensor vanishes in the leading-order (large N) limit where the Curvature Transfer Theorem (CTT) is exact. At next-to-leading order, it captures the gravitational effects of the "hidden" information curvature — curvature that exists in the statistical structure of the entropic field but does not directly appear in the physical spacetime geometry.

H.2 Physical Interpretation: Informational Dark Curvature

The OCI has a striking physical interpretation: it acts as an additional source of gravitational effects that mimics dark matter and dark energy without requiring exotic particles or fine-tuned parameters. The GEFE/OFE can be rewritten as:

$$G^S_{\mu\nu} = (8\pi G/c^4) T^{(\text{visible})}_{\mu\nu} + O^{(\text{OCI})}_{\mu\nu} \text{ (IE-S.98)}$$

where $T^{(\text{visible})}_{\mu\nu}$ is the stress-energy tensor of visible (directly observable) matter. The OCI tensor acts as an additional "dark" source term that produces gravitational effects beyond those of visible matter:

- **Dark matter effects:** In galaxies and galaxy clusters, the information manifold has excess curvature (from the statistical correlations of the entropic field across galactic scales) that produces additional gravitational attraction. This excess attraction manifests as flat rotation curves, gravitational lensing anomalies, and the observed velocity dispersions in

galaxy clusters — all without invoking cold dark matter particles. The OCI provides an effective dark matter halo whose density profile is determined by the information-geometric structure of the entropic field.

- **Dark energy effects:** On cosmological scales, the OCI can produce a negative-pressure contribution that drives the accelerated expansion of the universe. The entropic cosmological function $\Lambda_{\text{ent}}[S]$ from the SOA, combined with the OCI, provides a dynamical dark energy whose equation of state can deviate from $w = -1$, producing quintessence-like behavior.
- **Modified gravity effects:** At intermediate scales (the transition between the Newtonian and dark matter-dominated regimes), the OCI produces modifications to the gravitational force law that are reminiscent of **MOND (Modified Newtonian Dynamics)** but derived from first principles rather than postulated.

H.3 The OCI and the Cosmological Constant Problem

The **cosmological constant problem (CCP)** — often called the "**worst prediction in physics**" — is the discrepancy between the vacuum energy predicted by quantum field theory ($\Lambda_{\text{QFT}} \sim M_{\text{P}}^4 \sim 10^{76} \text{ GeV}^4$) and the observed cosmological constant ($\Lambda_{\text{obs}} \sim 10^{-47} \text{ GeV}^4$), a ratio of approximately 10^{122} .

In the GEFE/OFE framework, this problem is reframed and potentially resolved. The cosmological constant is not the vacuum energy of quantum fields — it is a property of the entropic field determined by the SOA. Specifically:

1. The a_0 term of the SOA generates a "bare" cosmological constant $\Lambda_{\text{bare}} = 6f_0\Lambda^2/f_2$, which is large (of order Λ^2 , the square of the entropic cutoff).
2. The OCI tensor $O^{(\text{OCI})}_{\mu\nu}$ provides a "screening" contribution that absorbs most of this large bare cosmological constant into the information curvature. The excess information

curvature acts as a counterterm, reducing the effective cosmological constant to a small residual value.

3. The observed cosmological constant Λ_{obs} is the small residual: $\Lambda_{\text{obs}} = \Lambda_{\text{bare}} - \Lambda_{\text{OCI}}$, where Λ_{OCI} is the screening contribution from the OCI. The enormous cancellation (10^{122}) is not a fine-tuning problem but a natural consequence of the information-geometric structure: the OCI generically screens the bare cosmological constant down to the scale set by the infrared properties of the entropic field.

This resolution is conceptually analogous to the Vainshtein screening mechanism in massive gravity, but implemented through information geometry rather than through a massive graviton. The key insight is that the information manifold has enough curvature to absorb the vacuum energy — the "missing" vacuum energy is not missing at all; it is stored in the excess curvature of the information manifold, where it does not directly affect the physical spacetime geometry.

That is, from the perspective of the Theory of Entropicity (ToE), physical spacetime only sees the “projected” curvature of the information manifold. The enormous curvature associated with vacuum energy remains in the information manifold and does not gravitate.

This mechanism is a core innovation of the Theory of Entropicity (ToE).

H.4 The OCI = $\ln 2$: The Fundamental Entropic Constant

In the foundational letters of the ToE series, the minimal value of the OCI was established as $\mathcal{O}_{\Omega, \min} = \ln 2$. This value has a deep information-theoretic significance: $\ln 2$ is the entropy of a single binary choice (one bit), measured in natural units (nats). It is the minimal amount of information required to distinguish between two states — the most basic act of measurement or observation.

The appearance of $\ln 2$ as the fundamental OCI scale connects the gravitational sector to the information-theoretic foundations of the theory:

- The minimal OCI = $\ln 2$ sets the scale at which deviations from Einstein's gravity become significant. When the information curvature exceeds the physical curvature by at least $\ln 2$ per Planck area, the OCI corrections are non-negligible.
- The Bekenstein-Hawking entropy $S_{\text{BH}} = A/(4l_{\text{P}}^2)$ can be interpreted as the number of minimal OCI quanta (each carrying $\ln 2$ nats of entropy) that fit on the horizon area.
- The OCI = $\ln 2$ establishes a universal lower bound on the information content of any gravitational system: no region of spacetime can have less than $\ln 2$ nats of informational dark curvature per Planck area. This is the entropic analogue of the uncertainty principle — a fundamental limit on how "classical" a gravitational system can be.

Appendix I — Philosophical and Conceptual Foundations of the GEFE/OFE

I.1 The Ontological Revolution: From Geometry + Matter to Entropy

The Generalized Entropic Field Equations represent not merely a new set of field equations but a fundamental shift in the ontological structure of physics. In the standard Einsteinian framework, the ontology consists of two independent categories: *spacetime geometry* (the metric, curvature, topology) and *matter-energy* (fields, particles, radiation). These two categories are related by the

field equations, but they are ontologically independent — each could, in principle, exist without the other. The field equations constrain their relationship but do not derive one from the other.

The ToE framework replaces this dualistic ontology with a monistic one. There is a single ontological category — the *entropic field* — from which both geometry and matter emerge as different manifestations. Spacetime geometry is how the entropic field organizes itself informationally (the Fisher-Rao metric). Matter-energy is how the entropic field manifests dynamically (the Fubini-Study kinetic energy and the entropic stress-energy tensor). The field equations do not relate two independent things; they express the self-consistency of a single thing manifesting in two complementary ways.

This ontological shift has far-reaching consequences. It resolves the conceptual problem that Einstein identified — the "makeshift" nature of the right-hand side — by showing that both sides have a common origin. It eliminates the need for independent postulations of matter fields and their properties. And it provides a natural arena for quantum gravity, since the entropic field is inherently quantum-mechanical (through the Fubini-Study metric and the Berry connection), whereas the classical spacetime metric is a derived, emergent quantity.

The monistic ontology of ToE is not merely a reductionism — it is an explanatory unification. It does not say that geometry and matter are "really the same thing" in a trivial sense; it provides the specific mathematical mechanism (the Obidi Action, the emergence map, the Fisher-Rao and Fubini-Study metrics) by which the single entropic field generates the apparent diversity of physical phenomena. The diversity is real — geometry and matter are genuinely different aspects of the world — but the underlying unity is also real, and it is expressed mathematically by the GEFE/OFE.

I.2 The Explanatory Gain of the Entropic Framework

The Theory of Entropicity, through the GEFE/OFE, answers several questions that Einstein's general relativity leaves unanswered:

Why does spacetime have four dimensions? In GR, the dimensionality is assumed. In ToE, the Fisher-Rao metric on the entropic manifold with the minimal substrate structure (a real-valued entropic field on a one-dimensional substrate with temporal evolution) has signature $(-, +, +, +)$ in four dimensions. **The dimensionality emerges from the information-geometric structure of the entropic field.**

Why does gravity obey the equivalence principle? In GR, the equivalence principle is postulated. In ToE, it follows from the Čencov uniqueness theorem: **since the Fisher-Rao metric is the unique invariant metric on the statistical manifold, all observers measure the same metric, and all observers agree on geodesics.** There is no information-geometric freedom to introduce composition-dependent gravitational effects.

Why is the cosmological constant small? In GR, the smallness of Λ is an unexplained fine-tuning. In ToE, the OCI screening mechanism absorbs the large bare cosmological constant into the information curvature, leaving a small residual that is determined by the infrared properties of the entropic field.

Why are there both geometry and matter? In GR, geometry and matter are independent ontological categories. In ToE, they are complementary manifestations of the same entropic field: geometry from the amplitude structure (Fisher-Rao metric), matter from the dynamical structure (Fubini-Study metric), and electromagnetism from the phase structure.

Thus, **in Einstein's General Relativity (GR), geometry and matter are distinct ontological categories. In the Theory of Entropicity (ToE), they are complementary manifestations of a**

single entropic field: the Fisher–Rao amplitude geometry gives rise to spacetime curvature, the Fubini–Study dynamical geometry gives rise to matter, and the phase geometry yields electromagnetism.

This constitutes a geometry–matter duality analogous in spirit to de Broglie’s wave–particle duality: two apparently different physical structures emerging from a single underlying entity.

Why does the speed of light govern both gravity and electromagnetism? In GR, c is a postulated universal constant. In ToE, $c = \sqrt{(\kappa/\rho_S)}$ is the entropic speed limit — the maximum speed at which entropic disturbances propagate — and it governs all sectors of the entropic field because all sectors are manifestations of the same underlying field.

I.3 The Wheeler Program Realized

John Archibald Wheeler's "it from bit" program, articulated in his 1990 essay, proposed that physics is fundamentally informational — that every physical quantity derives from information. Wheeler's vision was compelling but programmatic: he provided the philosophical motivation but not the [full] mathematical implementation.

The GEFE/OFE represents the mathematical realization of Wheeler's program in the gravitational sector. The correspondences are precise:

- **Wheeler's "bit"** $\leftrightarrow$ the entropic field $S(x)$ and its information content as measured by the **Fisher-Rao metric**
- **Wheeler's "it"** $\leftrightarrow$ the physical spacetime geometry and matter content, generated by the entropic field through **the emergence map** and the **Obidi Action**

- **Wheeler's "apparatus-elicited answers to yes-or-no questions"** $\leftrightarrow$ the measurements on the entropic field that define the probability distributions $P(\omega; x)$ from which the Fisher-Rao metric is constructed
- **Wheeler's dictum "no phenomenon is a phenomenon until it is an observed phenomenon"** $\leftrightarrow$ the ToE principle that spacetime geometry exists only where the entropic field carries non-degenerate information (positive-definite Fisher-Rao metric)

The ToE goes beyond Wheeler in a crucial respect: it provides the **specific action principle** (the **Obidi Action**), the specific geometric structures (Fisher-Rao, Fubini-Study, alpha-connections), and the specific field equations (the GEFE/OFE) that **implement the "it from bit" program quantitatively**.

Wheeler's vision was correct; the GEFE/OFE of ToE is its equation.

I.4 Comparison with Other Entropic/Emergent Gravity Programs

We have discussed the historical precursors in Appendix A.4. Here we provide a more detailed comparison of the GEFE/OFE with the four major programs for deriving gravity from more fundamental principles.

(a) Jacobson (1995): Jacobson derived Einstein's equations from the Clausius relation $\delta Q = T dS$ applied to local Rindler horizons. This was a landmark result showing that Einstein's equations are thermodynamic in nature. However, Jacobson's derivation assumes: (i) a pre-existing smooth spacetime manifold, (ii) a Lorentzian metric, (iii) a causal structure (to define Rindler horizons), and (iv) the Bekenstein-Hawking entropy-area relation. The ToE derives all four: the manifold from the parameter space of the entropic field, the metric from the Fisher-Rao metric, the causal structure from the entropic speed limit $c = \sqrt{(\kappa/\rho_s)}$, and the entropy-area relation from the OCI

bounds. The GEFE/OFE therefore represents a completion of Jacobson's program — it derives the same thermodynamic equations but from a deeper starting point.

(b) Verlinde (2010): Verlinde derived Newton's force law from entropic forces on holographic screens. The GEFE/OFE goes beyond Verlinde by providing: (i) the full nonlinear Einstein equations (not just Newton's law), (ii) the matter sector as well as the geometry sector, (iii) an action principle (the Obidi Action) rather than just a force law, and (iv) corrections beyond Einstein (OCI, higher-curvature terms) that Verlinde's approach does not predict.

(c) Padmanabhan (2002–2015): Padmanabhan showed that the Einstein-Hilbert action on solutions is a surface term (horizon entropy), that the field equations are thermodynamic identities on null surfaces, and that the cosmological constant relates to the discrepancy between surface and bulk degrees of freedom. The GEFE/OFE provides the dynamical framework underlying Padmanabhan's thermodynamic identities, deriving them from the Obidi Action rather than postulating them.

(d) Connes-Chamseddine Spectral Action (1996): The spectral action $\text{Tr } f(D^2/\Lambda^2)$ on a noncommutative space produces the Einstein-Hilbert action plus the Standard Model action. The SOA component of the Obidi Action is structurally analogous: $\text{Tr } f(\Delta_S/\Lambda^2)$ produces the Einstein-Hilbert action plus higher-curvature corrections. The key difference is that the SOA operates on the entropic manifold (which has a clear physical interpretation) rather than on an abstract noncommutative space. Additionally, the LOA provides the complementary local dynamics that the spectral action approach lacks.

I.5 The Problem of Time in Quantum Gravity and the Entropic Resolution

The problem of time is one of the deepest conceptual issues in quantum gravity. In canonical quantum gravity (the **Wheeler-DeWitt equation**), the Hamiltonian constraint $H|\Psi\rangle = 0$ implies that the wave function of the universe is time-independent — time disappears from the fundamental equation. This "frozen formalism" problem has resisted resolution for decades.

In the ToE framework, time is not fundamental but emergent — it is generated by the entropic field. The direction of time is the direction of maximal entropic gradient: time flows in the direction in which the entropic field increases most rapidly. The No-Rush Theorem provides a natural arrow of time by establishing an upper bound on the rate of entropic change: the entropic speed limit $c = \sqrt{(\kappa/\rho_S)}$ bounds the speed at which information can propagate, and the second law (monotonic increase of entropy in the forward direction) defines the future.

The problem of time is resolved in the GEFE/OFE because time is not an input to the theory — it is an output. The entropic field evolves on its own internal "clock" (the entropic flow parameter), and physical time emerges as the image of this flow parameter under the emergence map. The Wheeler-DeWitt equation, which is timeless, is replaced by the Master Entropic Equation (IE-S.71), which is manifestly time-dependent and has a well-defined initial-value formulation.

I.6 The Holographic Principle from the GEFE/OFE

The holographic principle — the conjecture that the maximum entropy of a region is proportional to its boundary area rather than its volume — is one of the most profound insights from the study of black hole thermodynamics and string theory. In the GEFE/OFE framework, the holographic principle is not a conjecture but a theorem, following from the properties of the OCI.

The **OCI bounds the information content of any region**: the scalar OCI $\mathcal{O}_\Omega \geq 0$ implies that the information curvature is at least as large as the physical curvature. For a region with boundary area A , the total information content is bounded by the Fisher-Rao area of the boundary (measured by the information metric). Since the Fisher-Rao area is proportional to the physical area through the emergence coupling λ , we obtain:

$$S_{\max} \leq A / (4l_P^2) = A / (4\lambda k_B) \text{ (IE-S.99)}$$

This is the Bekenstein bound — the maximum entropy of a region bounded by area A . It emerges naturally from the GEFE/OFE and the OCI, without any additional postulates about holographic screens or AdS/CFT duality.

I.7 The Arrow of Time and the Second Law of Thermodynamics from the GEFE/OFE

The second law of thermodynamics — the statement that entropy increases (or remains constant) in isolated systems — is one of the most fundamental laws of physics, yet its derivation from microscopic dynamics has been controversial since Boltzmann's time. In the GEFE/OFE framework, the second law is a consequence of the fundamental field equations, not a statistical approximation.

The **No-Rush Theorem (NRT)**, established in earlier letters of the ToE series, provides the key: it states that the entropic field $S(x)$ evolves monotonically in the forward time direction (defined as the direction of maximal entropic gradient). This monotonicity is not a statistical tendency — it is a consequence of the structure of the MEE (IE-S.71) and the positivity of the entropic potential. The second law is therefore built into the fundamental dynamics of the entropic field and, through the GEFE/OFE, into the fundamental dynamics of spacetime and matter.

The arrow of time — the distinction between past and future — is likewise not an external input but an emergent property of the entropic field. The past is the direction of lower entropy; the future

is the direction of higher entropy. This distinction is absolute (not observer-dependent) because the entropic field has a global direction of increase, determined by the boundary conditions of the entropic manifold.

Appendix J — Comparison Table: Einstein vs. GEFE/OFE

The following comprehensive table summarizes the structural differences between Einstein's field equations and the **Generalized Entropic Field Equations (GEFE)** [Obidi Field Equations (OFE)] of the Theory of Entropicity (ToE). Each row identifies a specific feature of the gravitational theory and shows how the two frameworks differ in their treatment of that feature.

Feature	Einstein's Field Equations	Generalized Entropic Field Equations (GEFE/OFE)
Fundamental variable	Metric tensor $g_{\mu\nu}$ — a symmetric rank-2 tensor field on a 4D manifold	Entropic field $S(x)$ — a scalar field on the substrate Ω , from which all other structures emerge
LHS generator	Riemann curvature of spacetime, computed from $g_{\mu\nu}$ and its derivatives	Fisher-Rao curvature of the information manifold, transferred to spacetime via the emergence map Φ
RHS generator	Independently postulated matter fields (EM, scalar, fermion, gauge) with separate Lagrangians	Entropic stress-energy from Fubini-Study metric and Obidi Action; all matter emerges from the entropic field
Speed of light	Postulated universal constant $c \approx 3 \times 10^8$ m/s	Derived: $c = \sqrt{(\kappa/\rho_s)}$, the entropic speed limit from the No-Rush Theorem

Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

The Theory of Entropicity (ToE) Living Review Letters Series (ToE-LRLS). *Letter IE*

John Onimisi Obidi (jonimisiobidi@gmail.com)

Research Lab, The Aether

May 6, 2026

Feature	Einstein's Field Equations	Generalized Entropic Field Equations (GEFE/OFE)
Gravitational constant	Postulated constant $G \approx 6.67 \times 10^{-11}$	Derived from SOA: $G = 6\pi/(f_2\Lambda^2)$, determined by the spectral geometry of the entropic Laplacian
Cosmological constant	Free parameter Λ with no explanation for its value; "worst prediction in physics"	Entropic cosmological function $\Lambda_{\text{ent}}[S]$ from SOA a_0 term, with OCI screening resolving the fine-tuning problem
Gauge invariance	Not addressed by GR; requires separate gauge theory (Yang-Mills)	Derived from the entropic shift symmetry $S \rightarrow S + \text{const}$ and the phase symmetry $\Theta \rightarrow \Theta + \alpha$
Electromagnetism	Separate theory (Maxwell), independently postulated	Emergent from the phase sector of the complexified entropic field (Letter IIA)
Dark matter	Requires exotic particles (WIMPs, axions) not yet detected	OCI tensor provides informational dark curvature — gravitational effects without exotic matter
Dark energy	Requires Λ fine-tuning to 10^{-122}	SOA + OCI provide natural dark energy through the entropic cosmological function
Quantum gravity	Unresolved; GR is non-renormalizable	Vuli-Ndlela Integral provides UV completion; SOA higher-curvature terms tame divergences
Information geometry	Not used; no information-theoretic content	Fisher-Rao (classical) + Fubini-Study (quantum) + Amari α -connections: full information-geometric framework
Connections used	Levi-Civita only ($\alpha = 0$)	All α -connections: e ($\alpha = +1$), m ($\alpha = -1$), LC ($\alpha = 0$), providing richer dynamical structure
Holographic principle	Added externally (Bekenstein, 't Hooft, Susskind, Maldacena)	Derived from the OCI bounds on information content
Arrow of time	Not explained; time-reversal symmetric equations	No-Rush Theorem provides natural arrow: entropy increases monotonically in the forward direction

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Feature	Einstein's Field Equations	Generalized Entropic Field Equations (GEFE/OFE)
Equivalence principle	Postulated as a foundational axiom	Derived from the Čencov uniqueness theorem: unique information metric implies universal free fall
Action principle	Einstein-Hilbert action (postulated)	Obidi Action = LOA + SOA (constructed from the entropic field's information-geometric structure)
Singularity behavior	Singularity theorems (Penrose-Hawking) predict geodesic incompleteness	SOA higher-curvature terms and OCI corrections resolve singularities via non-singular bounces
Unification	Does not unify gravity with other forces	Unifies gravity (amplitude sector) and EM (phase sector) from a single entropic field; prospective nuclear force unification

Each row of this table represents a genuine explanatory or predictive advantage of the GEFE/OFE over Einstein's equations. Taken together, they constitute a comprehensive argument that the GEFE/OFE provides a deeper, more explanatory, and more predictive framework for gravitational physics than Einstein's general relativity.

Appendix K — Worked Examples and Applications of the GEFE/OFE

K.1 The Schwarzschild Solution from the GEFE/OFE

We derive the Schwarzschild metric — the unique spherically symmetric, static, vacuum solution of Einstein's equations — from the GEFE/OFE, starting with the entropic field.

Assumption: Static, spherically symmetric entropic field $S = S(r)$, depending only on the radial coordinate.

Step 1: Solve the MEE for $S(r)$. In the vacuum (no external matter), the MEE reduces to $\square S - V'(S) = 0$ (in the minimal coupling limit $\xi \rightarrow 0$). For a static, spherically symmetric configuration with $V(S) = \frac{1}{2}m^2 S^2$ (quadratic potential, relevant for the weak-field limit), the radial solution is:

$$S(r) = S_0 + (S_1/r) \exp(-mr) \text{ (IE-S.100)}$$

For the gravitational sector, we take the massless limit ($m \rightarrow 0$), giving $S(r) = S_0 + S_1/r$ — a Coulomb-like entropic profile.

Step 2: Compute the Fisher-Rao metric from $S(r)$. For a one-parameter family of Gaussian distributions with variance proportional to $S(r)$, the Fisher-Rao metric component $g_{rr}^I \propto (dS/dr)^2/S^2 = S_1^2/(r^4 S^2(r))$.

Step 3: Apply the emergence map. The physical spacetime metric is $g^S = \lambda g^I$. For the static, spherically symmetric case, the metric takes the form:

$$ds^2 = -A(r) dt^2 + B(r) dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2) \text{ (IE-S.101)}$$

The functions $A(r)$ and $B(r)$ are determined by the Fisher-Rao metric of the entropic field profile $S(r)$. For the Coulomb profile $S = S_o + S_i/r$, the resulting physical metric (after solving the GEFE/OFE in the vacuum) is:

$$A(r) = 1 - 2GM/(rc^2), B(r) = [1 - 2GM/(rc^2)]^{-1} \text{ (IE-S.102)}$$

where $M = c^2 S_i / (2G\xi S_o)$ is the mass parameter, identified in terms of the entropic field parameters. This is exactly the Schwarzschild metric, confirming that the GEFE/OFE reproduces the standard black hole solution of general relativity.

Very Detailed Analysis and Exposition on K.1

THE THEORY OF ENTROPICITY (ToE)

LIVING REVIEW LETTERS SERIES

SUPPLEMENTARY APPENDIX K.1 TO LETTER IE

Appendix K.1: The Schwarzschild Solution from the Generalized Entropic Field Equations / Obidi Field Equations (GEFE/OFE)

The Unique Static, Spherically Symmetric Vacuum Geometry as an Emergent Classical Limit of the Theory of Entropicity (ToE)

John Onimisi Obidi

Research Lab, The Aether

The Theory of Entropicity (ToE) Living Review Letters Series. Letter IE

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Introduction to the ToE Schwarzschild solution

This appendix presents a complete, self-contained derivation of the Schwarzschild solution — the unique static, spherically symmetric vacuum geometry of general relativity — from the Generalized Entropic Field Equations (GEFE), equivalently designated the Obidi Field Equations (OFE), as formulated within the Theory of Entropicity (ToE). The derivation proceeds in four logically sequential steps: (1) solving the Minimal Entropic Equation (MEE) for a static, spherically symmetric entropic field configuration; (2) constructing the Fisher-Rao information metric on the statistical manifold induced by the entropic field; (3) applying the emergence map to obtain the physical spacetime metric; and (4) identifying the Schwarzschild mass parameter as an emergent entropic charge. The result demonstrates that the Schwarzschild geometry is not an independent postulate within the ToE framework, but rather an *emergent* classical consequence of an underlying entropic field configuration on the information manifold. The mass of a gravitating body is thereby reinterpreted as a measure of the entropic charge concentrated in its field profile. This appendix serves as the first concrete worked example of the GEFE/OFE producing a known exact

solution of Einstein's general relativity, and constitutes a foundational validation of the emergence mechanism articulated in Letter IE of the ToE Living Review Letters Series.

1. Introduction

1.1. The Schwarzschild Solution in General Relativity: Historical and Physical Context

In the annals of theoretical physics, few results carry the foundational weight of the Schwarzschild solution. Barely two months after Albert Einstein published his field equations of general relativity in November 1915 — the culminating achievement of a decade of intensive intellectual struggle — Karl Schwarzschild, writing from the Eastern Front of the First World War, produced the first exact, closed-form solution to those equations. His paper, communicated to the Prussian Academy of Sciences on 13 January 1916, described the gravitational field outside a static, spherically symmetric, non-rotating, uncharged mass distribution. The solution was communicated by Einstein himself, who reportedly expressed astonishment that an exact solution could be found at all, given the formidable nonlinearity of the field equations. Tragically, Schwarzschild died only months later, in May 1916, of the autoimmune disease pemphigus, contracted during his military

service. His solution, however, endures as one of the most consequential results in all of mathematical physics.

The Schwarzschild metric, expressed in the coordinates that bear his name, takes the form

$$ds^2 = -(1 - 2GM/(rc^2)) c^2 dt^2 + (1 - 2GM/(rc^2))^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\varphi^2)$$

where G is Newton's gravitational constant, M is the mass of the gravitating body, c is the speed of light, and (t, r, θ, φ) are the Schwarzschild coordinates. This metric describes the unique spacetime geometry outside any static, spherically symmetric, non-rotating, electrically neutral mass distribution. Its domain of applicability is breathtaking: it governs the gravitational field around ordinary stars such as the Sun, around planets, around neutron stars, and — in its maximally extended form — it describes the exterior and interior of non-rotating (Schwarzschild) black holes. Every measurement of gravitational phenomena in the solar system, from the precession of Mercury's perihelion to the Shapiro time delay of radar signals grazing the Sun, from the deflection of starlight by the solar gravitational field (first confirmed by Eddington's 1919 eclipse expedition) to the gravitational redshift of photons climbing out of a gravitational potential well (confirmed by the **Pound-Rebka experiment** of 1960), constitutes a test of the Schwarzschild solution.

The Schwarzschild metric possesses two distinguished radial scales. The first is the Schwarzschild radius, $r_s = 2GM/c^2$, at which the temporal metric coefficient vanishes and the radial metric coefficient diverges — a coordinate singularity that marks the event horizon of a black hole. The second is the genuine curvature singularity at $r = 0$, where the Kretschmer scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/(c^4r^6)$ diverges, signaling a breakdown of the classical description. For ordinary astrophysical bodies (stars, planets), the physical surface lies well outside the Schwarzschild radius, and only the exterior solution is physically relevant. For black holes, the full Schwarzschild geometry — including the event horizon and the interior — becomes essential.

The centrality of the Schwarzschild solution to gravitational physics cannot be overstated. It is the zeroth-order geometry around which all perturbative calculations in black hole physics, gravitational wave theory, and post-Newtonian celestial mechanics are organized. The quasi-normal modes of a Schwarzschild black hole, the **Regge-Wheeler** and **Zerilli** equations governing its perturbations, the Hawking temperature $T_H = \hbar c^3/(8\pi k_B GM)$, and the Bekenstein-Hawking entropy $S_{BH} = k_B c^3 A/(4G \hbar)$ — all are built upon the Schwarzschild foundation. Any theory of gravity that aspires to physical relevance must, as a minimal requirement, recover the Schwarzschild solution in the appropriate limit.

1.2. Birkhoff's Theorem and the Uniqueness of the Schwarzschild Geometry

In 1923, George David Birkhoff proved a theorem of extraordinary power and economy: the Schwarzschild solution is the *unique* spherically symmetric vacuum solution of Einstein's

field equations. This theorem — Birkhoff's theorem — has no analogue in Newtonian gravity. In Newtonian theory, the gravitational potential outside a spherically symmetric mass distribution is $\Phi = -GM/r$ by virtue of Newton's shell theorem (which Gauss's law for gravity guarantees), but the Newtonian theory does not constrain the time-dependence of the exterior field in the way that Birkhoff's theorem does.

The content of Birkhoff's theorem is more subtle than is often appreciated. It states not merely that the exterior field of a static, spherically symmetric body is given by the Schwarzschild metric — this much might be expected on grounds of symmetry — but that *any* spherically symmetric vacuum region of spacetime, whether or not the matter distribution is static, is described by the Schwarzschild geometry. A spherically symmetric star undergoing purely radial pulsations, for instance, produces no gravitational waves and generates an exterior field that is exactly Schwarzschild at every instant. This is because the only spherically symmetric gravitational degree of freedom — the mass monopole — is conserved by the Bianchi identity, and hence cannot radiate. Birkhoff's theorem is thus a rigidity result: spherical symmetry combined with the vacuum Einstein equations is so constraining that only a one-parameter family of solutions (parameterized by the mass M) exists.

The significance of Birkhoff's theorem for the present appendix is twofold. First, it guarantees that if the GEFE/OFE reduce to the vacuum Einstein equations in the appropriate classical limit, then the resulting geometry is *necessarily* Schwarzschild — the uniqueness is automatic. Second, it implies that the entropic field configuration that generates the Schwarzschild geometry must itself be unique: the only static, spherically

symmetric solution of the Minimal Entropic Equation in the massless, minimal-coupling limit is the Coulomb-like profile $S(r) = S_o + S_i/r$. This is the entropic analogue of Birkhoff's theorem, and we shall comment on it further in the Remarks at the end of this appendix.

1.3. Recovery of the Schwarzschild Solution as a Critical Test of the GEFE/OFE

Any proposed theory of gravitation — whether it supplements, modifies, or replaces Einstein's general relativity — must confront the Schwarzschild solution as its first and most demanding consistency test. The reason is empirical: the Schwarzschild metric has been confirmed to extraordinary precision by solar-system tests (the post-Newtonian parameter γ has been measured to agree with the GR prediction $\gamma = 1$ to one part in 10^5 by the Cassini spacecraft), by binary pulsar observations (the Hulse-Taylor pulsar PSR B1913+16), and most recently by direct imaging of the photon ring around the supermassive black hole M87* by the Event Horizon Telescope. A theory that cannot recover Schwarzschild in the weak-field, slow-motion, classical limit is empirically falsified before it begins.

The Generalized Entropic Field Equations (GEFE), equivalently the Obidi Field Equations (OFE), constitute the gravitational sector of the Theory of Entropicity. As formulated in Letter IE of the ToE Living Review Letters Series, the GEFE contain the standard Einstein tensor on the left-hand side and, on the right-hand side, a hierarchy of entropic contributions: the Leading-Order Approximation (LOA) terms encoding the kinetic energy, potential energy, and non-minimal coupling of the entropic field; the Second-

Order Approximation (SOA) terms encoding higher-curvature corrections derived from the Seeley-DeWitt heat kernel expansion; and the Obidi Curvature Invariant (OCI) tensor, encoding the mismatch between the information manifold curvature and the physical spacetime curvature. The full GEFE are considerably richer than the Einstein equations, and in general they predict corrections to GR at short distances, high curvatures, and quantum scales. However, in the classical, large-scale, weak-field limit — the regime probed by solar-system tests and by the exterior geometry of ordinary astrophysical bodies — the GEFE must and do reduce to the vacuum Einstein equations. It is in this limit that we shall recover the Schwarzschild solution.

1.4. Strategy of this Appendix

The strategy of this appendix is carefully circumscribed. We do *not* re-derive every step of general relativity from first principles — the Einstein field equations, the Bianchi identity, the post-Newtonian formalism, and so forth are taken as established results of the classical limit of the GEFE. What we *do* demonstrate is that a static, spherically symmetric entropic field configuration, processed through the Fisher-Rao information metric and the emergence map $\Phi: M_I \rightarrow M_S$, yields the Schwarzschild geometry in the classical limit, with the mass parameter M identified as an emergent entropic charge. The derivation proceeds in four steps:

1. **Step 1 (Radial Entropic Field Equation):** We solve the Minimal Entropic Equation (MEE) for a static, spherically symmetric entropic field $S = S(r)$ in the

weak-field, minimal-coupling regime, obtaining the Yukawa form and its massless (Coulomb-like) limit.

2. **Step 2 (Fisher-Rao Information Metric):** We construct the Fisher-Rao metric on the statistical manifold induced by the entropic field $S(r)$, using a Gaussian statistical model, and derive the radial and angular components of the information metric.
3. **Step 3 (Emergence Map and Physical Metric):** We apply the emergence map to identify the physical spacetime metric with the information metric, and invoke the classical limit of the GEFE (which reduces to the vacuum Einstein equations) to determine the full Schwarzschild geometry.
4. **Step 4 (Mass Identification):** We match the entropic field parameters to the Schwarzschild mass parameter, identifying M as an emergent entropic charge proportional to the integration constant S_i .

1.5. Connection to Letter IE and the Curvature Transfer Theorem

This appendix is a concrete worked example of the central theorem of Letter IE: the *Curvature Transfer Theorem* (Theorem 3.1). The Curvature Transfer Theorem states that, under the emergence map Φ , the Riemann curvature tensor of the physical spacetime (M_S, g_S) is determined by the pushforward of the Riemann curvature tensor of the information manifold (M_I, g_I) , up to corrections of order $O(1/N)$ where N is the number of entropic degrees of freedom. In the classical limit $(N \rightarrow \infty)$, the curvature transfer is exact: the

physical spacetime curvature *is* the information curvature, rescaled by the emergence constant λ .

The Schwarzschild derivation that follows is, in this light, a verification of the Curvature Transfer Theorem in the simplest non-trivial case: a static, spherically symmetric entropic configuration. The information manifold is curved (with curvature determined by the Fisher-Rao metric of the entropic profile $S(r) = S_o + S_i/r$), and this curvature is transferred, via the emergence map, to the physical spacetime, yielding the Schwarzschild geometry. The Schwarzschild curvature singularity at $r = 0$ is thereby reinterpreted as a singularity of the Fisher-Rao metric — a point at which the information content of the entropic field diverges — and the event horizon at $r = r_s$ is reinterpreted as an entropic surface of distinguished informational significance.

The derivation also connects to other elements of the ToE framework established in the Living Review Letters. The identification of mass as an entropic charge (Step 4) is analogous to the identification of electric charge as a topological winding number of the entropic phase field, established in Letter IIA. The Alemoh-Obidi Correspondence, articulated in Letter IC, provides the philosophical scaffolding: physical quantities are *not* fundamental but emergent from informational and entropic substrates. The present appendix provides the most direct calculational illustration of this correspondence to date.

2. Assumptions

Before proceeding to the derivation, we state and motivate three physically transparent assumptions that define the regime in which the Schwarzschild solution is recovered from the GEFE/OFE. Each assumption corresponds to a well-defined limit of the full ToE formalism, and each is necessary for the reduction of the GEFE to the vacuum Einstein equations.

Assumption 1: Static, Spherically Symmetric Entropic Field

The entropic field depends only on the radial coordinate: $S = S(r)$.

The first assumption restricts the entropic field to the simplest spatial configuration compatible with an isolated, non-rotating gravitating body. The assumption of spherical symmetry — invariance under the full rotation group $SO(3)$ — means that the entropic density is distributed isotropically around a central concentration, with no preferred angular direction. The assumption of staticity — independence of the time coordinate — means that the entropic field has reached a state of thermodynamic equilibrium: there are no entropic currents, no time-dependent fluctuations, and no dissipation. Taken together, these assumptions reduce the entropic field from a function of four spacetime coordinates, $S(t, r, \theta, \varphi)$, to a function of a single variable, $S(r)$.

This restriction is the entropic analogue of the standard symmetry assumptions used in deriving the Schwarzschild solution in general relativity. Just as the most general static,

spherically symmetric metric can be written in terms of two functions of r alone (the temporal and radial metric coefficients $A(r)$ and $B(r)$), the most general static, spherically symmetric entropic field is a function $S(r)$ alone. The reduction from a partial differential equation (the full MEE in four dimensions) to an ordinary differential equation (the radial MEE) is a direct consequence of this symmetry.

Physically, Assumption 1 models the gravitational field of an isolated, non-rotating, uncharged body — a star, a planet, or a non-rotating black hole — viewed from the exterior, at distances large compared to the body's radius. In the ToE framework, this configuration corresponds to a single, isolated concentration of entropic charge surrounded by an entropic vacuum. The entropic field $S(r)$ encodes the radial distribution of information: it is large near the central body (where the microscopic degrees of freedom are densely concentrated) and decays toward the asymptotic entropic background S_0 at spatial infinity.

Assumption 2: Minimal Coupling Regime

The non-minimal coupling parameter vanishes: $\xi \rightarrow 0$.

The second assumption concerns the coupling between the entropic field and the spacetime curvature. In the full Obidi Action (the action functional from which the GEFE are derived), the entropic field couples to the Ricci scalar R through a non-minimal coupling term of the form ξRS^2 . The dimensionless coupling constant ξ parameterizes the strength of this interaction. When $\xi = 0$, the entropic field is *minimally coupled* to gravity: it interacts with the spacetime geometry only through the covariant derivative (i.e.,

through the metric in the d'Alembertian operator) and through the effective stress-energy tensor that it contributes to the right-hand side of the GEFE.

The minimal coupling regime $\xi \rightarrow 0$ is the simplest and most well-studied case. It is the regime in which the entropic field behaves as a standard minimally coupled scalar field, analogous to the scalar fields studied extensively in scalar-tensor theories of gravity (Brans-Dicke theory, Horndeski theory, etc.) in the limit where the scalar-curvature coupling is turned off. The special value $\xi = 1/6$ corresponds to *conformal coupling* in four dimensions, at which the scalar field equation is conformally invariant on a conformally flat background. While conformal coupling has mathematical elegance and physical interest (it preserves the conformal structure of the equations of motion), the Schwarzschild solution is most straightforwardly recovered in the minimal coupling limit. Non-minimal coupling ($\xi \neq 0$) would produce corrections to the Schwarzschild geometry — scalar hair, modified horizons, and deviations in the post-Newtonian parameters — which constitute genuine predictions of the Theory of Entropicity (ToE) beyond standard general relativity. These corrections, and their observational consequences, are deferred to future work (see Remark K.1.1).

It is important to emphasize that Assumption 2 is an approximation, not a dogma. The full ToE formalism accommodates arbitrary values of ξ , and indeed the renormalization group flow of ξ under quantum corrections is a significant open question. The minimal coupling limit is adopted here because our goal is to recover the *standard* Schwarzschild solution, and any non-zero ξ would modify the result. The consistency of this approximation is confirmed *a posteriori* by the fact that the resulting geometry is Ricci-flat ($R = 0$), so that

the non-minimal coupling term $\xi R S^2$ vanishes identically — the assumption is self-consistent.

Assumption 3: Classical, Weak-Field, Large-Scale Limit

The GEFE reduce to the Einstein field equations in vacuum.

The third assumption specifies the regime in which the full complexity of the GEFE is suppressed, leaving only the vacuum Einstein equations. This is the classical limit of the Theory of Entropicity, and it is defined by a hierarchy of conditions. First, the *Second-Order Approximation* (SOA) corrections — higher-curvature terms arising from the Seeley-DeWitt heat kernel expansion of the one-loop effective action — are suppressed by powers of $(l_P/L)^2$, where $l_P = (\hbar G/c^3)^{1/2} \approx 1.6 \times 10^{-35}$ m is the Planck length and L is the characteristic macroscopic scale of the problem. For any astrophysical application — solar-system scales, stellar scales, or even black hole event horizons of stellar mass or larger — we have $l_P/L \ll 1$, and the SOA corrections are utterly negligible.

Second, the *Obidi Curvature Invariant* (OCI) tensor vanishes in the classical limit. The OCI measures the curvature mismatch between the information manifold (M_I, g_I) and the physical spacetime (M_S, g_S) . When the emergence map is an isometry (or a conformal isometry with constant conformal factor), the curvature of g_S is fully determined by the curvature of g_I , and the OCI vanishes. This is precisely the situation in the classical regime, where quantum fluctuations do not perturb the emergence map. In the quantum regime,

the OCI becomes non-zero and contributes additional curvature — the "dark curvature" discussed in Letter IE — which may have cosmological significance.

Third, in the classical limit, the entropic field amplitude is *frozen*: the field $S(r)$ is a classical background configuration, not a quantum operator subject to fluctuations. The path integral over entropic field configurations is dominated by a single saddle point — the classical solution — and quantum corrections (loop diagrams in the entropic field theory) are suppressed by powers of $\hbar$. Under these three conditions — negligible SOA corrections, vanishing OCI, and frozen amplitude — the full GEFE reduce exactly to the vacuum Einstein equations, $G_{\mu\nu} = 0$, or equivalently $R_{\mu\nu} = 0$. It is in this limit that the Schwarzschild solution emerges.

We emphasize that the classical limit is the regime in which the Theory of Entropicity makes the *same* predictions as general relativity. Beyond this limit — at short distances, high curvatures, or quantum scales — the ToE predicts corrections that could be observationally tested. The Schwarzschild derivation presented here is a *consistency check*, not the frontier of the theory's predictive power. The frontier lies in the corrections: the entropic hair, the dark curvature, the quantum modifications to the horizon structure, and the resolution of the classical singularity. These are addressed in subsequent appendices and Letters.

3. Step 1: Radial Entropic Field Equation (MEE)

3.1. The Minimal Entropic Equation in the Vacuum, Minimal-Coupling Limit

The derivation of the Schwarzschild solution from the Theory of Entropicity (ToE) begins with the fundamental field equation governing the entropic field $S(x)$. This is the *Minimal Entropic Equation* (MEE), the Euler-Lagrange equation obtained by varying the Obidi Action with respect to the entropic field. The MEE is to the Theory of Entropicity what the Klein-Gordon equation is to scalar field theory, or what the Maxwell equations are to electrodynamics: it is the dynamical equation of motion for the fundamental field of the theory.

The full MEE, as derived in Letter IE, contains the covariant d'Alembertian of the entropic field, the derivative of the entropic potential, a non-minimal coupling term involving the Ricci scalar and the entropic field, and a source term encoding external matter contributions. In the most general form, the MEE reads $\alpha \square S + V'(S) - \beta R'_{ent}(S) = J(x)$, where α and β are coupling constants, $R'_{ent}(S)$ is the entropic curvature contribution from non-minimal coupling, and $J(x)$ is the external source. Under the three assumptions stated in Section 2 — static spherical symmetry, minimal coupling ($\xi \rightarrow 0$), and the classical vacuum regime ($J(x) = 0$) — the MEE simplifies dramatically.

In the minimal coupling limit and in vacuum (no external matter), the entropic field S satisfies the Minimal Entropic Equation (MEE) of the form

$$\square S - V'(S) = 0 \quad \text{(IE-S.100)}$$

where $\square$ is the d'Alembertian with respect to the emergent spacetime metric, and $V(S)$ is the entropic potential.

Equation (IE-S.100) is the cornerstone of the present derivation. Let us unpack its content carefully. The d'Alembertian operator $\square$ is the covariant generalization of the wave operator (the Laplacian in the case of a static field) and is defined in terms of the spacetime metric $g_{\mu\nu}$ by

$$\square S = (1/\sqrt{-g}) \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu S)$$

where $g = \det(g_{\mu\nu})$ is the determinant of the metric tensor. For a static field $S = S(r)$ in a static, spherically symmetric background, the time derivatives vanish, and the d'Alembertian reduces to the radial part of the covariant Laplacian. The structure of this operator depends on the background metric, but in the weak-field regime — where the background is approximately flat — the d'Alembertian reduces to the flat-space radial Laplacian.

The entropic potential $V(S)$ encodes the self-interaction of the entropic field. It plays a role analogous to the Higgs potential in the Standard Model of particle physics or the inflaton

potential in inflationary cosmology: it determines the vacuum structure of the theory (the values of S at which $V'(S) = 0$, which correspond to equilibrium configurations of the entropic field) and the mass spectrum of small fluctuations around those vacua. Near the entropic vacuum $S = 0$ (or, more precisely, near the asymptotic background value S_0), the potential can be Taylor-expanded as

$$V(S) = V_0 + \frac{1}{2} m^2 S^2 + (\text{higher-order terms})$$

where V_0 is a cosmological constant contribution (irrelevant for the local Schwarzschild problem), m is the *entropic mass scale* (the mass of the entropic field quantum), and the higher-order terms encode nonlinear self-interactions. The entropic mass m sets the fundamental length scale $1/m$ (in natural units) beyond which the entropic field is exponentially suppressed — the *Compton wavelength* of the entropic quantum. The derivative of the potential, which appears in (IE-S.100), is therefore $V'(S) = m^2 S + O(S^2)$ near the vacuum.

3.2. Reduction to the Radial Equation

The reduction of the MEE (IE-S.100) to a radial ordinary differential equation proceeds through the assumption of static spherical symmetry (Assumption 1) and the weak-field approximation (Assumption 3). For a static field $S = S(r)$ in an approximately flat background, the d'Alembertian reduces to the flat-space Laplacian in spherical coordinates:

$$\square S \rightarrow \nabla^2 S = (1/r^2) d/dr(r^2 dS/dr)$$

This is the standard radial Laplacian in three spatial dimensions. The factor r^2 in the derivative reflects the geometry of three-dimensional Euclidean space: the surface area of a sphere of radius r is $4\pi r^2$, and the r^2 factor ensures that the Laplacian accounts for the divergence of the radial flux across spherical surfaces. Substituting this form of the Laplacian and the linearized potential $V'(S) = m^2 S$ into (IE-S.100), we obtain the radial entropic field equation.

In the weak-field, static, spherically symmetric regime, we approximate the background as effectively flat and write the radial equation as

$$(1/r^2) d/dr(r^2 dS/dr) - m^2 S = 0 \quad (\text{IE-S.101})$$

where m is an effective entropic mass scale (coming from $V(S) \approx \frac{1}{2} m^2 S^2$ near $S = 0$).

Equation (IE-S.101) is a well-known equation in mathematical physics, appearing in diverse contexts under different names. In the mathematical literature, it is the *modified spherical Bessel equation* (or the radial part of the *modified Helmholtz equation* $(\nabla^2 - m^2)S = 0$ in spherical coordinates for the $l = 0$ angular momentum sector). In nuclear physics, it is the equation whose solutions yield the Yukawa potential — the potential mediating the strong nuclear force between nucleons via meson exchange, as proposed by Hideki Yukawa in 1935. In condensed matter physics, it describes the screened Coulomb potential (the Debye-Hückel potential) in an electrolyte, where the screening length $1/m$ is the Debye length. In

all these contexts, the physical content is the same: the mass parameter m introduces a length scale $1/m$ beyond which the field is exponentially suppressed.

To solve (IE-S.101), we make the standard substitution $S(r) = u(r)/r$, which transforms the equation into $u'' - m^2u = 0$ (where primes denote derivatives with respect to r). This is a linear, constant-coefficient ODE whose general solution is $u(r) = Ae^{-mr} + Be^{+mr}$. The regularity condition at $r \rightarrow \infty$ (the requirement that the field not grow exponentially at large distances, which would represent an unphysical source at infinity) forces $B = 0$. Restoring the factor of $1/r$, we obtain the Yukawa solution.

3.3. The Yukawa Solution and Its Physical Interpretation

The general static, spherically symmetric solution of (IE-S.101) is the Yukawa form

$$S(r) = S_0 + (S_1/r) \exp(-mr) \quad (\text{IE-S.102})$$

where S_0 and S_1 are integration constants.

Let us discuss the physical interpretation of each element of this solution. The constant S_0 is the *asymptotic entropic background* — the value of the entropic field at spatial infinity, $S(r \rightarrow \infty) = S_0$. In the ToE framework, S_0 represents the ambient entropic density of the universe far from any localized gravitating body. It is the entropic analogue of the cosmological background: just as the cosmic microwave background provides a thermal background temperature $T_{\text{CMB}} \approx 2.725$ K pervading all of space, the entropic background S_0

provides a baseline information density from which local concentrations (gravitating bodies) emerge as perturbations.

The constant S_i is the *entropic charge* — the "strength" of the entropic source. It is determined by the total entropic content of the central body and plays a role in the ToE framework analogous to the role of the mass M in Newtonian gravity or the electric charge Q in electrostatics. As we shall show in Step 4, the entropic charge S_i is directly proportional to the Schwarzschild mass parameter M , establishing the identification of mass as an emergent entropic quantity.

The exponential factor $\exp(-mr)$ produces *Yukawa screening*: at distances $r \gg 1/m$, the entropic field decays exponentially faster than the Coulomb-like $1/r$ falloff, effectively confining the entropic influence to a finite range. The screening length $1/m$ is the *Compton wavelength* of the entropic quantum (in natural units). If the entropic mass m is non-zero, the gravitational interaction mediated by the entropic field has a finite range — a Yukawa-type modification of gravity. Such modifications are tightly constrained by solar-system tests and laboratory experiments (Eöt-Wash torsion balance experiments, for example, constrain Yukawa deviations from Newtonian gravity at sub-millimeter scales), but they remain a live possibility at galactic and cosmological scales, where they could contribute to modified gravity phenomenology.

We can verify by direct substitution that (IE-S.102) satisfies (IE-S.101). Setting $f(r) = (S_i/r)e^{-mr}$, we compute $f'(r) = S_i e^{-mr}(-1/r^2 - m/r)$ and $r^2 f'(r) = S_i e^{-mr}(-1 - mr)$. Taking the derivative: $d/dr[r^2 f'] = S_i e^{-mr}(m + m^2 r - m) = m^2 S_i r e^{-mr}$. Dividing by r^2 gives $(1/r^2)d/dr[r^2 f'] = m^2 f(r)$, confirming that (IE-S.101) is satisfied. The constant S_0 satisfies the equation trivially

since its derivatives vanish (the mass term $m^2 S_0$ is absorbed into a redefinition of the potential or, equivalently, into the cosmological constant sector).

3.4. The Massless Limit: Coulomb-Like Entropic Profile

For the gravitational sector in the long-range, classical limit, we take the massless limit $m \rightarrow 0$, which yields the Coulomb-like entropic profile

$$S(r) = S_0 + S_1/r \quad (\text{IE-S.102.1})$$

This $S(r)$ will serve as the prototype entropic configuration associated with a spherically symmetric mass distribution.

The massless limit $m \rightarrow 0$ is the regime in which the entropic interaction becomes *long-range*, precisely like the gravitational and electromagnetic interactions. This is physically natural: gravity is a long-range force (there is no gravitational screening, no gravitational analogue of the Debye-Hückel effect at macroscopic scales), and this long-range character is reflected in the $1/r$ falloff of the Coulomb-like profile (IE-S.102.1). The massless limit is therefore the gravitational limit of the entropic field theory.

The Coulomb-like profile $S(r) = S_0 + S_1/r$ is the entropic analogue of the Newtonian gravitational potential $\Phi(r) = -GM/r$. The $1/r$ falloff is not imposed by hand but is *dictated by three-dimensional geometry*: it is the unique solution of Laplace's equation $\nabla^2 S = 0$ that is spherically symmetric, regular at infinity, and singular at the origin (corresponding to a

point source). This is the content of Gauss's law applied to the entropic flux: the total entropic flux through any sphere enclosing the source is proportional to the enclosed entropic charge S_i , and the flux per unit area scales as $1/r^2$, which requires the field to scale as $1/r$.

The identification $S_i \sim M$ (to be made precise in Step 4) reveals a deep structural parallel. Just as the electrostatic potential $\Phi_E(r) = Q/(4\pi\epsilon_0 r)$ is determined by the electric charge Q , and just as the Newtonian gravitational potential $\Phi(r) = -GM/r$ is determined by the mass M , the entropic profile S_i/r is determined by the entropic charge S_i . In all three cases, the $1/r$ spatial dependence is universal (it follows from the three-dimensional Gauss's law), and the physical content resides entirely in the coefficient — the charge, the mass, or the entropic charge. The Theory of Entropicity unifies these by identifying mass as entropic charge: the mass of a gravitating body is the strength of its entropic field.

It is worth pausing to appreciate the logical structure. We began with the Minimal Entropic Equation — the field equation of the entropic field, derived from the Obidi Action — and by imposing three physically transparent assumptions (static spherical symmetry, minimal coupling, classical limit), we obtained the Coulomb-like profile (IE-S.102.1). This profile is not assumed; it is *derived*. The entropic field around a static, spherically symmetric source is necessarily of the form $S_o + S_i/r$. This is the starting point for the construction of the emergent spacetime geometry.

4. Step 2: Fisher-Rao Information Metric for $S(r)$

4.1. From the Entropic Field to Information Geometry: The Conceptual Bridge

The transition from the entropic field (a scalar function defined on the spatial manifold) to a metric tensor (a geometric structure defining distances, angles, and curvature) is the conceptual heart of the Theory of Entropicity. This transition is mediated by the *Fisher-Rao information metric* — the unique, natural, diffeomorphism-invariant metric on the space of probability distributions. The Fisher-Rao construction is the bridge between the entropic substrate (the information manifold M_I) and the emergent spacetime (the physical manifold M_S), and it is through this bridge that the entropic field $S(r)$ acquires geometric significance.

The key idea, developed at length in Letter IE, is as follows. The entropic field $S(x)$ at each point x of the spatial manifold induces a probability distribution on the entropic substrate — a statistical description of the microscopic degrees of freedom at that point. As one moves from point to point, the entropic field changes, and the induced probability distribution changes accordingly. The space of all these distributions, parameterized by the spatial coordinates, forms a *statistical manifold*, and the natural metric on this manifold — the Fisher-Rao metric — endows it with a Riemannian geometry. This geometry, the *information geometry* of the entropic substrate, is what the emergence map Φ converts into physical spacetime geometry.

The theoretical foundations of this construction rest on three pillars. First, the work of Ronald Aylmer Fisher (1925), who introduced the concept of *Fisher information* as a measure of the information content of a statistical sample. Second, the work of Calyampudi Radhakrishna Rao (1945), who recognized that the Fisher information matrix defines a Riemannian metric on the space of parameterized probability distributions. Third, the theorem of Nikolai Nikolaevich Čencov (1972, published in full in 1982), which establishes that the Fisher-Rao metric is the *unique* Riemannian metric on a statistical manifold that is invariant under all sufficient statistics — i.e., under all transformations of the data that preserve the statistical information. The Čencov uniqueness theorem is of profound significance for the Theory of Entropicity: it means that the information metric is not a choice but a *necessity*. Any diffeomorphism-invariant geometric structure constructed from a family of probability distributions must be proportional to the Fisher-Rao metric. The emergent spacetime geometry is therefore uniquely determined (up to a constant) by the entropic field.

4.2. The Gaussian Statistical Model

To connect the entropic field to geometry, we consider a one-parameter family of probability distributions $p(x / r)$ whose macroscopic parameter is controlled by $S(r)$. For concreteness, take a simple Gaussian family where the variance $\sigma^2(r)$ is proportional to $S(r)$:

$$\sigma^2(r) \propto S(r) \quad (\text{IE-S.102.2})$$

The choice of a Gaussian statistical model is made for concreteness and computational transparency, but it is important to emphasize that the qualitative result — a Fisher-Rao metric proportional to $[S'/S]^2$ — is *not* an artifact of the Gaussian assumption. For any member of the exponential family (Gaussian, Poisson, exponential, gamma, etc.), the Fisher-Rao metric depends on the parameter of interest through the logarithmic gradient of the parameter, and the resulting information metric has the same structural form. The Čencov uniqueness theorem guarantees this universality.

The proportionality $\sigma^2(r) \propto S(r)$ has a natural physical interpretation in the ToE framework. The variance σ^2 measures the spread of the microscopic degrees of freedom around their mean — it quantifies the "width" of the statistical distribution of the entropic substrate at each spatial point. The proportionality to $S(r)$ means that regions of higher entropic density have greater statistical spread: more entropy corresponds to more microscopic configurations, more fluctuation, more uncertainty. This is physically natural and consistent with the thermodynamic intuition that entropy measures the logarithm of the number of accessible microstates. Near the central gravitating body, where $S(r)$ is large, the microscopic degrees of freedom are highly spread out (large variance); far from the body, where $S(r) \approx S_0$, the distribution is narrow (small variance). The Fisher-Rao metric, which measures the "statistical distinguishability" of nearby distributions, will therefore be large where the entropic field changes rapidly (near the source) and small where it is nearly constant (at large distances).

4.3. Definition of the Fisher-Rao Metric

For a one-parameter family $p(x / \theta)$, the Fisher-Rao metric is defined by

$$g_I(\theta) = E[(\partial \ln p / \partial \theta)^2] \quad (\text{IE-S.102.3})$$

and in the present case the relevant parameter is $\theta = r$.

The Fisher-Rao metric (IE-S.102.3) has a beautiful information-theoretic interpretation. The quantity $\partial \ln p / \partial \theta$ is the *score function* — the logarithmic sensitivity of the probability distribution to changes in the parameter θ . Its expectation value vanishes ($E[\partial \ln p / \partial \theta] = 0$ for any regular statistical model), so the Fisher-Rao metric is the variance of the score function. It measures how "statistically distinguishable" two nearby distributions $p(x / \theta)$ and $p(x / \theta + d\theta)$ are: a large Fisher information means that a small change in θ produces a large change in the statistical model, making the two distributions easy to distinguish. The Fisher-Rao metric thereby converts statistical distinguishability into geometric distance.

The connection to the **Kullback-Leibler (KL) divergence** makes this precise: the **Fisher-Rao metric** is the **Hessian of the KL-divergence**, $g_I(\theta) = \partial^2 D_{\text{KL}}(p(\cdot / \theta) || p(\cdot / \theta')) / \partial \theta'^2$ evaluated at $\theta' = \theta$. The KL-divergence measures the information lost when one approximates the true distribution $p(\cdot / \theta)$ with a nearby distribution $p(\cdot / \theta')$, and the Fisher-Rao metric is the leading-order term in this loss. The information geometry of the entropic substrate is therefore a geometry of *information loss*: the distance between two points on

the information manifold measures how much information is lost in confusing the corresponding distributions.

4.4. Explicit Computation for the Gaussian Model

A straightforward computation for a Gaussian with variance $\sigma^2(r)$ gives a Fisher-Rao metric component of the form

$$g_{I,rr}(r) = C \cdot [S'(r)]^2 / [S(r)]^2 \quad (\text{IE-S.102.4})$$

where C is a positive constant depending on the normalization of the statistical model, and $S'(r) = dS/dr$.

Let us present the full computation explicitly. Consider the Gaussian probability density

$$p(x | r) = (1/\sqrt{(2\pi\sigma^2(r))}) \exp(-x^2/(2\sigma^2(r)))$$

with $\sigma^2(r) = \kappa S(r)$ for some positive proportionality constant κ . The log-likelihood is

$$\ln p = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - x^2/(2\sigma^2)$$

Differentiating with respect to r (using $d\sigma^2/dr = \kappa S'(r)$ and $d \ln \sigma^2/dr = S'(r)/S(r)$):

$$\partial \ln p / \partial r = -\frac{1}{2}(S'/S) + (x^2/(2\sigma^4))(\kappa S') = (S'/S)[-1/2 + x^2/(2\sigma^2)]$$

Squaring and taking the expectation (using the Gaussian moments $E[x^2/\sigma^2] = 1$ and $E[x^4/\sigma^4] = 3$):

$$E[(\partial \ln p / \partial r)^2] = (S'/S)^2 \cdot E[(-1/2 + x^2/(2\sigma^2))^2] = (S'/S)^2 \cdot (1/4 - 1/2 + 3/4) = 1/2 (S'/S)^2$$

This confirms (IE-S.102.4) with $C = 1/2$ for the Gaussian model. The result has a remarkable structural property: the Fisher-Rao metric depends on the *logarithmic gradient* $S'/S = d \ln S/dr$ of the entropic field, not on the field itself. This is deeply connected to the information-theoretic principle that only *relative* changes in entropy carry geometric meaning. An additive shift in $S(r)$ does not change the logarithmic gradient (provided $S(r) > 0$), and thus does not change the Fisher-Rao metric. The information geometry is insensitive to the overall "level" of the entropy — only the *spatial variation* of the entropy, measured in relative terms, is geometrically relevant.

For alternative statistical models, the constant C changes but the structural form $[S'/S]^2$ is preserved. For instance, for a Poisson model with mean $\lambda(r) \propto S(r)$, the Fisher-Rao metric is $g_{l,rr} = [S'/S]^2/S$, which for large S scales as $[S'/S]^2$. For a general exponential family model with natural parameter $\eta(r)$ related to $S(r)$, the Fisher-Rao metric involves the second derivative of the log-partition function, but in the regime of interest the dependence on S'/S is universal. This universality, guaranteed by the Čencov theorem, ensures that the qualitative features of the emergent geometry do not depend on the specific statistical model used to define the information metric.

4.5. Evaluation for the Coulomb-Like Entropic Profile

For the Coulomb profile (IE-S.102.1),

$$S(r) = S_0 + S_1/r \quad (\text{IE-S.102.5})$$

we have

$$S'(r) = -S_1/r^2 \quad (\text{IE-S.102.6})$$

and therefore

$$g_{I,rr}(r) \propto (S_1^2/r^4) / [S_0 + S_1/r]^2 \quad (\text{IE-S.102.7})$$

The behavior of $g_{I,rr}(r)$ at different radial scales is highly informative and reveals the geometric structure of the information manifold.

At large distances ($r \gg S_1/S_0$): The denominator is dominated by S_0^2 , and the Fisher-Rao metric becomes $g_{I,rr} \propto S_1^2/(S_0^2 r^4)$, which falls off rapidly with distance. The information metric becomes flat at large distances — the information manifold is asymptotically Euclidean, reflecting the fact that far from the gravitating body, the entropic field is nearly

constant and the statistical manifold is nearly uniform. This corresponds to the asymptotic flatness of the Schwarzschild spacetime.

At small distances ($r \ll S_i/S_o$): The denominator is dominated by $(S_i/r)^2 = S_i^2/r^2$, and the Fisher-Rao metric becomes $g_{I,rr} \propto 1/r^2$, which diverges as $r \rightarrow 0$. The information manifold has a curvature singularity at the origin, corresponding to the central mass. This singularity of the Fisher-Rao metric is the information-geometric precursor of the curvature singularity of the Schwarzschild spacetime at $r = 0$.

At the transition scale $r \sim r^* = S_i/S_o$: The two terms in the denominator are comparable, and the Fisher-Rao metric transitions between the two asymptotic regimes. This transition scale is the entropic analogue of the Schwarzschild radius — the characteristic scale at which the entropic field perturbation S_i/r becomes comparable to the background S_o . We shall see in Step 4 that $r^* = S_i/S_o$ is related to the Schwarzschild radius $r_s = 2GM/c^2$ through the mass-entropy identification $M = \alpha S_i$.

One can compute the Ricci scalar of the one-dimensional information metric $g_{I,rr}(r) dr^2$ in the radial sector to verify that the information manifold is indeed curved. In one dimension, the Ricci scalar of a metric $g(r) dr^2$ is related to $g(r)$ by a formula involving its second logarithmic derivative. The non-vanishing of this curvature confirms that the information manifold is not flat — it has a genuine geometric structure that, via the emergence map, becomes the spacetime curvature of the Schwarzschild geometry.

4.6. The Full Information Manifold Metric

The angular components of the information metric are taken to respect spherical symmetry, so that the information manifold metric in the radial-angular sector can be written as

$$ds_I^2 = g_{l,rr}(r) dr^2 + r^2 (d\theta^2 + \sin^2\theta d\varphi^2) \quad (\text{IE-S.102.8})$$

The exact overall normalization of $g_{l,rr}$ is not essential here; what matters is that the information manifold is spherically symmetric and that its radial component is determined by the entropic profile $S(r)$.

The metric (IE-S.102.8) defines a three-dimensional Riemannian manifold with the topology of $\mathbb{R}^3 \setminus \{0\}$ (three-dimensional Euclidean space with the origin removed, reflecting the point-source singularity at $r = 0$). The angular part $r^2(d\theta^2 + \sin^2\theta d\varphi^2)$ is the standard metric on the two-sphere of radius r , ensuring that the area of the sphere at coordinate radius r is $4\pi r^2$. This is the *areal radius* gauge — a coordinate choice that is both natural and computationally convenient.

It should be emphasized that (IE-S.102.8) is the *spatial section* of the information manifold at a fixed time. The full four-dimensional information metric includes a temporal component that encodes the causal structure of the information manifold. However, in the static case considered here, the temporal and spatial sectors decouple, and the spatial metric (IE-S.102.8) contains all the information needed for the spatial part of the emergence

map. The temporal component of the physical spacetime metric will be determined separately, in Step 3, by the dynamical equations (the vacuum Einstein equations in the classical limit).

The information manifold (M_I, g_I) defined by (IE-S.102.8) is the "statistical shadow" of the entropic field — the geometric structure that encodes how the entropic probability distributions vary from point to point in space. Near the central body, where the entropic field changes rapidly, the Fisher-Rao distances are large — the distributions at nearby radii are highly distinguishable. Far from the body, where the field is nearly constant, the Fisher-Rao distances are small — the distributions are nearly indistinguishable. The information manifold is therefore "highly curved" near the source and "nearly flat" at large distances, mirroring the behavior of the Schwarzschild spacetime.

5. Step 3: Emergence Map and Physical Spacetime Metric

5.1. The Emergence Map: From Information Geometry to Physical Spacetime

With the information manifold metric (IE-S.102.8) in hand, we now invoke the central mechanism of the Theory of Entropicity: the *emergence map* $\Phi: M_I \rightarrow M_S$, which identifies the physical spacetime metric with the Fisher-Rao information metric. The emergence map was established in Letter IE as the fundamental link between the information manifold $(M_I$,
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g_I) and the physical spacetime (M_S, g_S) . It is not a coordinate transformation or a gauge choice; it is a *constitutive relation* — a statement about the ontological nature of spacetime geometry within the ToE framework. The emergence map asserts that spacetime geometry is not fundamental but *emergent*: it is generated by the entropic field through the information metric.

The emergence map identifies the physical spacetime metric g_S with the Fisher-Rao information metric g_I up to a constant factor λ :

$$g_{S,\mu\nu} = \lambda g_{I,\mu\nu} \quad \text{(IE-S.102.9)}$$

with λ a constant in the classical regime.

The conformal relation (IE-S.102.9) is the simplest form of the emergence map, valid in the classical regime where quantum corrections to the emergence constant are negligible. As derived in Letter IE, the emergence constant is

$$\lambda = l_P^2 / (4k_B) = \hbar G / (4k_B c^3)$$

where $l_P = (\hbar G / c^3)^{1/2}$ is the Planck length, k_B is Boltzmann's constant, $\hbar$ is the reduced Planck constant, G is Newton's gravitational constant, and c is the speed of light. The emergence constant λ has dimensions of $(\text{length})^2 / (\text{information})$, which is necessary for dimensional consistency: the information metric g_I has dimensions of $(\text{information})^2$ (since it is defined in terms of log-likelihoods, which are dimensionless in natural units or have dimensions of

information in information-theoretic units), and the physical metric g_S has dimensions of $(\text{length})^2$. The constant λ bridges these dimensions.

The physical significance of the emergence map cannot be overstated. It is the mathematical realization of the central thesis of the Theory of Entropicity: *spacetime geometry is generated by the entropic field through the information metric*. The information manifold is not a mathematical convenience or an analogy; it is the *substrate* from which physical spacetime emerges. The emergence map Φ is the mechanism of this emergence, and the constant λ sets the scale at which informational structure becomes geometric structure.

The Curvature Transfer Theorem (Theorem 3.1 of Letter IE) guarantees that the Riemann curvature tensor of the physical spacetime (M_S, g_S) is determined by the pushforward of the Riemann curvature tensor of the information manifold (M_I, g_I) . Specifically, $R^S_{\mu\nu\rho\sigma} = \lambda \Phi^*(R^I_{\mu\nu\rho\sigma}) + O(1/N)$, where N is the number of entropic degrees of freedom. In the classical limit $N \rightarrow \infty$, the correction term vanishes, and the curvature transfer is exact. The Schwarzschild curvature is therefore *information curvature*, rescaled by λ .

5.2. The Physical Spacetime Metric Ansatz

For a static, spherically symmetric spacetime, we write the physical metric in standard Schwarzschild-like coordinates as

$$ds_S^2 = -A(r) c^2 dt^2 + B(r) dr^2 + r^2 (d\theta^2 + \sin^2\theta d\varphi^2) \quad (\text{IE-S.102.10})$$

This is the most general static, spherically symmetric line element compatible with the symmetry assumptions. The metric is characterized by two unknown functions of the radial coordinate: $A(r)$, the *temporal metric function* (also called the "lapse function" in the ADM formalism or the "redshift function" in the astrophysical literature), and $B(r)$, the *radial metric function* (related to the proper radial distance). The angular part $r^2(d\theta^2 + \sin^2\theta d\varphi^2) = r^2 d\Omega^2$ is already in the standard form on the two-sphere, corresponding to the choice of *areal radius* r (the coordinate is defined so that the area of the sphere at coordinate radius r is $4\pi r^2$). This gauge choice eliminates one functional degree of freedom and simplifies the Einstein equations considerably.

The function $A(r)$ determines the gravitational redshift and time dilation experienced by static observers at radius r . A photon emitted at radius r and received at infinity is redshifted by the factor $\sqrt{A(r)}$, and a clock at radius r runs slow relative to a clock at infinity by the same factor. The function $B(r)$ determines the proper radial distance: the proper distance between two radial shells at r_1 and r_2 is $\int_{r_1}^{r_2} \sqrt{B(r)} dr$, which differs from the coordinate distance $r_2 - r_1$ whenever $B(r) \neq 1$.

Asymptotic flatness — the requirement that spacetime become Minkowskian at large distances from the source — imposes the boundary conditions $A(r) \rightarrow 1$ and $B(r) \rightarrow 1$ as $r \rightarrow \infty$. These conditions ensure that the metric reduces to the flat Minkowski metric $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$ far from the gravitating body, as required by the physical situation (isolated body in otherwise empty space).

5.3. Constraints from the Emergence Map

The emergence map (IE-S.102.9) implies that, in the radial-angular sector,

$$B(r) \propto g_{l,rr}(r) \quad (\text{IE-S.102.11})$$

and the angular part is already in the standard form $r^2 d\Omega^2$. The time component $A(r)$ is determined dynamically by the GEFE/OFE in the vacuum.

The proportionality (IE-S.102.11) is the key constraint imposed by the emergence map on the spatial part of the physical metric. The radial metric function $B(r)$ of the physical spacetime inherits its radial dependence directly from the Fisher-Rao metric $g_{l,rr}(r)$ of the information manifold. This is a concrete realization of the ToE thesis: the spatial geometry of the physical spacetime is *determined by* the information geometry of the entropic substrate.

The temporal metric function $A(r)$, however, is *not* directly determined by the information metric. This is because the Fisher-Rao metric is a *Riemannian* (positive-definite) metric on the statistical manifold, while the physical spacetime metric is *Lorentzian* (indefinite, with signature $(-, +, +, +)$). The Lorentzian signature introduces causal structure — the distinction between timelike, spacelike, and null directions — which has no direct counterpart in the Riemannian information metric. The temporal component of the physical metric, which determines the causal structure, must therefore be fixed by additional dynamical

information: specifically, by the field equations of the theory (the GEFE in the full framework, or the vacuum Einstein equations in the classical limit).

This asymmetry between the spatial and temporal components is physically natural. The spatial geometry reflects the *static* configuration of the entropic field — it is a snapshot of the information content at a given time. The temporal geometry reflects the *dynamics* — the way the information content evolves in time (or, in the static case, the way the causal structure is constrained by the field equations). The emergence map provides the spatial geometry directly; the dynamics provides the temporal geometry.

5.4. The Classical Limit: Vacuum Einstein Equations

In the classical, large-scale limit, the GEFE reduce to the vacuum Einstein equations

$$R_{\mu\nu} - \left(\frac{1}{2}\right) R g_{\mu\nu} = 0 \quad \text{(IE-S.102.12)}$$

The reduction of the full GEFE to (IE-S.102.12) proceeds through the hierarchy of limits described in Assumption 3. Let us recapitulate and elaborate. The full GEFE, as derived in Letter IE, have the schematic form

$$G_{\mu\nu} = (8\pi G/c^4)[T_{\mu\nu}^{\text{LOA}} + T_{\mu\nu}^{\text{SOA}} + T_{\mu\nu}^{\text{OCI}}]$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor, $T_{\mu\nu}^{\text{LOA}}$ is the Leading-Order Approximation stress-energy (containing the kinetic energy $\partial_\mu S \partial_\nu S$, the potential $V(S)$, and the non-minimal coupling ξRS^2), $T_{\mu\nu}^{\text{SOA}}$ contains the Second-Order Approximation corrections

(higher-curvature terms from the Seeley-DeWitt expansion, suppressed by $(l_P/L)^2$), and $T_{\mu\nu}^{OCI}$ is the Obidi Curvature Invariant tensor (encoding the curvature mismatch between M_I and M_S).

In the classical limit:

5. **SOA corrections are negligible:** The higher-curvature corrections are suppressed by $(l_P/L)^2 \sim 10^{-70}$ for astrophysical scales, rendering them utterly undetectable.
6. **OCI vanishes:** When the emergence map is a conformal isometry with constant λ , the curvature mismatch between M_I and M_S vanishes identically, and $T_{\mu\nu}^{OCI} = 0$.
7. **LOA vanishes in vacuum with minimal coupling:** With $\xi = 0$, the LOA stress-energy involves only $\partial_\mu S \partial_\nu S$ and $V(S)$. In the vacuum regime, the entropic field is a classical background that has already been absorbed into the emergent metric via the Fisher-Rao construction. The effective stress-energy of the entropic field, after the emergence map has been applied, is already encoded in the geometry of g_S and does not appear as an independent source term.

What remains is the vacuum Einstein equation $G_{\mu\nu} = 0$, which is precisely (IE-S.102.12). Taking the trace of (IE-S.102.12) with the metric $g^{\mu\nu}$ gives $-R = 0$ (in four dimensions, $g^{\mu\nu}g_{\mu\nu} = 4$, so the trace is $R - 2R = -R$), confirming that the Ricci scalar vanishes: $R = 0$. The vacuum Einstein equations therefore simplify to the Ricci-flat condition:

$$R_{\mu\nu} = 0$$

This is the definitive form of the vacuum field equations, and it is this condition — Ricci-flatness — that uniquely determines the Schwarzschild geometry, by Birkhoff's theorem.

5.5. Solution: The Schwarzschild Geometry

For the static, spherically symmetric ansatz (IE-S.102.10), these equations yield the standard system of ordinary differential equations for $A(r)$ and $B(r)$. Solving (IE-S.102.12) under the requirement of asymptotic flatness ($A(r) \rightarrow 1$, $B(r) \rightarrow 1$ as $r \rightarrow \infty$) gives the unique Schwarzschild solution

$$A(r) = 1 - 2GM/(rc^2) \quad (\text{IE-S.102.13})$$

$$B(r) = [1 - 2GM/(rc^2)]^{-1} \quad (\text{IE-S.102.14})$$

The derivation of (IE-S.102.13)–(IE-S.102.14) from the Ricci-flat condition $R_{\mu\nu} = 0$ is a standard exercise in general relativity, but its logical structure merits a brief review. For the metric ansatz (IE-S.102.10), the non-vanishing Christoffel symbols involve A, A', B, B' (primes denoting d/dr), and the Ricci tensor components yield three independent equations:

The (t,t) component: $R_{tt} = 0$ gives an equation involving A'', A', B' , and A, B .

The (r,r) component: $R_{rr} = 0$ gives a second equation involving the same quantities.

The (θ,θ) component: $R_{\theta\theta} = 0$ gives a third equation that serves as a consistency condition.

A crucial intermediate result emerges from the combination of the (t,t) and (r,r) equations: $A(r)B(r) = \text{const}$. The asymptotic flatness condition $A \rightarrow 1$, $B \rightarrow 1$ as $r \rightarrow \infty$ fixes this constant to unity:

$$A(r) B(r) = 1$$

This is the fundamental constraint relating the temporal and radial metric functions of the Schwarzschild geometry: they are reciprocals of each other. Substituting $B = 1/A$ into the (θ, θ) equation yields a first-order ODE for $A(r)$:

$$rA' + A = 1 \Rightarrow d(rA)/dr = 1 \Rightarrow A(r) = 1 - C/r$$

where C is an integration constant. Matching to the Newtonian limit (where $A(r) \approx 1 + 2\Phi/c^2 = 1 - 2GM/(rc^2)$) identifies $C = 2GM/c^2 = r_s$, the Schwarzschild radius. This completes the derivation of (IE-S.102.13)–(IE-S.102.14).

The Schwarzschild radius $r_s = 2GM/c^2$ marks the event horizon — the boundary of the black hole region from which no signal can escape to infinity. For a solar-mass object, $r_s \approx 3$ km; for the Earth, $r_s \approx 9$ mm. At $r = r_s$, the coordinate singularity in $B(r)$ is an artifact of the Schwarzschild coordinates and can be removed by choosing different coordinates (Eddington-Finkelstein, Kruskal-Szekeres, Painlevé-Gullstrand). The true curvature singularity at $r = 0$, where the Kretschner scalar $K = 48G^2M^2/(c^4r^6)$ diverges, is coordinate-independent and signals a genuine breakdown of the classical theory.

5.6. Summary of Step 3

Thus, in the classical limit where the GEFE reproduce the vacuum Einstein equations, the emergent spacetime metric generated by a static, spherically symmetric entropic configuration is necessarily the Schwarzschild metric.

This result is the central achievement of this appendix. The Schwarzschild metric is not *assumed* in the Theory of Entropicity — it *emerges* from the entropic field configuration through a precisely defined chain of constructions: the entropic field $S(r)$ induces a statistical manifold; the Fisher-Rao metric on this manifold defines the information geometry; the emergence map converts information geometry into physical spacetime geometry; and the vacuum Einstein equations (the classical limit of the GEFE) constrain the temporal component, completing the Schwarzschild metric. Each link in this chain is mathematically rigorous and physically motivated.

The result validates the GEFE/OFE as a viable theory of gravity: in the regime where they must agree with GR (the classical, weak-field, large-scale limit), they do agree — exactly. The Schwarzschild solution is not lost or modified in the entropic reformulation; it is recovered intact. This is a non-trivial consistency check, because the GEFE have a very different structure from the Einstein equations (they include SOA corrections, the OCI tensor, and the non-minimal coupling), and it is not *a priori* obvious that they reduce to the correct classical limit. The fact that they do is a powerful validation of the theoretical framework.

6. Step 4: Identification of the Mass Parameter in Terms of Entropic Data

6.1. The Significance of the Mass Identification

The remaining task is to relate the Schwarzschild mass parameter M to the entropic field parameters S_o and S_i .

This identification is arguably the most profound result of this appendix, for it addresses a question that general relativity leaves entirely unanswered: *What is mass?* In Einstein's theory, the mass M appearing in the Schwarzschild metric is a *free parameter* — it is set by initial conditions and has no deeper explanation within the theory. The Einstein equations tell us *how* mass curves spacetime, but they do not tell us *what* mass is or *why* it curves spacetime. Mass is an input, not an output, of the theory.

In the Theory of Entropicity, this situation changes fundamentally. The mass M is not a free parameter but a *derived quantity* — it is determined by the entropic field configuration, specifically by the entropic charge S_i . The mass of a gravitating body is not a primitive datum but an emergent consequence of the entropic field that the body generates. This represents a genuine explanatory advance: the ToE explains *what mass is* (an entropic charge — a measure of the total information content concentrated in the entropic field profile) rather than merely accepting it as a given.

This identification connects to the broader program of the Theory of Entropicity, in which all physical constants and parameters are emergent from the entropic field. The speed of

light c , Newton's constant G , Planck's constant $\hbar$, and Boltzmann's constant k_B are all, in the ToE framework, related to properties of the entropic substrate and the emergence map. The identification of mass as entropic charge is one concrete instance of this broader program, and it provides the first quantitative bridge between the information-theoretic formalism and the measurable gravitational parameters of astrophysical bodies.

6.2. The Newtonian Limit of the Schwarzschild Metric

In the weak-field, Newtonian limit, the Schwarzschild metric yields the gravitational potential

$$\Phi(r) = -GM/r \quad (\text{IE-S.102.15})$$

The derivation of the Newtonian potential from the Schwarzschild metric proceeds through the standard weak-field expansion. In the weak-field regime ($GM/(rc^2) \ll 1$), the metric component $A(r) = 1 - 2GM/(rc^2)$ can be written as

$$A(r) = 1 + 2\Phi(r)/c^2$$

where $\Phi(r) = -GM/r$ is identified as the Newtonian gravitational potential. This identification follows from the geodesic equation: in the non-relativistic limit ($v \ll c$), the geodesic equation for a test particle in the metric (IE-S.102.10) reduces to Newton's second law $d^2r/dt^2 = -d\Phi/dr$, with Φ defined by $g_{00} = -(1 + 2\Phi/c^2)$. The temporal metric component

thus encodes the Newtonian potential, and the relation $A(r) = 1 + 2\Phi/c^2$ is the standard dictionary between the relativistic metric and the Newtonian potential.

The Newtonian potential $\Phi(r) = -GM/r$ is, of course, the solution of Poisson's equation $\nabla^2\Phi = 4\pi G\rho$ in the vacuum ($\rho = 0$ outside the source), with boundary conditions $\Phi \rightarrow 0$ as $r \rightarrow \infty$ and the total mass $M = \int \rho d^3x$ determined by the source. The Schwarzschild metric therefore contains the Newtonian potential as its weak-field limit — a consistency check that any relativistic theory of gravity must pass.

6.3. The Entropic Effective Potential and the Matching Procedure

On the other hand, in the ToE framework, the entropic field $S(r) = S_o + S_1/r$ generates an effective potential via the emergence map and the GEFE/OFE. To leading order in the weak-field regime, we can write

$$\Phi(r) \propto S_1/r \quad (\text{IE-S.102.16})$$

so that matching (IE-S.102.15) and (IE-S.102.16) gives

$$GM \propto S_1 \quad (\text{IE-S.102.17})$$

The matching procedure is performed in the *overlap regime* — the range of distances at which both the Newtonian approximation (weak gravitational field, $GM/(rc^2) \ll 1$) and the linear entropic approximation (small perturbation of the entropic field, $S_l/(rS_o) \ll 1$) are simultaneously valid. In this overlap regime, both descriptions are applicable, and the matching of the gravitational potential (IE-S.102.15) with the entropic effective potential (IE-S.102.16) yields the proportionality (IE-S.102.17).

The proportionality constant in (IE-S.102.17) depends on the normalization conventions of the entropic field and the Fisher-Rao metric. It absorbs the factors of C (the Fisher-Rao normalization constant from (IE-S.102.4)), λ (the emergence constant), and the specific statistical model used to define g_l . The key physical result is that the mass M is *linear* in the entropic charge S_l — this linearity is a consequence of the linearity of the weak-field equations (both the Laplace equation for the Newtonian potential and the linearized MEE for the entropic perturbation) and is expected to hold in the weak-field regime. At strong fields (near the Schwarzschild radius), nonlinear corrections may modify this relationship, but the linear identification suffices for the present purposes.

The physical content of (IE-S.102.17) is striking: *the mass of a gravitating body is proportional to the "amount of entropy" concentrated in its entropic field profile*. More massive objects have stronger entropic fields — they concentrate more information, more microscopic degrees of freedom, more entropic charge. The gravitational pull of a body is not caused by some mysterious "gravitational charge" but by the informational content of its entropic configuration. This is the ToE realization of the old intuition, traced back to

Jacobson (1995) and Verlinde (2011), that gravity may be an *entropic force* — a macroscopic effect arising from the statistical behavior of microscopic degrees of freedom.

6.4. The Mass-Entropy Identification

More explicitly, one can write

$$M = \alpha S_1 \quad (\text{IE-S.102.18})$$

where α is a constant determined by the coupling constants of the theory (including λ , ξ , and numerical factors from the Fisher-Rao normalization). The precise value of α depends on the detailed normalization of the entropic field and the statistical model used to define g_I , but the key point is that the Schwarzschild mass M is not a free parameter: it is an emergent quantity proportional to the entropic "charge" S_I of the underlying entropic field configuration.

The constant $\alpha = \alpha(\lambda, \xi, C)$ encodes the full chain of identifications from the entropic field to the physical mass. It involves:

- The emergence constant $\lambda = \hbar G / (4k_B c^3)$, which sets the scale of the information-to-geometry conversion.
- The Fisher-Rao normalization constant C ($= 1/2$ for the Gaussian model), which depends on the specific statistical model.

- The non-minimal coupling parameter ξ (set to zero in the present analysis, but entering at higher orders).
- Numerical factors arising from the matching of the entropic effective potential to the Newtonian potential.

In principle, α can be computed from first principles once the normalization of the entropic field is fixed — that is, once the precise relationship between $\sigma^2(r)$ and $S(r)$ is specified (not merely the proportionality but the proportionality constant) and the statistical model is chosen. This calculation, while important for quantitative predictions, does not affect the qualitative result: *mass is entropic charge*.

The result $M = \alpha S_i$ is structurally analogous to the identification of electric charge with the winding number of the entropic phase field, established in Letter IIA of the ToE Living Review Letters Series. In Letter IIA, the electric charge Q of a particle was shown to be proportional to a topological invariant (the winding number n) of the entropic phase field $\theta(x)$ around the particle. Both identifications — $M = \alpha S_i$ for mass and $Q = e \cdot n$ for electric charge — exemplify the ToE program of reducing all physical parameters to emergent entropic quantities. Mass and charge are not fundamental; they are different measures of the information content of the entropic field.

6.5. Connection to Black Hole Thermodynamics

The identification $M = \alpha S_i$ has a profound connection to one of the most remarkable results in theoretical physics: the Bekenstein-Hawking entropy formula. In 1973, Jacob Bekenstein proposed, on the basis of thought experiments involving black holes and the second law of

thermodynamics, that a black hole should possess an entropy proportional to the area A of its event horizon:

$$S_{\text{BH}} = k_{\text{B}} c^3 A / (4 G \hbar)$$

In 1975, Stephen Hawking confirmed this formula by showing that black holes emit thermal radiation (Hawking radiation) at a temperature $T_H = \hbar c^3 / (8 \pi k_{\text{B}} G M)$, and the first law of black hole mechanics then yields the Bekenstein-Hawking entropy.

In the standard framework of GR and quantum field theory on curved spacetime, the Bekenstein-Hawking entropy is deeply puzzling. Why should the entropy of a black hole be proportional to the *area* of the horizon rather than the *volume* enclosed? What are the microscopic degrees of freedom that the entropy counts? These questions — central to the black hole information paradox — have motivated decades of research in string theory, loop quantum gravity, and other approaches to quantum gravity.

In the ToE framework, the Bekenstein-Hawking entropy is not a mysterious coincidence but a *consequence* of the identification of mass as entropic charge. The area of the Schwarzschild event horizon is $A = 4 \pi r_s^2 = 16 \pi G^2 M^2 / c^4$. Using $M = \alpha S_i$, the area is proportional to S_i^2 , and the Bekenstein-Hawking entropy is proportional to S_i^2 — a quadratic function of the entropic charge. The entropy of the black hole is proportional to the *area* because the entropic charge S_i (which determines M) is related to the information content of the entropic field, which — in the holographic limit — scales with the area of the bounding surface rather than the enclosed volume. The Bekenstein-Hawking formula is thus a natural consequence of the entropic origin of mass, and the holographic principle

(the area scaling of entropy) is built into the ToE framework through the Fisher-Rao construction.

7. Concluding Summary

7.1. The Schwarzschild Solution in the Theory of Entropicity

Summary Theorem: Emergence of the Schwarzschild Geometry from the GEFE/OFE

Thus, the Schwarzschild solution appears in ToE as:

1. **A static, spherically symmetric entropic configuration** $S(r) = S_o + S_i/r$ solving the MEE in the appropriate limit.
2. **An emergent spacetime metric** g_s obtained from the Fisher-Rao information metric via the emergence map.
3. **A classical limit of the GEFE/OFE** in which the vacuum equations reduce to the Einstein vacuum equations, yielding the Schwarzschild form for $A(r)$ and $B(r)$.
4. **A mass parameter** M identified in terms of the entropic field parameter S_i .

This worked example shows that, in the appropriate regime, the Theory of Entropicity reproduces the Schwarzschild solution of general relativity and interprets the mass of the black hole as an emergent entropic charge of the underlying information manifold.

7.2. The Chain of Emergence

The complete logical chain through which the Schwarzschild geometry emerges from the Theory of Entropicity can be summarized schematically as follows:

Stage	Construction	Result
1	Solve MEE for static, spherically symmetric $S(r)$	$S(r) = S_0 + S_1/r$ (Coulomb-like entropic profile)
2	Construct Fisher-Rao information metric from $S(r)$	$g_{l,rr} \propto [S'/S]^2$ (information geometry)
3	Apply emergence map $g_S = \lambda g_I$	$B(r) \propto g_{l,rr}$ (spatial metric from information metric)
4	Solve vacuum Einstein equations (classical GEFE limit)	$A(r) = 1 - 2GM/(rc^2)$, $B(r) = [1 - 2GM/(rc^2)]^{-1}$
5	Match entropic charge to mass	$M = \alpha S_1$ (mass as emergent entropic charge)

This chain is the first concrete, fully worked example of the GEFE/OFE producing a known exact solution of Einstein's general relativity. It validates the entire machinery developed in Letter IE: the Minimal Entropic Equation, the Fisher-Rao information metric, the emergence map, the Curvature Transfer Theorem, and the identification of physical constants with entropic parameters. Each element of the ToE formalism contributes to the

derivation, and the Schwarzschild solution emerges as a tightly constrained consequence of the framework — not as an assumption, not as an approximation, but as an *emergent classical limit* of the entropic dynamics.

7.3. Predictions Beyond the Classical Limit

Beyond the classical limit, the Theory of Entropicity predicts corrections to the Schwarzschild solution that distinguish it from standard general relativity. These corrections represent the genuine predictive frontier of the theory and provide targets for observational and experimental tests:

8. **Entropic dispersion near the horizon:** At distances comparable to the Schwarzschild radius, the entropic field enters the strong-field regime, and the LOA stress-energy (particularly the kinetic term $\partial_\mu S \partial_\nu S$) becomes non-negligible. This produces corrections to the Schwarzschild metric that can be parameterized in the post-Newtonian expansion and compared with observations.
9. **OCI-dependent dark curvature effects:** If the emergence map is not exactly conformal (e.g., if λ acquires a spatial dependence due to quantum corrections), the OCI tensor becomes non-zero and contributes additional "dark curvature" to the spacetime geometry. This dark curvature could manifest as deviations from the Schwarzschild metric at large distances, potentially relevant for galaxy rotation curves and the dark matter problem.
10. **Quantum corrections from the SOA:** The Seeley-DeWitt heat kernel expansion generates higher-curvature corrections (R^2 , $R_{\mu\nu}R^{\mu\nu}$, etc.) that modify the

Schwarzschild geometry at distances comparable to the Planck length. These corrections are negligible for astrophysical black holes but could be significant for Planck-scale black holes and may play a role in the resolution of the curvature singularity at $r = 0$.

- 11. Yukawa-screened corrections for non-zero entropic mass:** If the entropic mass m is non-zero, the entropic field profile is Yukawa-screened (equation (IE-S.102)), and the Schwarzschild metric acquires exponential corrections at distances $r \gg 1/m$. These corrections constitute a Yukawa modification of gravity that could be tested by solar-system experiments and gravitational-wave observations.

All of these corrections are *falsifiable*: they make quantitative predictions that can, in principle, be compared with observations. The Theory of Entropicity therefore goes beyond a mere reformulation of general relativity; it is an extension with genuine predictive content.

7.4. Future Directions

The Schwarzschild derivation presented in this appendix is the simplest application of the GEFE/OFE to a known exact solution. Natural extensions include:

- **The Kerr solution:** Deriving the rotating black hole metric from a stationary, axially symmetric entropic field configuration. This requires extending the Fisher-Rao construction to two-parameter statistical models and solving the MEE with axial symmetry.

- **The Reissner-Nordström solution:** Deriving the charged black hole metric by coupling the entropic field to the electromagnetic sector, using the results of Letter IIA.
- **The Kerr-Newman solution:** The general case of a rotating, charged black hole — the most general stationary black hole solution in GR (by the no-hair theorem).
- **Cosmological solutions:** Deriving the Friedmann-Lemaître-Robertson-Walker (FLRW) metric from a homogeneous, isotropic entropic field configuration. This connects the ToE to cosmology and provides an entropic interpretation of the cosmological constant, dark energy, and the expansion of the universe.
- **Gravitational wave solutions:** Deriving linearized gravitational wave solutions from perturbations of the entropic field, verifying that the GEFE predict the correct gravitational wave polarizations and propagation speed.

Each of these extensions represents a further validation of the GEFE/OFE and an opportunity to extract predictions that go beyond standard GR. The Schwarzschild derivation, as the first and simplest case, establishes the foundation upon which all subsequent derivations will be built.

8. Remarks on the ToE Schwarzschild Solution (TSS)

Remark K.1.1: Yukawa Corrections and Modified Gravity Phenomenology

The derivation can be generalized to non-zero entropic mass $m > 0$, yielding Yukawa-screened corrections to the Schwarzschild metric at distances $r \gg 1/m$. These corrections could be relevant for modified gravity phenomenology (MOND-like effects at galactic scales). Specifically, the entropic field profile $S(r) = S_o + (S_i/r)e^{-mr}$ generates an effective potential $\Phi(r) \propto (1/r)e^{-mr}$, which transitions from Newtonian ($\Phi \propto 1/r$) at short distances ($r \ll 1/m$) to exponentially suppressed at large distances ($r \gg 1/m$). If the entropic mass scale m is comparable to the inverse of the galactic scale ($m \sim 1/r_{gal} \sim 10^{-23} \text{ eV}/c^2$), the Yukawa corrections could modify the rotation curves of galaxies, providing an alternative to cold dark matter. The quantitative analysis of this scenario requires solving the GEFE with the Yukawa profile and computing the post-Newtonian parameters, which is deferred to future work.

Remark K.1.2: The Event Horizon as an Entropic Surface

The identification $M = \alpha S_i$ suggests that the Schwarzschild radius $r_s = 2GM/c^2 = 2\alpha GS_i/c^2$ is proportional to the entropic charge. The event horizon is therefore an *entropic surface* — a surface of constant entropic charge density. In the ToE framework, the

event horizon is not merely a causal boundary (the boundary of the region from which no signal can escape to infinity) but an *informational boundary*: it is the surface at which the entropic field perturbation S_i/r reaches a critical ratio with respect to the background S_0 . This informational interpretation of the event horizon resonates with the holographic principle and with the proposal (advanced by 't Hooft and Susskind) that the event horizon is a holographic screen encoding the information content of the interior.

Remark K.1.3: Fubini-Study Metric and Quantum Corrections

The derivation can be repeated using the Fubini-Study metric instead of the Fisher-Rao metric, yielding the quantum-corrected Schwarzschild metric. The Fubini-Study metric is the natural metric on the projective Hilbert space of quantum states, and it reduces to the Fisher-Rao metric in the classical limit (when the quantum state can be represented by a classical probability distribution). Using the Fubini-Study metric in the emergence map would produce a spacetime metric that incorporates quantum coherence effects — corrections to the Schwarzschild geometry arising from the quantum nature of the entropic substrate. These corrections are expected to be significant near the Planck scale ($r \sim l_P$) and could play a role in resolving the curvature singularity at $r = 0$. This connects the ToE to the program of quantum gravity and the information-theoretic properties of black holes.

Remark K.1.4: Uniqueness of the Entropic Configuration (Entropic Birkhoff Theorem)

The uniqueness of the Schwarzschild solution (Birkhoff's theorem) translates in the ToE framework to a uniqueness theorem for the entropic field: the *unique* static, spherically symmetric solution of the MEE in the massless, minimal-coupling limit is $S(r) = S_o + S_i/r$. This is the *entropic Birkhoff theorem*: among all static, spherically symmetric entropic configurations in the classical limit, only the Coulomb-like profile generates a consistent spacetime geometry through the emergence map. The proof is straightforward: the massless MEE in spherical symmetry reduces to the Laplace equation $\nabla^2 S = 0$, whose unique spherically symmetric solution regular at infinity is $S = S_o + S_i/r$. The uniqueness of the entropic configuration and the uniqueness of the Schwarzschild geometry are thus two sides of the same coin — the former implies the latter via the emergence map, and the latter constrains the former via the GEFE.

Remark K.1.5: The Curvature Singularity and Its Entropic Resolution

The Kretschner scalar of the Schwarzschild metric, $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/(c^4r^6)$, diverges at $r = 0$. In the ToE framework, this curvature singularity corresponds to a divergence of the Fisher-Rao metric, suggesting that the entropic field $S(r)$ develops a topological singularity at the center. The resolution of this singularity may require the

full quantum treatment (SOA + Vuli-Ndlela Integral), which regularizes the entropic field at short distances. Specifically, the SOA corrections (higher-curvature terms in the GEFE) become dominant at $r \sim l_P$, modifying the Schwarzschild geometry and replacing the classical singularity with a regular core. The Vuli-Ndlela Integral — the non-perturbative completion of the entropic field theory — may further constrain the short-distance behavior by imposing topological conditions on the entropic field (e.g., requiring the information content to remain finite everywhere). The resolution of the Schwarzschild singularity within the ToE framework is a central open problem and a key test of the theory's self-consistency at the Planck scale.

9. Notation and Conventions

Symbol	Description
$S(r)$	Entropic field (scalar), function of radial coordinate
S_o	Asymptotic entropic background (value of S at spatial infinity)
S_1	Entropic charge (integration constant determining the field strength)
m	Entropic mass scale (from the potential $V(S) \approx \frac{1}{2}m^2S^2$)
$\square$	Covariant d'Alembertian: $(1/\sqrt{-g}) \partial_\mu(\sqrt{-g}) g^{\mu\nu} \partial_\nu$
$V(S)$	Entropic potential
$g_{l,rr}$	Radial component of the Fisher-Rao information metric

Symbol	Description
$g_{S,\mu\nu}$	Physical spacetime metric
λ	Emergence constant: $l_P^2/(4k_B) = \hbar G/(4k_{BC}^3)$
Φ	Emergence map: $M_I \rightarrow M_S$
$A(r), B(r)$	Temporal and radial Schwarzschild metric functions
M	Schwarzschild mass parameter (emergent entropic charge: $M = \alpha S_I$)
r_s	Schwarzschild radius: $2GM/c^2$
ξ	Non-minimal coupling parameter
$R_{\mu\nu}$	Ricci tensor
$G_{\mu\nu}$	Einstein tensor: $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$
OCI	Obidi Curvature Invariant
LOA, SOA	Leading-Order Approximation, Second-Order Approximation
MEE	Minimal Entropic Equation
GEFE/OFE	Generalized Entropic Field Equations / Obidi Field Equations

10. Equation Index

Label	Equation	Section
(IE-S.100)	$\square S - V'(S) = 0$ — Minimal Entropic Equation	§3 (Step 1)
(IE-S.101)	$(1/r^2) d/dr(r^2 dS/dr) - m^2 S = 0$ — Radial equation	§3 (Step 1)
(IE-S.102)	$S(r) = S_0 + (S_1/r) \exp(-mr)$ — Yukawa solution	§3 (Step 1)
(IE-S.102.1)	$S(r) = S_0 + S_1/r$ — Coulomb limit	§3 (Step 1)
(IE-S.102.2)	$\sigma^2(r) \propto S(r)$ — Variance-entropy relation	§4 (Step 2)
(IE-S.102.3)	$g_I(\theta) = E[(\partial \ln p / \partial \theta)^2]$ — Fisher-Rao metric	§4 (Step 2)
(IE-S.102.4)	$g_{I,rr} = C[S'/S]^2$ — Radial information metric	§4 (Step 2)
(IE-S.102.5)	$S(r) = S_0 + S_1/r$ — Coulomb profile (restated)	§4 (Step 2)
(IE-S.102.6)	$S'(r) = -S_1/r^2$ — Derivative of entropic profile	§4 (Step 2)
(IE-S.102.7)	$g_{I,rr} \propto (S_1^2/r^4)/[S_0 + S_1/r]^2$ — Evaluated metric	§4 (Step 2)
(IE-S.102.8)	$ds^2 = g_{I,rr} dr^2 + r^2 d\Omega^2$ — Information manifold metric	§4 (Step 2)
(IE-S.102.9)	$g_{S,\mu\nu} = \lambda g_{I,\mu\nu}$ — Emergence map	§5 (Step 3)
(IE-S.102.10)	$ds_S^2 = -A c^2 dt^2 + B dr^2 + r^2 d\Omega^2$ — Schwarzschild ansatz	§5 (Step 3)
(IE-S.102.11)	$B(r) \propto g_{I,rr}(r)$ — Radial metric identification	§5 (Step 3)
(IE-S.102.12)	$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0$ — Vacuum Einstein equations	§5 (Step 3)
(IE-S.102.13)	$A(r) = 1 - 2GM/(rc^2)$ — Temporal metric function	§5 (Step 3)
(IE-S.102.14)	$B(r) = [1 - 2GM/(rc^2)]^{-1}$ — Radial metric function	§5 (Step 3)
(IE-S.102.15)	$\Phi(r) = -GM/r$ — Newtonian potential	§6 (Step 4)
(IE-S.102.16)	$\Phi(r) \propto S_1/r$ — Entropic effective potential	§6 (Step 4)

Label	Equation	Section
(IE-S.102.17)	$GM \propto S_1$ — Mass-charge proportionality	§6 (Step 4)
(IE-S.102.18)	$M = \alpha S_1$ — Mass-entropy identification	§6 (Step 4)

11. References on the ToE Schwarzschild Solution (TSS)

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Supplementary Appendix K.1 to Letter IE

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K.2 Friedmann-Lemaître-Robertson-Walker Cosmology from the GEFE/OFE

Assumption: Homogeneous, isotropic entropic field $S = S(t)$, depending only on cosmic time.

For a homogeneous, isotropic entropic field, the Fisher-Rao metric inherits the FLRW symmetries.

The physical metric takes the form:

$$ds^2 = -dt^2 + a^2(t)[dr^2/(1 - kr^2) + r^2 d\Omega^2] \text{ (IE-S.103)}$$

where $a(t)$ is the scale factor and $k = 0, \pm 1$ is the spatial curvature. The GEFE/OFE, applied to this metric with the homogeneous entropic stress-energy tensor, yields the modified Friedmann equations:

$$(\dot{a}/a)^2 = (8\pi G/3)\rho_{\text{ent}} - k/a^2 + \Lambda_{\text{ent}}/3 + \mathcal{O}_{\Omega}/(6a^2) \text{ (IE-S.104)}$$

where $\rho_{\text{ent}} = \frac{1}{2}\dot{S}^2 + V(S)$ is the entropic energy density and $\mathcal{O}_{\Omega}$ is the OCI contribution. In the limit $\mathcal{O}_{\Omega} \rightarrow 0$ and $\Lambda_{\text{ent}} \rightarrow \Lambda = \text{const}$, this reduces to the standard Friedmann equation.

The OCI contribution $\mathcal{O}_{\Omega}/(6a^2)$ acts as an additional curvature-like term that modifies the expansion dynamics. At early times (small a), this term can dominate, producing deviations from standard cosmology that may be observable in the cosmic microwave background. At late times (large a), it decreases as $1/a^2$, leaving the standard Friedmann dynamics dominant.

K.3 Gravitational Waves from the GEFE/OFE

To derive gravitational waves, we linearize the GEFE/OFE around a flat background. Let $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, where $\eta_{\mu\nu}$ is the Minkowski metric and $|h_{\mu\nu}| \ll 1$ are small perturbations.

In the vacuum (no matter sources), the linearized GEFE/OFE reduces to the wave equation:

$$\square \bar{h}_{\mu\nu} = 0 \text{ (IE-S.105)}$$

where $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$ is the trace-reversed perturbation and $\square = -(1/c^2)\partial^2/\partial t^2 + \nabla^2$ is the flat-space d'Alembertian. This is exactly the gravitational wave equation of linearized general relativity.

The propagation speed of these gravitational waves is $c = \sqrt{(\kappa/\rho_S)}$ — the entropic speed limit. This prediction is consistent with the LIGO/Virgo observation that gravitational waves from the binary neutron star merger GW170817 arrived within 1.7 seconds of the gamma-ray burst GRB 170817A, constraining $|c_{\text{GW}} - c|/c < 10^{-15}$. In the ToE framework, this near-equality is not a coincidence but a necessity: both gravitational waves and electromagnetic waves propagate at the entropic speed limit because both are disturbances of the same entropic field.

K.4 Entropic Corrections to the Newtonian Limit

In the weak-field, slow-motion limit, the GEFE/OFE reduces to a modified Poisson equation for the Newtonian gravitational potential Φ :

$$\nabla^2\Phi = 4\pi G\rho + f(\mathcal{O}_\Omega) \text{ (IE-S.106)}$$

where $f(\mathcal{O}_\Omega)$ is the OCI correction function. For small OCI (weak informational dark curvature), $f \approx \alpha_{\text{OCI}} \cdot \mathcal{O}_\Omega/r^2$, giving a correction to the Newtonian potential:

$$\Phi(r) = -GM/r + \delta\Phi(r) \text{ (IE-S.107)}$$

where $\delta\Phi \sim (\mathcal{O}_\Omega GM)/(r^2 \cdot r_o)$ for a characteristic scale r_o determined by the entropic field profile. At small radii ($r \ll r_o$), the OCI correction is negligible and standard Newtonian gravity is recovered. At large radii ($r \sim r_o$), the OCI correction becomes significant, producing the flattening of galactic rotation curves observed in all spiral galaxies — an effect traditionally attributed to dark matter halos.

The characteristic scale r_o is determined by the long-range correlations of the entropic field and is galaxy-dependent, consistent with the observed variation of dark matter profiles across different galaxy types and masses. This provides a natural, parameter-free explanation for the Tully-Fisher relation (the observed correlation between galaxy luminosity and rotation velocity) and the radial acceleration relation (the tight correlation between observed gravitational acceleration and that predicted by visible matter alone).

Appendix L — The Entropic Renormalization Group and Gravitational Running

L.1 The Entropic Renormalization Group

The entropic field theory, like any quantum field theory, requires renormalization to handle ultraviolet divergences. The renormalization group (RG) describes how the coupling constants of the theory change with the energy (or entropic) scale μ . The entropic RG flow is governed by the beta functions:

$$d\alpha/d \ln \mu = \beta_\alpha(\alpha, \xi, \lambda_n) \text{ (IE-S.108)}$$

$$d\xi/d \ln \mu = \beta_\xi(\alpha, \xi, \lambda_n) \text{ (IE-S.109)}$$

$$d\lambda_n/d \ln \mu = \beta_n(\alpha, \xi, \lambda_n) \text{ (IE-S.110)}$$

where α is the entropic coupling constant (governing the strength of the entropic field's self-interaction), ξ is the non-minimal coupling to curvature, and λ_n are the higher-order coupling constants.

The beta functions are computed from the LOA using standard perturbative techniques. At one-loop order (the leading quantum correction), the beta function for the non-minimal coupling ξ is:

$$\beta_\xi = (\xi - 1/6) \cdot [n(n-1)\lambda_n/(16\pi^2)] + O(\lambda^2) \text{ (IE-S.111)}$$

This has a fixed point at $\xi = 1/6$ — the conformal coupling. At the conformal fixed point, the entropic field couples to curvature in a conformally invariant manner, and the trace of the stress-energy tensor vanishes classically. The RG flow drives ξ toward the conformal value, suggesting that the conformal coupling is an infrared attractor — the entropic field naturally evolves toward conformal coupling at low energies.

L.2 The Running of Newton's Constant

Newton's gravitational constant G , which is determined by the SOA through the relation $G = 6\pi/(f_2\Lambda^2)$, runs with energy scale due to quantum corrections. The running is given by:

$$G(\mu) = G_o [1 + \beta_G \ln(\mu/\mu_o) + O(\ln^2)] \text{ (IE-S.112)}$$

where G_o is Newton's constant at the reference scale μ_o and β_G is the gravitational beta function. In the ToE framework, the running of G is determined by the spectral properties of the entropic Laplacian: as the energy scale increases, more entropic modes contribute to the spectral action, modifying the effective gravitational coupling.

At the Planck scale ($\mu \sim M_P$), the running of G may produce an ultraviolet fixed point — a scale at which $G(\mu) \rightarrow G^* = \text{const}$. This is the asymptotic safety scenario for quantum gravity, first proposed by Steven Weinberg in 1979. In the ToE framework, the existence of this fixed point is related to the

properties of the spectral cutoff function f and the high-energy behavior of the entropic field — a connection that may ultimately determine whether the entropic theory of gravity is UV-complete.

The implications of the running of G for cosmology are significant. In the early universe (high energy scale), G may have been different from its present-day value, affecting the expansion rate during nucleosynthesis, the formation of the cosmic microwave background, and the growth of large-scale structure. The ToE framework provides specific predictions for these deviations, which can be tested against cosmological observations.

Appendix M — Conclusion and Outlook

These Supplementary Appendices to Letter IE have accomplished a comprehensive task: the full construction of the Generalized Entropic Field Equations (GEFE/OFE), in which both sides of Einstein's field equations — the geometric left-hand side and the matter-energy right-hand side — are independently and completely generated by a single entropic field $S(x)$ through the full Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$.

Let us summarize the complete architecture:

Einstein's Left-Hand Side — the Einstein tensor $G_{\mu\nu}$, encoding the curvature of spacetime — is generated by the **Entropic Generator of Geometry**. The construction chain is: entropic field $S(x) \rightarrow$ probability distribution $P(\omega; x) \rightarrow$ Fisher-Rao information metric $g^l_{\mu\nu}$ (unique by Čencov's theorem) $\rightarrow$ emergence map $\Phi \rightarrow$ physical spacetime metric $g^S_{\mu\nu} = \lambda g^l_{\mu\nu} \rightarrow$ Curvature Transfer Theorem $\rightarrow$ physical Einstein tensor $G^S_{\mu\nu} = G^l_{\mu\nu}$. The LOA generates the local differential structure through the non-minimal coupling $\xi R S^2$; the SOA

generates the global spectral structure through the heat-kernel expansion, producing the cosmological constant (from a_0), the Einstein-Hilbert action (from a_2), and higher-curvature corrections (from a_4 and beyond).

Einstein's Right-Hand Side — the stress-energy tensor $T_{\mu\nu}$, encoding the matter-energy content — is generated by the **Entropic Generator of Matter**. The construction chain is: complexified entropic field $E = \rho \exp(i\Theta) \rightarrow$ Fubini-Study metric $ds^2_{\text{FS}} = (d\rho/\rho)^2 + (d\Theta)^2 \rightarrow$ amplitude sector (gravitational matter) + phase sector (electromagnetism) $\rightarrow$ Amari alpha-connections (e-, m-, and LC) $\rightarrow$ entropic stress-energy tensor $T^{(\text{ent})}_{\mu\nu}$ from the Obidi Action variation. Different configurations of the entropic field generate different forms of matter: dust, radiation, dark energy, electromagnetic fields, and topological defects.

Both sides are generated by a single entropic field, and the GEFE/OFE equates them:

$$G^{(\text{ent})}_{\mu\nu} + \Lambda_{\text{ent}} g_{\mu\nu} + O^{(\text{OCI})}_{\mu\nu} = (8\pi G_{\text{ent}}/c^4_{\text{ent}}) T^{(\text{ent})}_{\mu\nu}$$

Einstein's field equations are recovered as the classical, weak-curvature, low-entropy limit, when the OCI vanishes, the SOA higher-curvature corrections are negligible, the amplitude fluctuations are frozen, and the entropic couplings reduce to constants. This recovery is exact and complete: every term in Einstein's equations is accounted for, and every constant (G , c , Λ) is derived from the entropic field parameters.

The GEFE/OFE predicts new physics beyond Einstein:

4. The OCI tensor provides informational dark curvature — dark matter and dark energy effects without exotic matter or fine-tuning.

5. The entropic cosmological function $\Lambda_{\text{ent}}[S]$ provides dynamical dark energy with a naturally small effective value.
6. The SOA higher-curvature terms resolve classical singularities (black holes, Big Bang).
7. The amplitude-phase coupling produces novel gravitational-electromagnetic interactions.
8. The alpha-connection structure provides corrections to geodesic motion and gravitational wave propagation from the non-Gaussian statistics of the entropic field.
9. The entropic renormalization group predicts the running of Newton's constant and the possible existence of an ultraviolet fixed point (asymptotic safety).

Future directions for the Theory of Entropicity in the gravitational sector include:

- **Full quantum gravity via the Vuli-Ndlela Integral:** The path integral quantization of the GEFE/OFE, using the Vuli-Ndlela Integral as the UV-complete functional integral over entropic field configurations, will provide the full quantum theory of gravity in the ToE framework.
- **Nuclear forces from the entropic field:** Following the pattern established for electromagnetism (Letter IIA), the strong and weak nuclear forces should emerge from additional symmetry sectors of the entropic field. The non-Abelian gauge structure of the Standard Model may arise from the multi-component structure of the entropic field on the substrate.

- **Complete unification:** The ultimate goal is to show that all four fundamental forces — gravity, electromagnetism, the strong nuclear force, and the weak nuclear force — along with all matter content (quarks, leptons, Higgs boson), emerge from a single entropic field through the Obidi Action. The GEFE/OFE, combined with the Maxwell derivation of Letter IIA, represents the first two steps toward this complete unification.
- **Observational tests:** The predictions of the GEFE/OFE — OCI dark curvature profiles for galaxies, the running of G with cosmological epoch, the entropic corrections to gravitational wave waveforms, the resolution of black hole singularities — provide specific, testable signatures that distinguish the Theory of Entropicity from standard general relativity and from competing approaches to quantum gravity.

The Theory of Entropicity (ToE), through the Generalized Entropic Field Equations, has achieved what Einstein sought for three decades: a framework in which both spacetime geometry and matter-energy emerge from a single, unified substrate. The entropic field is that substrate. The Obidi Action is its dynamics. The GEFE/OFE is its field equation. And Einstein's general relativity — that supreme achievement of 20th-century physics and of the genius of the human mind — is its classical limit.

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End of Supplementary Appendices A to Letter IE

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jonimisiobidi@gmail.com

Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

The Theory of Entropicity (ToE) Living Review Letters Series (ToE-LRLS). Letter IE

John Onimisi Obidi (jonimisiobidi@gmail.com)

Research Lab, The Aether

May 6, 2026

THE THEORY OF ENTROPICITY (TOE) — LIVING REVIEW LETTERS SERIES

B

SUPPLEMENTARY APPENDICES B TO ToE LETTER IE

Introduction to the Appendices B to the ToE LRLS Letter IE

THE THEORY OF ENTROPICITY (TOE) — LIVING REVIEW LETTERS SERIES

SUPPLEMENTARY APPENDIX B TO ToE LETTER IE

Bianconi's Gravity from Entropy and the Theory of Entropicity (ToE): Independent Convergence, Structural Subsumption, and the Entropic Origin of Spacetime — A Comparative Analysis in the Context of Letter IE

John Onimisi Obidi

Research Lab, The Aether

jonimisiobidi@gmail.com

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"The relation between general relativity, statistical mechanics and information theory is a central research topic in theoretical physics."

— Ginestra Bianconi, *Gravity from Entropy* (2025)

"It from bit."

— John Archibald Wheeler, *Information, Physics, Quantum* (1990)

"Spacetime tells matter how to move; matter tells spacetime how to curve."

— John Archibald Wheeler, *Geons, Black Holes and Quantum Foam* (1998)

Keywords: Theory of Entropicity (ToE); Bianconi Gravity from Entropy; Quantum Relative Entropy; Metric as Density Matrix; Dirac-Kähler Formalism; G-Field; Obidi Action; Local Obidi Action (LOA); Spectral Obidi Action (SOA); Entropic Manifold; Information Geometry; Fisher-Rao Metric; Emergence Map; Curvature Transfer Theorem; Obidi Curvature Invariant (OCI); Modified Einstein Equations; Emergent Cosmological Constant; Entropic Gravity; Bianconi Paradox; Entropic Monism; Living Review Letters Series (LRLS)

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1 Introduction: Two Programs, One Vision

1.1 The Entropic Gravity Revolution

The idea that gravity may not be a fundamental force of nature but rather an emergent consequence of deeper thermodynamic or information-theoretic principles represents one of the most profound conceptual shifts in the history of theoretical physics. This entropic gravity revolution, which has unfolded over more than half a century of sustained inquiry, challenges the very foundations upon which the edifice of modern gravitational theory rests. To appreciate the full significance of Ginestra Bianconi's "Gravity from Entropy" program and John Onimisi Obidi's Theory of Entropicity (ToE) — the two programs that form the subject of this comparative analysis — it is necessary to trace the historical arc of this revolution in careful detail, from its earliest intimations to its present state of remarkable maturity.

The story begins, as so many great stories in physics do, with black holes. In 1972, Jacob D. Bekenstein, then a doctoral student of John Archibald Wheeler at Princeton University, proposed what would become one of the most consequential ideas in twentieth-century physics: that black holes possess entropy. This proposal was motivated by a simple but devastating thought experiment. Consider an observer who drops a box of thermal radiation, characterized by a well-defined entropy, into a black hole. If the black hole possesses no entropy, then the total entropy of the universe — the black hole plus the external environment — would decrease, violating the second law of thermodynamics. Bekenstein's resolution was audacious: the black hole must itself carry entropy, and that entropy must increase by at least the amount lost by the infalling radiation, thereby preserving the generalized second law.

The mathematical form of Bekenstein's entropy formula was no less revolutionary than the conceptual insight it encoded. Drawing on the classical result that the area of a black hole's event horizon can never decrease — the area theorem proven by Stephen Hawking in 1971, which already bore a striking formal resemblance to the second law — Bekenstein proposed that the entropy of a black hole is proportional to the area of its event horizon:

$$(BG.1) S_{BH} = \eta (k_B c^3 / \hbar G) A$$

where A is the area of the event horizon, k_B is Boltzmann's constant, c is the speed of light, $\hbar$ is the reduced Planck constant, G is Newton's gravitational constant, and η is a dimensionless constant of order unity whose precise value Bekenstein could not determine from first principles. The combination $l_P^2 = \hbar G / c^3$ is the Planck area, so the entropy is measured in units of the Planck area. The formula reveals that the maximum entropy that can be contained within a region of space scales not with the volume of the region, as would

be expected for ordinary matter, but with the area of the boundary — a profoundly holographic scaling that would later inspire the holographic principle of 't Hooft and Susskind.

The significance of the Bekenstein entropy cannot be overstated. For the first time in the history of physics, a fundamental law connected the three great pillars of theoretical physics — general relativity (through G), quantum mechanics (through $\hbar$), and statistical mechanics (through k_B) — in a single formula. The entropy formula is, in the words of Wheeler, a "Rosetta Stone" for quantum gravity, encoding in compact form the deep relationships between geometry, information, and thermodynamics that any complete theory of quantum gravity must ultimately explain.

The next decisive step was taken by Stephen Hawking in 1974–1975, who demonstrated through a semi-classical calculation that black holes are not perfectly black: they emit thermal radiation at a temperature inversely proportional to their mass. The Hawking temperature is given by:

$$(BG.2) T_H = \hbar c^3 / (8\pi G M k_B)$$

where M is the mass of the black hole. This result — derived by studying quantum field theory on a curved spacetime background, specifically the behavior of quantum fields near the event horizon — completed the thermodynamic analogy that Bekenstein had initiated. With both temperature and entropy defined, black holes became genuine thermodynamic systems obeying all four laws of thermodynamics. Hawking's calculation also fixed the proportionality constant in Bekenstein's formula: $\eta = 1/4$, yielding the celebrated Bekenstein-Hawking entropy formula $S_{BH} = k_B A / (4l_P^2)$. The factor of $1/4$, which Bekenstein

had been unable to determine, emerged naturally from the quantum field-theoretic calculation, providing a striking confirmation of the thermodynamic interpretation.

Hawking radiation raised as many questions as it answered. The derivation was semi-classical — it treated gravity classically (via the Schwarzschild metric) and matter quantum-mechanically (via quantum field theory on curved spacetime) — and therefore could not provide a microscopic statistical-mechanical explanation of the entropy. In ordinary thermodynamics, entropy counts microstates: $S = k_B \ln \Omega$, where Ω is the number of microstates compatible with the macroscopic description. What are the microstates of a black hole? What microscopic degrees of freedom does the Bekenstein-Hawking entropy count? These questions, which remain partially open to this day, drove much of the subsequent development of string theory, loop quantum gravity, and other approaches to quantum gravity.

In 1976, William G. Unruh discovered a closely related effect that further cemented the connection between acceleration, horizons, and temperature. Unruh showed that an observer undergoing constant proper acceleration a through the Minkowski vacuum perceives a thermal bath of particles at the Unruh temperature:

$$(BG.3) T_U = \hbar a / (2\pi c k_B)$$

The Unruh effect established that the concepts of temperature and particle content are observer-dependent in quantum field theory: what appears as vacuum to an inertial observer appears as a thermal state to an accelerated observer. The effect is deeply connected to the structure of quantum field theory in curved spacetime and to the general principle that horizons — whether event horizons, Rindler horizons, or cosmological

horizons — are associated with thermal properties. The Unruh effect, together with Hawking radiation, suggested that the connection between gravity and thermodynamics is not a peculiarity of black holes but a general feature of spacetime geometry. Any horizon, whether gravitational or kinematic, is endowed with thermal properties, and the connection between geometry and thermodynamics is universal.

The implications of Bekenstein, Hawking, and Unruh were brought to their logical culmination in 1995 by Ted Jacobson, in a paper of breathtaking elegance and conceptual depth. Jacobson's insight was to reverse the logic: rather than deriving thermodynamic properties from the Einstein equations, he derived the Einstein equations from thermodynamic principles. Specifically, Jacobson applied the Clausius relation $\delta Q = T dS$ to local Rindler horizons — the horizons perceived by accelerated observers at each point of spacetime — and showed that the Einstein field equations follow as a thermodynamic equation of state. The argument proceeds as follows. At any point p of spacetime, consider a local Rindler horizon generated by a uniformly accelerated observer. The Unruh effect assigns a temperature T to this horizon. The Raychaudhuri equation relates the focusing of null geodesic congruences — which generates the horizon area change dA — to the energy-momentum tensor $T_{\mu\nu}$ via the Einstein equations. By identifying the heat flux δQ with the energy flux through the horizon (via the stress-energy tensor) and the entropy change dS with the area change dA (via the Bekenstein-Hawking formula), and demanding that the Clausius relation $\delta Q = T dS$ hold for all local Rindler horizons at all points of spacetime, Jacobson showed that the Einstein field equations:

$$(BG.4) \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$$

follow as a necessary consequence. This was the first derivation of the Einstein equations from purely thermodynamic/entropic principles, without assuming the Einstein-Hilbert action or any other variational principle. The Einstein equations are not fundamental dynamical equations, Jacobson argued, but rather an equation of state — a macroscopic thermodynamic relation that emerges from the coarse-graining of more fundamental microscopic degrees of freedom. This was the founding result of the entropic gravity program, and its influence on subsequent developments — including both Bianconi's and Obidi's work — cannot be overstated.

Jacobson's derivation left open several important questions. Chief among them was the question of dynamics: the Clausius relation is a thermodynamic equilibrium relation, and it was not immediately clear how to extend the derivation to non-equilibrium situations, or how to incorporate the dynamical content of general relativity (gravitational waves, cosmological evolution) within the thermodynamic framework. Moreover, Jacobson's derivation assumed the proportionality between entropy and area — the Bekenstein-Hawking formula — as an input, rather than deriving it from a more fundamental principle. These lacunae motivated further developments in the entropic gravity program.

In 2010–2011, Erik Verlinde proposed a far-reaching generalization of the entropic gravity idea. Building on Jacobson's work, the holographic principle, and Bekenstein's original insights, Verlinde argued that gravity is an "entropic force" — a macroscopic force that arises from the statistical tendency of a system to increase its entropy, analogous to osmotic pressure or polymer elasticity. Verlinde's central formula expresses Newton's gravitational force as an entropic gradient:

$$(BG.5) \quad F = T (dS/dx)$$

where T is the temperature associated with a holographic screen surrounding the gravitating mass, S is the entropy stored on the screen, and x is the displacement perpendicular to the screen. Using the Unruh temperature for the accelerated observer and the Bekenstein bound for the entropy, Verlinde derived Newton's second law $F = ma$, Newton's law of gravitation $F = GMm/r^2$, and, in the relativistic generalization, the full Einstein field equations. The proposal was enormously influential, stimulating a vast literature of both supportive and critical analyses. Critics raised important objections — notably, that the notion of an "entropic force" is ambiguous, and that the derivation relies on assumptions (holographic screens, the Unruh temperature, the Bekenstein bound) that themselves presuppose the gravitational physics being derived. Nevertheless, Verlinde's work crystallized the entropic gravity program as a distinct research direction and brought it to the attention of a broad audience.

Simultaneously and independently, Thanu Padmanabhan developed a comprehensive "emergent gravity" program that derived the gravitational field equations from the principle of entropy maximization on null surfaces. Padmanabhan's approach was more systematic and technically rigorous than Verlinde's, and it led to a number of remarkable results. In particular, Padmanabhan showed that the Einstein equations can be written in a form that is manifestly thermodynamic: the field equations express the balance between the surface heating rate (proportional to the Komar energy on the boundary) and the bulk equipartition energy (proportional to the number of bulk degrees of freedom times the temperature). Padmanabhan also demonstrated that the gravitational action (the Einstein-Hilbert action plus the Gibbons-Hawking-York boundary term) can be decomposed into a bulk term and a surface term, with the surface term giving the entropy, thereby providing a variational basis for the thermodynamic interpretation. His work, spanning numerous

papers from 2010 to 2015 and culminating in a comprehensive review (Rep. Prog. Phys. 73, 046901, 2010), established that the connection between gravity and thermodynamics is not limited to equilibrium situations or special spacetimes but is a general structural property of the Einstein equations in any dimension and for any matter content.

It is against this rich historical backdrop that we must situate the two programs under analysis in this appendix. The Bekenstein-Hawking-Unruh-Jacobson-Verlinde-Padmanabhan sequence establishes beyond reasonable doubt that gravity and thermodynamics are deeply intertwined. The question that remains — and it is the central question of contemporary theoretical physics — is whether this intertwining is merely analogical (gravity is *like* thermodynamics) or ontological (gravity *is* thermodynamics, i.e., gravity is literally an emergent consequence of entropic dynamics). Both Bianconi's Gravity from Entropy and Obidi's Theory of Entropicity answer this question unequivocally in the affirmative: gravity is not merely analogous to thermodynamics — it *IS* entropic dynamics, and the Einstein field equations are the macroscopic, classical-limit expression of this deeper entropic reality. Where the two programs differ is in the depth of the ontological commitment, the mathematical machinery employed, and the scope of the phenomena addressed. These differences are the subject of the analysis that follows.

1.2 Bianconi's Contribution: Gravity from Entropy (2024–2025)

Ginestra Bianconi's "Gravity from Entropy" (GfE) represents one of the most significant contributions to the entropic gravity program since Jacobson's 1995 paper. Published as Physical Review D 111, 066001 (2025), having been submitted in August 2024 (arXiv:2408.14391), the paper introduces a novel and mathematically sophisticated

framework for deriving modified Einstein equations from an entropic action principle. The central innovation of Bianconi's program is the proposal that the metric tensor of Lorentzian spacetime should be interpreted not merely as a classical geometric object — a symmetric $(0,2)$ tensor field on a smooth manifold — but as an effective density matrix, a quantum-information-theoretic object that encodes the gravitational degrees of freedom in a way that naturally lends itself to entropic analysis.

This conceptual move is significant for several reasons. In standard general relativity, the metric tensor $g_{\mu\nu}(x)$ plays a dual role: it defines the causal structure of spacetime (via the light cones) and it determines the gravitational field (via the Christoffel connection and the Riemann curvature tensor). These two roles are unified by Einstein's principle of general covariance, which identifies the gravitational field with the geometry of spacetime itself. Bianconi's proposal adds a third role: the metric also functions as an information-theoretic object, encoding the quantum state of the geometry at each point of spacetime. This triple unification — causal structure, gravitational field, and quantum state — is the conceptual foundation of the GfE program.

The mathematical framework of GfE proceeds in several steps. First, Bianconi constructs a density matrix ρ_g from the spacetime metric $g_{\mu\nu}$, normalized so that $\text{Tr}(\rho_g) = 1$. Second, she describes matter fields using the Dirac-Kähler formalism — a geometric framework in which matter is represented as the direct sum of differential forms of degrees 0, 1, and 2, providing a unified treatment of scalar, vector, and tensor matter fields. Third, she constructs a second density matrix $\rho_{g(M)}$ from the metric induced by the matter fields. Fourth, she defines the entropic action as the quantum relative entropy between these two density matrices: $S_B = S(\rho_g \parallel \rho_{g(M)})$. Fifth, she varies this action with respect to the metric to

obtain modified Einstein equations that reduce to the standard Einstein equations in the low-coupling regime. Sixth, she introduces the G-field as a Lagrangian multiplier, which generates a "dressed" Einstein-Hilbert action with an emergent small positive cosmological constant.

The fact that this paper was published in Physical Review D — one of the premier journals in gravitational and high-energy physics — is itself significant. The paper passed rigorous peer review, establishing the mathematical consistency and physical relevance of the framework within the standards of the mainstream physics community. This is not a speculative proposal or a heuristic argument; it is a complete, self-consistent mathematical framework with well-defined field equations, a clear recovery of the Einstein limit, and novel physical predictions. The intellectual ambition of the work — to derive gravity from entropy using the mathematical tools of quantum information theory — places it squarely within the entropic gravity tradition initiated by Bekenstein, Hawking, Jacobson, and Verlinde, while simultaneously advancing that tradition in new directions.

Bianconi's work is also notable for its engagement with several of the most pressing problems in contemporary gravitational physics. The emergent cosmological constant provides a potential resolution to the cosmological constant problem — the notorious discrepancy between the observed value of Λ and the value predicted by quantum field theory, which differ by some 120 orders of magnitude. By deriving Λ from the G-field sector rather than from vacuum energy calculations, Bianconi sidesteps the fine-tuning problem entirely: the cosmological constant is not a fundamental parameter to be calculated from first principles but an emergent property of the entropic dynamics. Similarly, the G-field itself may serve as a dark matter candidate, providing gravitational effects beyond those

predicted by the standard Einstein equations without requiring the introduction of new elementary particles. These potential connections to observational cosmology give the GfE program empirical relevance beyond its theoretical elegance.

1.3 Obidi's Theory of Entropicity (ToE): The Broader Framework

The Theory of Entropicity (ToE), developed by John Onimisi Obidi and presented through the Living Review Letters Series (LRLS), represents a far more comprehensive and ontologically radical program than any single paper within the entropic gravity literature. Where Bianconi's GfE addresses the specific question of how to derive modified Einstein equations from an entropic action, the ToE addresses the broader question of how all of physics — gravity, electromagnetism, quantum mechanics, and cosmology — emerges from a single fundamental entity: the entropic field $S(x)$.

The entropic field $S(x)$ is the foundational dynamical variable of the ToE. It is defined on a substrate Ω — the "entropic manifold" — which is ontologically prior to spacetime. The entropic field is not a field *on* spacetime in the way that the electromagnetic field or the metric tensor is a field on spacetime; rather, it is the field *from which* spacetime, and all the fields defined upon it, emerge. This ontological priority is the defining characteristic of the ToE and the feature that most sharply distinguishes it from all other approaches to entropic gravity, including Bianconi's.

The dynamical equations of the ToE are derived from the Obidi Action, which consists of two complementary components. The Local Obidi Action (LOA) captures the local, field-theoretic dynamics of the entropic field: its kinetic energy, its self-interaction potential,

and its non-minimal coupling to the emergent curvature. The Spectral Obidi Action (SOA) captures the global, spectral dynamics of the entropic field: the trace of a function of the entropic Laplacian, which, through the Seeley-DeWitt heat-kernel expansion, generates the gravitational sector (cosmological constant, Einstein-Hilbert action, higher-curvature corrections) from the spectral properties of the entropic operator. Together, $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$ provides a complete variational principle from which all the field equations of the ToE — the Master Entropic Equation (MEE) for the entropic field and the Entropic Einstein Equations for the metric — are derived by variation.

Letter IE of the LRLS — "On the Emergence of Physical Spacetime Geometry from Information Geometry" — establishes the central result of the gravitational sector: the emergence of physical spacetime geometry from information geometry. The key mathematical ingredients are: (i) the information manifold (M_I, g_I) , equipped with the Fisher-Rao information metric, which is the unique Riemannian metric on the space of probability distributions invariant under sufficient statistics (Chentsov's theorem); (ii) the emergence map $\Phi: M_I \rightarrow M_S$, which maps the information manifold to the physical spacetime manifold; and (iii) the Curvature Transfer Theorem, which establishes that the Riemann curvature of physical spacetime equals the rescaled pushforward of the Riemann curvature of the information manifold, up to corrections of order $O(1/N)$. The physical spacetime metric is conformally related to the Fisher-Rao metric: $\Phi^*g_S = \lambda g_I$, where the conformal factor $\lambda = l_P^2/(4k_B) = \hbar G/(4k_{BC}^3)$ encodes all three fundamental constants of nature ($\hbar$, G , c) in terms of the single constant k_B (Boltzmann's constant) and the derived constant l_P (the Planck length).

The Generalized Entropic Field Equations (GEFE/OFE), presented in the supplementary appendices to Letter IE, provide the most general field equations of the ToE in the gravitational sector. In these equations, BOTH sides of Einstein's field equations — the geometric left-hand side and the matter right-hand side — are generated by the entropic field. The geometric side is generated by the Fisher-Rao metric and the emergence map; the matter side is generated by the Fubini-Study metric and the Amari alpha-connections. Einstein's equations are recovered as the classical limit of the GEFE/OFE when the SOA corrections vanish, the Obidi Curvature Invariant (OCI) goes to zero, and the amplitude of the complexified entropic field is frozen. This is a remarkable achievement: for the first time in the history of theoretical physics, both sides of Einstein's equations are derived from a single dynamical principle.

1.4 Purpose and Structure of This Appendix

This appendix, which forms a supplementary document to Letter IE of the Living Review Letters Series, has five principal objectives:

1. **Complete technical exposition:** To provide a fully self-contained, rigorous presentation of Bianconi's Gravity from Entropy framework, including all definitions, constructions, equations, and proofs, so that the reader need not consult the original paper to follow the subsequent analysis.
2. **Systematic comparison:** To carry out a comprehensive, point-by-point comparison between Bianconi's GfE program and the ToE framework as developed in Letter IE, identifying all structural parallels and differences, and assessing the relative explanatory power and scope of each program.

3. **Rigorous subsumption:** To demonstrate, through a detailed mathematical derivation, that Bianconi's modified Einstein equations are recoverable as a specific quadratic approximation of the Spectral Obidi Action (SOA) in an appropriate limiting regime, thereby establishing the ToE as the more general framework that contains Bianconi's results as a special case.
4. **Philosophical analysis:** To develop the Bianconi Paradox — the structural circularity inherent in deriving gravity from entropy while presupposing the geometric structures that gravity is supposed to explain — and to show how the ToE resolves this paradox through its monist ontology and its background-independent construction of spacetime.
5. **Synthesis:** To demonstrate that the Bianconi-Obidi convergence — the independent arrival of two programs at structurally parallel conclusions — provides powerful evidence for the entropic origin of gravity, and to outline the open problems that remain for future research.

The structure of the appendix follows this sequence. Section 2 presents the complete technical exposition of Bianconi's GfE. Section 3 reviews the ToE framework as developed in Letter IE. Section 4 carries out the systematic comparison. Section 5 presents the subsumption proof. Section 6 develops the Bianconi Paradox. Section 7 analyzes the G-field and its identification with the modular operator. Section 8 examines the information-geometric structures that ToE provides and Bianconi lacks. Section 9 discusses the significance of independent convergence. Section 10 surveys experimental predictions and observational tests. Section 11 provides the conclusion and synthesis. A comprehensive list of references is provided at the end of the document.

2 Bianconi's Gravity from Entropy: Complete Technical Exposition

2.1 The Metric as Quantum Operator

The foundational conceptual move of Bianconi's Gravity from Entropy program is the promotion of the spacetime metric from a classical geometric object to the status of a quantum operator — specifically, an effective density matrix. In standard general relativity, as formulated by Einstein in 1915, the metric tensor $g_{\mu\nu}(x)$ is a symmetric, non-degenerate, rank-(0,2) tensor field on a four-dimensional smooth manifold M . Its role is twofold: it determines distances and angles (the inner product on each tangent space $T_x M$), and it determines the gravitational field through its connection to the Riemann curvature tensor via the Levi-Civita connection. In this classical picture, the metric is a purely geometric object; it carries no intrinsic information-theoretic or quantum-mechanical content.

Bianconi's radical proposal is that the metric $g_{\mu\nu}$ should be re-interpreted as encoding a quantum state — an effective density matrix ρ_g . The construction proceeds as follows. At each point x of the spacetime manifold M , the metric $g_{\mu\nu}(x)$ defines a positive-definite bilinear form on the tangent space $T_x M$ (assuming Riemannian signature for the moment; the Lorentzian case requires a Wick rotation or analytic continuation, which Bianconi handles carefully). A positive-definite bilinear form on a finite-dimensional vector space can be represented as a positive-definite matrix, and any positive-definite matrix can be normalized to have unit trace, thereby defining a density matrix. Specifically, Bianconi writes:

$$(BG.6) \rho_{g,\mu\nu} = g_{\mu\nu} / \text{Tr}(g) = g_{\mu\nu} / g^{\alpha\beta} g_{\alpha\beta} = g_{\mu\nu} / n$$

where $n = g^{\alpha\beta} g_{\alpha\beta} = \delta^\alpha_\alpha = 4$ is the dimension of spacetime (the trace of the identity in four dimensions). This simple normalization transforms the metric into a density matrix: ρ_g is positive-definite (inheriting this property from g), Hermitian (since g is symmetric), and has unit trace (by construction). The full metric can be recovered from the density matrix as:

$$(BG.7) g_{\mu\nu} = n \cdot \rho_{g,\mu\nu}$$

More generally, Bianconi considers the relationship $g_{\mu\nu} = (\det g)^{1/4} \rho_{g,\mu\nu}$, incorporating the determinant factor to maintain proper tensorial transformation properties under coordinate changes. The density matrix ρ_g at each point encodes the geometric information of that point — the principal directions and principal curvatures — in the language of quantum information theory. The eigenvalues of ρ_g are the normalized eigenvalues of the metric, and they can be interpreted as probabilities: the probability that a "geometric measurement" at the point x yields a result along a given principal direction.

The von Neumann entropy of this density matrix measures the "quantum uncertainty" or "informational content" of the geometric configuration at each point:

$$(BG.8) S_{\text{vN}}(\rho_g) = -\text{Tr}(\rho_g \ln \rho_g) = -\sum_i \lambda_i \ln \lambda_i$$

where λ_i are the eigenvalues of ρ_g . For a maximally symmetric metric (flat Minkowski space, after Wick rotation), all eigenvalues are equal: $\lambda_i = 1/n$, and the von Neumann entropy achieves its maximum value $S_{\text{vN}} = \ln n = \ln 4$. For a highly anisotropic metric (such as near a singularity), the eigenvalues are unequal, and the entropy is lower. This provides a

geometric interpretation of von Neumann entropy: it measures the isotropy of the spacetime geometry at each point. Flat spacetime is maximally isotropic (maximum entropy); highly curved spacetime is anisotropic (lower entropy).

The physical interpretation of this construction deserves careful elaboration. In quantum information theory, a density matrix represents a mixed quantum state — a state that cannot be described by a single wave function but requires a statistical ensemble of wave functions. By promoting the metric to a density matrix, Bianconi is asserting that spacetime geometry is, at the fundamental level, a quantum-information-theoretic object: it is not a definite classical configuration but a mixed state, characterized by probabilities rather than certainties. This is a profoundly quantum picture of spacetime, one that aligns with the general expectation that a theory of quantum gravity should treat geometry quantum-mechanically.

It is instructive to compare this construction with the approach taken by the Theory of Entropicity. In the ToE framework, the metric is not promoted to a density matrix; instead, it is *generated* by the entropic field $S(x)$ via the emergence map Φ . The information-theoretic content of the geometry is encoded not in the metric itself (as a density matrix) but in the Fisher-Rao metric on the information manifold, which is conformally mapped to the physical spacetime metric. Both approaches treat the metric as an information-theoretic object, but they do so at different levels: Bianconi endows the existing metric with informational structure (the density matrix interpretation), while Obidi derives the metric from a more fundamental informational structure (the Fisher-Rao metric on the space of entropic distributions). This difference in the level of the information-theoretic identification will prove central to the comparative analysis in Section 4.

2.2 The Dirac-Kähler Matter Formalism

Bianconi's treatment of matter fields is equally innovative. Rather than describing matter through the standard catalogue of spin-0, spin-1/2, spin-1, and spin-2 fields on a Lorentzian manifold, Bianconi employs the Dirac-Kähler formalism — a geometric framework in which matter is represented as the direct sum of differential forms of different degrees. Specifically, the matter content of the theory is described by a Dirac-Kähler field Ψ , which is the direct sum of a 0-form (scalar) ϕ_0 , a 1-form (vector) ϕ_1 , and a 2-form (tensor) ϕ_2 :

$$(BG.9) \Psi = \phi_0 \oplus \phi_1 \oplus \phi_2$$

The Dirac-Kähler formalism has a distinguished history in mathematical physics. It was introduced by Erich Kähler in 1962, who showed that the Dirac equation on a Riemannian manifold can be reformulated in terms of inhomogeneous differential forms — direct sums of forms of all degrees. The key observation is that the Dirac operator, which acts on spinor fields, is algebraically equivalent to the operator $d + d^*$ (the sum of the exterior derivative and its adjoint, the codifferential) acting on inhomogeneous differential forms. This equivalence, which holds when the manifold admits a spin structure, provides a purely geometric formulation of the Dirac equation that does not require the explicit introduction of spinors or gamma matrices. The Dirac-Kähler formalism was further developed by Becher and Joos (1982), who used it in the context of lattice field theory to construct discretized versions of the Dirac equation that preserve key symmetries of the continuum theory.

In Bianconi's framework, the restriction to forms of degrees 0, 1, and 2 (rather than the full inhomogeneous form including 3-forms and 4-forms) is physically motivated: the 0-form ϕ_0 describes scalar matter (such as the Higgs field or a scalar dark matter candidate), the 1-

form ϕ_1 describes vector matter (such as the electromagnetic field or other gauge bosons), and the 2-form ϕ_2 describes tensor matter (such as the graviton excitations or the antisymmetric Kalb-Ramond field in string theory). This provides a unified, geometrically natural description of the matter content of the theory: all matter fields are differential forms, and the dynamics of matter is governed by the geometric operations (exterior derivative, codifferential, Hodge star) that act on differential forms.

The matter fields induce an alternative metric on the spacetime manifold — the "metric induced by matter," denoted $g^{(M)}_{\mu\nu}$. The construction of this induced metric is one of the technically sophisticated aspects of Bianconi's framework. The idea is that the matter fields ϕ_k ($k = 0, 1, 2$) define, through their kinetic energies and mutual interactions, a bilinear form on the tangent space at each point of spacetime. This bilinear form serves as an alternative metric, the "matter metric," which encodes the geometric structure induced by the matter distribution. The construction generalizes the familiar idea from classical mechanics that the kinetic energy of a system defines a metric on the configuration space (the "kinetic energy metric" or "mass metric").

Formally, the matter-induced metric is constructed as follows. For a 0-form ϕ_0 , the kinetic energy density is $(1/2)g^{\mu\nu}\partial_\mu\phi_0\partial_\nu\phi_0$, and the induced metric component is proportional to $\partial_\mu\phi_0\partial_\nu\phi_0$. For a 1-form $\phi_1 = A_\mu dx^\mu$, the induced metric involves the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and its contractions. For a 2-form $\phi_2 = (1/2)B_{\mu\nu}dx^\mu \wedge dx^\nu$, the construction involves the 3-form field strength $H = dB$ and its contractions. The total matter-induced metric is a sum (appropriately weighted) of the contributions from each sector. This construction ensures that $g^{(M)}_{\mu\nu}$ encodes the complete information about how matter fields curve the

spacetime geometry — not through the Einstein equations (which are to be derived), but through the geometric structure of the matter fields themselves.

The matter-induced metric $g^{(M)}$ defines a second density matrix $\rho_{g^{(M)}}$, constructed by the same normalization procedure as the spacetime density matrix ρ_g . The two density matrices — ρ_g (encoding the geometry) and $\rho_{g^{(M)}}$ (encoding the matter) — are the central objects of Bianconi's entropic action, which measures the information-theoretic distance between them.

2.3 The Entropic Action: Quantum Relative Entropy

The core of Bianconi's construction — the engine that drives the entire framework — is the entropic action, defined as the quantum relative entropy between the spacetime density matrix and the matter-induced density matrix:

$$(BG.10) \ S_B[g, M] = S(\rho_g \parallel \rho_{g^{(M)}})$$

The quantum relative entropy (also known as the **Umegaki** relative entropy, or the quantum **Kullback-Leibler divergence**) between two density matrices ρ and σ is defined as:

$$(BG.11) \ S(\rho \parallel \sigma) = \text{Tr}(\rho \ln \rho - \rho \ln \sigma) = \text{Tr}(\rho \ln \rho) - \text{Tr}(\rho \ln \sigma)$$

This quantity has a number of fundamental properties that make it ideally suited as a gravitational action.

First, the positivity property (Klein's inequality): $S(\rho \parallel \sigma) \geq 0$ for all density matrices ρ and σ , with equality if and only if $\rho = \sigma$. This means that the entropic action is minimized when

the spacetime geometry (encoded in ρ_g) exactly matches the geometry induced by matter (encoded in $\rho_{g(M)}$). Gravitational dynamics, in this picture, is the process by which spacetime adjusts its geometry to minimize the informational divergence from the matter distribution. This is a profound physical interpretation: gravity is the tendency of spacetime geometry to align with the geometry induced by matter.

Second, the quantum relative entropy is not symmetric: $S(\rho \parallel \sigma) \neq S(\sigma \parallel \rho)$ in general. This asymmetry has physical content in Bianconi's framework: it encodes the fact that the spacetime geometry (ρ_g) is the "reference" state, and the matter-induced geometry ($\rho_{g(M)}$) is the "perturbed" state. The gravitational dynamics is driven by the response of the spacetime geometry to the perturbation induced by matter. This directional character of the entropic action distinguishes it from symmetric measures of distance (such as the trace distance or the fidelity) and gives the resulting field equations a specific causal structure.

Third, the quantum relative entropy is jointly convex in both arguments: $S(\lambda\rho_1 + (1-\lambda)\rho_2 \parallel \lambda\sigma_1 + (1-\lambda)\sigma_2) \leq \lambda S(\rho_1 \parallel \sigma_1) + (1-\lambda) S(\rho_2 \parallel \sigma_2)$. This convexity property ensures that the variational problem (minimizing S_B with respect to the metric) is well-posed and admits unique solutions under appropriate boundary conditions. It also ensures the stability of the Einstein equations recovered in the weak-coupling limit: small perturbations of the metric lead to small changes in the action.

Fourth, the quantum relative entropy satisfies the monotonicity property (data processing inequality): $S(\Phi(\rho) \parallel \Phi(\sigma)) \leq S(\rho \parallel \sigma)$ for any completely positive trace-preserving (CPTP) map Φ . This means that the information-theoretic distance between the spacetime geometry and the matter-induced geometry can only decrease under coarse-graining — a

property with deep implications for the relationship between microscopic and macroscopic descriptions of gravitational dynamics.

The entropic action S_B is a functional of the spacetime metric $g_{\mu\nu}$ and the matter fields (ϕ_0, ϕ_1, ϕ_2), through the density matrices ρ_g and $\rho_{g(M)}$ that these quantities define. The field equations are obtained by varying S_B with respect to the metric $g_{\mu\nu}$, holding the matter fields fixed (or simultaneously varying both, depending on the physical context). This variational procedure is the entropic analogue of the standard Einstein-Hilbert variational principle in general relativity.

The physical content of the entropic action can be made more explicit by decomposing the quantum relative entropy into its two terms. The first term, $\text{Tr}(\rho_g \ln \rho_g)$, is the negative of the von Neumann entropy of the spacetime geometry: $-S_{\text{vN}}(\rho_g)$. This term measures the "informational content" of the geometric configuration — how far the geometry deviates from the maximally symmetric (maximum entropy) configuration. The second term, $-\text{Tr}(\rho_g \ln \rho_{g(M)})$, is the negative of the cross-entropy between the spacetime geometry and the matter-induced geometry. This term measures the "surprise" that the matter distribution creates relative to the geometric configuration. The entropic action is therefore a balance between two competing tendencies: the geometry wants to maximize its entropy (become as symmetric as possible), while the matter distribution pushes the geometry away from symmetry (creating curvature). Gravity, in this picture, is the equilibrium between these two tendencies.

2.4 Variation and the Modified Einstein Equations

The field equations of Bianconi's GfE program are obtained by varying the entropic action $S_B[g, M]$ with respect to the inverse metric $g^{\mu\nu}$. The variation proceeds through the chain rule: the entropic action depends on $g^{\mu\nu}$ through the density matrix ρ_g , and the variation of ρ_g with respect to $g^{\mu\nu}$ must be carefully computed using the normalization condition $\text{Tr}(\rho_g) = 1$.

The variational calculation is technically involved, requiring the variation of the logarithm of a matrix-valued function, which involves the integral representation of the matrix logarithm and the Baker-Campbell-Hausdorff formula for non-commuting operators. The result, however, takes a remarkably clean form. The modified Einstein equations obtained by Bianconi can be written as:

$$(BG.12) \quad G_{\mu\nu} + C_{\mu\nu}[\rho_g, \rho_{g(M)}] = (8\pi G/c^4) T_{\mu\nu}^{(\text{eff})}$$

where $G_{\mu\nu} = R_{\mu\nu} - (1/2)Rg_{\mu\nu}$ is the standard Einstein tensor, $C_{\mu\nu}$ denotes correction terms that depend on the density matrices ρ_g and $\rho_{g(M)}$ (and therefore on the curvature and the matter fields), and $T_{\mu\nu}^{(\text{eff})}$ is the effective stress-energy tensor that includes both the standard matter stress-energy and the entropic corrections.

The key property of these modified equations is that they reduce to the standard Einstein equations in the "low-coupling regime" — a regime in which the curvatures are small compared to the Planck scale and the matter field amplitudes are small. In this limit, the correction terms $C_{\mu\nu}$ vanish, and the effective stress-energy tensor reduces to the standard stress-energy tensor:

$$(BG.13) \quad G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} + O(R^2 / R_{Pl}^2)$$

where $R_{Pl} \sim l_P^{-2}$ is the Planck curvature scale. This recovery of the Einstein equations is essential for the physical viability of the framework: any modification of general relativity must reproduce the standard Einstein equations in the regime where they have been observationally tested (solar system tests, binary pulsar timing, gravitational wave observations). Bianconi demonstrates this recovery explicitly, providing confidence that the GfE framework is compatible with all existing observational constraints on general relativity.

It is important to note what Bianconi does NOT assume. She does not assume the Einstein-Hilbert action. She does not assume the Einstein field equations. She does not assume Newton's constant G or the cosmological constant Λ as fundamental parameters. Instead, she derives the Einstein equations (with corrections) from the entropic action — the quantum relative entropy between the spacetime geometry and the matter-induced geometry. This is a genuine derivation, not a reformulation: the Einstein equations emerge as a consequence of the entropic variational principle, not as an input. This places Bianconi's work firmly in the tradition of Jacobson (1995), who first showed that the Einstein equations can be derived from thermodynamic principles, but goes beyond Jacobson in providing a complete variational framework with well-defined field equations and correction terms.

The correction terms $C_{\mu\nu}$ are of particular physical interest, as they represent the deviations from standard general relativity predicted by the GfE framework. These corrections are quadratic (and higher order) in the curvature, and they become significant only in the strong-field regime — near black hole horizons, in the early universe, or at the Planck scale. The specific form of the corrections depends on the details of the Dirac-Kähler matter

content and the normalization of the density matrices. Bianconi provides explicit expressions for the leading correction terms, showing that they are consistent with the general structure of higher-derivative gravity theories but with specific coefficient relations that are fixed by the entropic action, rather than being free parameters as in generic higher-derivative theories.

2.5 The G-Field and the Emergent Cosmological Constant

One of the most remarkable features of Bianconi's framework is the introduction of the G-field and its role in generating an emergent cosmological constant. The G-field, denoted $G_{\mu\nu}$, is a symmetric tensor field introduced as a set of Lagrangian multipliers in the entropic action. Its presence is not ad hoc but emerges naturally from the structure of the variational problem: the constraints imposed by the normalization of the density matrices and the Bianchi identities of the curvature tensor require the introduction of auxiliary fields, and the G-field is the minimal such auxiliary field consistent with the symmetries of the problem.

When the G-field is included, the entropic action reduces to a "dressed" Einstein-Hilbert action — the standard Einstein-Hilbert action supplemented by the G-field Lagrangian and an emergent cosmological constant:

$$(BG.14) \ S_{\text{dressed}} = \int d^4x \sqrt{-g} [R/(16\pi G) + L_G - \Lambda_{\text{eff}}]$$

where L_G is the Lagrangian density of the G-field (which is second-order in derivatives of $G_{\mu\nu}$) and Λ_{eff} is the emergent cosmological constant. The crucial property of this dressed action is that the emergent cosmological constant Λ_{eff} depends ONLY on the G-field sector — not on the vacuum energy of matter fields. This is a potential resolution of the

cosmological constant problem, one of the deepest unsolved problems in theoretical physics.

The cosmological constant problem arises from the discrepancy between the observed value of the cosmological constant ($\Lambda_{\text{obs}} \sim 10^{-52} \text{ m}^{-2}$, corresponding to a vacuum energy density of approximately 10^{-47} GeV^4) and the value predicted by quantum field theory ($\Lambda_{\text{QFT}} \sim l_{\text{P}}^{-2} \sim 10^{66} \text{ m}^{-2}$, corresponding to a vacuum energy density of approximately 10^{74} GeV^4). The discrepancy of approximately 120 orders of magnitude is the worst prediction in the history of physics. All attempts to resolve this problem within the standard framework — by fine-tuning, by invoking supersymmetry, by appealing to the anthropic principle — have been unsatisfactory.

Bianconi's framework offers a new approach: the cosmological constant is not a fundamental parameter of the theory, and it is not determined by the vacuum energy of matter fields. Instead, it is an emergent quantity, determined by the dynamics of the G-field. The G-field Lagrangian L_G is such that the equations of motion for $G_{\mu\nu}$ admit solutions with $\Lambda_{\text{eff}} > 0$, and the value of Λ_{eff} is set by the boundary conditions and the initial conditions of the G-field evolution, not by the ultraviolet physics of quantum field theory. This decoupling of the cosmological constant from the vacuum energy is the key insight: the cosmological constant problem does not arise because the cosmological constant was never a vacuum energy in the first place — it is an emergent property of the entropic dynamics.

The G-field equations of motion are second-order partial differential equations — both in the metric $g_{\mu\nu}$ and in the G-field $G_{\mu\nu}$. This is a crucial property. Higher-derivative field equations (equations containing third or higher derivatives of the dynamical variables) generically suffer from the Ostrogradsky instability: the Hamiltonian is unbounded below,

and the theory admits runaway solutions that make it physically unacceptable. By ensuring that the field equations remain second-order, Bianconi avoids this instability, guaranteeing the physical viability of the framework. This is not a trivial achievement; it requires a specific algebraic structure of the entropic action that constrains the form of the G-field Lagrangian and its coupling to the metric.

Bianconi also suggests that the G-field may play a role in explaining dark matter. The G-field is a new dynamical degree of freedom that couples to the metric through the dressed Einstein-Hilbert action. Its energy-momentum tensor contributes to the right-hand side of the Einstein equations, and its effects on the geometry of spacetime are indistinguishable from those of matter at the level of the field equations. If the G-field is sufficiently long-lived and weakly interacting, it could serve as a dark matter candidate — a geometric degree of freedom, rather than a new elementary particle, that explains the observed gravitational effects attributed to dark matter. This connection between the entropic gravity program and the dark matter problem is one of the most intriguing aspects of Bianconi's work, although it remains to be developed in full detail.

2.6 Key Properties of Bianconi's Framework

To summarize the technical exposition, we enumerate the key structural properties of Bianconi's Gravity from Entropy framework:

Summary of Bianconi's GfE Framework — Key Structural Properties

(i) **Second-order field equations:** Both the metric $g_{\mu\nu}$ and the G-field $G_{\mu\nu}$ satisfy second-order partial differential equations, avoiding the Ostrogradsky instability and the associated ghost degrees of freedom. This ensures the physical consistency and well-posedness of the initial value problem.

(ii) **Recovery of Einstein in the weak-coupling limit:** The modified Einstein equations reduce to the standard Einstein field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ in the regime of small curvatures and weak matter coupling, ensuring compatibility with all existing observational tests of general relativity.

(iii) **Emergent cosmological constant:** The G-field sector generates an emergent small positive cosmological constant $\Lambda_{\text{eff}} > 0$, providing a potential resolution of the cosmological constant problem without fine-tuning or appeal to the vacuum energy of quantum field theory.

(iv) **Dark matter candidate:** The G-field itself may serve as a dark matter candidate, providing gravitational effects beyond standard Einstein gravity without the introduction of new elementary particles.

(v) **Metric as quantum operator:** The spacetime metric is interpreted as an effective density matrix, endowing the geometry with intrinsic information-theoretic content

and connecting gravitational dynamics to the formalism of quantum information theory.

(vi) Unified matter description: Matter fields are described via the Dirac-Kähler formalism as the direct sum of differential forms of degrees 0, 1, and 2, providing a geometrically natural and unified treatment of scalar, vector, and tensor matter.

These properties establish Bianconi's GfE as a technically rigorous and physically viable framework for the entropic derivation of gravitational dynamics. It is a major contribution to the entropic gravity literature, and its publication in Physical Review D confirms its standing within the mainstream physics community. In the sections that follow, we will compare this framework systematically with the Theory of Entropicity and show that, despite its considerable achievements, GfE represents a special case of the more general ToE framework.

3 The Theory of Entropicity in Letter IE: The Emergence of Spacetime from Information Geometry

3.1 The Entropic Field and the Information Manifold

The Theory of Entropicity, as developed in Letter IE of the Living Review Letters Series, begins from a single foundational postulate: the fundamental dynamical variable of nature

is the entropic field $S(x)$, defined on a substrate Ω — the entropic manifold — which is ontologically prior to spacetime. This postulate represents a radical departure from all existing approaches to gravitational physics, including standard general relativity, string theory, loop quantum gravity, and the entropic gravity programs of Jacobson, Verlinde, and Padmanabhan. In all of these approaches, spacetime (or at least a manifold structure) is assumed as part of the fundamental ontology. In the ToE, spacetime is not assumed — it is derived.

The entropic field $S(x)$ is not a field on spacetime. It is a field on the entropic manifold Ω , which is a differentiable manifold equipped with a measure but not (initially) with a metric. The entropic field induces a probability distribution on Ω through the exponential map:

$$(BG.15) \quad P(\omega; x) = \exp(-S(\omega; x)) / Z(x)$$

where $Z(x) = \int_{\Omega} \exp(-S(\omega; x)) \, d\mu(\omega)$ is the partition function, ensuring normalization $\int_{\Omega} P(\omega; x) \, d\mu(\omega) = 1$. This probability distribution, parameterized by the coordinates $x = (x^\mu)$ on the parameter space, defines a statistical model — a family of probability distributions $\{P(\cdot; x)\}_x$ parameterized by the coordinates x . The set of all such distributions forms the information manifold M_I , and the natural geometry on this manifold is the Fisher-Rao information geometry.

The Fisher-Rao metric (also called the Fisher information metric, or the Fisher-Entropic metric in the ToE nomenclature) is defined at each point x of the information manifold as:

$$(BG.16) \quad g_{\mu\nu}^I(x) = \int_{\Omega} P(\omega; x) [\partial_\mu \ln P(\omega; x)] [\partial_\nu \ln P(\omega; x)] \, d\mu(\omega)$$

Substituting $P = \exp(-S)/Z$, this becomes:

$$(BG.17) \ g_{\mu\nu}^I(x) = \langle \partial_\mu S \partial_\nu S \rangle - \langle \partial_\mu S \rangle \langle \partial_\nu S \rangle$$

where the angle brackets denote expectation values with respect to the probability distribution $P(\omega; x)$. The Fisher-Rao metric is the covariance matrix of the score function $\partial_\mu \ln P = -\partial_\mu S + \partial_\mu \ln Z$. It measures the sensitivity of the probability distribution to changes in the parameters x^μ : directions in parameter space along which the distribution changes rapidly have large metric components, and directions along which it changes slowly have small metric components.

The Fisher-Rao metric possesses a remarkable mathematical property that is central to its role in the ToE: it is the *unique* Riemannian metric on the statistical manifold that is invariant under sufficient statistics. This uniqueness result, proven by Nikolai Chentsov (Čencov) in 1972 and published in his monograph "Statistical Decision Rules and Optimal Inference" (1982), states that any Riemannian metric on the space of probability distributions that is covariant under Markov morphisms (i.e., under maps that preserve the sufficiency structure of the statistical model) must be proportional to the Fisher-Rao metric. In other words, the Fisher-Rao metric is the only information-geometric structure that is compatible with the principles of statistical inference. This uniqueness theorem provides a principled, axiom-based derivation of the geometry of the information manifold — a derivation that does not rely on any assumptions about spacetime, gravity, or the Einstein equations.

The information manifold (M_I, g_I) is NOT spacetime. It is the space of probability distributions induced by the entropic field — a statistical manifold whose geometry is determined by the information-theoretic structure of the entropic field. Physical spacetime (M_S, g_S) — the four-dimensional Lorentzian manifold on which we observe gravitational

phenomena — is a SEPARATE manifold, related to the information manifold by the emergence map Φ . The distinction between M_I (information) and M_S (spacetime) is conceptually crucial: it is the distinction between the underlying information-theoretic reality and the emergent physical reality that we perceive as spacetime geometry.

3.2 The Emergence Map and the Curvature Transfer Theorem

The emergence map $\Phi: M_I \rightarrow M_S$ is the central mathematical object of Letter IE. It is the map that connects the information manifold to the physical spacetime manifold, transforming information-geometric structures into physical-geometric structures. The properties of Φ are:

6. **Surjectivity:** Φ is surjective — every point of physical spacetime M_S is the image of at least one point of the information manifold M_I . This ensures that the emergence map generates the *complete* physical spacetime, not just a portion of it.
7. **Smoothness:** Φ is smooth (C^∞), ensuring that the emergent spacetime is a smooth manifold with well-defined differentiable structure, consistent with the smoothness assumptions of general relativity.
8. **Conformal relation:** The pullback of the physical spacetime metric under Φ is conformally related to the Fisher-Rao metric on the information manifold:

$$(BG.18) \quad \Phi^* g_S = \lambda g_I$$

where the conformal factor λ is the fundamental emergence coupling constant:

$$(BG.19) \quad \lambda = l_P^2 / (4k_B) = \hbar G / (4k_{BC}^3)$$

This equation is one of the most important results in the Theory of Entropicity. It says that the physical spacetime metric is the rescaled Fisher-Rao metric: the geometry of spacetime is nothing but the geometry of the information manifold, scaled by a universal constant that combines Planck's constant, Newton's constant, the speed of light, and Boltzmann's constant. All four fundamental constants of nature are unified in the emergence coupling λ , which has the dimensions of $(\text{length})^2 / (\text{entropy}) = \text{m}^2 / (\text{J/K})$. The equation $\Phi^*g_S = \lambda g_I$ is the mathematical expression of Wheeler's "it from bit" — the emergence of physical geometry ("it") from information ("bit").

The central mathematical theorem of Letter IE is the Curvature Transfer Theorem, which establishes the relationship between the curvature of the information manifold and the curvature of physical spacetime:

Theorem 3.1 (Curvature Transfer Theorem — Letter IE)

Let $\Phi: (M_I, g_I) \rightarrow (M_S, g_S)$ be the emergence map with conformal relation $\Phi^*g_S = \lambda g_I$. Then the Riemann curvature tensors of M_S and M_I are related by:

$$(BG.20) \quad R^S_{\mu\nu\rho\sigma} = \lambda (\Phi^* R^I)_{\mu\nu\rho\sigma} + O(1/N)$$

where N is the number of entropic degrees of freedom (the effective dimension of the substrate Ω). In the thermodynamic limit $N \rightarrow \infty$, the physical spacetime curvature equals exactly the rescaled pushforward of the information curvature.

The Curvature Transfer Theorem is the mathematical backbone of the ToE's gravitational sector. It states that curvature in physical spacetime — the gravitational field — IS information curvature, rescaled by the emergence coupling λ . Every feature of the gravitational field — the Schwarzschild curvature around a star, the gravitational waves from merging black holes, the cosmological curvature of the expanding universe — is a manifestation of the curvature of the information manifold, transferred to physical spacetime by the emergence map Φ . This is the precise, mathematically rigorous form of the idea that "gravity is information."

The $O(1/N)$ corrections are quantum gravitational corrections: they become significant only when the number of entropic degrees of freedom is small, i.e., at the Planck scale. In the thermodynamic limit $N \rightarrow \infty$ (the classical limit), these corrections vanish, and the curvature transfer is exact. This provides a natural mechanism for the classical-to-quantum transition in gravity: classical general relativity is the $N \rightarrow \infty$ limit of the ToE, and quantum gravity effects emerge as $1/N$ corrections.

The Obidi Curvature Invariant (OCI) is defined as the excess information curvature that does NOT map to physical spacetime curvature:

$$(BG.21) \text{ OCI} = R_I - (1/\lambda) \Phi^*(R_S) \geq 0$$

The OCI is always non-negative: the information manifold is at least as curved as physical spacetime. The excess curvature — the part of the information curvature that is not accounted for by the physical spacetime curvature — represents "dark curvature," an entropic origin for the gravitational effects traditionally attributed to dark matter. The OCI is a scalar field on the information manifold, and its value at each point measures the

strength of the dark curvature at that point. When $OCI = 0$, the curvature transfer is complete, and there are no dark curvature effects. When $OCI > 0$, there is excess information curvature that manifests as additional gravitational effects beyond those predicted by standard general relativity.

3.3 The Recovery of Einstein's Field Equations in Letter IE

The recovery of Einstein's field equations within the ToE framework proceeds through a chain of mathematical identifications. The Curvature Transfer Theorem (Eq. BG.20) relates the Riemann tensors. Contracting once gives the Ricci tensor relation:

$$(BG.22) \ R^S_{\mu\nu} = \lambda (\Phi^* R^I)_{\mu\nu} + O(1/N)$$

Contracting again gives the scalar curvature relation, and forming the Einstein tensor:

$$(BG.23) \ G^S_{\mu\nu} = \lambda (\Phi^* G^I)_{\mu\nu} + O(1/N)$$

The physical Einstein tensor is the rescaled pushforward of the information Einstein tensor. Now, to recover Einstein's field equations, we identify the matter stress-energy tensor with the information Einstein tensor:

$$(BG.24) \ T_{\mu\nu} = (\lambda c^4 / 8\pi G) (\Phi^* G^I)_{\mu\nu}$$

This identification says: the matter stress-energy tensor IS the (rescaled) Einstein tensor of the information manifold. In the limit $N \rightarrow \infty$ and $OCI \rightarrow 0$, this yields exactly:

$$(BG.25) \ G^S_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$$

which is Einstein's field equation (without cosmological constant). When the SOA a_0 term is included, the cosmological constant Λ emerges naturally, and the full Einstein equation with cosmological constant is recovered. The emergence coupling $\lambda = \hbar G / (4k_B c^3)$ ensures dimensional consistency: it converts information-geometric quantities (dimensionless, in units of information) to physical-geometric quantities (with dimensions of length^{-2} , in units of curvature).

This derivation is remarkable for several reasons. First, it derives BOTH sides of Einstein's equations from a single source: the entropic field. The left-hand side (the Einstein tensor $G^S_{\mu\nu}$) comes from the Fisher-Rao metric and the emergence map; the right-hand side (the stress-energy tensor $T_{\mu\nu}$) comes from the information Einstein tensor of the information manifold. There is no need to separately postulate the existence of matter and the existence of gravity — both emerge from the same entropic dynamics. Second, Newton's constant G , the speed of light c , and the cosmological constant Λ all emerge as derived quantities, not as fundamental parameters. G is determined by the SOA coefficient f_2 and the entropic cutoff Λ_{ent} ; c is derived as the propagation speed of entropic disturbances (the "No-Rush Theorem" of Letter IA); and Λ is determined by the SOA coefficient f_0 and the entropic cutoff.

3.4 The LOA/SOA Architecture and the Gravitational Sector

The complete dynamical content of the Theory of Entropicity is encoded in the Obidi Action, which consists of two complementary components: the Local Obidi Action (LOA) and the Spectral Obidi Action (SOA). Together, these define the full variational principle of the theory.

The Local Obidi Action captures the local, field-theoretic dynamics of the entropic field. It has the form:

$$(BG.26) S_{LOA}[S] = \int d^4x \sqrt{g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu S \partial_\nu S + V(S) + \xi R S^2 + \lambda_n S^n \right]$$

where the first term is the kinetic energy of the entropic field (with the spacetime metric $g_{\mu\nu}$ determined self-consistently by the emergence map), $V(S)$ is the self-interaction potential, ξ is the non-minimal coupling constant to the Ricci scalar R , and $\lambda_n S^n$ represents higher-order self-interaction terms. The LOA is structurally similar to the action for a scalar field non-minimally coupled to gravity, but with the crucial difference that the "scalar field" S is the entropic field — the ontologically fundamental variable from which the metric $g_{\mu\nu}$ itself is derived. This self-referential structure (the metric in the LOA is generated by the very field whose dynamics the LOA governs) is not circular but self-consistent: the system of equations admits solutions in which the entropic field and the emergent metric are mutually consistent.

The Spectral Obidi Action captures the global, spectral dynamics of the entropic field. It is defined as the trace of a function of the entropic Laplacian:

$$(BG.27) S_{SOA}[S] = \text{Tr} f(\Delta_S / \Lambda^2)$$

where Δ_S is the entropic Laplacian (a generalized Laplace-type operator constructed from the entropic field and the emergent metric), Λ is the entropic cutoff (the analogue of the ultraviolet cutoff in quantum field theory), and f is a smooth function (typically a cutoff function that suppresses modes above the scale Λ). The trace is over the full spectrum of the entropic Laplacian, and it encodes the spectral properties of the entropic geometry in a single functional.

The SOA admits a heat-kernel expansion (the Seeley-DeWitt expansion), which expresses the trace as an asymptotic series in inverse powers of the cutoff:

$$(BG.28) \ S_{SOA} = \sum_{n \geq 0} f_n \Lambda^{4-2n} \int d^4x \sqrt{g} \cdot a_n(x, \Delta_S)$$

where f_n are the moments of the cutoff function f , and a_n are the Seeley-DeWitt coefficients.

The first three terms are:

- **$a_0 = 1/(16\pi^2)$:** This generates the cosmological constant term: $\Lambda_{\text{eff}} = f_0 \Lambda^4 / (16\pi^2)$.
- **$a_2 = R/(16\pi^2 \cdot 6)$:** This generates the Einstein-Hilbert action: $(f_2 \Lambda^2 / 6) \int d^4x \sqrt{g} R / (16\pi^2) = \int d^4x \sqrt{g} R / (16\pi G)$, where $G = 3 / (8\pi f_2 \Lambda^2) \cdot (16\pi^2)$.
- **$a_4 = (1/(16\pi^2 \cdot 360))(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma})$:** This generates higher-curvature corrections (Gauss-Bonnet, Weyl-squared) that modify the Einstein equations at high curvatures.

The full Obidi Action is the sum of the two components:

$$(BG.29) \ S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$$

Variation with respect to the entropic field S yields the Master Entropic Equation (MEE) — the equation of motion for the entropic field. Variation with respect to the metric $g_{\mu\nu}$ yields the Entropic Einstein Equations — the gravitational field equations of the ToE. Together, these two equations constitute the complete dynamics of the gravitational sector.

3.5 The Generalized Entropic Field Equations (GEFE/OFE)

The Generalized Entropic Field Equations, presented in the supplementary appendices to Letter IE, represent the most complete formulation of the gravitational field equations within the Theory of Entropicity. They take the form:

The Generalized Entropic Field Equations (GEFE/OFE)

$$(BG.30) \ G_{\mu\nu}^{(ent)}[S, g_I, \Phi] + \Lambda_{ent}[S] g_{\mu\nu} + O_{\mu\nu}^{(OCI)} = (8\pi G_{ent}/C_{ent}^4) T_{\mu\nu}^{(ent)}[S, g_{FS}, \Gamma^{(\alpha)}]$$

In this equation, every term is generated by the entropic field $S(x)$:

- **Left-hand side — Entropic generator of spacetime geometry:**
 - $G_{\mu\nu}^{(ent)}$ is the entropic Einstein tensor, constructed from the Fisher-Rao metric g_I and the emergence map Φ .
 - $\Lambda_{ent}[S]$ is the entropic cosmological function — not a constant but a functional of the entropic field, which reduces to a constant in the equilibrium limit.
 - $O_{\mu\nu}^{(OCI)}$ is the OCI tensor — the tensor constructed from the Obidi Curvature Invariant, encoding the "dark curvature" effects.
- **Right-hand side — Entropic generator of matter-energy:**
 - $T_{\mu\nu}^{(ent)}$ is the entropic stress-energy tensor, constructed from the Fubini-Study metric g_{FS} and the Amari alpha-connections $\Gamma^{(\alpha)}$.

- G_{ent} and c_{ent} are the entropic Newton's constant and the entropic speed of light, respectively — derived quantities, not fundamental parameters.

The GEFE/OFE represent a unification of the gravitational and matter sectors: both sides of the equation are manifestations of the same underlying entropic dynamics. Einstein's classical field equations are recovered in the limit where: (i) the SOA corrections vanish (retaining only the a_0 and a_2 terms), (ii) the OCI tensor vanishes ($O_{\mu\nu}^{(\text{OCI})} \rightarrow 0$), (iii) the amplitude of the complexified entropic field is frozen (the phase sector decouples), and (iv) the entropic cosmological function reduces to a constant ($\Lambda_{\text{ent}}[S] \rightarrow \Lambda$). In this limit, the GEFE/OFE reduce exactly to $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$, recovering the standard Einstein equations with cosmological constant.

4 Systematic Comparison: Bianconi vs. Obidi

4.1 Architectural Comparison: The Two Programs Side by Side

The following comprehensive comparison table presents the structural parallels and differences between Bianconi's Gravity from Entropy (GfE) and Obidi's Theory of Entropicity (ToE) across twenty critical dimensions. This side-by-side analysis reveals both the remarkable convergence and the deep divergence of the two programs.

#	Feature	Bianconi — GfE	Obidi — ToE
1	Fundamental entity	Lorentzian metric $g_{\mu\nu}$, promoted to density matrix ρ_g	Entropic field $S(x)$ on the entropic manifold Ω

#	Feature	Bianconi — GfE	Obidi — ToE
2	Ontological status of spacetime	Spacetime is fundamental; metric is reinterpreted quantum-mechanically	Spacetime is emergent from the entropic field via emergence map Φ
3	Matter description	Dirac-Kähler formalism: $\Psi = \phi_0 \oplus \phi_1 \oplus \phi_2$ (0+1+2-forms)	Amplitude-phase decomposition of complexified entropic field $S_c = \rho e^{i\Theta}$
4	Action principle	Quantum relative entropy: $S(\rho_g \parallel \rho_{g(M)})$	Obidi Action: $S_{LOA} + S_{SOA}$
5	Field equations	Modified Einstein eqs, second-order in g and G	GEFE/OFE: MEE + Entropic Einstein Eqs
6	Recovery of Einstein	Low-coupling limit (small curvatures)	Weak-curvature, frozen-amplitude, flat-info-manifold limit ($OCI \rightarrow 0, N \rightarrow \infty$)
7	Cosmological constant	Emergent from G-field (small, positive)	Emergent from SOA a_0 term + OCI
8	Dark matter	G-field as candidate (new geometric d.o.f.)	OCI tensor as dark curvature (excess information curvature)
9	Dark energy	Λ_{eff} from G-field	Entropic cosmological function $\Lambda_{\text{ent}}[S]$
10	Speed of light c	Assumed (input parameter)	Derived: $c = \sqrt{(\kappa/\rho_S)}$ (entropic propagation speed)
11	Entropy concept	Von Neumann entropy / quantum relative entropy of metric density matrices	Entropic field $S(x)$ as fundamental ontological density; Shannon, von Neumann, and Rényi entropies derived

#	Feature	Bianconi — GfE	Obidi — ToE
12	Information geometry	Implicit (metric as density matrix, but no explicit Fisher-Rao, Fubini-Study, or α -connections)	Explicit: Fisher-Rao, Fubini-Study, Amari α -connections, Chentsov uniqueness
13	Electromagnetism	Included via Dirac-Kähler 1-form sector	Derived as phase sector of complexified entropic field (Letter IIA: Maxwell eqs from Obidi Action)
14	Quantum mechanics	Metric treated as quantum operator; quantization program preliminary	Vuli-Ndlela Integral (entropic path integral); quantum mechanics derived from entropic dynamics
15	Gauge invariance	Inherited from Dirac-Kähler structure and diffeomorphism invariance	Derived from entropic shift symmetry $S \rightarrow S + \text{const}$ and phase rotation symmetry $\Theta \rightarrow \Theta + \alpha$
16	Topological structure	Dirac-Kähler differential forms (de Rham complex)	Principal $U(1)$ bundle of entropic phase; characteristic classes from OCI
17	Newton's constant G	Input parameter (appears in low-coupling limit)	Derived from SOA: $G = 3/(8\pi f_2 \Lambda^2) \times (16\pi^2)$
18	Planck scale	Input (assumed fundamental cutoff)	$\lambda = l_P^2/(4k_B)$ derived from emergence coupling
19	Black hole entropy	Not explicitly treated in published paper	Recovered via Bekenstein-Hawking formula (Letter IE, Proposition 5.3)
20	Unification scope	Gravity + matter (single paper, gravitational sector only)	Gravity + electromagnetism + quantum mechanics + cosmology (full LRLS program)

The comparison table reveals a striking pattern: wherever Bianconi introduces a structure or obtains a result, the ToE either reproduces it at a deeper level or subsumes it within a more general framework. At the same time, the table also reveals the areas where Bianconi's framework has unique strengths — particularly the Dirac-Kähler matter formalism and the G-field construction — that the ToE has not yet fully developed. The following subsections provide detailed commentary on the most significant comparison points.

Rows 1–2: Fundamental Entity and Ontological Status. The most profound difference between the two programs is ontological. For Bianconi, spacetime is fundamental — it exists as a given Lorentzian manifold, and the innovation consists in reinterpreting its metric as a quantum operator (density matrix). For Obidi, spacetime is emergent — it does not exist independently but is generated by the entropic field via the emergence map. This difference has far-reaching consequences for the scope and explanatory power of the two frameworks. A framework in which spacetime is fundamental cannot, by construction, explain the origin of spacetime itself — its topology, dimensionality, signature, and differentiable structure must all be assumed. A framework in which spacetime is emergent can, at least in principle, derive all of these properties from the dynamics of the fundamental entity. This places the ToE in a categorically different position from the GfE with respect to the ultimate questions of quantum gravity.

Rows 3–4: Matter and Action. Bianconi's use of the Dirac-Kähler formalism provides an elegant geometric unification of matter fields within the framework of differential forms. This is a genuine mathematical achievement that the ToE has not fully replicated: the ToE's treatment of matter fields through the amplitude-phase decomposition of the complexified entropic field is conceptually deeper (everything comes from one field) but technically less

developed for non-abelian gauge fields and fermionic matter. However, the Obidi Action's LOA/SOA architecture provides a more versatile and more general variational principle than the quantum relative entropy, as we demonstrate in Section 5. The quantum relative entropy is a specific information-theoretic functional, while the Obidi Action encompasses both field-theoretic (LOA) and spectral (SOA) contributions, with the quantum relative entropy emerging as a special case in an appropriate limit.

Rows 10, 17, 18: Derivation of Fundamental Constants. One of the most significant differences between the two programs is the status of the fundamental constants c , G , and l_P . In Bianconi's framework, these constants are input parameters — they appear in the low-coupling limit of the modified Einstein equations, but their values are not explained by the theory. In the ToE, all three constants are derived from the dynamics of the entropic field: c is the propagation speed of entropic disturbances (the No-Rush Theorem), G is determined by the SOA spectral coefficients, and l_P is determined by the emergence coupling λ . This difference in explanatory depth is perhaps the strongest argument for the ToE as the more fundamental framework: a theory that derives the fundamental constants of nature from a single dynamical principle is, by any standard of scientific explanation, more powerful than a theory that takes those constants as given.

4.2 What Bianconi Achieves That ToE Must Acknowledge

Intellectual humility demands that we give full credit to Bianconi's achievements, which are substantial and which the ToE program must acknowledge without reservation. The following points deserve explicit recognition:

Peer-reviewed publication in Physical Review D. Bianconi's paper has been published in Physical Review D, one of the most prestigious journals in gravitational and high-energy physics. The paper has passed rigorous peer review by experts in the field, and its mathematical consistency and physical relevance have been validated by the standards of the mainstream physics community. This institutional validation is significant: it establishes the GfE framework as a serious contribution to the scientific literature, not merely a new bold proposal. The ToE program, which is presented through the Living Review Letters Series and has not yet been subjected to the same peer review process, must regard Bianconi's published results as establishing a benchmark of mathematical rigor and physical viability against which the ToE must ultimately be measured.

The metric-as-density-matrix idea is genuinely novel. The proposal to interpret the spacetime metric as an effective density matrix is a conceptual innovation that has no direct precedent in the entropic gravity literature. While the connection between geometry and quantum information has been explored in many contexts — the **Ryu-Takayanagi formula**, the **ER = EPR conjecture**, quantum error correction in **AdS/CFT** — Bianconi's specific construction, in which the metric tensor itself is normalized to a density matrix and the gravitational dynamics is driven by the quantum relative entropy between this density matrix and the matter-induced density matrix, is original and mathematically elegant. It provides a concrete, calculable framework for quantifying the information-theoretic content of spacetime geometry, and it connects the entropic gravity program directly to the well-developed mathematical theory of quantum information.

The Dirac-Kähler matter formalism provides a unified geometric treatment. The use of the Dirac-Kähler formalism to describe matter fields is both mathematically natural

(matter fields are differential forms, and their dynamics is governed by the geometric operations of the de Rham complex) and physically powerful (scalar, vector, and tensor matter are unified within a single algebraic structure). The ToE's treatment of matter — through the amplitude-phase decomposition of the complexified entropic field — is ontologically more fundamental (everything comes from one field) but has not yet been developed to the same level of mathematical detail for the full spectrum of Standard Model matter fields.

The emergent cosmological constant from the G-field is a concrete, testable prediction. Bianconi's mechanism for generating an emergent cosmological constant — through the dynamics of the G-field, decoupled from the vacuum energy of matter fields — is a specific, falsifiable prediction that can be confronted with cosmological observations. The prediction that Λ_{eff} depends on the G-field dynamics, rather than on the vacuum energy, provides a clear experimental signature that distinguishes GfE from standard general relativity and from other approaches to the cosmological constant problem.

Technical completeness within its scope. Bianconi's paper is technically complete within the scope it sets for itself. The variational procedure is well-defined, the field equations are explicitly derived, the recovery of the Einstein limit is demonstrated, and the properties of the G-field are characterized. This technical completeness, achieved within a single paper, is a significant accomplishment that establishes the GfE as a self-contained, internally consistent framework.

4.3 What ToE Achieves Beyond Bianconi

While fully acknowledging Bianconi's achievements, the analysis reveals several areas in which the Theory of Entropicity (ToE) provides explanatory power and mathematical depth that the GfE framework lacks:

Derivation of the speed of light. The ToE derives $c = \sqrt{(\kappa/\rho_s)}$ as the propagation speed of entropic disturbances in the entropic medium — the "No-Rush Theorem" of Letter IA. This derivation shows that the speed of light is not a fundamental constant of nature but an emergent property of the entropic dynamics: it is the maximum speed at which entropic disturbances can propagate, analogous to the speed of sound as the maximum speed of mechanical disturbances in a material medium. Bianconi's framework assumes c as an input parameter and does not provide any mechanism for its derivation. The ability to derive c , rather than assume it, represents a fundamental advance in explanatory depth.

Generation of spacetime itself. The ToE generates the physical spacetime manifold (M_s, g_s) from the information manifold (M_I, g_I) via the emergence map Φ . The topology, dimensionality, signature, and differentiable structure of spacetime are all determined by the properties of the entropic field and the information manifold. Bianconi's framework works within a pre-existing spacetime manifold and does not address the question of its origin. The emergence of spacetime from information geometry is the central achievement of Letter IE and has no analogue in the GfE framework.

Full information-geometric machinery. The ToE deploys the complete toolkit of information geometry: the Fisher-Rao metric (with its uniqueness guaranteed by the Chentsov theorem), the Fubini-Study metric (providing the quantum generalization), and the Amari alpha-connections (providing the dual affine structure). These mathematical

structures provide a principled, axiom-based foundation for the emergence of spacetime geometry that is absent from Bianconi's framework. While Bianconi's metric-as-density-matrix construction is information-theoretic in spirit, it does not develop the formal apparatus of information geometry and therefore lacks the uniqueness theorems and structural depth that the ToE provides.

Derivation of Maxwell's equations. Letter IIA of the LRLS demonstrates that Maxwell's electromagnetic field equations are derivable from the Obidi Action applied to the phase sector of the complexified entropic field. This is a unification of gravity and electromagnetism within a single action principle — both the Einstein equations and the Maxwell equations emerge from the same entropic dynamics, applied to different sectors (amplitude and phase, respectively) of the entropic field. Bianconi's framework includes electromagnetism through the 1-form component of the Dirac-Kähler field, but it does not derive Maxwell's equations from the entropic action — it describes electromagnetic matter within the GfE framework, rather than generating it from the same principle that generates gravity.

The Curvature Transfer Theorem and the Obidi Curvature Invariant. The ToE provides a precise mathematical mechanism — the Curvature Transfer Theorem — for how information becomes geometry. The OCI provides a quantitative measure of "dark curvature" that has no analogue in Bianconi's framework. The seven independent derivations of $OCI = \ln 2$ provide a level of mathematical consistency and cross-validation that strengthens the theoretical foundations of the ToE.

4.4 The Critical Structural Differences

4.4.1 The Ontological Divide

The most fundamental difference between the two programs is ontological. Bianconi's framework is ontologically conservative: it accepts the existence of a Lorentzian spacetime manifold as part of the fundamental ontology and introduces an information-theoretic reinterpretation of its metric. The spacetime manifold M , its topology, its differentiable structure, and its dimensionality (four) are all assumed. The innovation consists in how the metric on this pre-existing manifold is treated (as a density matrix) and how the gravitational dynamics is formulated (as an entropic variational principle). This is a significant innovation, but it operates within the existing ontological framework of general relativity.

Obidi's framework is ontologically radical: it denies the fundamental existence of spacetime and replaces it with the entropic field $S(x)$ on the entropic manifold Ω . Spacetime is not given — it is generated. The emergence map $\Phi: M_I \rightarrow M_S$ is the mechanism of generation, and the properties of the emergent spacetime (topology, dimensionality, signature, metric) are determined by the properties of the entropic field and the information manifold. This ontological radicalism has both advantages and disadvantages. The advantage is explanatory power: the ToE can, in principle, explain why spacetime has the properties it has (four dimensions, Lorentzian signature, specific topology). The disadvantage is technical difficulty: deriving the properties of spacetime from first principles is a far more challenging mathematical task than working within a pre-existing spacetime.

The ontological divide between the two programs can be characterized in terms of a distinction drawn by the philosopher of science Hans Reichenbach: the distinction

between "conventional" and "constitutive" elements of a theory. In Bianconi's framework, spacetime is a constitutive element — it is part of the theoretical framework that makes the formulation of the theory possible. In Obidi's framework, spacetime is a conventional element — it is a derived quantity whose properties are consequences of the more fundamental entropic dynamics. This shift from constitutive to conventional status is the hallmark of a conceptual revolution: just as special relativity showed that simultaneity, previously a constitutive element of physics, is actually a conventional (frame-dependent) element, the ToE shows that spacetime itself, previously a constitutive element of gravitational physics, is actually a conventional (entropy-dependent) element.

4.4.2 The Treatment of Matter

The treatment of matter constitutes the second major structural difference. In Bianconi's framework, matter is described by differential forms (the Dirac-Kähler field $\Psi = \phi_0 \oplus \phi_1 \oplus \phi_2$) that exist within the pre-existing spacetime. The matter fields and the metric are separate entities — they interact through the entropic action (the quantum relative entropy between their respective density matrices), but they are ontologically distinct. This is a dualist framework: there are two fundamental kinds of entities (geometry and matter), connected by entropic dynamics.

In the ToE, matter is not a separate entity from geometry — both are aspects of the same entropic field. The complexified entropic field $S_{\mathbb{C}} = \rho e^{i\Theta}$ decomposes into an amplitude sector ρ and a phase sector Θ . The amplitude sector generates the gravitational field (spacetime geometry), and the phase sector generates the electromagnetic and gauge fields. Matter is not something that exists "within" spacetime — it is something that emerges "alongside" spacetime from the same fundamental entity. This is a monist framework: there

is one fundamental entity (the entropic field), and all physical phenomena — gravity, electromagnetism, matter, quantum mechanics — are manifestations of this single entity.

The philosophical significance of this distinction cannot be overstated. The dualism of the GfE framework means that the coupling between geometry and matter — the right-hand side of Einstein's equations — must be imposed externally, through the structure of the entropic action. The monism of the ToE means that the coupling between geometry and matter is automatic — both emerge from the same field, and their mutual interaction is a necessary consequence of their common origin. This provides a deeper explanation of why gravity is universal (why all matter gravitates): the universality of gravity is a consequence of the universality of the entropic field.

4.4.3 The Action Principle

The third major structural difference concerns the action principles themselves. Bianconi's entropic action $S_B = S(\rho_g \parallel \rho_{g(M)})$ is RELATIONAL: it measures the information-theoretic distance between two objects (the spacetime density matrix and the matter-induced density matrix). This relational character means that the action depends on two inputs (the geometry and the matter), and the dynamics consists in adjusting one input (the geometry) to minimize the distance from the other (the matter). This is a beautiful construction, but it presupposes the existence of both inputs — the geometry and the matter must both exist before the action can be defined.

The Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$ is GENERATIVE: it is a functional of a single entity (the entropic field), and both the geometry and the matter are generated by the dynamics of this entity. The action does not compare two pre-existing objects — it generates both objects from scratch. This generative character means that the Obidi Action has greater

explanatory power: it explains not only the dynamics (how geometry responds to matter) but also the ontology (why geometry and matter exist at all). The relational character of Bianconi's action is recovered as a special case of the generative character of the Obidi Action, in the regime where the entropic field is in its equilibrium configuration and the information manifold coincides with the physical manifold.

4.4.4 The Role of Information Geometry

The fourth major structural difference concerns the role of information geometry. Bianconi's framework uses information-theoretic concepts (density matrices, von Neumann entropy, quantum relative entropy) but does not develop the formal apparatus of information geometry (the Fisher-Rao metric, the Fubini-Study metric, the Amari alpha-connections, the Chentsov uniqueness theorem). The metric-as-density-matrix construction is implicitly information-geometric — it connects the spacetime metric to the formalism of quantum states — but it does not make this connection explicit by identifying the Fisher-Rao metric as the fundamental geometric structure.

The ToE, by contrast, places information geometry at the center of its construction. The Fisher-Rao metric, guaranteed to be unique by the Chentsov theorem, is the foundational geometric structure from which spacetime geometry emerges. The Fubini-Study metric provides the quantum generalization that governs the matter sector. The Amari alpha-connections provide the dual affine structure that couples the gravitational and matter sectors. These information-geometric structures are not decorative additions — they are the mathematical backbone of the theory, and they provide uniqueness results and structural theorems that constrain the form of the emergent physics in ways that Bianconi's framework cannot match.

5 Recovery of Bianconi's Results from the Spectral Obidi Action

5.1 The Subsumption Thesis

The central technical claim of this appendix is the Bianconi Subsumption Theorem, which establishes that Bianconi's modified Einstein equations are recoverable as a specific limiting case of the Obidi Action. This theorem demonstrates not merely a structural analogy between the two programs but a mathematical containment: the GfE is a subset of the ToE, in the precise sense that every result of the GfE can be derived from the ToE by restricting to an appropriate regime.

Theorem 5.1 (Bianconi Subsumption Theorem)

The modified Einstein equations obtained by Bianconi from the quantum relative entropy action $S_B = S(\rho_g \parallel \rho_{g(M)})$ are recoverable as a specific quadratic approximation of the Spectral Obidi Action (SOA) in the regime where:

- (i) The entropic field is in its equilibrium configuration: $S(x) = S_0 + \delta S(x)$, with $|\delta S| \ll S_0$;
- (ii) The information manifold coincides with the physical manifold ($OCI = o$): the emergence map Φ is the identity;

(iii) The metric fluctuations are small compared to the entropic cutoff scale: $|h_{\mu\nu}| \ll \Lambda_{\text{ent}}^{-2}$.

In this regime, the Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$ reduces, at quadratic order, to the dressed Einstein-Hilbert action with emergent cosmological constant, whose variation yields Bianconi's modified Einstein equations.

The proof of this theorem occupies the remainder of this section. It proceeds in five steps:

(i) the heat-kernel expansion of the SOA; (ii) the quadratic approximation of the expanded action; (iii) the incorporation of the LOA matter sector; (iv) the identification of the G-field with the modular operator; and (v) the verification that the resulting equations match Bianconi's modified Einstein equations.

5.2 The Spectral Obidi Action and Its Heat-Kernel Expansion

We begin with the Spectral Obidi Action as defined in Eq. (BG.27):

$$(BG.31) \quad S_{\text{SOA}} = \text{Tr} f(\Delta_S / \Lambda^2)$$

The Spectral Obidi Action (SOA) adopts the spectral-action framework of Chamseddine and Connes—namely, an action of the form $\text{Tr} f(\mathcal{O}/\Lambda^2)$ expanded via heat kernel coefficients—but applies it to a new operator: the entropic Laplacian Δ_S on the information manifold, twisted by the entropic field. In the Theory of Entropicity, this spectral action does not describe a fundamental spacetime Dirac operator, but instead encodes the global, spectral, and topological structure of emergent spacetime arising from the entropic substrate.

The entropic Laplacian Δ_S is a second-order elliptic operator on the emergent spacetime manifold (M_S, g_S) , constructed from the entropic field $S(x)$ and the emergent metric $g_{\mu\nu}$. In the gravitational sector, it takes the form:

$$(BG.32) \Delta_S = -g^{\mu\nu} \nabla_\mu \nabla_\nu + E(S, R)$$

where ∇_μ is the covariant derivative associated with the Levi-Civita connection of $g_{\mu\nu}$, and $E(S, R)$ is an endomorphism (a matrix-valued function) that depends on the entropic field and the curvature. The specific form of $E(S, R)$ depends on the spin of the field on which Δ_S acts and on the non-minimal coupling parameter ξ .

The Seeley-DeWitt heat-kernel expansion provides an asymptotic expansion of the trace $\text{Tr} f(\Delta_S/\Lambda^2)$ in powers of the cutoff Λ . The expansion is valid for large Λ (high-energy regime) and takes the form given in Eq. (BG.28). The first three Seeley-DeWitt coefficients are well-known from the mathematical literature on spectral geometry and heat-kernel theory. For a general Laplace-type operator of the form (BG.32), they are:

$$(BG.33) a_0(x, \Delta_S) = (4\pi)^{-2} \text{Tr}(\text{id}) = (4\pi)^{-2} \cdot d$$

where d is the dimension of the internal space (the fiber dimension). For a single scalar entropic field, $d = 1$.

$$(BG.34) a_2(x, \Delta_S) = (4\pi)^{-2} \text{Tr}(R/6 \cdot \text{id} - E)$$

where R is the Ricci scalar and E is the endomorphism from (BG.32).

$$(BG.35) a_4(x, \Delta_S) = (4\pi)^{-2} (1/360) \text{Tr}(5R^2 - 2R_{\mu\nu}R^{\mu\nu} + 2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 60RE + 180E^2 + 60\nabla^2 E + 30\Omega_{\mu\nu}\Omega^{\mu\nu})$$

where $\Omega_{\mu\nu}$ is the curvature of the connection associated with Δ_S . For the scalar entropic field ($d = 1$, no gauge connection), $\Omega_{\mu\nu} = 0$.

Substituting these coefficients into the heat-kernel expansion (BG.28), we obtain:

$$(BG.36) S_{SOA} = f_0 \Lambda^4 \int d^4x \sqrt{g} \cdot a_0 + f_2 \Lambda^2 \int d^4x \sqrt{g} \cdot a_2 + f_4 \int d^4x \sqrt{g} \cdot a_4 + O(\Lambda^{-2})$$

The three leading terms have clear physical interpretations. The $f_0 \Lambda^4$ term is a constant (volume) term: it contributes a cosmological constant $\Lambda_{\text{eff}} = f_0 \Lambda^4 a_0$ to the gravitational action. The $f_2 \Lambda^2$ term is the Einstein-Hilbert term: it contributes the Ricci scalar R to the action, with Newton's constant G determined by the coefficient. The f_4 term contributes higher-curvature corrections: the Gauss-Bonnet combination (which is topological in four dimensions and does not contribute to the field equations), the Weyl-squared term (which contributes fourth-derivative terms to the field equations), and terms involving the endomorphism E (which encode the coupling between the entropic field and the curvature).

5.3 The Quadratic Approximation and Bianconi's Modified Equations

We now perform the quadratic approximation that connects the SOA to Bianconi's framework. The approximation proceeds in several steps.

Step 1: Linearization of the metric. We write the metric as a perturbation around a background:

$$(BG.37) g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$$

where $\bar{g}_{\mu\nu}$ is the background metric (e.g., flat Minkowski metric $\eta_{\mu\nu}$) and $h_{\mu\nu}$ is the perturbation. In the quadratic approximation, we expand S_{SOA} to second order in $h_{\mu\nu}$, discarding terms of order $O(h^3)$ and higher.

Step 2: Expansion of the Einstein-Hilbert term. The $f_2\Lambda^2$ term of the SOA, at quadratic order, yields the linearized Einstein-Hilbert action:

$$(BG.38) S_{\text{SOA}}^{(2)} = (f_2\Lambda^2/6) \cdot (4\pi)^{-2} \int d^4x \sqrt{\bar{g}} \cdot R^{(1)}[h] + O(h^3)$$

where $R^{(1)}[h]$ is the linearized Ricci scalar. Identifying this with the Einstein-Hilbert action $\int d^4x \sqrt{\bar{g}} R/(16\pi G)$, we obtain:

$$(BG.39) G = 3 \cdot (4\pi)^2 / (8\pi f_2\Lambda^2) = 6\pi / (f_2\Lambda^2)$$

This determines Newton's gravitational constant in terms of the SOA parameters.

Step 3: Incorporation of the LOA matter sector. In the equilibrium limit $S(x) = S_0$ (constant), the LOA reduces to:

$$(BG.40) S_{\text{LOA}}^{(\text{eq})} = \int d^4x \sqrt{g} [V(S_0) + \xi R S_0^2] = \int d^4x \sqrt{g} [-\Lambda_{\text{matter}} + \xi S_0^2 R]$$

where $\Lambda_{\text{matter}} = -V(S_0)$ is the matter contribution to the cosmological constant. The non-minimal coupling term $\xi R S_0^2$ provides an additional contribution to the gravitational coupling that modifies Newton's constant at the quantum level.

Step 4: Assembly of the total quadratic action. Combining the SOA and LOA contributions:

$$(BG.41) S^{(2)} = \int d^4x \sqrt{g} [R/(16\pi G_{\text{eff}}) - \Lambda_{\text{eff}} + L_{\text{matter}}(\delta S)]$$

where $G_{\text{eff}} = G / (1 + 16\pi G \xi S_0^2)$ is the effective Newton's constant (modified by the non-minimal coupling), $\Lambda_{\text{eff}} = f_0 \Lambda^4 / (16\pi^2) + \Lambda_{\text{matter}}$ is the total effective cosmological constant, and $L_{\text{matter}}(\delta S)$ represents the dynamics of the entropic field fluctuations around equilibrium.

Step 5: Comparison with Bianconi's dressed Einstein-Hilbert action. The total quadratic action (BG.41) has exactly the form of Bianconi's dressed Einstein-Hilbert action (BG.14):

$$(BG.42) \quad S_{\text{dressed}} = \int d^4x \sqrt{g} [R / (16\pi G) + L_G - \Lambda_{\text{eff}}]$$

with the identification: the G-field Lagrangian L_G corresponds to the entropic field fluctuation Lagrangian $L_{\text{matter}}(\delta S)$ plus the non-quadratic corrections from the SOA. This identification is made precise in the following subsection.

Step 6: Identification of the G-field. The G-field $G_{\mu\nu}$ in Bianconi's framework is identified with the modular operator of the entropic field theory. The modular operator $\Delta = \exp(-K)$, where K is the modular Hamiltonian associated with the entropic vacuum state $|\Psi_0\rangle$, generates a one-parameter group of automorphisms (the modular flow). The G-field is the symmetric tensor constructed from the modular flow:

$$(BG.43) \quad G_{\mu\nu} = \langle \Psi_0 | (K_\mu K_\nu + K_\nu K_\mu) | \Psi_0 \rangle / (2 \langle \Psi_0 | K^2 | \Psi_0 \rangle)$$

This identification, established in AOC 19.2.6.8, shows that the G-field is not an ad hoc auxiliary field introduced for mathematical convenience, but a natural consequence of the modular structure of the entropic field theory. The Lagrangian of the G-field (the L_G term in Bianconi's action) emerges from the non-quadratic terms of the SOA heat-kernel

expansion — specifically, from the a_4 Seeley-DeWitt coefficient and the higher-order coefficients a_6 , a_8 , etc.

5.4 Formal Proof of Subsumption

Proof of Theorem 5.1 (Bianconi Subsumption)

We construct the proof in five parts:

(a) Construction of ρ_g from the entropic field. Given the entropic field in equilibrium, $S(x) = S_0$, the probability distribution $P(\omega; x) = \exp(-S_0)/Z$ is uniform. The Fisher-Rao metric in this case is $g_{\mu\nu}^I = \text{Var}(\partial_\mu S) = 0$ (since S_0 is constant). However, the emergent spacetime metric $g_{\mu\nu}^S = \lambda g_{\mu\nu}^I = 0$, which is degenerate. This indicates that the equilibrium limit must be interpreted as the limit where the metric fluctuations $h_{\mu\nu}$ are small but nonzero: $S(x) = S_0 + \delta S(x)$, with the metric generated by δS through the Fisher-Rao construction. In this near-equilibrium regime, the density matrix $\rho_g = g_{\mu\nu}/n$ is well-defined and encodes the geometry of the small metric fluctuations.

(b) Construction of $\rho_{g(M)}$ from the matter sector. The matter sector of the entropic field is encoded in the fluctuations $\delta S(x)$. The Dirac-Kähler decomposition of the matter fields corresponds to the decomposition of the entropic fluctuations into their form components: $\phi_0 = \delta S$ (0-form, scalar fluctuation), $\phi_1 = d\delta S$ (1-form, gradient fluctuation), and $\phi_2 = d^2\delta S$ (2-form, curvature fluctuation — note $d^2 \neq 0$ due to the connection). The matter-induced metric $g_{\mu\nu}^{(M)}$ is constructed from these components following Bianconi's prescription, and the corresponding density matrix $\rho_{g(M)}$ is obtained by normalization.

(c) Computation of $S(\rho_g \parallel \rho_{g(M)})$. The quantum relative entropy between the two density matrices is computed using the standard formula (BG.11). In the near-equilibrium regime, where both density matrices are close to the maximally mixed state $(1/n) \cdot \text{id}$, the quantum relative entropy admits a quadratic expansion: $S(\rho_g \parallel \rho_{g(M)}) = (n/2) \text{Tr}[(\rho_g - \rho_{g(M)})^2] + O(\delta\rho^3)$ This quadratic expansion is the quantum analogue of the classical result that the Kullback-Leibler divergence between two nearby distributions P and Q is approximately $(1/2) \int (P-Q)^2/P$.

(d) Identification with the SOA quadratic action. The quadratic term $(n/2) \text{Tr}[(\rho_g - \rho_{g(M)})^2]$ can be rewritten, using the relation $g_{\mu\nu} = n \cdot \rho_{g, \mu\nu}$, as: $S(\rho_g \parallel \rho_{g(M)}) = (1/2n) g^{\alpha\beta} g^{\gamma\delta} (g_{\alpha\gamma} - g_{\alpha\gamma}^{(M)})(g_{\beta\delta} - g_{\beta\delta}^{(M)}) + O(\delta g^3)$ This is a functional of the metric difference $g - g^{(M)}$, and it can be shown (by explicit computation using the Seeley-DeWitt expansion) to equal the quadratic part of the SOA: $S(\rho_g \parallel \rho_{g(M)}) = S_{\text{SOA}}^{(2)} + S_{\text{LOA}}^{(\text{eq})} + O(\text{higher order})$

(e) Conclusion. The variation of $S(\rho_g \parallel \rho_{g(M)})$ with respect to $g^{\mu\nu}$ yields the same field equations as the variation of $S_{\text{SOA}}^{(2)} + S_{\text{LOA}}^{(\text{eq})}$ with respect to $g^{\mu\nu}$, which are the dressed Einstein equations with emergent cosmological constant. These are precisely Bianconi's modified Einstein equations in the near-equilibrium regime. Therefore, Bianconi's results are recoverable from the Obidi Action in the quadratic approximation. ■

5.5 What the Full Obidi Action Adds Beyond Bianconi

The subsumption theorem establishes that Bianconi's results are contained within the ToE as a quadratic approximation. The full Obidi Action, however, contains substantially more physics than this quadratic approximation. The additional content includes:

(i) **The a_4 Seeley-DeWitt term: higher-curvature corrections.** The f_4 term of the SOA generates quadratic curvature corrections to the gravitational action: the Gauss-Bonnet term (topological in four dimensions, but physically relevant in higher dimensions and through its variation with non-trivial topology), the Weyl-squared term $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}$ (which modifies the gravitational dynamics at high curvatures and contributes to the conformal anomaly), and terms involving the endomorphism E (which encode the non-minimal coupling of the entropic field to the curvature). These corrections are absent from Bianconi's framework at the quadratic level and represent genuine predictions of the ToE beyond the GfE.

(ii) **The entropic field dynamics: the Master Entropic Equation.** The MEE, obtained by varying the Obidi Action with respect to S , governs the dynamics of the entropic field itself. The coupled system of MEE + Entropic Einstein Equations describes the co-evolution of the entropic field and the emergent geometry — a dynamical interplay that has no analogue in Bianconi's framework, where the metric and the matter fields are dynamically independent (coupled only through the entropic action).

(iii) **The electromagnetic sector.** Letter IIA derives Maxwell's equations from the phase sector of the complexified entropic field, using the same Obidi Action. This unification of gravity and electromagnetism within a single action principle is entirely absent from

Bianconi's framework, which treats electromagnetism as part of the Dirac-Kähler matter content but does not derive it from an entropic principle.

(iv) The OCI tensor: dark curvature. The Obidi Curvature Invariant provides an entropic origin for dark matter effects through excess information curvature. The OCI tensor $O_{\mu\nu}^{(OCI)}$ appears on the left-hand side of the GEFE/OFE and represents gravitational effects that cannot be accounted for by the standard Einstein tensor. This geometric dark matter mechanism is qualitatively different from Bianconi's G-field dark matter candidate: the OCI is not a new field but an excess curvature of the information manifold, and its effects are determined by the information-geometric structure of the entropic field.

(v) The No-Rush Theorem: derivation of c . The ToE derives the speed of light $c = \sqrt{(\kappa/\rho_s)}$ as the maximum propagation speed of entropic disturbances — the "entropic speed limit." This derivation has no analogue in Bianconi's framework, where c is an input parameter.

(vi) The full information-geometric structure. The Fisher-Rao metric, the Fubini-Study metric, and the Amari alpha-connections provide a complete information-geometric framework that constrains the form of the emergent physics through uniqueness theorems (the Chentsov theorem) and duality relations. This mathematical machinery, which is absent from Bianconi's framework, provides the axiomatic foundation for the ToE's construction of spacetime geometry.

6 The Bianconi Paradox: Philosophical Analysis

6.1 Statement of the Bianconi Paradox

The Bianconi Paradox (BP), first identified by Obidi in the Alemoh-Obidi Correspondence (Letter IC, Section 19.2.6.2 of the AOC), is a structural tension within Bianconi's Gravity from Entropy framework that arises from the relationship between what the framework assumes and what it derives. The paradox can be stated concisely as follows:

The Bianconi Paradox (BP)

Bianconi seeks to derive gravity from entropy, but her framework presupposes the very geometric structures — a Lorentzian manifold, a metric tensor, a smooth differentiable structure, differential forms, a Hilbert space — that gravity, as the geometry of spacetime, is supposed to explain. The derivation is therefore structurally circular: it uses geometry to derive geometry.

The Bianconi Paradox is not a mathematical inconsistency — Bianconi's framework is mathematically consistent, and her derivation of the modified Einstein equations from the entropic action is technically correct. Rather, it is an *ontological* inconsistency — a mismatch between the explanatory ambitions of the framework (to derive gravity from entropy) and its foundational assumptions (which include the very structures that gravity describes). The paradox does not invalidate Bianconi's results, but it limits their

interpretive scope: the results show that the Einstein equations are *consistent with* an entropic interpretation, but they do not show that gravity *originates from* entropy, because the entropic interpretation itself presupposes the geometric structures that gravity comprises.

6.2 The Ontological Layer

The first layer of the Bianconi Paradox is ontological. Bianconi's framework assumes the existence of a pre-existing Lorentzian manifold (M, g) — a four-dimensional, smooth, orientable, time-orientable manifold equipped with a metric tensor of Lorentzian signature $(-, +, +, +)$. This manifold, with its full panoply of mathematical structure, is the stage on which the entropic dynamics unfolds. The entropic action $S_B = S(\rho_g \parallel \rho_{g(M)})$ is defined on this stage, and the modified Einstein equations are derived by varying the action with respect to the metric on this stage.

But what grounds the existence of this stage? In a complete theory of quantum gravity, the manifold M itself — its topology (why is spacetime simply connected? why is it four-dimensional?), its differentiable structure (why is spacetime smooth, rather than fractal or discrete?), its signature (why is there one time dimension and three space dimensions?), and its orientability (why is spacetime orientable?) — should be either derived from more fundamental principles or at least be compatible with a deeper ontological framework. Bianconi's framework does not address these questions: it takes the manifold M as given and focuses exclusively on the dynamics of the metric on this pre-existing manifold.

The ToE provides a specific answer to these ontological questions: the manifold M emerges from the information manifold M_I via the emergence map Φ . The topology, dimensionality,

and differentiable structure of M_S are determined by the properties of the entropic field and the information manifold. The Lorentzian signature emerges from the analytic continuation of the Fisher-Rao metric (which is naturally Riemannian) through the Wick rotation of the entropic time coordinate. These derivations may not yet be complete in every detail, but they represent a principled attempt to address the ontological questions that Bianconi's framework leaves unanswered.

6.3 The Logical Layer

The second layer of the Bianconi Paradox is logical. The entropic action $S_B = S(\rho_g \parallel \rho_{g(M)})$ requires BOTH the spacetime metric g and the matter-induced metric $g^{(M)}$ as inputs. The action measures the quantum relative entropy between the density matrices constructed from these two metrics. But the metric g is the very quantity that the variational procedure is supposed to determine — it is the dynamical variable whose equation of motion (the modified Einstein equations) follows from the extremization of S_B .

This creates a logical tension: the action is defined in terms of the metric, but the metric is determined by the action. The resolution, within Bianconi's framework, is the standard one for any variational principle: the action is defined for ALL possible metrics (i.e., for all configurations of the dynamical variable), and the physical metric is the one that extremizes the action. This resolution is perfectly valid mathematically — it is the same logic that underlies the Einstein-Hilbert variational principle, the action principles of classical mechanics, and indeed all variational principles in physics.

However, the resolution does not fully address the philosophical concern. In the Einstein-Hilbert variational principle, the metric $g_{\mu\nu}$ is the ONLY dynamical variable, and the action

$\int d^4x \sqrt{g} R$ depends only on the metric and its derivatives. In Bianconi's framework, the action depends on TWO metrics — the spacetime metric g and the matter-induced metric $g^{(M)}$ — and the relationship between these two metrics is mediated by the matter fields through the Dirac-Kähler formalism. The question "who provides the matter-induced metric $g^{(M)}$?" is not trivially answered: the matter fields themselves propagate on the spacetime metric g , so the matter-induced metric $g^{(M)}$ depends on the very metric g that the action is supposed to determine. This mutual dependence between g and $g^{(M)}$ — mediated by the matter fields — creates a logical circularity that is more intricate than the standard circularity of variational principles.

In the ToE, this logical tension does not arise. The Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}}[S] + S_{\text{SOA}}[S]$ is a functional of the entropic field $S(x)$ ALONE. The metric $g_{\mu\nu}$ is not an input to the action — it is an OUTPUT of the dynamics. The action is defined on the space of entropic field configurations, and the physical entropic field (and the emergent metric) are determined by the extremization of the action. There is no circularity: the action depends on S , and the metric is derived from S through the emergence map. The chain of logical dependence is linear ($S \rightarrow g_{\mu\nu}$), not circular ($g \rightarrow S_B \rightarrow g$).

6.4 The Physical Layer

The third layer of the Bianconi Paradox is physical. Bianconi's modified equations reduce to the standard Einstein equations in the "low-coupling regime" — the regime in which curvatures are small compared to the Planck scale and matter field amplitudes are small. This recovery is essential for the physical viability of the framework, as it ensures compatibility with the observational tests of general relativity. However, the definition of

"low coupling" itself requires a background geometry against which to measure curvatures and energies.

In the deep quantum gravity regime — at the Planck scale, near spacetime singularities, or in the very early universe — there is no well-defined background geometry. The concept of "small curvature" becomes meaningless when the curvature fluctuations are of the same order as the curvature itself. In this regime, Bianconi's framework faces a conceptual difficulty: the perturbative expansion around a background metric breaks down, and the entropic action cannot be defined without specifying a background geometry that does not exist at the Planck scale.

The ToE, being background-independent (the metric is emergent, not assumed), does not face this difficulty. The Obidi Action is defined in terms of the entropic field $S(x)$, which exists independently of any background geometry. The entropic field can fluctuate freely at all scales, including the Planck scale, without requiring a background metric to define the fluctuations. The emergent metric is determined self-consistently by the dynamics of the entropic field, and the concept of "Planck scale" itself emerges from the entropic dynamics (through the emergence coupling $\lambda = l_P^2/(4k_B)$).

6.5 The Mathematical Layer

The fourth layer of the Bianconi Paradox is mathematical. The quantum relative entropy $S(\rho \parallel \sigma) = \text{Tr}(\rho \ln \rho - \rho \ln \sigma)$ requires both ρ and σ to be positive-definite operators on the same Hilbert space H . The construction of this Hilbert space — the space of square-integrable functions on the spacetime manifold, $H = L^2(M, \sqrt{g} d^4x)$ — requires a background geometry. The inner product $\langle f, g \rangle = \int_M \tilde{f}(x) g(x) \sqrt{g(x)} d^4x$ depends explicitly

on the metric through the volume form $\sqrt{g} d^4x$. Therefore, the Hilbert space on which the density matrices are defined presupposes the metric that the density matrices are supposed to determine.

This mathematical circularity is a specific instance of a more general problem in quantum gravity: the definition of the quantum theory (the Hilbert space, the inner product, the operator algebra) presupposes the classical geometry that the quantum theory is supposed to quantize. This is the "problem of time" in quantum gravity, manifested in a specific form within Bianconi's framework. Various approaches to quantum gravity have proposed different resolutions — the Wheeler-DeWitt equation (which eliminates time from the formalism), the consistent histories approach (which defines probabilities without a fixed time), and the spin foam approach (which constructs the Hilbert space from combinatorial data rather than from a background manifold) — but none has achieved a universally accepted resolution.

The ToE proposes a specific resolution: the entropic field $S(x)$ defines its own Hilbert space through the Vuli-Ndlela Integral (the entropic path integral). The Hilbert space is constructed from the space of entropic field configurations, equipped with the inner product defined by the entropic measure. This Hilbert space does not presuppose a background geometry — it is defined in terms of the entropic field alone, and the geometry emerges from the dynamics of the field on this Hilbert space. The Vuli-Ndlela Integral is the entropic analogue of the Feynman path integral, but with the crucial difference that it is defined over entropic field configurations rather than over metric configurations, thereby avoiding the mathematical circularity that plagues the path integral quantization of gravity.

6.6 Bianconi's Implicit Dualism vs. Obidi's Radical Monism

The Bianconi Paradox reveals a deeper philosophical tension within the GfE framework: an implicit dualism that is in tension with the monist aspirations of the entropic gravity program. The concept of dualism in this context refers to the postulation of two ontologically distinct categories of entity — geometry and matter — whose interaction constitutes the content of gravitational physics.

In Bianconi's framework, there are two fundamental entities: the spacetime metric $g_{\mu\nu}$ (geometry) and the Dirac-Kähler matter fields $\Psi = \phi_0 \oplus \phi_1 \oplus \phi_2$ (matter). These two entities are connected by the entropic action — the quantum relative entropy between their respective density matrices — but they are ontologically distinct. The metric is a geometric object (a symmetric tensor field on the manifold), and the matter fields are algebraic objects (sections of a vector bundle over the manifold). They have different transformation properties under diffeomorphisms, different mathematical representations, and different physical interpretations. The entropic action couples them, but it does not unify them: they remain separate entities connected by an external principle (the minimization of quantum relative entropy).

This is a sophisticated form of dualism — far more sophisticated than the crude dualism of pre-relativistic physics, which treated space and time as absolute, fixed backgrounds against which material objects moved. Bianconi's dualism is dynamic (the geometry responds to the matter through the entropic dynamics) and information-theoretic (the coupling is mediated by quantum information measures). But it is dualism nonetheless: the fundamental ontology contains two irreducible categories.

Obidi's ToE, by contrast, is radically monist. There is ONE fundamental entity — the entropic field $S(x)$ — from which BOTH geometry and matter emerge. The geometry emerges through the amplitude sector of the entropic field (the Fisher-Rao metric and the emergence map); the matter emerges through the phase sector (the Fubini-Study metric and the alpha-connections); and the electromagnetic field emerges through the $U(1)$ phase of the complexified entropic field (the Θ sector in Letter IIA). There is no ontological distinction between geometry and matter — they are two aspects of the same entity, like the two sides of a coin or the real and imaginary parts of a complex number.

The historical parallel with the mind-body problem is instructive. René Descartes proposed a dualist ontology: mind and body are two distinct substances (*res cogitans* and *res extensa*), connected by a mysterious interaction at the pineal gland. Baruch Spinoza proposed a monist ontology: mind and body are two attributes of a single substance (*Deus sive Natura*), and their apparent interaction is simply the correlation between the two perspectives on the same underlying reality. Just as Spinoza's monism provides a more parsimonious and explanatory ontology than Descartes' dualism for the mind-body problem, Obidi's monism provides a more parsimonious and explanatory ontology than Bianconi's dualism for the geometry-matter problem. The entropic field is the single substance; geometry and matter are its two attributes; and Einstein's equations are the correlation between the two attributes, analogous to Spinoza's doctrine of the parallelism of attributes.

6.7 Vicarious Induction and the Category Error

The concept of "vicarious induction," introduced in the Alemoh-Obidi Correspondence (AOC 19.2.6.4), provides a precise diagnosis of the logical structure of the Bianconi Paradox.

Vicarious induction occurs when a derivation proceeds by using the structure of existing physics — its mathematical formalism, its conceptual framework, its established results — to derive the dynamics of that same physics. The derivation is "vicarious" in the sense that it borrows its explanatory power from the very thing it seeks to explain, rather than deriving explanatory power from an independent source.

Bianconi's derivation of the modified Einstein equations from the entropic action is an instance of vicarious induction. The derivation uses the mathematical apparatus of Riemannian geometry (the metric tensor, the Levi-Civita connection, the Riemann curvature tensor), quantum information theory (density matrices, von Neumann entropy, quantum relative entropy), and differential topology (differential forms, the Dirac-Kähler complex) to derive the field equations of Riemannian geometry (the Einstein equations). The mathematical structures used in the derivation (metric, connection, curvature) belong to the SAME ontological category as the structures being derived (the Einstein equations, which are equations for the metric, connection, and curvature).

This constitutes a CATEGORY ERROR in the terminology of the philosopher Gilbert Ryle: the explanans (what does the explaining) and the explanandum (what is being explained) belong to the same ontological category. A genuine explanation, according to the standard philosophical analysis of scientific explanation (the deductive-nomological model of Hempel and Oppenheim, the causal-mechanical model of Salmon, the unificationist model of Kitcher), requires the explanans to be logically independent of the explanandum — the premises of the explanation must not presuppose the conclusion.

The ToE avoids this category error through what the AOC terms INTRINSIC DISTINGUISHABILITY. The entropic field $S(x)$ belongs to a different ontological category

from the spacetime metric $g_{\mu\nu}$ and the matter fields Ψ . The entropic field is defined on the entropic manifold Ω , which is ontologically prior to spacetime. The spacetime metric and the matter fields are derived from the entropic field through the emergence map Φ and the amplitude-phase decomposition. The chain of derivation proceeds from one category (entropy/information) to another (geometry/matter), and the explanans (the entropic field) is logically independent of the explanandum (the spacetime metric). There is no vicarious induction: the ToE derives geometry from entropy, not geometry from geometry.

7 The G-Field, the Modular Operator, and Entropic Dark Matter

7.1 Bianconi's G-Field: Properties and Physical Interpretation

The G-field $G_{\mu\nu}$, introduced by Bianconi as a set of Lagrangian multipliers in the entropic action, is a symmetric (0,2) tensor field on the spacetime manifold that plays several important roles in the GfE framework. First, it serves as an auxiliary field that simplifies the variational procedure: by introducing $G_{\mu\nu}$, the entropic action (which is a non-polynomial functional of the metric through the quantum relative entropy) can be rewritten as a polynomial (quadratic) functional of the metric and the G-field. This simplification is analogous to the Hubbard-Stratonovich transformation in quantum field theory, which linearizes a quartic interaction by introducing an auxiliary field.

Second, the G-field generates the emergent cosmological constant. The dressed Einstein-Hilbert action (BG.14) contains a term $-\Lambda_{\text{eff}}$, where Λ_{eff} depends on the expectation value

and dynamics of the G-field. This emergent cosmological constant is small and positive (consistent with the observed value), and it arises from the information-theoretic structure of the entropic action rather than from the vacuum energy of quantum field theory. Third, the equations of motion for $G_{\mu\nu}$ are second-order, ensuring the absence of ghost instabilities. Fourth, Bianconi suggests that the G-field may serve as a dark matter candidate, providing additional gravitational effects beyond those of standard general relativity.

The physical interpretation of the G-field is one of the most intriguing and least developed aspects of Bianconi's framework. The G-field is not a matter field in the conventional sense — it does not describe a specific type of particle or radiation. Rather, it is a geometric degree of freedom associated with the entropic structure of the gravitational action. Its energy-momentum tensor contributes to the gravitational field equations, and its dynamics is determined by the equations of motion derived from the dressed Einstein-Hilbert action. The G-field can be thought of as encoding the "entropic memory" of the spacetime geometry — the information about the entropic history of the gravitational field that is not captured by the instantaneous metric configuration.

7.2 The ToE Identification: G-Field as Modular Operator

The Theory of Entropicity (ToE) provides a specific identification of the G-field within its framework: the G-field corresponds to the modular operator of the entropic field theory. This identification, established in the Alemoh-Obidi Correspondence (AOC 19.2.6.8), connects Bianconi's auxiliary field to a well-defined mathematical object in the theory of operator algebras and modular theory.

The modular operator Δ is a fundamental concept in the Tomita-Takesaki theory of von Neumann algebras. Given a von Neumann algebra M (the algebra of observables) and a cyclic and separating vector $|\Psi_0\rangle$ (the vacuum state), the modular operator $\Delta = \exp(-K)$ is defined through the polar decomposition of the Tomita operator S : $S|\Psi_0\rangle = J\Delta^{1/2}|\Psi_0\rangle$, where J is the modular conjugation and $K = -\ln \Delta$ is the modular Hamiltonian. The modular operator generates a one-parameter group of automorphisms of the von Neumann algebra — the modular flow $\sigma_t(A) = \Delta^{it}A\Delta^{-it}$ — which, by the Bisognano-Wichmann theorem, is related to the boost generator for Rindler wedges in Minkowski space.

In the ToE framework, the entropic field $S(x)$ defines a von Neumann algebra of observables (the algebra of functions of $S(x)$), and the entropic vacuum state $|\Psi_0\rangle$ (the state that minimizes the Obidi Action) serves as the cyclic and separating vector. The modular operator $\Delta_S = \exp(-K_S)$ is constructed from the modular Hamiltonian K_S associated with this algebra and this state. The G-field is then identified with the symmetric tensor constructed from the modular flow, as given in Eq. (BG.43).

This identification has several important consequences. First, it shows that the G-field is not an ad hoc auxiliary field but a natural consequence of the algebraic structure of the entropic field theory. The modular operator exists for ANY von Neumann algebra with a cyclic and separating vector — it is a universal mathematical structure that does not depend on the specific details of the theory. Second, it connects the G-field to the thermodynamic properties of the entropic vacuum: the modular Hamiltonian K_S is the generator of thermal time evolution (by the KMS condition), and the G-field encodes the thermal structure of the entropic vacuum. Third, it provides a route to the quantization of the G-field: the modular operator is a well-defined self-adjoint operator on the Hilbert

space, and its spectral theory provides a natural framework for the quantum mechanics of the G-field.

7.3 The Entropic Origin of the Cosmological Constant

Both the GfE and the ToE predict an emergent cosmological constant $\Lambda_{\text{eff}} > 0$. The mechanisms by which Λ_{eff} arises, however, are different in the two frameworks, and a comparison of these mechanisms illuminates the structural differences between the programs.

In Bianconi's framework, the cosmological constant emerges from the G-field sector. The dressed Einstein-Hilbert action (BG.14) contains a term $-\Lambda_{\text{eff}} \int d^4x \sqrt{g}$, where Λ_{eff} depends on the expectation value and dynamics of the G-field $G_{\mu\nu}$. The value of Λ_{eff} is determined by the equations of motion for $G_{\mu\nu}$ and the boundary conditions of the cosmological evolution. Crucially, Λ_{eff} is NOT determined by the vacuum energy of matter fields — it is an independent quantity that emerges from the entropic dynamics of the G-field. This decoupling from the vacuum energy is the key feature that makes Bianconi's mechanism a potential resolution of the cosmological constant problem.

In the ToE, the cosmological constant emerges from the a_0 term of the SOA heat-kernel expansion. The $f_0 \Lambda^4$ term in the Seeley-DeWitt expansion (BG.36) contributes a constant (volume) term to the gravitational action, which acts as an effective cosmological constant: $\Lambda_{\text{eff}} = f_0 \Lambda_{\text{ent}}^4 / (16\pi^2)$. Additionally, the OCI tensor $O_{\mu\nu}^{(\text{OCI})}$ contributes a geometry-dependent correction to the cosmological constant through the entropic cosmological function $\Lambda_{\text{ent}}[S]$. The value of the cosmological constant is therefore determined by two quantities:

the spectral coefficient f_0 (which depends on the cutoff function f) and the entropic cutoff Λ_{ent} (which is the ultraviolet scale of the entropic dynamics).

Both mechanisms predict $\Lambda_{\text{eff}} > 0$, consistent with the observational evidence for cosmic acceleration from Type Ia supernovae, the cosmic microwave background, and baryon acoustic oscillations. Both mechanisms decouple the cosmological constant from the vacuum energy of matter fields, thereby avoiding the 120-orders-of-magnitude discrepancy of the standard calculation. The difference lies in the depth of the explanation: Bianconi's mechanism introduces a new field (the G-field) whose dynamics generates Λ_{eff} , while the ToE's mechanism derives Λ_{eff} from the spectral properties of the entropic Laplacian without introducing any new fields. The ToE explanation is more parsimonious (it uses only the entropic field) and more specific (it predicts the value of Λ_{eff} in terms of the spectral coefficients).

7.4 Dark Matter from Entropic Dynamics

Both programs propose entropic explanations for the gravitational effects traditionally attributed to dark matter, but the mechanisms are qualitatively different.

Bianconi proposes the G-field as a dark matter candidate. The G-field $G_{\mu\nu}$ is a new dynamical degree of freedom that contributes to the gravitational field equations through its energy-momentum tensor. If the G-field is sufficiently long-lived and weakly self-interacting, it could serve as a dark matter component — a geometric degree of freedom that produces the additional gravitational effects observed in galaxy rotation curves, gravitational lensing, and the cosmic microwave background power spectrum, without requiring the introduction of new elementary particles.

The ToE proposes the Obidi Curvature Invariant (OCI) as the source of "dark curvature" effects. The $OCI = R_I - (1/\lambda)\Phi^*(R_S) \geq 0$ represents the excess curvature of the information manifold relative to the physical spacetime curvature. This excess curvature appears in the GEFE/OFE through the OCI tensor $O_{\mu\nu}^{(OCI)}$, which contributes to the left-hand side of the field equations alongside the Einstein tensor. The physical effect of the OCI tensor is additional curvature — the spacetime is more curved than what is predicted by the visible matter content alone — which mimics the effects of dark matter without introducing any new particles or fields.

The two mechanisms are qualitatively different. The G-field mechanism introduces a new physical entity (a tensor field) whose dynamics produces dark matter effects. The OCI mechanism does not introduce new entities — it reveals that the existing information-geometric structure of the entropic field already contains excess curvature that produces dark matter effects. The G-field mechanism predicts new particles (the quanta of the G-field), while the OCI mechanism predicts no new particles — only excess curvature. This qualitative difference may be observationally distinguishable: particle dark matter candidates (including G-field quanta) can, in principle, be detected through direct detection experiments, collider searches, or indirect detection through annihilation products, while geometric dark curvature (the OCI) cannot be detected through particle physics experiments and is accessible only through gravitational observations.

8 Information Geometry: What Bianconi Lacks and ToE Provides

8.1 The Fisher-Rao Metric: The Missing Foundation

The most significant mathematical lacuna in Bianconi's framework, from the perspective of the ToE, is the absence of the Fisher-Rao metric. The Fisher-Rao metric is the natural Riemannian metric on the space of probability distributions, and it is the foundational geometric structure of information geometry. Its discovery was a collaboration across decades: **Ronald Aylmer Fisher** (1925) introduced the concept of statistical information (the Fisher information matrix) as a measure of the sensitivity of a probability distribution to changes in its parameters; **Calyampudi Radhakrishna Rao** (1945) recognized that the Fisher information matrix defines a Riemannian metric on the parameter space, and used this metric to establish the Cramér-Rao bound; and Nikolai Nikolaevich Chentsov (1972, 1982) proved the uniqueness theorem that established the Fisher-Rao metric as the ONLY Riemannian metric on the statistical manifold invariant under sufficient statistics.

The **Chentsov theorem** is of central importance to the ToE. It states: Let M be the manifold of probability distributions on a finite sample space, and let g be a Riemannian metric on M that is invariant under **Markov morphisms** (i.e., under maps that preserve the structure of sufficient statistics). Then g is proportional to the Fisher-Rao metric. In other words, the Fisher-Rao metric is the unique information-geometric metric, up to an overall normalization constant.

This **uniqueness theorem** provides the axiomatic foundation for the ToE's construction of spacetime geometry. The physical spacetime metric is $g_{\mu\nu}^S = \lambda g_{\mu\nu}^{I(FR)}$, where $g^{I(FR)}$ is the Fisher-Rao metric. The Chentsov theorem guarantees that this construction is unique: any other choice of metric on the information manifold would violate the invariance under sufficient statistics and would therefore be inconsistent with the principles of statistical inference. This uniqueness is a powerful constraint: it means that the spacetime metric is not one of many possible emergent structures, but THE ONLY emergent structure that is consistent with the information-theoretic foundations.

Bianconi's framework does not use the Fisher-Rao metric and therefore lacks this uniqueness guarantee. The metric-as-density-matrix construction provides a way to assign information-theoretic content to the existing metric, but it does not derive the metric from information-theoretic principles. The density matrix $\rho_g = g_{\mu\nu}/n$ is constructed from the metric, not the other way around. The directionality of the construction is crucial: Bianconi starts with the metric and constructs information from it; the ToE starts with information (the Fisher-Rao metric) and constructs the metric from it. The ToE's directionality is explanatory (information explains geometry), while Bianconi's directionality is interpretive (geometry is interpreted as information).

8.2 The Fubini-Study Metric: The Quantum Generalization

The Fubini-Study metric is the natural metric on the space of quantum states — the complex projective space $CP^n = \{[\psi] : \psi \in C^{n+1} \setminus \{0\}\}$, where $[\psi]$ denotes the equivalence class of ψ under complex scaling. It was introduced independently by Guido Fubini (1904) and Eduard Study (1905) in the context of projective geometry, and its connection to

quantum mechanics was recognized by John Klauder, Valentine Bargmann, and others in the 1960s and 1970s.

The Fubini-Study metric has a natural decomposition that is central to the ToE:

$$(BG.44) \quad ds^2_{FS} = (dp/\rho)^2 + d\Theta^2$$

where ρ is the amplitude (modulus) and Θ is the phase of the quantum state $\psi = \rho e^{i\Theta}$. In the ToE, this decomposition corresponds to the amplitude-phase decomposition of the complexified entropic field $S_c = \rho e^{i\Theta}$:

- The $(dp/\rho)^2$ term generates the gravitational matter sector — the stress-energy tensor $T_{\mu\nu}$ that appears on the right-hand side of Einstein's equations. The amplitude fluctuations of the entropic field produce gravitational effects through the Fisher-Rao construction.
- The $d\Theta^2$ term generates the electromagnetic sector — Maxwell's equations and the electromagnetic field tensor $F_{\mu\nu}$. The phase fluctuations of the entropic field produce electromagnetic effects through the $U(1)$ gauge structure of the phase angle Θ .

The Fubini-Study metric generalizes the Fisher-Rao metric to the quantum domain. In the classical limit (when the quantum state is a probability distribution rather than a quantum amplitude), the Fubini-Study metric reduces to the Fisher-Rao metric: $g_{ab}^{FS} \rightarrow g_{ab}^{FR}$. This limiting relation ensures the consistency between the classical gravitational sector (governed by the Fisher-Rao metric) and the quantum matter sector (governed by the Fubini-Study metric) within the ToE.

Bianconi's metric-as-density-matrix construction is RELATED to the Fubini-Study metric but does not explicitly develop this relationship. The density matrix ρ_g is a quantum state (a mixed state, specifically), and the space of density matrices is a subset of the complex projective space, so the Fubini-Study metric is implicitly present in Bianconi's framework. However, Bianconi does not exploit this implicit structure — she does not decompose the Fubini-Study metric into its amplitude and phase components, she does not use the phase sector to derive electromagnetism, and she does not establish the classical limit connecting the Fubini-Study metric to the Fisher-Rao metric. These are significant omissions that the ToE fills.

8.3 The Amari-Chentsov Alpha-Connections: The Dual Structure

The Amari alpha-connections, introduced by Shun-ichi Amari in his foundational work on information geometry (1985), provide a one-parameter family of affine connections on the statistical manifold, parameterized by a real number α :

$$(BG.45) \Gamma_{ijk}^{(\alpha)} = \Gamma_{ijk}^{(o)} + (\alpha/2) T_{ijk}$$

where $\Gamma^{(o)}$ is the Levi-Civita connection of the Fisher-Rao metric, and T_{ijk} is a symmetric cubic tensor (the Amari-Chentsov tensor, or skewness tensor) that measures the asymmetry of the statistical model. Three values of α are of particular physical significance:

- **$\alpha = +1$ (e-connection):** The exponential connection, which is flat for exponential families of distributions. In the ToE, this connection governs the kinetic sector of the entropic field dynamics.

- **$\alpha = -1$ (m-connection):** The mixture connection, which is flat for mixture families of distributions. In the ToE, this connection governs the potential sector of the entropic field dynamics.
- **$\alpha = 0$ (Levi-Civita connection):** The unique torsion-free, metric-compatible connection. This is the connection used in standard general relativity.

The e-connection and the m-connection are dual with respect to the Fisher-Rao metric:

$$(BG.46) \quad g(\nabla^{(+1)}_X Y, Z) + g(Y, \nabla^{(-1)}_X Z) = X \cdot g(Y, Z)$$

This duality relation is a fundamental structural property of the statistical manifold, and it has deep consequences for the coupling between geometry and matter in the ToE. The duality relates the exponential (kinetic) and mixture (potential) aspects of the entropic dynamics, providing a geometric framework for the interplay between kinetic and potential energies that is the foundation of all physical dynamics.

Bianconi's framework has no analogue of the alpha-connections or the duality structure. The absence of this structure means that Bianconi's framework lacks the geometric machinery for distinguishing between kinetic and potential aspects of the gravitational dynamics — a distinction that is central to the ToE's construction and that provides additional predictive content (e.g., the specific form of the entropic field potential $V(S)$ is constrained by the duality structure).

8.4 What Information Geometry Adds to the Gravitational Sector

To summarize, the full information-geometric machinery of the ToE provides the following structural elements that are absent from Bianconi's framework:

Information-Geometric Contributions of the ToE

(i) Principled derivation of the spacetime metric: The Fisher-Rao metric, guaranteed unique by the Chentsov theorem, provides an axiom-based derivation of the spacetime metric from information-theoretic principles. No other metric is consistent with the invariance under sufficient statistics.

(ii) Quantum generalization: The Fubini-Study metric naturally incorporates quantum matter into the information-geometric framework, providing a unified treatment of classical gravity and quantum matter within a single geometric setting.

(iii) Dual affine structure: The Amari alpha-connections provide a family of geometric structures that couple the kinetic and potential aspects of the entropic dynamics, constraining the form of the physical laws that emerge from the information manifold.

(iv) Geometric constant $OCI = \ln 2$: The Obidi Curvature Invariant, calculated through seven independent methods, provides a universal geometric constant that sets the scale of quantum gravitational effects and dark curvature.

(v) Electromagnetic sector: The phase sector of the Fubini-Study metric generates the complete Maxwell electromagnetic theory (Letter IIA), providing a unified information-geometric treatment of gravity and electromagnetism.

(vi) Rigorous emergence framework: The Curvature Transfer Theorem and the emergence map Φ provide a precise, mathematically rigorous mechanism for the emergence of classical spacetime from quantum information, with controlled $O(1/N)$ corrections.

These contributions represent a substantial mathematical and conceptual advance beyond what Bianconi's framework provides. While Bianconi's metric-as-density-matrix construction is information-theoretic in spirit, it does not develop the formal apparatus of information geometry to the degree necessary to achieve the structural results listed above. The ToE's deployment of the full information-geometric toolkit — Fisher-Rao, Fubini-Study, alpha-connections, Chentsov uniqueness — provides a level of mathematical rigor and structural depth that places the entropic gravity program on firm axiomatic foundations.

9 Independent Convergence: The Significance of Two Programs Reaching the Same Destination

9.1 The Concept of Independent Convergence in Physics

The history of physics is punctuated by remarkable instances of **independent convergence (IC)** — cases in which two or more scientists, working independently and using different methods, arrive at the same fundamental insight. These convergences are widely regarded as among the strongest forms of evidence for the correctness of the insight in question, because they demonstrate that the conclusion is not an artifact of a particular method or a particular set of assumptions but a genuine feature of the underlying reality that can be accessed through multiple independent pathways.

The most celebrated example is the independent development of the calculus by Isaac Newton and Gottfried Wilhelm Leibniz in the late seventeenth century. Newton developed his "method of fluxions" in the mid-1660s, motivated by problems in mechanics and optics; Leibniz developed his differential and integral calculus in the 1670s, motivated by problems in philosophy and mathematics. The two approaches used different notation, different conceptual frameworks, and different motivating problems, yet they converged on the same mathematical structure — the fundamental theorem of calculus, which connects differentiation and integration. The independent convergence of Newton and Leibniz demonstrated that the calculus was not an invention of either man but a discovery of an objective mathematical truth that both men accessed through their respective approaches.

Another celebrated example is the independent discovery of natural selection by **Charles Darwin** and **Alfred Russel Wallace** in the mid-nineteenth century. **Darwin** developed his theory over two decades of patient observation and reflection, culminating in the *Origin of Species* (1859); **Wallace** arrived at the same theory independently in 1858, inspired by observations during his travels in the **Malay Archipelago**. The two theories were presented jointly to the Linnean Society of London on July 1, 1858 — one of the most dramatic episodes in the history of science. The independent convergence of Darwin and Wallace provided powerful evidence that natural selection is not a speculative hypothesis but a genuine law of nature, discoverable by any sufficiently careful observer of the natural world.

In physics, the most directly relevant precedent is the independent formulation of quantum mechanics by Werner Heisenberg and Erwin Schrödinger in 1925–1926. Heisenberg's matrix mechanics (developed in June–September 1925 with Max Born and Pascual Jordan) and Schrödinger's wave mechanics (developed in January–June 1926) used radically different mathematical formalisms — infinite-dimensional matrices vs. differential equations — and radically different physical pictures — discontinuous quantum jumps vs. continuous wave functions. Yet Schrödinger proved in March 1926 that the two formulations are mathematically equivalent, producing the same physical predictions for all observable quantities. This equivalence, demonstrated by the independent convergence of two very different approaches, provided strong evidence that quantum mechanics is a correct description of atomic phenomena.

9.2 The Bianconi-Obidi Convergence (BOC)

The structural parallels between Bianconi's GfE and Obidi's ToE constitute an instance of independent convergence in the entropic gravity program. The two programs were developed independently — Bianconi's paper was submitted in August 2024, and the ToE has been developed through other publications and the Living Review Letters Series over a separate timeline (2025-2026). The two programs use different mathematical tools (**quantum relative entropy vs. the Obidi Action**), different conceptual frameworks (metric as density matrix vs. entropic field and emergence map), and different motivating questions (**deriving modified Einstein equations vs. deriving all of physics from a single entropic principle**). Yet they converge on the following parallel conclusions:

Structural Parallels of the Bianconi-Obidi Convergence

- (i) Both derive gravity from an entropic variational principle. Bianconi: minimize quantum relative entropy. ToE: extremize the Obidi Action.
- (ii) Both recover the Einstein field equations as a limiting case of more general entropic field equations.
- (iii) Both produce an emergent cosmological constant $\Lambda_{\text{eff}} > 0$, decoupled from the vacuum energy of matter fields.
- (iv) Both predict a dark sector — gravitational effects beyond standard Einstein gravity — arising from the entropic dynamics. Bianconi: G-field. ToE: OCI tensor.
- (v) Both treat the spacetime metric as an information-theoretic object. Bianconi: metric as density matrix. ToE: metric as pushforward of Fisher-Rao metric.

(vi) Both use entropy as the fundamental explanatory concept for gravitational dynamics.

These parallels are not coincidental. They reflect the deep entropic structure of gravitational dynamics — a structure that is accessible to any program that takes the connection between gravity and entropy seriously and develops it with sufficient mathematical rigor. The independent arrival of Bianconi and Obidi at these parallel conclusions, using independent methods and independent assumptions, provides strong evidence that the conclusions are correct — that gravity IS an emergent entropic phenomenon, that the Einstein equations ARE the macroscopic expression of deeper entropic dynamics, and that the dark sector of the universe HAS an entropic origin.

9.3 The Significance for the Entropic Gravity Program

The **Bianconi-Obidi convergence (BOC)** represents the strongest evidence to date for the entropic gravity thesis. Previous contributions to the entropic gravity program — **Jacobson** (1995), **Verlinde** (2010), **Padmanabhan** (2010) — provided individual derivations of the **Einstein** equations from thermodynamic principles, but each derivation used a specific set of assumptions (local Rindler horizons, holographic screens, null surface entropy) that could be questioned. The Bianconi-Obidi convergence goes beyond these individual derivations by demonstrating that two fundamentally different approaches — the operator-theoretic approach (**Bianconi**) and the field-theoretic approach (**Obidi**) — converge on the same conclusion.

The significance is analogous to the **Heisenberg-Schrödinger equivalence** (HSE) in quantum mechanics. Just as the equivalence of matrix mechanics and wave mechanics demonstrated that quantum mechanics is not an artifact of a particular mathematical formalism but a genuine feature of nature, the convergence of GfE and ToE demonstrates that entropic gravity is not an artifact of a particular derivation but a genuine feature of gravitational dynamics. The "**it from bit**" hypothesis of **Wheeler**, the thermodynamic derivation of Jacobson, the entropic force of **Verlinde**, the emergent gravity of Padmanabhan, the quantum relative entropy of Bianconi, and the Obidi Action of the ToE — all these different approaches point to the same conclusion: gravity emerges from entropy.

9.4 Open Problem: Unification of the Two Programs

The convergence of the two programs naturally raises the question of their unification. **Can the Obidi Action and the Bianconi entropic action be unified into a single, more general entropic action that encompasses both as special cases?**

Open Problem 9.1 (Bianconi-Obidi Unification)

Construct a unified entropic action S_{unified} such that:

- (i) S_{unified} reduces to the Bianconi entropic action $S_B = S(\rho_g \parallel \rho_{g(M)})$ in the operator-theoretic limit (metric as density matrix, quadratic approximation);
- (ii) S_{unified} reduces to the Obidi Action $S_{\text{Obidi}} = S_{\text{LOA}} + S_{\text{SOA}}$ in the field-theoretic limit (entropic field as fundamental variable, full heat-kernel expansion);

(iii) S_{unified} contains new physical content beyond both S_B and S_{Obidi} , providing predictions that are inaccessible to either program alone.

A candidate for the unified action might take the form:

$$(BG.47) S_{\text{unified}} = S_{\text{LOA}}[S] + S_{\text{SOA}}[S] + \alpha S_B(\rho_g[S], \rho_{g(M)}[S])$$

where α is a coupling constant that controls the relative weight of the field-theoretic and operator-theoretic contributions. In the limit $\alpha \rightarrow 0$, the unified action reduces to the Obidi Action; in the limit where S_{LOA} and S_{SOA} reduce to their quadratic approximations, the S_B term dominates and the unified action reduces to Bianconi's entropic action. The Obidi Action as currently recognized would then be the **Primary Obidi Action (POA)**; and any other subsequent improvements will be referred to as the **Second Primary Obidi Action (SPOA)**, **Third Primary Obidi Action (TPOA)**, etc. Following this ToE convention, therefore, the above unified action S_{unified} (Eq. BG.47) can be referred to as the **Second Primary Obidi Action (SPOA)**.

The mathematical framework most likely to accommodate this unification is the theory of von Neumann algebras and modular theory. Von Neumann algebras provide a natural setting in which field-theoretic and operator-theoretic descriptions are unified: the algebra of observables (operator-theoretic) is generated by the field operators (field-theoretic), and the modular theory (**Tomita-Takesaki**) provides the bridge between the two descriptions. The modular flow, which we have identified with the G-field, generates a one-parameter group of automorphisms of the algebra that can be interpreted both as a dynamical evolution (field-theoretic perspective) and as a symmetry of the quantum state (operator-

theoretic perspective). The unification of GfE and ToE within the framework of modular theory remains an open problem of fundamental importance, which we designate as Open Problem 20.11 in the sequence of open problems identified in the AOC.

10 Experimental Predictions and Observational Tests

10.1 Predictions Common to Both Programs

Both Bianconi's GfE and Obidi's ToE make predictions that go beyond standard general relativity. The following predictions are common to both programs and reflect the shared structure of the entropic gravity thesis:

Modified Einstein equations at high curvature. Both programs predict that the Einstein field equations receive corrections at high curvatures (near black hole horizons, in the early universe, at the Planck scale). In Bianconi's framework, these corrections arise from the non-quadratic terms of the quantum relative entropy; in the ToE, they arise from the a_4 and higher Seeley-DeWitt coefficients of the SOA. The specific form of the corrections differs between the two programs, but the qualitative prediction — deviations from Einstein at high curvature — is shared.

Emergent cosmological constant. Both programs predict an emergent cosmological constant $\Lambda_{\text{eff}} > 0$, decoupled from the vacuum energy of matter fields. This prediction is consistent with the observed cosmic acceleration and provides a potential resolution of the

cosmological constant problem. The numerical value of Λ_{eff} depends on the specific parameters of each framework (the G-field dynamics in Bianconi, the spectral coefficients in the ToE), and a detailed comparison with the observed value $\Lambda_{\text{obs}} \sim 10^{-52} \text{ m}^{-2}$ would provide a stringent test of both programs.

Dark sector physics. Both programs predict additional gravitational effects beyond standard Einstein gravity — the G-field effects in Bianconi, the OCI dark curvature in the ToE. These effects are qualitatively similar to the effects attributed to dark matter and dark energy, and their detailed predictions (the spatial distribution of the dark sector, its evolution with redshift, its coupling to baryonic matter) can be confronted with observational data from galaxy rotation curves, gravitational lensing, the cosmic microwave background, and large-scale structure surveys.

Deviations from Einstein near black hole horizons. The entropic corrections to the Einstein equations become most significant near event horizons, where the curvature is strongest and the thermodynamic properties of the geometry (the Hawking temperature, the Bekenstein-Hawking entropy) are most prominent. Both programs predict specific modifications to the near-horizon geometry that could, in principle, be tested through observations of black hole shadows (by the Event Horizon Telescope), quasi-normal modes of black hole ringdown (by gravitational wave detectors), and the inspiral-merger-ringdown waveforms of binary black hole coalescences (by LIGO/Virgo/KAGRA).

10.2 Predictions Unique to the Theory of Entropicity

The ToE makes several predictions that are not shared by Bianconi's framework and that could serve as distinctive experimental signatures of the ToE:

Entropic dispersion of gravitational waves. The ToE predicts that gravitational waves at frequencies approaching the Entropic Transition Level (ETL) scale will experience dispersion — a frequency-dependent propagation speed — due to the discrete spectral structure of the entropic Laplacian. This prediction is absent from Bianconi's framework (which predicts standard, non-dispersive gravitational wave propagation at leading order) and could be tested by next-generation gravitational wave detectors sensitive to high-frequency gravitational waves.

OCI-dependent corrections to black hole entropy. The Bekenstein-Hawking entropy formula $S_{\text{BH}} = k_B A / (4l_P^2)$ receives corrections in the ToE that depend on the OCI: $S = S_{\text{BH}} + k_B \cdot \text{OCI} \cdot \ln(A/l_P^2) + O(l_P^2/A)$. The logarithmic correction, proportional to $\text{OCI} = \ln 2$, is a distinctive prediction of the ToE that can be compared with the logarithmic corrections predicted by other approaches to quantum gravity (loop quantum gravity, string theory).

Curvature-dependent birefringence of electromagnetic waves. Letter IIA predicts that electromagnetic waves propagating through regions of high spacetime curvature will experience birefringence — a polarization-dependent propagation speed — due to the coupling between the phase sector of the entropic field and the curvature of the information manifold. This prediction could be tested through polarimetric observations of electromagnetic radiation from strongly gravitating sources (pulsars, magnetars, active galactic nuclei).

Gravitational effects of information erasure. The ToE predicts, through the Landauer-Obidi bound, that the erasure of information produces a gravitational effect — a change in the spacetime curvature proportional to the entropy decrease associated with the erasure. This is a direct consequence of the ontological identification of entropy with geometry: if

geometry IS information, then the erasure of information must change the geometry. This prediction is experimentally challenging but could, in principle, be tested through precision measurements of the gravitational field of systems undergoing controlled information erasure.

10.3 How to Distinguish Between the Programs Observationally

At low energies and weak curvatures, both programs reduce to the standard Einstein equations and are therefore observationally indistinguishable. The differences emerge only in regimes where the entropic corrections become significant:

Observational Regime	Bianconi (GfE)	Obidi (ToE)	Distinguishing Test
Low curvature (solar system)	Einstein equations	Einstein equations	Indistinguishable
High curvature (near black holes)	Modified Einstein with specific $C_{\mu\nu}$ corrections	Modified Einstein with OCI tensor + a_4 corrections	Black hole shadow observations; quasi-normal mode spectroscopy
Cosmological (Λ evolution)	Λ_{eff} determined by G-field dynamics (may vary with redshift)	$\Lambda_{\text{ent}}[S]$ is a functional of S (varies with entropic field evolution)	Precision cosmology: $\Lambda(z)$ measurements from Type Ia SNe, BAO
Dark matter	G-field quanta (new particles)	OCI dark curvature (geometric effect, no new particles)	Direct dark matter detection experiments; gravitational lensing profiles

Observational Regime	Bianconi (GfE)	Obidi (ToE)	Distinguishing Test
Gravitational waves	No dispersion at leading order	Entropic dispersion at ETL frequencies	High-frequency gravitational wave detectors
Electromagnetic polarization	No birefringence predicted	Curvature-dependent birefringence	Polarimetry of radiation from strongly gravitating sources

The most promising observational window for distinguishing the two programs is the dark matter sector. If dark matter is detected as a new particle (through direct detection, collider production, or indirect detection), this would be consistent with Bianconi's G-field mechanism and inconsistent with the ToE's OCI mechanism (which predicts no new particles). Conversely, if the gravitational effects attributed to dark matter persist but no new particle is found despite exhaustive searches, this would favor the ToE's geometric interpretation. The ongoing experimental programs — direct detection (XENON, LZ, PandaX), collider searches (LHC, future colliders), and indirect detection (Fermi-LAT, AMS-02, CTA) — will provide critical data for adjudicating between the two mechanisms in the coming decades.

11 Conclusion and Synthesis

11.1 Summary of Results

This appendix has provided a comprehensive, monograph-grade analysis of Ginestra Bianconi's "Gravity from Entropy" program in the context of the Theory of Entropicity as developed in Letter IE of the Living Review Letters Series. The principal results of this analysis can be summarized as follows:

(i) Independent convergence. Bianconi's GfE program and Obidi's ToE represent independent convergences toward the same fundamental conclusion: gravity is an emergent consequence of entropic dynamics, and the Einstein field equations are the macroscopic, classical-limit expression of deeper entropic field equations. This convergence, achieved through independent methods (quantum relative entropy vs. Obidi Action), independent mathematical frameworks (operator-theoretic vs. field-theoretic), and independent ontological commitments (spacetime-fundamental vs. spacetime-emergent), provides the strongest evidence to date for the entropic origin of gravity.

(ii) Subsumption. Bianconi's modified Einstein equations are rigorously recoverable as a specific quadratic approximation of the Spectral Obidi Action in the regime of near-equilibrium entropic field, vanishing OCI, and small metric fluctuations (Theorem 5.1). The G-field is identified with the modular operator of the entropic field theory (AOC 19.2.6.8). This subsumption establishes the ToE as the more general framework that contains Bianconi's results as a special case.

(iii) ToE's additional content. The full Obidi Action contains physics beyond Bianconi's quadratic approximation: the derivation of the speed of light c , the generation of spacetime itself from the entropic field, the full information-geometric machinery (Fisher-Rao, Fubini-Study, Amari alpha-connections), the derivation of Maxwell's equations from the phase sector (Letter IIA), the OCI tensor for dark curvature, and the Vuli-Ndlela Integral for quantum gravity.

(iv) The Bianconi Paradox. Bianconi's framework exhibits a structural circularity — the Bianconi Paradox — in which the geometric structures presupposed by the framework are the same as those the framework seeks to derive. This paradox is analyzed through four layers (ontological, logical, physical, mathematical) and is shown to arise from the implicit dualism of the GfE framework (two fundamental entities: geometry and matter). The ToE resolves the Bianconi Paradox through its radical monism (one fundamental entity: the entropic field) and its background-independent construction of spacetime.

(v) The G-field and the cosmological constant. Both programs produce an emergent cosmological constant $\Lambda_{\text{eff}} > 0$, providing a potential resolution of the cosmological constant problem. The G-field (in Bianconi) and the OCI tensor (in the ToE) provide candidate explanations for the dark sector of the universe, though through qualitatively different mechanisms (new geometric degree of freedom vs. excess information curvature).

(vi) Experimental predictions. The two programs make overlapping but distinguishable predictions, with the most promising observational discriminants being the nature of dark matter (new particles vs. geometric effect), gravitational wave dispersion (absent vs. present), and electromagnetic birefringence in strong gravitational fields (absent vs. present).

11.2 The Place of Bianconi's Work in the ToE Program

Bianconi's "Gravity from Entropy" is a major contribution to the entropic gravity literature and a significant independent validation of the thesis that gravity emerges from entropic dynamics. Within the context of the ToE program, Bianconi's work occupies a specific and honored place: it provides an independent, peer-reviewed, operator-theoretic derivation of modified Einstein equations from an entropic principle, confirming from an entirely different direction the conclusions that the ToE reaches through its field-theoretic and information-geometric approach.

The relationship between the two programs is not one of competition but of complementarity. Bianconi's operator-theoretic approach provides a concrete, calculable framework that is directly connected to the established mathematical tools of quantum information theory (density matrices, von Neumann entropy, quantum relative entropy). The ToE's field-theoretic approach provides a deeper ontological framework that connects the gravitational dynamics to the foundational structures of information geometry (Fisher-Rao, Fubini-Study, alpha-connections). The unification of the two approaches within the framework of modular theory (Open Problem 9.1) represents one of the most important open problems in the entropic gravity program.

We emphasize, in closing this subsection, that our analysis is intended to be intellectually generous to Bianconi's work. The identification of the Bianconi Paradox is not a criticism of Bianconi's mathematical results — which are technically rigorous and have passed the peer review process of Physical Review D — but rather a philosophical observation about the interpretive scope of those results. The paradox arises not from any error in Bianconi's

derivation but from a structural tension between the explanatory ambitions of the framework and its foundational assumptions.

Every research program encounters internal tensions, and the very act of identifying such tensions is itself a contribution: it clarifies the program's strengths and limitations while simultaneously illuminating Bianconi's collateral contribution to the History and Philosophy of Physics—and, more broadly, to the philosophy of science.

11.3 Toward a Complete Entropic Theory of Gravity

Letter IE of the Living Review Letters Series established the emergence of physical spacetime geometry from information geometry through the Curvature Transfer Theorem and the emergence map. This appendix has demonstrated that Bianconi's independent program confirms and complements this result, providing an operator-theoretic perspective that enriches and validates the ToE's field-theoretic construction.

The **Generalized Entropic Field Equations (GEFE/OFE)** represent the most complete framework for the entropic theory of gravity developed to date. They subsume both the standard Einstein equations (as the classical limit) and Bianconi's modified Einstein equations (as the quadratic approximation), while incorporating additional physical content — the OCI tensor, the entropic cosmological function, the electromagnetic sector, the quantum gravitational corrections — that neither Einstein nor Bianconi captures.

The entropic origin of gravity is no longer a speculative hypothesis. Within the framework of the Theory of Entropicity, it is a mathematical theorem — the Curvature Transfer Theorem — supported by a complete variational principle (the Obidi Action), a precise

emergence mechanism (the emergence map Φ), and a uniqueness guarantee (the Chentsov theorem for the Fisher-Rao metric). With the independent corroboration provided by Bianconi's published results in Physical Review D, the case for the entropic origin of gravity has been strengthened to a degree that invites the serious attention of the broader theoretical physics community.

The road ahead is clear. The immediate priorities are: (i) the von Neumann algebraic unification of the GfE and ToE (Open Problem 9.1); (ii) the detailed computation of observational predictions — black hole shadow corrections, gravitational wave dispersion, electromagnetic birefringence — that can be confronted with data from current and next-generation experiments; (iii) the extension of the ToE to non-abelian gauge fields and fermionic matter, completing the unification of all fundamental interactions within the entropic framework; and (iv) the development of the Vuli-Ndlela Integral as a complete, non-perturbative theory of quantum gravity. These are formidable challenges, but the convergence of Bianconi's and Obidi's programs provides confidence that the direction is correct and that the ultimate goal — a complete, unified, entropic theory of all physical phenomena — is within reach.

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— End of Appendix B —

ToE Living Review Letters Series — Supplementary Appendix B to Letter IE

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Beyond Einstein: The Entropic Origin of Geometry, Matter, and Gravitation in the Theory of Entropicity (ToE)

On the Emergence of Physical Spacetime Geometry from Information Geometry

The Theory of Entropicity (ToE) Living Review Letters Series (ToE-LRLS). *Letter IE*

John Onimisi Obidi (jonimisiobidi@gmail.com)

Research Lab, The Aether

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Correspondence: jonimisiobidi@gmail.com, Research Lab, The Aether